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Matheus Gallas Lopes

**EFEITOS DE ANTAGONISTAS NMDA NO COMPORTAMENTO SOCIAL:
RELEVÂNCIA PARA MODELAR FENÓTIPOS RELACIONADOS AOS SINTOMAS
NEGATIVOS DA ESQUIZOFRENIA EM ANIMAIS DE LABORATÓRIO**

Porto Alegre

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Tese apresentada ao Programa de Pós-Graduação em Ciências Biológicas: Farmacologia e Terapêutica do Instituto de Ciências Básicas da Saúde da Universidade Federal do Rio Grande do Sul como requisito parcial para a obtenção do título de doutor em Farmacologia e Terapêutica.

Orientadora: Prof.^a Dr.^a Ana Paula Herrmann

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RESUMO

A esquizofrenia é um transtorno psiquiátrico complexo, caracterizado por sintomas positivos, negativos e cognitivos. Apesar da eficácia dos antipsicóticos sobre sintomas positivos, os sintomas negativos, como prejuízos sociais, permanecem uma necessidade clínica não atendida. Modelos animais baseados na hipofunção de receptores *N*-metil-D-aspartato (NMDA) são amplamente utilizados para investigar fenótipos comportamentais relacionados à esquizofrenia e para a triagem de candidatos a fármacos com potencial antipsicótico. No entanto, a interpretação desses modelos é dificultada pela heterogeneidade metodológica, por potenciais confundidores comportamentais e por limitações relacionadas à validade interna e externa dos achados. Esta tese teve como objetivo avaliar os efeitos de antagonistas do receptor NMDA sobre o comportamento social em animais de laboratório e discutir sua relevância para modelar fenótipos relacionados aos sintomas negativos da esquizofrenia. Para isso, foram desenvolvidos dois estudos complementares. No primeiro estudo, realizou-se uma revisão sistemática e meta-análise de estudos pré-clínicos sobre os efeitos de dizocilpina (MK-801), cetamina e fenciclidina (PCP) sobre interação e preferência social em ratos, camundongos e peixes-zebra. Foram incluídos 264 estudos identificados nas bases Embase, PubMed e Web of Science. Os dados foram sintetizados por modelos de efeitos aleatórios multinível. A síntese dos dados indicou que antagonistas do receptor NMDA reduzem o comportamento social, embora a magnitude e a consistência desses efeitos variem conforme o paradigma experimental, com alta heterogeneidade e indícios de efeitos de estudos pequenos, especialmente nas análises de preferência social. Antipsicóticos clinicamente aprovados atenuaram alterações sociais induzidas por antagonistas NMDA; entretanto, esses efeitos frequentemente ocorreram junto a mudanças na locomoção, dificultando sua interpretação como restauração seletiva do comportamento social. No segundo estudo, foram avaliados os efeitos da cetamina em peixes-zebra adultos. Os animais foram expostos à cetamina por imersão durante 20 minutos. No primeiro experimento, as concentrações de 10, 20 e 40 mg/L foram avaliadas após exposição aguda, após cinco dias de exposição repetida e após reexposição no dia 7, precedida por 48 horas sem tratamento. Nos experimentos seguintes, a concentração de 40 mg/L foi utilizada para investigar se os efeitos persistiam 48 horas após protocolos de 5 ou 14 dias de exposição, sem nova administração antes dos testes. Os comportamentos foram avaliados nos testes de preferência social e de tanque aberto. A cetamina reduziu o comportamento social, aumentou a atividade locomotora no teste de preferência social e induziu alterações no padrão de nado no teste de tanque aberto. Esses efeitos, porém, não se intensificaram após exposições repetidas, indicando ausência de sensibilização comportamental, nem persistiram após o período sem exposição, revelando uma diferença significativa em relação ao que se observa em roedores. Em conjunto, os resultados indicam que antagonistas do receptor NMDA induzem alterações sociais em modelos animais, mas também evidenciam limitações relevantes para a interpretação translacional desses fenótipos. Esta tese contribui para uma avaliação crítica da utilidade e dos limites de modelos baseados em antagonismo NMDA no estudo de sintomas negativos da esquizofrenia.

Palavras-chave: esquizofrenia; sintomas negativos; comportamento social; antagonistas do receptor NMDA; cetamina; peixe-zebra; modelos animais.

ABSTRACT

Schizophrenia is a complex psychiatric disorder characterized by positive, negative, and cognitive symptoms. Although antipsychotics are effective in controlling positive symptoms, negative symptoms, such as social impairments, remain an important unmet clinical need. Animal models based on *N*-methyl-D-aspartate (NMDA) receptor hypofunction are widely used to investigate schizophrenia-related behavioral phenotypes and to screen drug candidates with potential antipsychotic effects. However, the interpretation of these models is hampered by methodological heterogeneity, potential behavioral confounders, and limitations related to the internal and external validity of the findings. This thesis aimed to evaluate the effects of NMDA receptor antagonists on social behavior in laboratory animals and to discuss their relevance for modeling phenotypes related to the negative symptoms of schizophrenia. To this end, two complementary studies were conducted. In the first study, a systematic review and meta-analysis of preclinical studies on the effects of dizocilpine (MK-801), ketamine, and phencyclidine (PCP) on social interaction and social preference in rats, mice, and zebrafish was conducted. A total of 264 studies identified in Embase, PubMed, and Web of Science were included. Data were synthesized using multilevel random-effects models. The data synthesis indicated that NMDA receptor antagonists reduce social behavior, although the magnitude and consistency of these effects vary according to the experimental paradigm, with high heterogeneity and evidence of small-study effects, especially in the social preference analyses. Clinically approved antipsychotics attenuated social alterations induced by NMDA antagonists; however, these effects frequently occurred alongside changes in locomotion, making it difficult to interpret them as selective restoration of social behavior. In the second study, the effects of ketamine were evaluated in adult zebrafish. Animals were exposed to ketamine by immersion for 20 minutes. In the first experiment, concentrations of 10, 20, and 40 mg/L were evaluated after acute exposure, after five days of repeated exposure, and after re-exposure on day 7, preceded by 48 hours without treatment. In the following experiments, the 40 mg/L concentration was used to investigate whether the effects persisted 48 hours after 5- or 14-day exposure protocols, without a new administration before testing. Behaviors were assessed in the social preference and open tank tests. Ketamine reduced social behavior, increased locomotor activity in the social preference test, and induced alterations in swimming patterns in the open tank test. However, these effects did not intensify after repeated exposures, indicating the absence of behavioral sensitization, nor did they persist after the period without exposure, revealing a significant difference compared to what is observed in rodents. Taken together, the results indicate that NMDA receptor antagonists induce social alterations in animal models, but also highlight relevant limitations for the translational interpretation of these phenotypes. This thesis contributes to a critical evaluation of the utility and limitations of models based on NMDA antagonism in the study of negative symptoms of schizophrenia.

Keywords: schizophrenia; negative symptoms; social behavior; NMDA receptor antagonists; ketamine; zebrafish; animal models.

LISTA DE ABREVIATURAS

5-HT_{2A} – receptor serotoninérgico do tipo 2A

D₂ – receptor dopaminérgico do tipo 2

DALYs – anos de vida ajustados por incapacidade (*disability-adjusted life years*)

DSM-5-TR – Manual Diagnóstico e Estatístico de Transtornos Mentais, quinta edição, texto revisado

FDA – U.S. Food and Drug Administration

GABA – ácido gama-aminobutírico

M1 – receptor muscarínico do tipo 1

M4 – receptor muscarínico do tipo 4

MK-801 – dizocilpina

NMDA – N-metil-D-aspartato

PCP – fenciclidina

PDE10A – fosfodiesterase 10A

TAAR1 – receptor associado a aminas-traço do tipo 1

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1 INTRODUÇÃO

1.1 ESQUIZOFRENIA

A esquizofrenia é um transtorno psiquiátrico complexo e heterogêneo, marcado por início relativamente precoce e por prejuízos que podem se estender a diferentes dimensões do cotidiano. A primeira apresentação clínica costuma aparecer no fim da adolescência ou no início da vida adulta, com um pico de incidência entre os 20 e 24 anos (Zhan; Wang; Shen, 2025). Em muitos casos, porém, as mudanças começam antes do primeiro episódio psicótico. Esse período prodromico pode envolver declínio cognitivo e alterações graduais no desempenho ocupacional e interpessoal, contribuindo para prejuízos funcionais ainda nas fases iniciais do transtorno (Leucht et al., 2025). Embora afete uma parcela pequena da população, sua relevância epidemiológica é expressiva considerando que estimativas recentes apontam uma prevalência ao longo da vida entre 0,4% e 0,62% na população geral (Leucht et al., 2025; Zhang et al., 2026).

Segundo as estimativas do *Global Burden of Disease* de 2019 e 2021, o número de pessoas vivendo com esquizofrenia aumentou de cerca de 14 milhões em 1990 para mais de 23 milhões (Solmi et al., 2023; Zhan; Wang; Shen, 2025). Os anos de vida ajustados por incapacidade (DALYs, do inglês *disability-adjusted life years*) também aumentaram, passando de valores próximos de 9 milhões para cerca de 15 milhões no mesmo período (Solmi et al., 2023; Zhan; Wang; Shen, 2025). Ambas as análises indicam relativa estabilidade das estimativas padronizadas por idade em escala global, sugerindo que o crescimento da carga se relaciona principalmente a mudanças demográficas, como o crescimento populacional e o envelhecimento, além de diferenças regionais na identificação dos casos (Solmi et al., 2023; Zhan; Wang; Shen, 2025).

Embora a esquizofrenia seja mais frequentemente lembrada pelos sintomas psicóticos, suas consequências se estendem para além da fase aguda do transtorno. Em uma meta-análise sobre a recuperação após o primeiro episódio de psicose, apenas cerca de 20% dos indivíduos avaliados alcançaram a recuperação clínica (Hansen et al., 2023). Essa baixa frequência de recuperação global não se restringe à persistência de sintomas psicóticos. Mesmo após a remissão parcial de

delírios e alucinações, a melhora clínica nem sempre se traduz em recuperação funcional. Muitos indivíduos continuam a apresentar prejuízos cognitivos e dificuldades de funcionamento social e ocupacional, que são pouco responsivos aos tratamentos disponíveis (Leucht et al., 2025). A inserção no mercado de trabalho também é frequentemente comprometida, com taxas de emprego consistentemente reduzidas (Holm et al., 2021; Marwaha; Johnson, 2004). Além dos prejuízos funcionais, a esquizofrenia está associada à mortalidade prematura. Em uma revisão sistemática e meta-análise que incluiu 135 estudos, pessoas com esquizofrenia apresentaram risco de mortalidade por todas as causas 2,52 vezes maior em comparação com grupos sem o transtorno, além de uma redução estimada de 15 a 20 anos na expectativa de vida (Correll et al., 2022). De forma convergente, uma meta-análise mais ampla sobre transtornos mentais estimou 15,37 anos de vida potencialmente perdidos devido a este transtorno, em comparação com a população geral (Chan et al., 2023). Esses dados indicam que o prejuízo funcional e a mortalidade prematura são componentes centrais da carga clínica e social associada a este transtorno mental.

As evidências disponíveis apontam para uma etiologia multifatorial, resultante da interação entre predisposição genética e exposição a fatores ambientais. Embora estudos com gêmeos indiquem uma contribuição genética importante para o transtorno, essa vulnerabilidade, por si só, não é suficiente para explicar sua ocorrência (Hilker et al., 2018; Sullivan; Kendler; Neale, 2003). A literatura também associa o transtorno a fatores presentes em diferentes fases do desenvolvimento, incluindo complicações obstétricas e infecções maternas (Davies et al., 2020), bem como adversidades na infância (Varese et al., 2012). Outros fatores ambientais e sociais, como residência em áreas urbanas, migração e uso frequente de cannabis, também são recorrentes em estudos epidemiológicos (Henssler et al., 2020; Marconi et al., 2016; Vassos et al., 2012). Esse conjunto de evidências reforça a heterogeneidade da esquizofrenia e ajuda a explicar a variabilidade observada em sua apresentação clínica.

1.1.1 Apresentação clínica

A apresentação clínica varia consideravelmente entre indivíduos e ao longo do curso do transtorno, envolvendo alterações no comportamento, na percepção, no pensamento, na motivação e na cognição. De acordo com o Manual Diagnóstico e Estatístico de Transtornos Mentais (DSM-5-TR), o diagnóstico depende da presença de sintomas como delírios, alucinações, discurso desorganizado, comportamento desorganizado ou catatônico e sintomas negativos, acompanhados de comprometimento funcional e de sinais contínuos do transtorno por pelo menos seis meses, com pelo menos um mês de sintomas da fase ativa (American Psychiatric Association, 2022). Embora essa definição seja útil para o diagnóstico clínico, a esquizofrenia costuma ser descrita na literatura em três grandes grupos de sintomas: positivos, negativos e cognitivos (Leucht et al., 2025).

Os sintomas positivos correspondem a manifestações que se acrescentam ao funcionamento habitual da pessoa, incluindo delírios, alucinações e discurso desorganizado. Como tendem a se tornar mais evidentes na fase ativa do transtorno, esses sintomas costumam ser prontamente associados à esquizofrenia e frequentemente motivam o primeiro contato com os serviços de saúde. No DSM-5-TR, os sintomas positivos ocupam posição central nos critérios diagnósticos, já que pelo menos um deles deve estar presente para o diagnóstico (American Psychiatric Association, 2022).

Os sintomas negativos, por outro lado, manifestam-se pela diminuição de aspectos centrais do funcionamento emocional, comunicativo, motivacional e social. A conceituação atual descreve cinco componentes principais: embotamento afetivo, alogia, anedonia, avolia e associalidade (Correll; Schooler, 2020; Marder; Galderisi, 2017). Esses componentes podem ser agrupados em dois eixos principais: um relacionado à expressão reduzida, que inclui embotamento afetivo e alogia, e outro relacionado à motivação e ao prazer, que inclui avolia, anedonia e associalidade (Marder; Galderisi, 2017). Além dessa organização, é importante considerar a origem desses sintomas. Eles podem ser primários, quando fazem parte do próprio transtorno, ou secundários, quando decorrem de outros fatores, como sintomas psicóticos persistentes, depressão, efeitos adversos de medicamentos, privação social ou outras comorbidades (Correll; Schooler, 2020; Marder; Galderisi, 2017).

A relação entre sintomas negativos e funcionamento social merece atenção particular, pois essas manifestações podem reduzir tanto o interesse por interações quanto a iniciativa necessária para mantê-las. A associalidade tende a manifestar-se

como menor busca por contato social, enquanto a avolia dificulta o engajamento em atividades orientadas a objetivos. A anedonia também pode contribuir para esse processo, ao reduzir a expectativa ou a experiência de prazer em situações sociais e cotidianas. Com isso, os sintomas negativos podem comprometer de forma persistente a interação social. Sua avaliação, porém, exige cautela, já que manifestações semelhantes podem decorrer de fatores potencialmente modificáveis, como sedação, depressão, ansiedade social ou sintomas psicóticos ativos (Correll; Schooler, 2020; Marder; Galderisi, 2017).

Os sintomas cognitivos, por sua vez, compreendem prejuízos em domínios como velocidade de processamento, atenção/vigilância, memória de trabalho, aprendizagem verbal e visual, raciocínio e resolução de problemas (McCutcheon; Keefe; McGuire, 2023; Nuechterlein et al., 2008). Embora não constituam critério diagnóstico central do transtorno, são reconhecidos como uma dimensão importante da esquizofrenia presente, muitas vezes desde o período prodromico (American Psychiatric Association, 2022; Leucht et al., 2025; McCutcheon; Keefe; McGuire, 2023). Esses prejuízos se relacionam à autonomia, ao desempenho ocupacional e à capacidade de lidar com demandas da vida cotidiana (Green et al., 2000; Leucht et al., 2025; McCutcheon; Keefe; McGuire, 2023). A cognição social, que também pode estar alterada, tem relevância particular para o funcionamento interpessoal, pois envolve processos como o reconhecimento de emoções e a interpretação de pistas sociais (Fett et al., 2011; Green; Horan; Lee, 2015).

A organização da esquizofrenia em sintomas positivos, negativos e cognitivos permite descrever a diversidade clínica do transtorno de forma mais objetiva, mas esses domínios frequentemente se sobrepõem ao longo da doença. O mesmo indivíduo pode apresentar diferentes combinações de sintomas, com impacto variável sobre a autonomia e as relações sociais. Essa distinção entre domínios também é importante porque sintomas positivos, negativos e cognitivos não apresentam a mesma evolução clínica nem a mesma resposta às intervenções disponíveis, o que é central para compreender as limitações dos tratamentos farmacológicos atuais.

1.1.2 Tratamento farmacológico e lacunas terapêuticas

O tratamento farmacológico da esquizofrenia é baseado principalmente em fármacos que modulam a neurotransmissão dopaminérgica (Coyle; Paul, 2026). Estudos clássicos mostraram que a potência clínica dos antipsicóticos se correlaciona com a capacidade de ligação aos receptores dopaminérgicos, especialmente do tipo 2 (D_2) (Creese; Burt; Snyder, 1976; Seeman et al., 1976). Além disso, estudos em humanos demonstraram que a anfetamina, ao aumentar a liberação de dopamina, intensifica sintomas psicóticos em pacientes com esquizofrenia, e que esse efeito está relacionado à liberação dopaminérgica estriatal (Laruelle et al., 1996). Mais recentemente, estudos de neuroimagem voltados ao sistema dopaminérgico indicaram que essa alteração envolve principalmente mecanismos pré-sinápticos no estriado, com maior participação de regiões dorsais e associativas (Howes et al., 2024; Howes; Kapur, 2009; Kegeles et al., 2010; McCutcheon; Abi-Dargham; Howes, 2019).

Os antipsicóticos de primeira geração, também chamados de típicos, constituem a classe mais antiga desses medicamentos, sendo os primeiros a serem introduzidos no mercado, e caracterizam-se por uma ação predominantemente dopaminérgica, especialmente pelo antagonismo dos receptores D_2 . Embora sejam importantes no controle dos sintomas positivos, o bloqueio de receptores D_2 na via nigroestriatal está associado a um maior risco de efeitos adversos motores, como sintomas extrapiramidais e discinesia tardia (Carbon et al., 2018; Farde et al., 1992; Kapur et al., 2000; Leucht et al., 2025). Posteriormente, consolidou-se a distinção entre esses fármacos e os chamados antipsicóticos de segunda geração, ou atípicos, que apresentam perfis farmacológicos mais heterogêneos e combinam a modulação dopaminérgica com ações sobre outros sistemas de neurotransmissão, especialmente o antagonismo dos receptores serotoninérgicos do tipo 2A ($5-HT_{2A}$) (Howes et al., 2024; Leucht et al., 2025; Meltzer; Matsubara; Lee, 1989; Miyamoto et al., 2005). Essa ampliação farmacológica, porém, não eliminou as limitações do tratamento, pois a menor propensão a efeitos motores em alguns fármacos veio acompanhada de outros problemas, incluindo sedação, ganho de peso, alterações metabólicas e efeitos cardiovasculares (Huhn et al., 2019; Pillinger et al., 2020). Na prática, esses perfis distintos fazem com que a decisão terapêutica envolva a tentativa de equilibrar a resposta clínica com a tolerabilidade e as características individuais do paciente (Huhn et al., 2019; Leucht et al., 2025).

Embora os antipsicóticos sejam relativamente eficazes no controle dos sintomas positivos, seus efeitos sobre os sintomas negativos e déficits cognitivos permanecem bastante limitados (Coyle; Paul, 2026; Huhn et al., 2019; Leucht et al., 2012). Em estudos que avaliam especificamente os sintomas negativos, as reduções observadas em escalas clínicas tendem a ser pequenas e, muitas vezes, não alcançam magnitude considerada clinicamente relevante (Fusar-Poli et al., 2015). A interpretação desses resultados é dificultada pela heterogeneidade dos sintomas negativos e pela complexidade em distinguir sintomas primários de sintomas secundários (Correll; Schooler, 2020; Krause et al., 2018). No domínio cognitivo, observa-se um problema semelhante, uma vez que os déficits cognitivos respondem pouco aos tratamentos farmacológicos disponíveis (McCutcheon; Keefe; McGuire, 2023).

Além disso, estima-se que entre 25% e 33% dos pacientes apresentem esquizofrenia resistente ao tratamento, embora essa proporção varie conforme os critérios utilizados (Howes et al., 2017, 2024; Siskind; Siskind; Kisely, 2017). Para esses casos, a clozapina permanece como a principal alternativa farmacológica, mas parte dos pacientes apresenta sintomas persistentes mesmo após sua introdução (Siskind; Siskind; Kisely, 2017). As dificuldades terapêuticas, no entanto, não se limitam aos casos classificados como resistentes. Mesmo quando há resposta inicial aos antipsicóticos, a manutenção da resposta clínica ao longo do tempo permanece limitada. Em uma coorte nacional sueca, 43,7% dos pacientes tiveram nova internação psiquiátrica durante o seguimento, e 71,7% preencheram critérios de falha terapêutica, incluindo internação psiquiátrica, tentativa de suicídio, descontinuação ou troca do antipsicótico, ou morte (Tiihonen et al., 2017). A descontinuação do tratamento também é frequente em estudos de maior duração, frequentemente associada à eficácia insuficiente, efeitos adversos ou baixa aceitabilidade do tratamento (Kahn et al., 2008; Lieberman et al., 2005; Velligan et al., 2017). Ou seja, as limitações do tratamento não se restringem aos sintomas negativos e cognitivos, mas também abrangem pacientes para os quais o controle da psicose permanece incompleto.

A persistência dessas lacunas levou à busca por tratamentos capazes de atuar além do bloqueio dopaminérgico. Entre elas estão abordagens centradas na modulação de sistemas de neurotransmissão relacionados ao glutamato, ao ácido gama-aminobutírico (GABA), à serotonina e à acetilcolina, bem como intervenções

com ação anti-inflamatória e abordagens que interferem na regulação da sinalização dopaminérgica, incluindo receptores associados a aminas-traço do tipo 1 (TAAR1) e vias intracelulares moduladas pela fosfodiesterase 10A (PDE10A) (Coyle; Paul, 2026; Howes et al., 2024). Apesar de resultados promissores em estudos pré-clínicos, muitos fármacos não demonstraram eficácia em fases posteriores do desenvolvimento clínico (Coyle; Paul, 2026; Howes et al., 2024).

A aprovação, em 2024, pela U.S. Food and Drug Administration (FDA), nos Estados Unidos, da combinação de xanomelina e cloreto de tróspio, comercializada pela biofarmacêutica Bristol Myers Squibb®, marcou uma mudança nesse cenário. Diferentemente dos antipsicóticos tradicionais, essa estratégia não atua por meio do bloqueio direto dos receptores dopaminérgicos D₂. Seu componente central, a xanomelina, atua como agonista dos receptores muscarínicos do tipo 1 e 4 (M1 e M4, respectivamente), enquanto o cloreto de tróspio é incluído para reduzir efeitos colinérgicos periféricos (Howes et al., 2024; Kaul et al., 2024; U.S. Food and Drug Administration, 2024). Embora amplie as possibilidades de tratamento, a disponibilização deste novo medicamento não elimina as principais lacunas terapêuticas da esquizofrenia. Persistem limitações importantes, especialmente em relação aos sintomas negativos e cognitivos, que continuam a responder de forma insuficiente às intervenções disponíveis, bem como ao desafio representado pelos pacientes refratários. Esses desafios indicam que a busca por novas intervenções depende de uma compreensão mais refinada dos mecanismos neurobiológicos da esquizofrenia e de modelos pré-clínicos com maior relevância translacional.

1.2 MODELOS ANIMAIS EM ESQUIZOFRENIA

Os modelos animais desempenham um papel importante na pesquisa translacional e na prospecção de candidatos a fármacos antipsicóticos. Eles permitem examinar, em condições experimentais controladas, como alterações neurobiológicas específicas podem se relacionar a desfechos neuroquímicos e comportamentais, bem como à resposta a intervenções farmacológicas (Jones; Watson; Fone, 2011; Nestler; Hyman, 2010). Essa abordagem, porém, apresenta limitações importantes nos transtornos psiquiátricos, em que muitos sintomas envolvem experiências subjetivas e dependem de relatos verbais. No caso da

esquizofrenia, essa dificuldade é ainda mais evidente, já que delírios e alucinações, fenômenos centrais do transtorno, não podem ser avaliados diretamente em animais (Feifel; Shilling, 2010; Lipska; Weinberger, 2000; Powell; Miyakawa, 2006). Assim, esses modelos são mais adequadamente compreendidos como ferramentas experimentais voltadas à investigação de dimensões específicas do transtorno, permitindo o estudo de processos associados a endofenótipos relevantes para a esquizofrenia (Lipska; Weinberger, 2000; Nestler; Hyman, 2010).

A pesquisa em esquizofrenia tem se concentrado especialmente no desenvolvimento e na aplicação de modelos desenvolvimentais, genéticos e farmacológicos. Modelos desenvolvimentais investigam como perturbações em períodos sensíveis, como a ativação imune materna e o estresse pré-natal, podem levar a alterações na fase adulta (Jones; Watson; Fone, 2011; Meyer; Feldon, 2010; Percelay et al., 2024). Modelos genéticos, por outro lado, exploram intervenções em genes e vias associados ao risco de esquizofrenia (Jones; Watson; Fone, 2011; Nestler; Hyman, 2010). Já os modelos farmacológicos interferem de forma mais direta nos sistemas de neurotransmissores envolvidos no transtorno (Jones; Watson; Fone, 2011; Lee; Zhou, 2019). Essas diferentes abordagens têm sido utilizadas para investigar mecanismos neurobiológicos envolvidos em diversos domínios da esquizofrenia (Jones; Watson; Fone, 2011; Lee; Zhou, 2019; Spark et al., 2022).

Historicamente, essa literatura tem se concentrado no uso de roedores como organismos modelo. Essa concentração favoreceu avanços relevantes, em parte porque camundongos e ratos dispõem de paradigmas comportamentais consolidados, linhagens genéticas e intervenções farmacológicas amplamente caracterizadas (Jones; Watson; Fone, 2011; Lee; Zhou, 2019; Powell; Miyakawa, 2006). Ao mesmo tempo, a tradução de achados pré-clínicos não depende apenas da validade interna dos experimentos, mas também da capacidade de verificar se um efeito se mantém quando a hipótese é testada em diferentes contextos biológicos e experimentais. Estratégias voltadas ao aumento da validade externa, como a implementação de testes de generalização e de desenhos multicêntricos, têm sido propostas justamente para ampliar a base de evidências que orienta decisões translacionais (Carneiro et al., 2023). Esse ponto é particularmente relevante na esquizofrenia, já que, como apontado anteriormente, muitos candidatos terapêuticos com resultados promissores em estudos pré-clínicos não demonstraram eficácia em ensaios clínicos (Leenaars et al., 2019; Pound; Ritskes-Hoitinga, 2018;

Spark et al., 2022). Nesse contexto, a incorporação de espécies complementares pode contribuir para avaliar a conservação de mecanismos neurobiológicos e de respostas farmacológicas entre vertebrados e favorecer triagens farmacológicas.

1.2.1 Peixes-zebra como organismo modelo em esquizofrenia

Entre os organismos utilizados em neurociência translacional, o peixe-zebra tem se consolidado em estudos de neuropsicofarmacologia (Kalueff; Stewart; Gerlai, 2014; Kalueff; Echevarria; Stewart, 2014; Kari; Rodeck; Dicker, 2007; Stewart et al., 2014). Seu uso se deve a características práticas e biológicas, como baixo custo relativo, alta fecundidade, desenvolvimento rápido e a possibilidade de manipulações genéticas e farmacológicas em larga escala (Howe et al., 2013; Nasiadka; Clark, 2012; Stewart et al., 2014). Além disso, trata-se de um vertebrado com sistemas neuroquímicos relevantes conservados, incluindo vias glutamatérgicas, dopaminérgicas, serotoninérgicas, colinérgicas e endocanabinoides (Maximino; Herculano, 2010; Oltrabella et al., 2017; Panula et al., 2010, 2006; Rico et al., 2011). O modelo também permite a avaliação de repertórios comportamentais quantificáveis em larvas e adultos, como locomoção, exploração, resposta ao estresse, aprendizagem, memória, preferência social e comportamento de cardume (Benvenuti et al., 2021b; Cachat et al., 2010; Kalueff et al., 2013, 2025; Miller; Gerlai, 2012; Norton; Bally-Cuif, 2010; Ogi et al., 2021).

No contexto da esquizofrenia, o uso do peixe-zebra também se baseia na presença de sistemas neurobiológicos relevantes para o transtorno (Kalueff; Stewart; Gerlai, 2014; Langova et al., 2020). Embora o encéfalo de teleósteos apresente diferenças importantes em relação ao de mamíferos, algumas regiões telencefálicas do peixe-zebra têm sido associadas a circuitos envolvidos em memória, processamento emocional, motivação e controle motor (Silva et al., 2015; Wullimann; Rink, 2002). O pálio, por exemplo, é frequentemente associado a funções comparáveis às do hipocampo e da amígdala, enquanto as regiões do subpálio apresentam características compatíveis com as do estriado (Silva et al., 2015; Wullimann; Rink, 2002). Além disso, populações dopaminérgicas localizadas no diencéfalo basal, especialmente no tubérculo posterior, projetam-se para regiões do telencéfalo basal e do subpálio, o que sustenta a comparação com circuitos

dopaminérgico-estriatais relevantes para modelos de esquizofrenia (Langova et al., 2020; Rink; Wullimann, 2002; Wullimann; Rink, 2002).

Do ponto de vista comportamental, o peixe-zebra permite investigar domínios frequentemente explorados em modelos de roedores de esquizofrenia, incluindo atividade locomotora, comportamento social, resposta ao sobressalto, aprendizagem e memória (Benvenuti et al., 2022; Langova et al., 2020). Em modelos farmacológicos, manipulações de sistemas dopaminérgicos e glutamatérgicos podem alterar a locomoção, a exploração do ambiente, padrões repetitivos de nado e respostas a estímulos, medidas usadas como aproximações experimentais de fenótipos relacionados a sintomas positivos e a alterações sensório-motoras do transtorno (Benvenuti et al., 2022; Kalueff; Echevarria; Stewart, 2014; Langova et al., 2020). Outros paradigmas também descrevem prejuízos de habituação, aprendizagem e comportamento social, incluindo alterações de preferência social e comportamento de cardume, que se aproximam de domínios cognitivos e negativos investigados em modelos pré-clínicos (Benvenuti et al., 2022; Langova et al., 2020). Assim, os padrões comportamentais observados permitem comparação com achados em roedores e ajudam a avaliar a consistência de respostas induzidas por perturbações neuroquímicas relevantes para a esquizofrenia (Benvenuti et al., 2022; Langova et al., 2020).

1.3 MODELOS FARMACOLÓGICOS BASEADOS EM ANTAGONISTAS NMDA

Entre os modelos farmacológicos, a hipótese dopaminérgica exerceu influência central na pesquisa pré-clínica sobre a esquizofrenia. Os primeiros paradigmas foram desenvolvidos com base no uso de psicoestimulantes, como a anfetamina, ou de agonistas dopaminérgicos como a apomorfina, capazes de induzir hiperlocomoção, estereotipias e outros comportamentos sensíveis ao bloqueio dos receptores D_2 (Jones; Watson; Fone, 2011; Lipska; Weinberger, 2000). Esses paradigmas foram importantes para a triagem de compostos com perfil antipsicótico e para o estudo de alterações relacionadas à hiperatividade dopaminérgica, mas também restringiram parte da pesquisa pré-clínica a desfechos mais sensíveis à modulação dos receptores D_2 (Lipska; Weinberger, 2000; Spark et al., 2022). Com o tempo, esse foco começou a se ampliar para abordagens farmacológicas voltadas a

outros mecanismos, especialmente à hipofunção glutamatérgica e ao antagonismo dos receptores *N*-metil-D-aspartato (NMDA).

O interesse por antagonistas NMDA na pesquisa sobre esquizofrenia surgiu a partir de achados farmacológicos em humanos. A fenciclidina (PCP) e a cetamina, em doses subanestésicas, induzem alterações perceptuais, cognitivas e comportamentais que se aproximam de aspectos do transtorno, tanto em voluntários saudáveis quanto em indivíduos com esquizofrenia (Javitt; Zukin, 1991; Krystal et al., 1994; Lahti et al., 1995; Newcomer et al., 1999). Essa observação foi relevante porque esses fármacos não reproduziam apenas alterações relacionadas aos sintomas positivos, mas também mostravam impacto sobre os domínios cognitivos e negativos da esquizofrenia (Jentsch; Roth, 1999; Neill et al., 2010).

A partir desses achados, a hipofunção de receptores NMDA passou a ser considerada uma via importante para investigar mecanismos relacionados à esquizofrenia (Coyle, 2012; Moghaddam; Javitt, 2012). Essa hipótese propõe que alterações glutamatérgicas podem modificar o funcionamento de circuitos corticais e sua influência sobre sistemas dopaminérgicos subcorticais (Howes; Shatalina, 2022; Moghaddam; Javitt, 2012). Um dos mecanismos propostos envolve a redução da função NMDA em interneurônios GABAérgicos positivos para parvalbumina, o que pode comprometer o controle inibitório cortical e favorecer desequilíbrios entre atividade excitatória e inibitória (Gonzalez-Burgos; Lewis, 2012; Nakazawa; Jeevakumar; Nakao, 2017). Alterações desse tipo podem afetar circuitos corticais e hipocampus que modulam a atividade dopaminérgica, contribuindo para a hiperatividade dopaminérgica subcortical associada à psicose (Grace, 2010, 2016; Howes; Shatalina, 2022). Essa hipótese também é compatível com achados *post mortem* que indicam alterações em marcadores GABAérgicos e parvalbumina no córtex pré-frontal de indivíduos com esquizofrenia (Beasley; Reynolds, 1997; Hashimoto et al., 2003; Kaar et al., 2019; Lewis; Hashimoto; Volk, 2005).

Com base nessa hipótese, antagonistas do receptor NMDA passaram a ser amplamente utilizados em modelos animais de esquizofrenia. O receptor NMDA é um receptor ionotrópico de glutamato que funciona como canal catiônico dependente de ligante e de voltagem (Paoletti; Bellone; Zhou, 2013; Traynelis et al., 2010). PCP, cetamina e dizocilpina (MK-801) atuam como antagonistas não competitivos desse receptor, bloqueando o canal. Essa propriedade permite induzir farmacologicamente uma redução funcional da sinalização mediada por receptores NMDA e avaliar suas

consequências comportamentais e neuroquímicas (Lee; Zhou, 2019; Mouri et al., 2007; Neill et al., 2010).

Em roedores, esses fármacos podem causar hiperlocomoção, estereotipias, déficits de inibição por pré-pulso, prejuízos em tarefas de aprendizagem e memória e déficits no comportamento social (Jentsch; Roth, 1999; Lee; Zhou, 2019; Mouri et al., 2007; Neill et al., 2010). Esses efeitos são descritos em protocolos que variam quanto ao momento e à duração da exposição. Nos protocolos agudos, os animais são avaliados após uma única administração ou após exposições pontuais ao fármaco (Jentsch; Roth, 1999; Lee; Zhou, 2019; Neill et al., 2010). Nos protocolos repetidos ou subcrônicos, a exposição ocorre ao longo de vários dias, e pode ser seguida de avaliação durante o tratamento ou após um período sem administração do fármaco, permitindo investigar a persistência dos efeitos comportamentais e a ocorrência de sensibilização (Lee; Zhou, 2019; Neill et al., 2010, 2014). Há ainda protocolos desenvolvimentais, nos quais antagonistas NMDA são administrados em janelas específicas, como o período neonatal ou a adolescência, e os desfechos comportamentais são avaliados posteriormente, muitas vezes na vida adulta (Jeevakumar et al., 2015; Mouri et al., 2007; Nakazawa; Jeevakumar; Nakao, 2017; Wang et al., 2001).

1.3.1 Antagonistas NMDA em peixes-zebra

O uso de antagonistas de receptores NMDA em peixe-zebra surgiu como extensão natural dos paradigmas farmacológicos consolidados em roedores, permitindo investigar, nessa espécie, alterações comportamentais com relevância para a esquizofrenia. Entre os compostos empregados nessa espécie, MK-801 e cetamina concentram a maior parte da literatura disponível, enquanto PCP e outros antagonistas aparecem de forma mais esparsa e com protocolos menos padronizados (Benvenuti et al., 2022). Além de induzir fenótipos de interesse, esses fármacos também têm sido utilizados em contextos de avaliação farmacológica, nos quais a reversão ou atenuação das alterações induzidas pelo antagonismo NMDA é usada como indicador da sensibilidade do modelo a antipsicóticos ou compostos candidatos (Benvenuti et al., 2022; Langova et al., 2020).

Dentre os antagonistas, o MK-801 é o mais utilizado, com efeitos descritos em tarefas de aprendizagem e memória, atividade locomotora e medidas sociais, enquanto a cetamina aparece com mais frequência em protocolos voltados a alterações do padrão natatório e exploração (Benvenuti et al., 2021a, 2022; Cognato et al., 2012; Maaswinkel; Zhu; Weng, 2013; Riehl et al., 2011; Seibt et al., 2010, 2011; Zakhary et al., 2011; Zimmermann et al., 2016). Alguns desses fenótipos também foram avaliados quanto à sensibilidade aos antipsicóticos. Em estudos com MK-801, haloperidol, sulpirida e olanzapina foram testados quanto a desfechos locomotores, cognitivos e sociais, prevenindo ou revertendo parte das alterações induzidas. Esses achados são interpretados como evidências iniciais de validade preditiva para ensaios comportamentais em peixe-zebra, embora ainda estejam restritas a poucos estudos (Benvenuti et al., 2022; Seibt et al., 2011). Ainda assim, a alta heterogeneidade entre desenhos experimentais, regimes de exposição, comportamentos analisados e formas de reportar os resultados dificulta comparações diretas entre estudos.

1.4 COMPORTAMENTO SOCIAL COMO DESFECHO TRANSLACIONAL

O comportamento social é uma medida recorrente em modelos animais relacionados à esquizofrenia, pois permite operacionalizar aspectos do domínio negativo, especialmente associalidade e retraimento social, por meio de parâmetros como aproximação, permanência ou interação com coespecíficos (Marder; Galderisi, 2017; Wilson; Koenig, 2014). Alterações nessas medidas após manipulações experimentais são amplamente utilizadas para investigar mecanismos neurobiológicos associados ao transtorno e avaliar intervenções com potencial relevância translacional (Neill et al., 2010, 2014; Wilson; Koenig, 2014). No entanto, ainda não está claro em que medida essas alterações sociais mimetizam, de fato, sintomas negativos observados em humanos ou têm validade translacional consistente.

Embora o comportamento social seja frequentemente tratado como um desfecho único, sua avaliação em animais envolve medidas que capturam componentes distintos do repertório comportamental. Alguns paradigmas quantificam encontros sociais livres, nos quais o animal pode investigar, seguir,

evitar ou interagir diretamente com um coespecífico. Outros estimam a aproximação social de forma indireta, a partir do tempo gasto próximo a um estímulo social ou da preferência por uma área do aparato associada a esse estímulo (File; Seth, 2003; Yang; Silverman; Crawley, 2011). Essa diversidade metodológica torna a interpretação menos direta, pois uma redução em medidas sociais pode refletir menor motivação para contato com coespecíficos, mas também menor exploração do aparato, aumento de evitação, alteração na resposta à novidade ou mudanças na distribuição do tempo entre diferentes comportamentos (Toth; Neumann, 2013; Wilson; Koenig, 2014).

A interpretação se torna mais delicada quando medidas de comportamento social são usadas como indicadores de efeitos sobre sintomas negativos. Em estudos pré-clínicos, antipsicóticos são frequentemente avaliados pela capacidade de prevenir ou reverter alterações sociais induzidas por manipulações experimentais, incluindo antagonistas NMDA. No entanto, essa leitura exige cautela, já que os antipsicóticos atuais apresentam eficácia limitada em sintomas negativos primários e no funcionamento social em pacientes (Correll; Schooler, 2020; Fusar-Poli et al., 2015; Krause et al., 2018). Assim, não está claro se a atenuação de déficits sociais em animais representa uma melhora seletiva de processos relacionados à sociabilidade ou se reflete efeitos inespecíficos sobre locomoção, sedação, ansiedade, exploração ou alocação comportamental dentro do teste (Hagan; Jones, 2005; Spark et al., 2022; Wilson; Koenig, 2014). Além disso, considerando a heterogeneidade de espécies, drogas, doses, esquemas de administração e paradigmas comportamentais utilizados, permanece incerta qual é a magnitude e a consistência dos efeitos dos antagonistas NMDA sobre o comportamento social, bem como se a literatura pré-clínica, quando analisada em conjunto, sustenta uma reversão desses efeitos por fármacos antipsicóticos aprovados na clínica.

1.4.1 Antagonistas NMDA e comportamento social em peixes-zebra

Em peixe-zebra adulto, o comportamento social costuma ser avaliado por paradigmas que exploram a tendência inata da espécie a se aproximar de coespecíficos e formar grupos (Miller; Gerlai, 2007; Spence et al., 2008). Uma das

abordagens envolve testes de preferência social, nos quais o animal focal é colocado em uma arena com acesso visual a estímulos sociais, que podem consistir em coespecíficos vivos separados fisicamente por barreiras transparentes, vídeos de coespecíficos ou imagens computadorizadas/animadas de peixes. Nesses paradigmas, o comportamento social pode ser medido pelo tempo em zonas próximas ao estímulo social, pela preferência por compartimentos associados a esse estímulo ou pela distância do animal focal em relação ao estímulo (Barba-Escobedo; Gould, 2012; Ogi et al., 2021; Qin et al., 2014; Saverino; Gerlai, 2008). Essa lógica foi adaptada dos testes de sociabilidade em três câmaras usados em roedores (Barba-Escobedo; Gould, 2012; Langova et al., 2020). Outra abordagem avalia interação direta ou coesão entre indivíduos, utilizando medidas como distância interindividual, distância do vizinho mais próximo e área ocupada pelo cardume (Miller; Gerlai, 2007).

Nos paradigmas sociais, a literatura em peixe-zebra permanece concentrada principalmente no MK-801, enquanto os dados com cetamina, PCP e outros antagonistas NMDA ainda são escassos (Benvenutti et al., 2022). Para o MK-801, o achado mais recorrente é a redução de medidas de interação social, preferência por coespecíficos ou coesão social, sustentando seu uso na investigação de fenótipos relacionados ao domínio negativo da esquizofrenia em peixe-zebra (Maaswinkel; Zhu; Weng, 2013; Seibt et al., 2011; Sison; Gerlai, 2011; Zimmermann et al., 2016). Esses efeitos, porém, não ocorrem de forma isolada, já que o MK-801 também pode modificar parâmetros locomotores e prejudicar o desempenho em tarefas cognitivas, o que dificulta interpretar alterações sociais como medidas exclusivamente relacionadas à sociabilidade (Seibt et al., 2010, 2011; Sison; Gerlai, 2011). Além disso, a avaliação da reversão desses fenótipos por antipsicóticos ou compostos candidatos ainda é pontual, embora estudos com sulpirida e olanzapina indiquem reversão de déficits sociais induzidos por MK-801 (Seibt et al., 2011). Assim, embora o peixe-zebra apresente potencial para ampliar a investigação pré-clínica de fenótipos sociais relacionados à esquizofrenia, essa área ainda depende de maior replicação e de comparações sistemáticas entre antagonistas, regimes de exposição e desfechos ou testes comportamentais.

Essa lacuna é particularmente relevante para a cetamina. Embora esse antagonista NMDA seja amplamente utilizado em modelos farmacológicos de esquizofrenia em roedores, incluindo protocolos agudos e repetidos, sua

caracterização em peixe-zebra ainda é limitada quando comparada ao MK-801 (Benvenuti et al., 2022; Frohlich; Van Horn, 2014; Lee; Zhou, 2019; Neill et al., 2010). Os estudos disponíveis com cetamina em peixe-zebra descrevem principalmente alterações locomotoras e rotacionais, além de efeitos sobre agressividade, enquanto medidas mais diretamente relacionadas à preferência social, interação com coespecíficos ou retraimento social permanecem pouco exploradas (Benvenuti et al., 2022; Michelotti et al., 2018, 2022; Riehl et al., 2011). Além disso, a literatura em peixe-zebra é baseada sobretudo em exposições agudas, com pouca informação sobre efeitos de exposições repetidas ou sobre a persistência das alterações após o término das exposições. Essa limitação dificulta a comparação com achados obtidos em roedores, nos quais protocolos repetidos com antagonistas NMDA são usados para investigar alterações comportamentais mais duradouras e, potencialmente, com maior relevância translacional (Frohlich; Van Horn, 2014; Lee; Zhou, 2019; Neill et al., 2010). Assim, ainda é necessário caracterizar melhor os efeitos da cetamina em peixes-zebra adultos, integrando desfechos sociais, locomotores e exploratórios em protocolos que considerem tanto respostas agudas quanto efeitos após exposição repetida.

1.5 JUSTIFICATIVA

Apesar dos avanços da pesquisa pré-clínica, ainda há um descompasso entre os efeitos observados em modelos animais e a eficácia clínica dos tratamentos farmacológicos, especialmente para sintomas negativos e cognitivos. Nos modelos baseados em antagonistas NMDA, parte desse problema passa pela própria interpretação dos desfechos comportamentais. A literatura carece de uma síntese integrada que permita estimar a magnitude dos efeitos desses fármacos sobre o comportamento social, sua consistência entre estudos, sua relação com alterações locomotoras e a reversão desses efeitos por antipsicóticos.

Uma forma de fortalecer essa interpretação é ampliar a base comparativa dos modelos experimentais, incluindo espécies complementares aos roedores. O peixe-zebra se insere nesse cenário, mas a literatura com antagonistas NMDA nessa espécie ainda é concentrada principalmente no MK-801. A cetamina, embora amplamente utilizada em roedores, permanece pouco explorada em peixes-zebra

adultos, sobretudo em relação ao comportamento social, à locomoção, aos efeitos de exposições repetidas e à persistência das respostas após o término da exposição. Assim, esta tese parte dessas lacunas para integrar evidências pré-clínicas sobre antagonistas NMDA e comportamento social e caracterizar os efeitos da cetamina em peixes-zebra adultos.

2 OBJETIVOS

2.1 OBJETIVO GERAL

Avaliar os efeitos de antagonistas do receptor NMDA sobre o comportamento social em animais de laboratório e sua relevância para modelar fenótipos relacionados aos sintomas negativos da esquizofrenia, com ênfase na interpretação translacional.

2.2 OBJETIVOS ESPECÍFICOS

- a) Realizar uma revisão sistemática dos efeitos de antagonistas do receptor NMDA no comportamento social em animais de laboratório;
- b) Quantificar, por meta-análise, os efeitos de antagonistas do receptor NMDA sobre a interação social e a preferência social em estudos pré-clínicos;
- c) Investigar se antipsicóticos clinicamente aprovados modificam as alterações de comportamento social induzidas por antagonistas do receptor NMDA;
- d) Analisar alterações locomotoras associadas ao uso de antagonistas do receptor NMDA, considerando seu potencial papel como fator de confusão na interpretação de testes sociais;
- e) Caracterizar os efeitos agudos da cetamina sobre o comportamento social e locomotor em peixe-zebra adulto;
- f) Avaliar se a exposição repetida à cetamina induz sensibilização comportamental em peixe-zebra adulto;
- g) Investigar se os efeitos comportamentais da cetamina persistem após um período sem exposição em peixe-zebra adulto.

3 ARTIGOS CIENTÍFICOS

A presente tese está organizada no formato de coletânea de artigos científicos e é composta por dois capítulos. O primeiro capítulo apresenta uma revisão sistemática e meta-análise sobre os efeitos de antagonistas do receptor NMDA no comportamento social em animais de laboratório. O segundo capítulo apresenta um estudo experimental sobre os efeitos da cetamina no comportamento social e locomotor em peixes-zebra adultos.

3.1 CAPÍTULO 1

Effects of NMDA antagonists on social behaviour: a systematic review and meta-analysis of preclinical studies

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Effects of NMDA antagonists on social behaviour: a systematic review and meta-analysis of preclinical studies

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ABSTRACT

Social withdrawal is a key component of the negative symptom domain of schizophrenia, and pharmacological blockade of the *N*-methyl-D-aspartate receptor (NMDAR) is widely used to model schizophrenia-relevant phenotypes in animals. However, findings on social behaviour are inconsistent across paradigms and laboratories. We therefore conducted a systematic review and meta-analysis to synthesise the effects of dizocilpine, ketamine, and phencyclidine on social interaction and social preference, to evaluate whether clinically approved antipsychotics modify these outcomes, and to examine locomotor activity measured within the same social tests to aid interpretation. We searched Embase, PubMed and Web of Science without language or date restrictions. Controlled *in vivo* studies in laboratory animals administering an eligible NMDAR antagonist and reporting social interaction and/or social preference outcomes were included. Two reviewers independently screened records, extracted data and assessed risk of bias. Effect sizes were computed as standardised mean differences and synthesised using correlated multilevel random-effects models with cluster-robust variance estimation. In total, 264 studies met the inclusion criteria. Overall, NMDAR antagonists were associated with reduced social interaction and reduced social preference relative to controls, although the social preference literature appeared vulnerable to small-study effects and imprecision. Locomotor activity measured during social interaction tests tended to be higher following NMDAR antagonists, whereas during social preference no consistent overall change was observed. In animals exposed to NMDAR antagonists, antipsychotics increased social behaviour, but these changes commonly co-occurred with reduced locomotion during social interaction tests, suggesting that improvements in social measures may partly reflect altered behavioural competition and time allocation rather than selective restoration of social functioning. Taken together, the evidence supports an overall link between NMDAR antagonism and reduced social behaviour, but the strength and interpretability of this signal depend on the paradigm and are constrained by heterogeneity and limitations in reporting.

Keywords: NMDA receptor antagonists, ketamine, PCP, MK-801, social interaction, social preference, locomotor activity, antipsychotics, preclinical animal models, schizophrenia.

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I. INTRODUCTION

Social withdrawal and impaired social functioning are core components of the negative symptom domain of schizophrenia and are closely associated with long-term functional disability and reduced quality of life (Galderisi *et al.*, 2018; Harvey, Strassnig & Silberstein, 2019; Correll & Schooler, 2020; Mosolov & Yaltonskaya, 2022). In contrast to positive symptoms, which are responsive to antipsychotic treatment in most patients, negative symptoms demonstrate limited and inconsistent improvement with currently available pharmacotherapies, with reported benefits often attributable to changes in secondary negative symptoms (Fusar-Poli *et al.*, 2015; Leucht *et al.*, 2017; Krause *et al.*, 2018; Correll & Schooler, 2020; Sabe *et al.*, 2021). Social dysfunction, therefore, remains a persistent clinical problem and an important target for translational research (Millan *et al.*, 2014; Spark *et al.*, 2022; Howes, Fusar-Poli & Osugo, 2023).

Among the neurobiological hypotheses of schizophrenia, *N*-methyl-D-aspartate receptor (NMDAR) hypofunction has received substantial empirical support from convergent *post-mortem* findings (Vrajová *et al.*, 2010; Weickert *et al.*, 2013; Catts *et al.*, 2016) and pharmacological evidence in humans (Allen & Young, 1978; Newcomer *et al.*, 1999; Adler *et al.*, 1999; Lahti *et al.*, 2001; Pomarol-Clotet *et al.*, 2006). Consistent with this framework, NMDAR antagonist-based animal models using non-competitive antagonists such as dizocilpine (MK-801), ketamine, and phencyclidine (PCP) are well established in preclinical research, being employed to elicit behavioural and neurochemical alterations related to the positive, cognitive, and negative symptoms of schizophrenia (Neill *et al.*, 2010; Frohlich & Van Horn, 2014; Miyamoto & Nitta, 2014; Lee & Zhou, 2019; Benvenuti *et al.*, 2022; Li *et al.*, 2024).

Although social behaviour has been extensively examined in these models, reported findings remain inconsistent across the literature, with studies reporting null effects (Becker & Grecksch, 2004; Hanks *et al.*, 2013) and even unexpected increases in sociability (Dunn, Corbett & Fielding, 1989; Jenkins *et al.*, 2008; Rawat *et al.*, 2022). The latter has also been observed in our laboratory, where acute MK-801 administration increased social behaviour in adult C57BL/6 mice (Giongo *et al.*, 2023). Social outcomes are measured using heterogeneous paradigms, most commonly free social interaction and social preference tests (Toth & Neumann, 2013; Wilson & Koenig, 2014).

Social interaction paradigms quantify direct reciprocal behaviours between freely moving conspecifics (File & Hyde, 1978), whereas social preference paradigms measure approach or proximity to a constrained social stimulus (Landauer & Balster, 1982; Wee *et al.*, 1995; Haller & Bakos, 2002; Nadler *et al.*, 2004), also capturing motivational or exploratory components besides interaction *per se*. Furthermore, substantial variation exists in apparatus design, outcome definitions, species, developmental stage, dosing schedules, and administration routes (Neill *et al.*, 2010; Benvenutti *et al.*, 2022). As a result, it remains unclear to what extent the observed differences reflect biological effects of NMDAR blockade rather than methodological inconsistencies.

Interpretation is further complicated by two recurring issues. First, antipsychotics are routinely used in animal studies to test the pharmacological sensitivity of NMDAR antagonist-induced phenotypes (Spark *et al.*, 2022), yet clinically available antipsychotics provide little to no benefit for primary negative symptoms in patients and have a limited impact on their social functioning (Correll & Schooler, 2020; Howes *et al.*, 2023). For this reason, the apparent normalisation of social alterations in animals cannot, on its own, be taken as evidence of translational relevance (Hagan & Jones, 2005; Large, 2007; Wilson & Koenig, 2014). Therefore, it remains crucial to determine whether the reported antipsychotic effects on social outcomes reflect specific modulation of social behaviour or instead arise from non-specific changes in arousal, anxiety, or motor output, and to assess how robust these effects are across species, paradigms, and laboratories. Second, NMDAR antagonists are also known to cause hyperlocomotion (Lee & Zhou, 2019; Benvenutti *et al.*, 2022). Locomotor changes can directly influence measures of both reciprocal interaction and proximity-based social behaviour (Yang, Silverman & Crawley, 2011; Mitchell *et al.*, 2020), separating social effects from non-specific motor effects is thus not always straightforward.

Beyond heterogeneity in experimental design and insufficient reporting of key methodological practices, this area of research may also be affected by inadequate handling of non-independent observations (Sena *et al.*, 2010; Landis *et al.*, 2012; Hirst *et al.*, 2014; Macleod *et al.*, 2015; Wilson *et al.*, 2023; Townsend *et al.*, 2025). In social behaviour experiments, non-independence can arise when multiple measurements are obtained from the same animal, when interacting animals within

a dyad are treated as independent experimental units, or when outcomes are clustered within cages or litters. Failure to account for such dependencies can artificially increase precision and inflate apparent effect sizes (Lazic, Clarke-Williams & Munafò, 2018; Lazic *et al.*, 2020). These issues extend beyond social paradigms and reflect broader challenges in preclinical research, limiting the interpretability and reproducibility of individual studies (Wilson *et al.*, 2023; Eleftheriou *et al.*, 2025).

Although the existing literature suggests that NMDAR antagonists induce social behaviour deficits, the magnitude, consistency, and moderators of these effects have not been established. A quantitative synthesis is therefore required to integrate the available evidence, characterise heterogeneity, evaluate potential sources of bias, and assess the extent to which experimental factors and pharmacological interventions modulate reported effects (Bahadoran *et al.*, 2020; Ineichen *et al.*, 2024b). This systematic review and meta-analysis addresses the following question: what are the effects of NMDAR antagonists on social behaviour in laboratory animals? Secondary objectives are to evaluate the effects of clinically approved antipsychotics on social outcomes in NMDAR antagonist-exposed animals and to assess locomotor activity as a control outcome to aid interpretation of social measures.

II. METHODS

(1) Protocol

The protocol for this systematic review and meta-analysis was registered in the International Prospective Register of Systematic Reviews (PROSPERO, registration number: CRD42023402129) prior to study screening and data extraction (Gallas-Lopes *et al.*, 2024). The preregistered protocol specified the research questions, eligibility criteria, search strategy, and planned analyses.

The primary objective of the review was to evaluate the effects of NMDAR antagonists (MK-801, ketamine, or PCP) on social behaviour in laboratory animals, with social behaviour operationalised as social interaction or social preference outcomes. Secondary objectives were to assess the effects of clinically approved antipsychotics on social behaviour in animals exposed to NMDAR

antagonists, and to evaluate locomotor activity in both comparisons (NMDAR antagonist vs control, and NMDAR antagonist plus antipsychotic vs NMDAR antagonist alone) as a control outcome to determine whether observed social behaviour changes could be confounded by alterations in general activity levels.

This review adheres to the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) guidelines (Page *et al.*, 2021a, 2021b). All data and materials associated with this study, including extracted datasets and analysis code, are publicly available via the Open Science Framework (OSF) at osf.io/98fau (Gallas-Lopes *et al.*, 2023a) and GitHub at github.com/matheusglls/NMDA-sb (Gallas-Lopes, 2026).

(2) Study selection

Systematic searches were conducted in Embase, PubMed, and Web of Science on February 25th, 2023. The search strategy was designed to capture studies evaluating the effects of NMDAR antagonists on social behaviour in laboratory animals and combined controlled vocabulary and free-text terms relating to the population (laboratory animals), intervention (NMDAR antagonists), and outcomes (social interaction and social preference). For the population component, validated SYRCLE animal filters were used to ensure broad coverage of laboratory animal studies across species (van der Mierden *et al.*, 2022). No restrictions were applied on language or publication date. Full database-specific search strategies were preregistered and are available at <https://osf.io/5k2br> (Gallas-Lopes *et al.*, 2023b).

Records retrieved from the database searches were exported and imported to Rayyan (Ouzzani *et al.*, 2016), where duplicate entries were identified and removed using a combination of automated detection and manual verification. Study selection was carried out in two phases: an initial screening based on titles and abstracts, followed by a full-text eligibility assessment. In both phases, each record was evaluated independently by two reviewers, and decisions were made blinded to the other reviewer's judgment.

During the title and abstract screening stage, records were excluded only when they clearly failed to meet eligibility criteria related to study design, population, or intervention. Specifically, studies

were excluded if they were not original controlled experiments, did not involve live laboratory animals, or did not include administration of eligible NMDAR antagonists (MK-801, ketamine, or PCP). Records for which eligibility could not be confidently determined were retained for full-text assessment. In the second stage, full texts of potentially eligible articles were assessed in detail using the same criteria applied in the first stage, with the addition of criteria related to the presence of an appropriate control group and the assessment of relevant social behaviour outcomes (social interaction and/or social preference). Any disagreements were resolved through discussion, and when consensus was not achieved, a third reviewer was consulted.

All screening decisions followed a standardised reviewer guide detailing operational definitions of inclusion and exclusion criteria and specifying hierarchical exclusion rules available at osf.io/98fau/files/3pm2n. This guide was developed before screening and applied throughout both screening phases. All screening decisions were managed and archived within Rayyan.

(3) Data collection

Data extraction was performed in two sequential stages: an initial qualitative extraction to characterise study and experimental features, followed by quantitative extraction of outcome data for meta-analysis. Qualitative and quantitative extractions were conducted following predefined guides, available at osf.io/98fau/files/k928m and osf.io/98fau/files/bhdp3, respectively. Both stages were conducted independently by two reviewers using standardised extraction forms developed before data collection. When discrepancies occurred or when differences between extracted quantitative values exceeded 10%, reconciliation was performed by a third reviewer who re-extracted the discrepant information to establish final values.

In the qualitative extraction stage, descriptive information was collected for each experiment, including publication details, animal characteristics (species, strain, sex, age, or developmental stage at both induction and outcome assessment), and intervention parameters (NMDAR antagonist used, dose or concentration, route of administration, frequency and duration of exposure, and interval between drug administration and behavioural testing). Characteristics of the

behavioural assessment were also recorded, including the type of test (social interaction or social preference) and the specific outcome measure.

Following qualitative characterisation, quantitative extraction was performed for all eligible social behaviour outcomes. For each comparison, group-level summary statistics were extracted, including sample size, mean values, and measures of variability (standard deviation or standard error of the mean). When data were reported in text or tables, exact values were preferentially extracted from those sources. When numerical values were not provided, data were estimated from figures using WebPlotDigitizer (<https://automeris.io/>). If it was unclear whether error bars represented standard deviation or standard error, values were conservatively treated as standard error and later converted to standard deviation for effect size estimation. When the sample size was reported as a range, the lowest value was used for effect size estimation, whereas the highest value was used for calculating standard deviation when it had to be derived from the standard error of the mean. When pseudoreplication was suspected, sample sizes were adjusted to reflect the appropriate biological unit. For example, when social behaviour was quantified separately for each animal within interacting pairs, the effective sample size was divided by two to account for non-independence within dyads.

Experiments conducted in distinct sets of animals within the same study were entered as separate effect sizes, while dependence among effect sizes from the same study was accounted for using multilevel meta-analytic models. Studies employing cross-over designs or reusing the same animals across multiple experimental conditions were excluded when it was impossible to isolate results obtained exclusively from naive animals. When multiple experiments were conducted sequentially in the same animals, only data from the first behavioural assessment were retained. Locomotor activity data were extracted only when locomotion was measured in the same animals and within the same experimental context as the social behaviour assessment. When outcomes were assessed at multiple time points within the same experiment, only the earliest post-intervention time point was considered. Outcomes exclusively assessing social novelty preference were not considered eligible. When essential quantitative information, such as sample size or variability measures, was missing or could not be reliably obtained, corresponding authors

were contacted by email on up to two occasions. Studies were excluded when required data could not be retrieved. Studies reporting outcomes exclusively as medians with interquartile ranges were excluded from the quantitative synthesis when raw data were not available to calculate means and standard deviations, and when additional data were not provided by the authors.

(4) Risk of bias

Risk of bias was assessed using SYRCLE's risk of bias tool for animal studies (Hooijmans *et al.*, 2014b), with additional domains addressing pseudoreplication and procedural equivalence. Pseudoreplication refers to the inappropriate treatment of non-independent observations (e.g., multiple measurements from the same animal or cage) as independent experimental units, which can inflate precision and bias effect estimates (Lazic *et al.*, 2018). Procedural equivalence assesses whether experimental groups were handled identically aside from the intervention (e.g., same housing conditions, handling, route of administration, timing, and outcome assessment procedures), minimising performance and detection bias. Due to pervasive poor reporting in animal studies, the allocation concealment domain was omitted, as it cannot be objectively evaluated in most included studies (Hirst *et al.*, 2014). A detailed assessment guide with operational definitions and decision rules for each domain is publicly available at osf.io/98fau/files/5gkct.

Only experiments and behavioural outcomes relevant to this review were considered when assigning risk of bias judgments. When a study included more than one eligible experiment or behavioural test, judgments were made for each relevant experiment and then summarised at the study level. This summary followed a conservative approach: if any relevant experiment within a study was judged as high risk in a given domain, the study was classified as high risk for that domain. Domains included sequence generation, baseline characteristics, random housing, blinding during the experiment, random outcome assessment, blinding during outcome assessment, incomplete outcome data, selective outcome reporting, pseudoreplication, and procedural equivalence. For each domain, risk of bias was rated as low, unclear, or high. An overall study-level classification was then assigned and dichotomised as low or high, with studies rated as high overall risk of bias when at least one domain was judged as high risk. All assessments were

performed independently by two reviewers, and discrepancies were resolved by discussion, with conciliation by a third reviewer when necessary.

Inter-rater reliability was quantified based on initial independent assessments using observed agreement (i.e., the proportion of identical ratings between assessors), Cohen's kappa (Cohen, 1960), and prevalence-adjusted bias-adjusted kappa (PABAK) (Byrt, Bishop & Carlin, 1993), calculated overall and by domain. Risk of bias results were summarised descriptively across domains and studies using summary bar plots and traffic-light plots. All analyses and visualisations were conducted in R (version 4.3.1). The full analysis scripts, package versions, and raw data are publicly available via OSF (<https://osf.io/98fau>) and GitHub (github.com/matheusgills/NMDA-sb/tree/main/risk_of_bias).

(5) Outcome selection and effect size estimation

When more than one outcome assessing the same behavioural domain (social preference, social interaction, or locomotor activity) was extracted for the same group of animals within an experiment, only one outcome was selected for inclusion in the meta-analysis. This approach was adopted to avoid multiple dependent effect sizes arising from the same animals. Outcome selection followed a predefined hierarchy based on the frequency in which specific outcome types were reported across studies and on how directly each measure captured the behavioural construct of interest. Outcomes with different names but assessing the same underlying construct were grouped and ranked accordingly.

For social preference and social interaction, time-based measures reflecting the absolute duration of social behaviour were prioritised, as these were the most frequently reported and provided the most direct quantification of social behaviour. When time-based measures were not available, we selected the available measure that most directly represented the same construct, prioritising proportion- or index-based measures of social behaviour. Latency- or distance-based outcomes (e.g., latency to first contact, distance to the stimulus) were selected only when no other measures were reported.

For locomotor activity, absolute measures of locomotor activity based on total distance or movement trajectories were prioritised. When these were not available, count-based measures of activity, such as line crossings, sector crossings, beam breaks, or locomotor/motor activity counts, were selected, followed by time-based measures, such as activity time or duration of locomotion. Relative, normalised, or derived locomotor measures were included when no absolute locomotor outcomes were reported.

Effect sizes were calculated as standardised mean differences using the Hedges' *g* method, with corresponding sampling variances estimated using the *escalc* function of the *metafor* package in R (version 4.3.1) (Viechtbauer, 2010). To ensure consistent directionality across outcomes, effect sizes were reversed when lower values indicated higher levels of social behaviour, such as latency to first social contact or measures based on inter-individual distance.

The distribution of effect sizes included in the meta-analyses was summarised visually to characterise the structure of the available evidence. Alluvial plots were used to illustrate how effect sizes were distributed across key study characteristics and were constructed using the *ggalluvial* package (Brunson & Read, 2023). Evidence maps were constructed to summarise the number of effect sizes and their mean magnitude across combinations of experimental factors, providing an overview of data density and coverage.

(6) Data synthesis

Meta-analyses were conducted separately for two conceptually distinct domains of social behaviour: social interaction and social preference. Social interaction paradigms were defined as those allowing direct physical interaction between two or more conspecifics. In contrast, social preference paradigms assessed interest in or proximity to a conspecific whose movement or physical contact was restricted (e.g., wire cages or enclosed compartments). Locomotor activity outcomes assessed during social behaviour tests were analysed in parallel with each social domain.

Based on these distinctions and the experimental contrasts of interest, seven independent meta-analyses were performed:

1. Effects of NMDAR antagonists on social interaction;
2. Effects of NMDAR antagonists on locomotor activity measured during social interaction paradigms;
3. Effects of NMDAR antagonists on social preference;
4. Effects of NMDAR antagonists on locomotor activity measured during social preference paradigms;
5. Effects of antipsychotics on NMDAR antagonist-induced alterations in social interaction;
6. Effects of antipsychotics on locomotor activity measured during social interaction paradigms in animals exposed to NMDAR antagonists;
7. Effects of antipsychotics on NMDAR antagonist-induced alterations in social preference.

Effects of antipsychotics on locomotor activity measured during social preference paradigms in NMDAR antagonist-exposed animals were not meta-analysed, as no eligible comparisons reported this outcome within the same animals and behavioural context.

Effect sizes were synthesised using correlated multilevel random-effects models, in which effect sizes were nested within experiments and experiments were nested within studies, to properly account for dependencies. Random intercepts for both study and experiment were included to capture between-study and within-study variability. Models were fitted using restricted maximum likelihood estimation. In addition, cluster-robust variance estimation at the study level was applied to all models using the *clubSandwich* package (Pustejovsky, Pekofsky & Zhang, 2025), providing additional protection against residual dependence.

To account for shared control groups and resulting dependency between effect sizes derived from the same experiment, correlations were modelled using a variance-covariance matrix constructed with the *vca/c* function. A within-experiment correlation coefficient (ρ) of 0.5 was assumed.

Heterogeneity was examined to evaluate how much effect sizes varied beyond sampling error. Cochran's Q statistic was used to test whether heterogeneity was present. The magnitude of heterogeneity was quantified using multilevel I^2 estimates (Nakagawa *et al.*, 2023) and variance

components (σ^2) from the random-effects models, which allowed heterogeneity to be assessed separately at the study and experiment levels.

Pooled effects were summarised graphically using orchard plots constructed with the *orchaRd* package (Nakagawa *et al.*, 2023), which display study-level effect estimates alongside the pooled effect while accounting for multilevel heterogeneity and within-study dependency. To complement confidence intervals around pooled estimates, prediction intervals were calculated to indicate the expected range of true effects that might be observed in future studies, incorporating both sampling error and between-study heterogeneity.

Small-study effects, which may reflect publication bias, were examined using funnel plots and regression-based methods. Funnel plots were constructed using standard error and the inverse square root of the total sample size to assess asymmetry (Egger *et al.*, 1997; Sterne *et al.*, 2011; Zwetsloot *et al.*, 2017). Precision-effect test (PET) and precision-effect estimate with standard error (PEESE) models were fitted by regressing effect sizes on standard error and sampling variance, respectively, within the multilevel framework. Following recommended practice, when the PET indicated evidence of small-study effects (i.e., a significant slope), the PEESE estimate was interpreted as the primary bias-adjusted effect size; otherwise, the PET estimate was retained (Stanley, 2008; Stanley & Doucouliagos, 2014). As an additional sensitivity analysis, we performed a regression using a standard error adjusted for effective sample size. This approach reduces the artifactual correlation between standardized mean differences and their standard errors, providing a complementary assessment of small-study effects. Time-lag bias was assessed using meta-regression with publication year centred around its mean and included as a continuous moderator (Yang *et al.*, 2023).

Meta-regression analyses were conducted within the multilevel framework to examine whether the effects of NMDAR antagonists on social interaction and social preference varied across key experimental and biological characteristics and to assess whether these variables explained part of the between-study heterogeneity. A priori planned categorical moderators included NMDAR antagonist, species, developmental stage at induction, route of administration, dosing schedule, sex, and, when applicable, antipsychotic treatment. All moderator models were fitted; however,

only moderators for which each predictor level was informed by at least five independent studies were considered sufficiently powered and therefore reported and discussed in the main text. For the remaining moderators, sparse data within individual levels limited the precision and interpretability of estimates.

We also performed meta-regressions to examine dose–response patterns using cumulative NMDAR antagonist exposure as a continuous predictor. This analysis was restricted to studies conducted in rats and mice, where reporting of dosing parameters was sufficiently consistent to derive cumulative exposure estimates. When data were available, cumulative exposure was calculated as the product of the administered dose (mg/kg), intervention frequency (number of administrations per day), and duration of exposure (days). This allowed single-administration and repeated-dosing schedules to be expressed on a common exposure scale. To address potential non-linearity and reduce skewness, cumulative exposure was $\log_{10}$ -transformed before analysis. The meta-regression was fitted within the same correlated multilevel random-effects framework, with robust variance estimation.

Multivariate meta-regression models were used to evaluate whether variation in effect sizes could be explained by multiple experimental characteristics simultaneously. Moderators included species, type of NMDAR antagonist, dosing schedule, and developmental stage at induction. Including these predictors jointly allowed the estimation of the independent association of each characteristic while accounting for potential confounding due to imbalance in study designs.

For meta-regression and multivariable models, marginal and conditional R^2 values were calculated to quantify the proportion of heterogeneity explained by the moderators alone (marginal R^2) and by the full model including both moderators and random effects (conditional R^2) (Nakagawa & Schielzeth, 2013).

The full analytical pipeline was applied to the primary meta-analyses evaluating the effects of NMDAR antagonists on social interaction and social preference, which constituted the main confirmatory analyses of this review. For these outcomes, analyses included multilevel random-effects models, categorical moderator analyses, multivariable meta-regression, cumulative

exposure meta-regression, time-lag analyses, assessment of small-study effects, and sensitivity analyses. For locomotor activity outcomes and antipsychotic comparisons, analyses were restricted to the core meta-analytic model and selected sensitivity analyses because of more limited and heterogeneous data structures. The primary meta-analyses are therefore emphasised in the main text.

All analyses were conducted in R (version 4.3.1). The full analysis scripts, package versions, and raw data are publicly available via OSF (<https://osf.io/98fau>) and GitHub (github.com/matheusglls/NMDA-sb/tree/main/meta_analysis).

(7) Sensitivity analyses

Sensitivity analyses were conducted using the same correlated multilevel random-effects models with robust variance estimation applied in the primary analyses. First, to assess the robustness of pooled estimates to the assumed within-experiment correlation used in the variance–covariance matrix, models were re-estimated using alternative values of ρ across a plausible range (0.0, 0.3, 0.5, and 0.8). Second, to assess the influence of individual studies, leave-one-study-out analyses were performed by sequentially excluding each study and re-estimating the pooled effect, allowing evaluation of whether the overall results were disproportionately driven by any single study. Third, analyses were repeated after excluding studies classified as having high overall risk of bias, to examine whether conclusions were sensitive to the inclusion of studies with greater methodological limitations. Consistency between the primary and sensitivity analyses was interpreted as evidence of robustness.

(8) Deviations from protocol

Some qualitative variables were extracted as prespecified, yet were not analysed because reporting was unclear or incomplete in most studies. Locomotor activity was initially planned for extraction whenever reported; however, we ultimately extracted it only when it was measured during the social test and in the same animals, to better evaluate whether locomotor changes could confound social outcomes. Locomotion measured in separate apparatuses or in different animals was therefore not included in our analyses. In addition, the data synthesis approach and risk of

bias analyses were adapted to better accommodate dependent effect sizes and the multilevel structure of the included evidence. Although several subgroup and moderator analyses were prespecified, many predictor levels were informed by too few studies to allow reliable estimation. Finally, we conducted a non-prespecified dose-response meta-regression in rats and mice using log-transformed cumulative NMDAR antagonist exposure.

III. RESULTS

Visualisations of the primary and secondary analyses are provided at matheusglls.github.io/NMDA-sb.

(1) Study selection

A total of 2,261 records were identified through database searches, including Embase ($n = 913$), PubMed ($n = 708$), and Web of Science ($n = 640$). After the removal of 941 duplicate records, 1,320 records remained and were screened based on titles and abstracts. Of these, 401 records were excluded, primarily due to study design ($n = 204$), intervention ($n = 183$), and population ($n = 14$) not meeting the eligibility criteria. Consequently, 919 reports were sought for retrieval, of which 8 reports could not be retrieved. The remaining 911 reports underwent full-text assessment for eligibility. At this stage, 647 reports were excluded for not meeting the following criteria: study design ($n = 147$), population ($n = 11$), intervention ($n = 172$), comparison ($n = 12$), outcome of interest ($n = 288$), and insufficient or missing data ($n = 17$). Ultimately, 264 studies met the inclusion criteria and were included in the quantitative synthesis (Figure 1), including studies in rats (*Rattus* spp.), mice (*Mus* spp.), and zebrafish (*Danio rerio*). The complete list of included studies is available at <https://osf.io/98fau/files/5qevp>.

(2) Risk of bias

Across domains, most studies were judged as having an unclear risk of bias, reflecting widespread under-reporting of key methodological details in the preclinical literature (Figure 2). This was particularly evident for domains related to randomisation procedures. Random sequence generation was judged as unclear in 98% of studies. Similarly, random housing was rated as

unclear in 98% of studies, and random outcome assessment was unclear in 97%, indicating that these procedures were rarely reported in sufficient detail to allow confident judgment. Baseline characteristics were unclear in 61% of studies and judged as low risk in 39%.

Blinding procedures were also inconsistently reported. Blinding of caregivers or investigators was judged as unclear in 85% of studies, low risk in 4%, and high risk in 11%. Blinding of outcome assessment showed comparatively better reporting, with 40% of studies classified as low risk, 58% as unclear, and 2% as high risk.

Domains related to data handling and reporting showed greater variability. Incomplete outcome data were judged as low risk in 21% of studies, unclear in 58%, and high risk in 20%. Selective outcome reporting was more frequently judged as low risk (81%), although 8% of studies were classified as unclear and 11% as high risk. Because preregistered protocols were generally unavailable, these judgments were based primarily on whether outcomes described in the methods were reported in the results.

For design-related domains, pseudoreplication was rated as low risk in 68% of studies, unclear in 18%, and high risk in 14%. High-risk judgments were primarily driven by an inappropriate definition of the experimental unit, especially in social interaction paradigms where individual animals within the same interacting dyad were treated as independent observations. Procedural equivalence was judged as low risk in 75% of studies, unclear in 13%, and high risk in 12%, indicating that most studies applied comparable procedures across groups, although methodological concerns remained in a minority of cases. When summarised across all domains, 55% of studies were classified as overall low risk of bias and 45% as high risk.

Inter-rater reliability for the risk of bias assessment indicated an adequate level of consistency between reviewers, with a Cohen's kappa of 0.643 and a PABAK of 0.627. Of all ratings, 81.3% ($n = 2,147$) represented agreement between reviewers, while 18.7% ($n = 493$) reflected disagreement. Critical disagreement, defined as opposing judgments at the extremes of the scale (low vs high risk), accounted for 4.8% ($n = 128$) of all ratings.

Inter-rater reliability varied across individual risk of bias domains (Figure S1). Observed agreement was highest for sequence generation (0.981), random housing (0.981), and random outcome assessment (0.958); however, corresponding Cohen's kappa values were moderate (0.538, 0.437, and 0.333, respectively), likely reflecting the high prevalence of "unclear" ratings in these domains. Blinding of outcome assessment showed both high observed agreement (0.883) and high agreement beyond chance (kappa 0.765). Lower levels of agreement were observed for incomplete outcome data (observed agreement 0.682, kappa 0.439), selective outcome reporting (observed agreement 0.769, kappa 0.215), pseudoreplication (observed agreement 0.557, kappa 0.245), and procedural equivalence (observed agreement 0.712, kappa 0.179), indicating greater subjectivity in judging these items.

Study- and domain-level risk of bias assessments are reported in the online resource at matheusglls.github.io/NMDA-sb/risk_of_bias/rob_and_interrater.html, together with additional analyses of inter-rater reliability.

(3) Meta-analyses

(a) Effects of NMDAR antagonists on social interaction

All qualitative mappings of study characteristics and the complete results of the analyses conducted for the effects of NMDAR antagonists on social interaction are available at matheusglls.github.io/NMDA-sb/meta_analysis/nmda_antagonist_effects/social_interaction/nmda_effects_social_interaction.html.

(i) Qualitative synthesis

Studies with rats contributed the greatest number of effect sizes ($k = 341$ from 123 studies), followed by mice ($k = 141$ from 57 studies), with a small number obtained from zebrafish ($k = 9$ from 6 studies). For the NMDAR antagonists, PCP accounted for the largest number of effect sizes ($k = 227$ from 78 studies), followed by MK-801 ($k = 174$ from 71 studies) and ketamine ($k = 90$ from 43 studies). The distribution of effect sizes across species and NMDAR antagonists is summarised in Figure 3A, which illustrates the relative contribution of each species and antagonist combination.

Most effect sizes were obtained following intraperitoneal administration ($k = 240$ from 112 studies), representing the dominant administration route across species and antagonists (Figure S2-3). Subcutaneous administration was also frequently used ($k = 224$ from 62 studies). Other administration routes were sparsely represented, including immersion ($k = 7$ from 4 studies), minipump delivery ($k = 9$ from 4 studies), retro-orbital injection ($k = 1$ from 1 study), combined intraperitoneal and subcutaneous administration ($k = 1$ from 1 study), and unclear administration route ($k = 9$ from 4 studies).

When considering the dosing schedule, effect sizes were most commonly derived from repeated administrations ($k = 284$ from 110 studies) and acute administration ($k = 198$ from 79 studies), with continuous exposure observed in only a small number of studies ($k = 9$ from 4 studies) and restricted to MK-801 and PCP experiments (Figure S4).

The developmental timing of induction varied considerably across studies. Adult and juvenile/adolescent animals accounted for the majority of effect sizes (each $k = 123$, from 63 and 44 studies, respectively), while neonatal exposure contributed relatively few ($k = 28$ from 10 studies). In addition, a large number of effect sizes were associated with studies that did not clearly report the developmental stage at induction ($k = 217$ from 70 studies) (Figure S5).

There was a marked imbalance in sex representation. Most effect sizes were derived from male animals ($k = 460$, 93.7%; mouse: $k = 132$; rat: $k = 328$), whereas female-only samples ($k = 16$, 3.3%; mouse: $k = 2$; rat: $k = 10$; zebrafish: $k = 4$) and mixed-sex samples ($k = 11$, 2.2%; mouse: $k = 5$; rat: $k = 1$; zebrafish: $k = 5$) were rarely used. Only a small number of effect sizes were drawn from studies in which sex was unclear ($k = 4$, 0.8%; mouse: $k = 2$; rat: $k = 2$) (Figure S6).

(ii) Quantitative synthesis

The quantitative synthesis included 491 effect sizes from 186 independent studies. Exposure to NMDAR antagonists was associated with lower social interaction relative to controls (*Hedges' g* = -1.90 , 95% CI [-2.18 , -1.62], with a prediction interval of -5.56 to 1.76 ; Figure 3B). Heterogeneity was high ($I^2_{total} = 90.5\%$), with substantial between-study variability ($I^2_{study} = 84.3\%$) and a smaller contribution attributable to within-study clustering ($I^2_{study/exp} = 6.2\%$). The test for heterogeneity was

significant ($Q_{490} = 3230.49$, $p < 0.001$), indicating considerable dispersion in effects across studies. The estimated variance components were $\sigma^2_{study} = 3.179$ and $\sigma^2_{study/exp} = 0.233$.

Visual inspection of the funnel plots (Figure S7) suggested asymmetry, consistent with the presence of publication bias. PET indicated a significant association between effect size and study precision ($slope = -5.94$, $p < 0.001$; $marginal R^2 = 0.906$; $conditional R^2 = 1.000$), providing evidence of small-study effects. PEESE also identified a significant association between effect size and sampling variance ($slope = -1.77$, $p = 0.001$). The bias-adjusted intercept from the PEESE was -0.72 , 95% CI $[-1.16, -0.29]$, indicating that adjustment for small-study effects reduced the magnitude of the effect. Residual heterogeneity remained high ($QE_{489} = 2000.13$, $p < 0.001$; $marginal R^2 = 0.944$; $conditional R^2 = 0.998$). The regression using effective sample-size adjusted standard errors did not detect a significant association between effect size and precision ($slope = -0.66$, $p = 0.364$; $marginal R^2 = 0.002$; $conditional R^2 = 0.932$), suggesting that conclusions regarding small-study effects should be interpreted cautiously.

No significant temporal trend in effect sizes was detected ($slope = -0.037$, $p = 0.133$), indicating no evidence that reported effects systematically changed over time. Heterogeneity remained substantial ($QE_{489} = 3229.45$, $p < 0.001$; $marginal R^2 = 0.026$; $conditional R^2 = 0.935$).

There was evidence of a moderator effect of NMDAR antagonist ($F_{(3, 26.05)} = 56.50$, $p < 0.001$). Pooled estimates differed across antagonists, with significant effects observed for ketamine (-2.54 , 95% CI $[-3.33, -1.75]$), MK-801 (-1.56 , 95% CI $[-2.01, -1.11]$), and PCP (-1.88 , 95% CI $[-2.33, -1.43]$) (Figure 3C). Nonetheless, considerable residual heterogeneity persisted after inclusion of NMDAR antagonist as a moderator ($QE_{488} = 3207.037$, $p < 0.001$; $marginal R^2 = 0.033$; $conditional R^2 = 0.937$). A moderator effect of species was also detected ($F_{(3, 24.15)} = 54.06$, $p < 0.001$). Pooled estimates were significant for mice (-2.11 , 95% CI $[-2.64, -1.57]$) and rats (-1.82 , 95% CI $[-2.16, -1.47]$). Although zebrafish were included in the model, the corresponding estimate (-1.77 , 95% CI $[-3.81, 0.27]$) did not reach statistical significance under cluster-robust inference and was based on a small number of studies (Figure 3D). Residual heterogeneity remained present after accounting for species ($QE_{488} = 3215.040$, $p < 0.001$; $marginal R^2 = 0.005$; $conditional R^2 = 0.933$). A moderator effect of administration schedule was detected ($F_{(2, 29.21)} = 88.99$, $p < 0.001$). Significant

effects were observed across acute (-1.92 , 95% CI $[-2.27, -1.57]$) and repeated/continuous exposure (-1.89 , 95% CI $[-2.25, -1.52]$) paradigms (Figure S8). Substantial heterogeneity remained after inclusion of administration schedule ($QE_{489} = 3230.051$, $p < 0.001$; *marginal* $R^2 = 0.0001$; *conditional* $R^2 = 0.932$). Effect sizes also differed by developmental stage at induction ($F_{(4, 15.86)} = 39.19$, $p < 0.001$). Pooled estimates were significant for adult (-1.98 , 95% CI $[-2.49, -1.46]$), juvenile/adolescent (-1.92 , 95% CI $[-2.56, -1.27]$), and neonatal animals (-1.23 , 95% CI $[-2.09, -0.37]$). Estimates for studies with unclear developmental stage were included in the model but were not interpreted further (Figure S9). Inclusion of the developmental stage as a moderator did not substantially reduce heterogeneity ($QE_{487} = 3011.570$, $p < 0.001$; *marginal* $R^2 = 0.008$; *conditional* $R^2 = 0.931$).

In the multivariate meta-regression including species, NMDAR antagonist, dosing schedule, and developmental stage simultaneously, the overall test of moderators was statistically significant ($F_{(9, 26.73)} = 16.571$, $p < 0.001$). The model explained a modest proportion of the between-study heterogeneity ($QE_{482} = 2934.082$, $p < 0.001$; *marginal* $R^2 = 0.072$; *conditional* $R^2 = 0.942$). These findings indicate that, although part of the variability in effect sizes is associated with differences in experimental design characteristics, most heterogeneity remained unexplained after accounting for these moderators, suggesting that additional methodological and/or biological factors likely contribute to variability across studies.

Log-transformed cumulative NMDAR antagonist exposure was available for 453 effect sizes from 171 independent studies with rats and mice and was examined as a continuous moderator (Figure S10). A significant association between cumulative exposure and effect size was detected ($F_{(1, 8.29)} = 8.07$, $p = 0.021$). Each ten-fold increase in cumulative exposure was associated with an average change of -0.75 Hedges' g units (95% CI $[-1.36, -0.15]$ $p = 0.021$). Considerable unexplained heterogeneity persisted ($QE_{451} = 2780.93$, $p < 0.001$; *marginal* $R^2 = 0.165$; *conditional* $R^2 = 0.957$).

The robustness of the overall pooled effect was examined through a series of sensitivity analyses. Varying the assumed within-experiment correlation resulted in only minor changes in the pooled effect size, with point estimates ranging from -1.95 to -1.85 and overlapping confidence intervals throughout. Leave-one-study-out influence analyses confirmed the stability of the findings;

excluding individual studies minimally changed the pooled estimate, all confidence intervals remained statistically significant, and no single study exerted disproportionate influence on the overall result. The overall effect was also re-estimated after excluding studies classified as high overall risk of bias. This analysis, based on 262 effect sizes from 95 independent studies, yielded a pooled effect size of -2.17 (95% CI $[-2.61, -1.72]$), which was comparable in magnitude to the primary analysis and remained significant. Although heterogeneity remained present ($Q_{261} = 1847.083$, $p < 0.001$), the persistence of the effect after the exclusion of high overall risk of bias studies suggests the main findings are robust to study quality considerations.

(b) Effects of NMDAR antagonists on locomotor activity measured during social interaction paradigms

All qualitative mappings of study characteristics and the complete results of the analyses conducted for the effects of NMDAR antagonists on locomotor activity measured during social interaction paradigms are available at matheusglls.github.io/NMDA-sb/meta_analysis/nmda_antagonist_effects/locomotion_interaction/nmda_effects_social_interaction_loc.html.

(i) Qualitative synthesis

Effect sizes were predominantly derived from rat studies ($k = 150$ from 42 studies), with limited representation from mice ($k = 17$ from 8 studies) and zebrafish ($k = 3$ from 1 study). Among NMDAR antagonists, PCP and MK-801 were most frequently studied ($k = 78$ from 23 studies and $k = 72$ from 24 studies, respectively), whereas ketamine contributed fewer effect sizes ($k = 20$ from 8 studies). The combined distribution across species and antagonists is illustrated in Figure S11A.

(ii) Quantitative synthesis

The quantitative synthesis included 170 effect sizes from 51 independent studies. Overall, exposure to NMDAR antagonists was associated with a significant increase in locomotor activity measured during social interaction paradigms when compared with control animals (*Hedges' g* = 0.40 , 95% CI $[0.05, 0.75]$, with a prediction interval of -1.99 to 2.79 ; Figure S11B). Heterogeneity

was high ($I^2_{total} = 83.8\%$), with the majority attributable to between-study variability ($I^2_{study} = 71.6\%$), and a smaller proportion arising from within-study clustering of experiments ($I^2_{study/exp} = 12.2\%$). The test for heterogeneity was significant ($Q_{169} = 992.02$, $p < 0.001$). The estimated variance components were $\sigma^2_{study} = 1.180$ and $\sigma^2_{study/exp} = 0.201$.

(c) Effects of NMDAR antagonists on social preference

All qualitative mappings of study characteristics and the complete results of the analyses conducted for the effects of NMDAR antagonists on social preference are available at matheusglls.github.io/NMDA-sb/meta_analysis/nmda_antagonist_effects/social_preference/nmda_effects_social_preference.html.

(i) Qualitative synthesis

Mice contributed the largest number of effect sizes ($k = 98$ from 56 studies), followed by rats ($k = 42$ from 17 studies), with a smaller number obtained from zebrafish ($k = 22$ from 8 studies). Across NMDAR antagonists, MK-801 accounted for the largest number of effect sizes ($k = 71$ from 32 studies), followed by ketamine ($k = 57$ from 33 studies) and PCP ($k = 34$ from 18 studies). The distribution of effect sizes across species and NMDAR antagonists is illustrated in Figure 4A, which shows that ketamine and PCP were investigated almost exclusively in rodent models, whereas zebrafish studies were restricted to MK-801 exposure.

Intraperitoneal administration was the most frequent route overall ($k = 100$ from 54 studies; Figures S12–S13) and predominated in mouse and rat studies. Subcutaneous administration was also frequently used ($k = 32$ from 16 studies), particularly in rodent studies. Other administration routes were sparsely represented, including immersion ($k = 22$ from 8 studies), which was restricted to zebrafish experiments, minipump delivery ($k = 1$ from 1 study), and studies in which the route of administration was unclear ($k = 7$ from 4 studies).

In terms of dosing schedule, acute administration contributed the greatest number of effect sizes ($k = 88$ from 28 studies), whereas repeated dosing was also commonly used ($k = 71$ from 52 studies).

Continuous exposure was not frequent ($k = 3$ from 2 studies) and restricted to MK-801 studies (Figure S14).

Studies also varied in the developmental stage at induction. Most effect sizes were derived from adult animals exposed to NMDAR antagonists ($k = 72$ from 31 studies), followed by juvenile/adolescent animals ($k = 50$ from 31 studies). Fewer effect sizes were associated with neonatal exposure ($k = 12$ from 9 studies) or larval stages ($k = 2$ from 1 study). A substantial number of effect sizes originated from studies in which the developmental stage at induction was unclear ($k = 26$ from 9 studies) (Figure S15).

Reporting of sex was uneven across the literature. Most effect sizes were derived from male animals ($k = 128$, 79.0%; mouse: $k = 86$; rat: $k = 40$; zebrafish: $k = 2$). Female-only samples were rare ($k = 4$, 2.5%; mouse: $k = 4$), whereas mixed-sex samples contributed a smaller proportion of the evidence ($k = 23$, 14.2%; mouse: $k = 5$; zebrafish: $k = 18$). A limited number of effect sizes were derived from studies in which sex was unclear ($k = 7$, 4.3%; mouse: $k = 3$; rat: $k = 2$; zebrafish: $k = 2$) (Figure S16).

(ii) Quantitative synthesis

The quantitative synthesis comprised 162 effect sizes derived from 80 independent studies. Overall, exposure to NMDAR antagonists was associated with a reduction in social preference compared with animals in the control group (*Hedges' g* = -1.27, 95% CI [-1.59, -0.95], with a prediction interval of -3.95 to 1.41; Figure 4B). Heterogeneity was high ($I^2_{total} = 89.1\%$). The majority of this heterogeneity was attributable to between-study variability ($I^2_{study} = 88.2\%$), with only a minimal contribution from within-study clustering ($I^2_{study/exp} = 1.0\%$). The test for heterogeneity was significant ($Q_{161} = 719.16$, $p < 0.001$), confirming that observed variation in effect sizes exceeded what would be expected from sampling error alone. Variance components further supported this pattern, with substantial variance at the study level ($\sigma^2_{study} = 1.77$) and comparatively little variance attributable to clustering of multiple effect sizes within experiments ($\sigma^2_{study/exp} = 0.02$).

Visual inspection of the funnel plots suggested marked asymmetry, consistent with small-study effects, which may reflect publication bias (Figure S17). PET indicated a significant association

between effect size and study precision ($slope = -5.63$, $p < 0.001$; $marginal R^2 = 0.802$; $conditional R^2 = 0.938$), providing strong evidence of small-study effects. The PEESE model also identified a significant association between effect size and sampling variance ($slope = -3.50$, $p < 0.001$). The bias-adjusted intercept from the PEESE was small and not statistically significant (Hedges' $g = 0.04$, 95% CI $[-0.21, 0.29]$), indicating that adjustment for small-study effects eliminated the statistical significance of the overall effect estimate. Residual heterogeneity remained high following bias adjustment ($QE_{160} = 390.59$, $p < 0.001$; $marginal R^2 = 0.866$; $conditional R^2 = 0.984$). Using effective-sample-size adjusted standard errors, the regression did not identify a significant association between effect size and precision ($slope = -0.35$, $p = 0.625$; $marginal R^2 = 0.001$; $conditional R^2 = 0.982$), suggesting that evidence for small-study effects should be interpreted with caution.

No evidence of a temporal trend in effect sizes was observed. Meta-regression of effect size on publication year revealed no significant association ($slope = 0.01$, $p = 0.785$). Substantial residual heterogeneity persisted in the time-lag model ($QE_{160} = 712.51$, $p < 0.001$; $marginal R^2 = 0.001$; $conditional R^2 = 0.989$).

There was evidence of a moderator effect of NMDAR antagonist ($F_{(3, 4.14)} = 18.83$, $p = 0.007$). Pooled estimates differed across antagonists, with significant reductions in social preference observed for ketamine (Hedges' $g = -1.36$, 95% CI $[-1.79, -0.94]$), MK-801 (-1.10 , 95% CI $[-1.45, -0.75]$), and PCP (-1.44 , 95% CI $[-1.87, -1.01]$) (Figure 4C). However, substantial residual heterogeneity remained after accounting for NMDAR antagonist ($QE_{159} = 713.960$, $p < 0.001$; $marginal R^2 = 0.012$; $conditional R^2 = 0.984$). A moderator effect of species was also detected ($F_{(3, 14.29)} = 19.81$, $p < 0.001$) (Figure 4D). Pooled estimates were significant for mice (-1.24 , 95% CI $[-1.67, -0.80]$), rats (-0.84 , 95% CI $[-1.34, -0.34]$), and zebrafish (-2.52 , 95% CI $[-4.16, -0.89]$); however, the latter was based on a relatively small number of studies and has wide confidence intervals, and should therefore be interpreted with caution. Inclusion of species as a moderator reduced heterogeneity attributable to within-study clustering, but residual heterogeneity remained significant ($QE_{159} = 696.651$, $p < 0.001$; $marginal R^2 = 0.124$; $conditional R^2 = 1.000$), suggesting that species explained only a limited proportion of the overall variability. A moderator effect of

administration schedule was detected ($F_{(2, 14.93)} = 32.52, p < 0.001$). Significant effects were observed across acute ($-1.00, 95\% \text{ CI } [-1.51, -0.49]$) and repeated/continuous exposure paradigms ($-1.41, 95\% \text{ CI } [-1.84, -0.98]$) (Figure S18). Substantial heterogeneity remained after including the administration schedule as a moderator ($QE_{160} = 610.659, p < 0.001$; *marginal* $R^2 = 0.026$; *conditional* $R^2 = 0.962$). Effect sizes also differed by developmental stage at induction ($F_{(4, 16.02)} = 14.87, p < 0.001$). Pooled estimates were significant for adult animals ($-1.35, 95\% \text{ CI } [-1.80, -0.89]$) and juvenile/adolescent animals ($-1.33, 95\% \text{ CI } [-1.97, -0.68]$). The estimate for neonatal/larval animals was also negative but did not reach statistical significance under cluster-robust inference ($-1.22, 95\% \text{ CI } [-2.48, 0.04]$). Estimates for studies with unclear developmental stage were included in the model but were not interpreted further (Figure S19). Inclusion of the developmental stage as a moderator did not substantially reduce heterogeneity ($QE_{158} = 702.942, p < 0.001$; *marginal* $R^2 = 0.009$; *conditional* $R^2 = 0.990$).

When species, NMDAR antagonist, dosing schedule, and developmental stage were included simultaneously in a multivariate meta-regression model, the overall set of moderators was statistically significant ($F_{(9, 5.39)} = 4.99, p = 0.040$). Despite this, the proportion of heterogeneity accounted for by these variables was limited ($QE_{153} = 555.237, p < 0.001$; *marginal* $R^2 = 0.158$; *conditional* $R^2 = 0.993$), indicating that most between-study variability remained unexplained.

Log-transformed cumulative NMDAR antagonist exposure was available for 135 effect sizes from 68 independent studies conducted in rats and mice (Figure S20). No significant association between cumulative exposure and effect size was detected ($F_{(1, 3.83)} = 7.24, p = 0.057$). Although the estimated slope was negative, indicating larger reductions in social preference with increasing exposure (-0.24 Hedges' g units per ten-fold increase; $95\% \text{ CI } [-0.50, 0.01]$), this association did not reach statistical significance under cluster-robust inference. Substantial unexplained heterogeneity remained after inclusion of cumulative exposure as a moderator ($QE_{133} = 505.06, p < 0.001$; *marginal* $R^2 = 0.057$; *conditional* $R^2 = 0.989$).

Varying the assumed within-experiment correlation resulted in only minimal changes in the pooled effect size, with point estimates ranging from -1.28 to -1.26 and highly overlapping confidence intervals throughout. Leave-one-study-out influence analyses confirmed the stability of the findings;

excluding individual studies minimally changed the pooled estimate, and all confidence intervals remained statistically significant. When studies judged to be at high overall risk of bias were excluded, the meta-analysis comprised 82 effect sizes from 50 independent studies and yielded a pooled effect of -1.22 (95% CI $[-1.59, -0.85]$). This estimate was similar in magnitude to the primary analysis and remained significant. Despite this, heterogeneity remained present ($Q_{81} = 371.19, p < 0.001$).

(d) Effects of NMDAR antagonists on locomotor activity measured during social preference paradigms

All qualitative mappings of study characteristics and the complete results of the analyses conducted for the effects of NMDAR antagonists on locomotor activity measured during social preference paradigms are available at matheusqls.github.io/NMDA-sb/meta_analysis/nmda_antagonist_effects/locomotion_preference/nmda_effects_social_preference_loc.html.

(i) Qualitative synthesis

Most effect sizes were derived from mouse studies ($k = 24$ from 6 independent studies), with relatively few from rats ($k = 5$ from 2 studies) and zebrafish ($k = 5$ from 3 studies). Across NMDAR antagonists, MK-801 was most frequently examined ($k = 18$ from 8 studies), whereas PCP ($k = 9$ from 2 studies) and ketamine ($k = 7$ from 3 studies) were less commonly represented. These distributions are summarised in Figure S21A.

(ii) Quantitative synthesis

The quantitative synthesis included 34 effect sizes derived from 11 independent studies. Overall, exposure to NMDAR antagonists was not associated with a statistically significant change in locomotor activity measured during the social preference test (*Hedges' g* = 0.18, 95% CI $[-0.17, 0.53]$, with a prediction interval ranging from -0.62 to 0.99 ; Figure S21B). Heterogeneity was moderate ($I^2_{total} = 34.4\%$), with most of the variability attributable to between-study differences ($I^2_{study} = 26.8\%$) and a smaller proportion arising from within-study clustering of experiments ($I^2_{study/exp} =$

7.6%). The test for heterogeneity was statistically significant ($Q_{33} = 132.27$, $p < 0.001$). The estimated variance components were $\sigma^2_{study} = 0.076$ and $\sigma^2_{study/exp} = 0.021$.

(e) Effects of antipsychotics on NMDAR antagonist–induced alterations in social interaction

All qualitative mappings of study characteristics and the complete results of the analyses conducted for the effects of antipsychotics on NMDAR antagonist–induced alterations in social interaction are available at matheusglls.github.io/NMDA-sb/meta_analysis/antipsychotics_on_nmda/social_interaction/atp_effects_social_interaction.html.

(i) Qualitative synthesis

Rats contributed the majority of effect sizes ($k = 190$ from 42 independent studies), followed by mice ($k = 58$ from 24 studies). PCP accounted for the largest share of NMDAR antagonist–related effect sizes ($k = 140$ from 33 studies), followed by MK-801 ($k = 72$ from 27 studies), while ketamine contributed comparatively fewer effect sizes ($k = 36$ from 14 studies). A wide range of antipsychotics was investigated, with clozapine contributing the largest number of effect sizes ($k = 70$ from 33 studies), followed by haloperidol ($k = 50$ from 21 studies), risperidone ($k = 39$ from 17 studies), and olanzapine ($k = 28$ from 13 studies). Other antipsychotics, including aripiprazole ($k = 20$ from 8 studies), quetiapine ($k = 8$ from 1 study), sulpiride ($k = 8$ from 3 studies), ziprasidone ($k = 6$ from 3 studies), amisulpride ($k = 5$ from 2 studies), lurasidone ($k = 5$ from 2 studies), chlorpromazine ($k = 3$ from 2 studies), and cariprazine ($k = 6$ from 2 studies), were represented by a smaller number of effect sizes. The distribution of effect sizes across species, NMDAR antagonists, and antipsychotic drugs is illustrated in Figure 5A.

For PCP, the most frequently represented pairings with antipsychotics were with clozapine ($k = 35$), haloperidol ($k = 25$), and risperidone ($k = 25$) (Figure 5C). For MK-801, most estimates involved clozapine ($k = 29$) and haloperidol ($k = 14$). Ketamine-related estimates were less common, with the largest numbers reported for haloperidol ($k = 11$), followed by clozapine and olanzapine (each k

= 6); most other ketamine–antipsychotic pairings were represented by only a small number of effect sizes.

Across antipsychotics, most effect sizes were derived from studies using rats, with fewer estimates available in mice (Figure 5D). In rats, clozapine ($k = 58$), haloperidol ($k = 37$), risperidone ($k = 26$), and olanzapine ($k = 17$) contributed the largest numbers of effect sizes. In mice, the highest representation was for haloperidol ($k = 13$), risperidone ($k = 13$), and olanzapine ($k = 11$), with fewer effect sizes available for the remaining compounds.

(ii) Quantitative synthesis

The quantitative synthesis included 248 effect sizes derived from 66 independent studies. Overall, antipsychotic treatment was associated with increased social interaction in NMDAR antagonist–exposed animals (*Hedges' g* = 0.61, 95% CI [0.38, 0.83], with a prediction interval ranging from −1.06 to 2.27; Figure 5B). Heterogeneity was substantial ($I^2_{total} = 69.2\%$), with nearly half of the variability attributable to between-study differences ($I^2_{study} = 48.0\%$) and a further proportion arising from within-study clustering of experiments ($I^2_{study/exp} = 21.2\%$). The test for heterogeneity was statistically significant ($Q_{247} = 1191.89$, $p < 0.001$). The estimated variance components were $\sigma^2_{study} = 0.471$ and $\sigma^2_{study/exp} = 0.208$.

(f) Effects of antipsychotics on locomotor activity measured during social interaction paradigms in animals exposed to NMDAR antagonists

All qualitative mappings of study characteristics and the complete results of the analyses conducted for the effects of antipsychotics on locomotor activity measured during social interaction paradigms in animals exposed to NMDAR antagonists are available at matheusglls.github.io/NMDA-sb/meta_analysis/antipsychotics_on_nmda/locomotion_interaction/at_p_effects_social_interaction_loc.html.

(i) Qualitative synthesis

Studies using rats contributed almost all effect sizes ($k = 121$ from 18 independent studies), with only one effect size obtained from mice ($k = 1$ from 1 study). PCP accounted for the largest number

of effect sizes ($k = 69$ from 11 studies), followed by MK-801 ($k = 44$ from 7 studies), while ketamine contributed comparatively fewer effect sizes ($k = 9$ from 4 studies). A range of antipsychotics was examined, with clozapine contributing the largest number of effect sizes ($k = 39$ from 11 studies), followed by haloperidol ($k = 29$ from 7 studies), risperidone ($k = 21$ from 7 studies), and olanzapine ($k = 9$ from 3 studies). Other antipsychotics, including aripiprazole ($k = 7$ from 2 studies), quetiapine ($k = 8$ from 1 study), sulpiride ($k = 4$ from 1 study), chlorpromazine ($k = 2$ from 1 study), and cariprazine ($k = 3$ from 1 study), were represented by a smaller number of effect sizes. The distribution of effect sizes across species, NMDAR antagonists, and antipsychotic drugs is illustrated in Figure S22A.

(ii) Quantitative synthesis

The quantitative synthesis included 122 effect sizes derived from 19 independent studies. Overall, antipsychotic treatment was associated with a significant reduction in locomotor activity measured during social interaction paradigms when compared to animals treated with NMDAR antagonists alone (*Hedges' g* = -0.49, 95% CI [-0.70, -0.29], with a prediction interval ranging from -1.31 to 0.32; Figure S22B). Heterogeneity was moderate ($I^2_{total} = 33.7\%$), with variability arising from both between-study differences ($I^2_{study} = 11.0\%$) and within-study clustering of experiments ($I^2_{study/exp} = 22.7\%$). The test for heterogeneity was statistically significant ($Q_{121} = 678.70$, $p < 0.001$). The estimated variance components were $\sigma^2_{study} = 0.044$ and $\sigma^2_{study/exp} = 0.090$.

(g) Effects of antipsychotics on NMDAR antagonist-induced alterations in social preference

All qualitative mappings of study characteristics and the complete results of the analyses conducted for the effects of antipsychotics on NMDAR antagonist-induced alterations in social preference are available at matheusglls.github.io/NMDA-sb/meta_analysis/antipsychotics_on_nmda/social_preference/atp_effects_social_preference.html.

(i) Qualitative synthesis

Mice contributed the majority of effect sizes ($k = 29$ from 17 independent studies), followed by rats ($k = 8$ from 3 studies), with a small number obtained from zebrafish ($k = 3$ from 1 study). Ketamine accounted for the largest number of effect sizes ($k = 20$ from 10 studies), followed by MK-801 ($k = 14$ from 6 studies), while PCP contributed fewer effect sizes overall ($k = 6$ from 5 studies). Several antipsychotics were investigated, with risperidone contributing the largest number of effect sizes ($k = 14$ from 10 studies), followed by clozapine ($k = 7$ from 6 studies), aripiprazole ($k = 6$ from 4 studies), olanzapine ($k = 6$ from 5 studies), and haloperidol ($k = 5$ from 4 studies). Other antipsychotics, including sulpiride ($k = 1$ from 1 study) and amisulpride ($k = 1$ from 1 study), were represented by a single effect size each. The distribution of effect sizes across species, NMDAR antagonists, and antipsychotic drugs is illustrated in Figure 6A.

Within the ketamine studies, risperidone was the most frequently reported antipsychotic ($k = 11$), with fewer effect sizes for clozapine ($k = 3$), haloperidol ($k = 3$), olanzapine ($k = 2$), and a single estimate for amisulpride ($k = 1$) (Figure 6C). Under MK-801, most estimates involved aripiprazole ($k = 6$), followed by olanzapine ($k = 3$) and risperidone ($k = 3$), with single estimates for haloperidol ($k = 1$) and sulpiride ($k = 1$). PCP contributed comparatively few estimates overall and was largely represented by clozapine ($k = 4$), with single estimates for haloperidol ($k = 1$) and olanzapine ($k = 1$).

Across species, mouse studies contributed the bulk of the evidence for social preference outcomes, particularly for risperidone ($k = 12$), alongside clozapine ($k = 6$), haloperidol ($k = 4$), olanzapine ($k = 4$), and aripiprazole ($k = 3$) (Figure 6D). Rat studies contributed fewer effect sizes, mainly for aripiprazole ($k = 3$) and risperidone ($k = 2$), with single estimates for clozapine ($k = 1$), olanzapine ($k = 1$), and amisulpride ($k = 1$). Zebrafish data were limited to single estimates for olanzapine, haloperidol, and sulpiride ($k = 1$ each).

(ii) Quantitative synthesis

The quantitative synthesis included 40 effect sizes derived from 21 independent studies. Overall, antipsychotic treatment was associated with a significant increase in social preference in animals

exposed to NMDAR antagonists (*Hedges' g* = 0.94, 95% CI [0.31, 1.57], with a prediction interval ranging from -1.82 to 3.70; Figure 6B). Heterogeneity was high (I^2_{total} = 85.2%), with all observed variability attributable to between-study differences (I^2_{study} = 85.2%), and no additional contribution from within-study clustering of experiments ($I^2_{study/exp}$ = 0%). The test for heterogeneity was statistically significant (Q_{39} = 353.19, p < 0.001). The estimated variance components were σ^2_{study} = 1.66 and $\sigma^2_{study/exp}$ = 0.

IV. DISCUSSION

Across the included studies, exposure to NMDAR antagonists was associated with reduced social behaviour overall; however, the robustness of this pattern varied depending on the behavioural paradigm and on bias diagnostics. In social interaction tests, NMDAR antagonists significantly reduced social interaction relative to controls, with substantial heterogeneity and wide prediction intervals; although the effect size was attenuated after adjustment for small-study effects, it remained statistically significant. For social preference paradigms, the pooled estimate was also significant, yet this effect was eliminated after bias adjustment, indicating greater sensitivity to imprecise studies and potential publication bias. Locomotor activity assessed during social paradigms provided important interpretive context, as antagonists increased activity during social interaction tests but showed no clear effects during social preference. Finally, clinically approved antipsychotics were associated with higher social interaction and social preference in animals exposed to NMDAR antagonists, with reduced locomotion in social interaction paradigms, indicating that apparent normalisation of social outcomes co-occurred with changes in general activity.

In this systematic review, we analysed social interaction and social preference as distinct outcome domains, not only because they operationalise social behaviour in different ways, but also because the evidence bases underpinning each domain have different profiles. Social interaction outcomes were drawn from a much larger evidence base and covered a longer publication period (1989 to 2023, median 2013), whereas social preference outcomes came from a smaller and more recent literature (2009 to 2023, median 2016). Although these differences in publication era and reporting practices could, in principle, contribute to discrepancies between domains, the time-lag analyses

showed no systematic association between effect size and publication year in either domain, suggesting that simple temporal trends are unlikely to explain the contrast between paradigms.

Both domains were represented across the same species, including rats, mice, and zebrafish, but the distribution of studies across species and interventions differed between paradigms. In the extracted dataset, social interaction outcomes predominantly comprised studies with rats and administration of PCP or MK-801, and most frequently employed repeated dosing designs alongside acute exposures to antagonists. Social preference outcomes relied more heavily on mice, included a proportionally larger contribution from zebrafish studies, and were informed by a relatively greater share of acute paradigms. Even with this methodological diversity, our results are consistent with the view that NMDAR antagonism reduces social behaviour, whether measured as reciprocal interaction or as social preference (Neill *et al.*, 2010, 2014; Lee & Zhou, 2019). However, although the average effects in both domains pointed towards reduced sociability, the prediction intervals indicate that future studies could plausibly observe anything from reduced sociability to no detectable change and, in some contexts, increased sociability. Consistent with this, the literature includes both social interaction and social preference studies reporting little or no change (Becker & Grecksch, 2004; Egerton *et al.*, 2008; Seillier & Giuffrida, 2016; Herzog *et al.*, 2020) as well as studies reporting increased sociability following NMDAR antagonist exposure (Dunn *et al.*, 1989; Jenkins *et al.*, 2008; Rawat *et al.*, 2022; Giongo *et al.*, 2023).

This combination of methodological diversity and wide prediction intervals requires careful interpretation, with particular emphasis on bias diagnostics (Riley, Higgins & Deeks, 2011; Sterne *et al.*, 2011; Stanley & Doucouliagos, 2014; Zwetsloot *et al.*, 2017). Concerns about inflated effect estimates and the selective publication of clear behavioural findings are well documented in preclinical animal research (Sena *et al.*, 2010; Korevaar, Hooft & Ter Riet, 2011; Macleod *et al.*, 2015). In our data, this issue was particularly evident for social preference. Although the conventional pooled estimate suggested reduced preference, this signal was not robust to adjustment for small-study effects, and the estimate was compatible with no effect. By contrast, the social interaction effect remained significant after adjustment for small-study effects, suggesting a more robust signal in that domain. Notably, regressions using effective-sample-size adjusted

standard errors were not statistically significant for either primary outcome, indicating that evidence for small-study effects was not fully consistent across alternative modelling approaches and should therefore be interpreted cautiously. However, small-study effects are only one pathway through which effect sizes can be inflated in preclinical research (Button *et al.*, 2013; Munafò *et al.*, 2017). Low statistical power and selective reporting can exaggerate observed effects, with analytic flexibility yielding overconfident inferences even when individual decisions appear defensible (Simmons, Nelson & Simonsohn, 2011; Gelman & Loken, 2013; Button *et al.*, 2013). This concern is relevant to the present dataset, as extracted experiments generally used small sample sizes, typically below 10 animals per experimental group. In addition, incomplete reporting of key design and analytical features can make it difficult to distinguish bias from genuine heterogeneity across laboratories and protocol families (Landis *et al.*, 2012; Percie du Sert *et al.*, 2020). Taken together, these issues lead to cautious interpretation of the magnitude and generalisability of pooled effects, including those that remain significant after bias adjustment.

These concerns are amplified by the way social behaviour is operationalised in the primary literature. In the included studies, social interaction was most commonly assessed by measuring total interaction time, often complemented by measures of direct contact or investigatory behaviours. For social preference, sociability was most commonly indexed by time spent in a social zone or compartment. However, the operational definition of interaction and related endpoints varies significantly across studies, resulting in ambiguity regarding the observed variables and limiting comparability across experiments. This diversity raises the possibility that different studies are quantifying distinct behavioural components that are not directly comparable and may not carry the same ecological relevance or pharmacological sensitivity, even when they are discussed as evidence of a single social deficit (Saverino C & Gerlai R, 2008; Millan *et al.*, 2014; Arakawa, 2018; Kondrakiewicz *et al.*, 2019; Geng & Peterson, 2019; Ogi A *et al.*, 2020). A more comprehensive mapping of relevant social outcomes remains needed, alongside explicit specification of which component of social behaviour is being targeted in a given paradigm or outcome. Apart from that, consistent reporting of primary outcomes would help limit selective emphasis on favourable endpoints and reduce the scope for *post hoc* rationalisation, when multiple measures are available,

a concern that has also been documented in behavioural neuroscience (Simmons *et al.*, 2011; Rosso *et al.*, 2024).

Moderator analyses were used to test whether variation in effects across studies was consistent with identifiable experimental and biological differences and to provide a partial explanation for the substantial heterogeneity observed in both domains (Yang *et al.*, 2023). The moderator results were broadly in line with the literature by showing reduced social behaviour across NMDAR antagonists, species, schedule, and developmental stages (Neill *et al.*, 2010, 2014; Millan *et al.*, 2014; Lee & Zhou, 2019). For zebrafish, the interaction estimate was imprecise and did not remain significant under robust inference. Given the small number of zebrafish studies and the diversity of paradigms and designs used in this subset, this pattern is more consistent with limited evidence and heterogeneity than with a reliable species difference (Benvenuti *et al.*, 2022).

Even after accounting for these moderators, both in analyses evaluating individual moderators separately and in multivariable models combining multiple experimental characteristics simultaneously, heterogeneity remained high in both social interaction and social preference, suggesting that important sources of between-study variation were not captured by the moderators that could be analysed with adequate support. This is not unexpected in preclinical evidence synthesis, where studies are often small and highly diverse, and differences in species, dosing regimens, testing context, and endpoint definitions can generate wide dispersion in effect sizes (Hooijmans *et al.*, 2014a; Vesterinen *et al.*, 2014; Yang *et al.*, 2023; Ineichen *et al.*, 2024b). In our dataset, incomplete reporting and sparse data within many candidate moderator levels limited the ability to examine additional sources of heterogeneity with confidence, consistent with broader evidence that deficiencies in reporting are common in animal research and constrain what systematic reviews can explain (Leung *et al.*, 2018; Hunniford *et al.*, 2021; Ritskes-Hoitinga & Pound, 2022; Wilson *et al.*, 2023).

To examine whether dosing intensity contributed to variation in effect sizes and residual heterogeneity, we modelled cumulative exposure in rodents, as dosing information was sufficiently consistent in rat and mouse studies. Higher cumulative exposure to NMDAR antagonists was associated with larger reductions in social interaction, while the corresponding analysis in social

preference did not show a clear association in our dataset. This direction is consistent with reviews of rodent NMDAR hypofunction models, which describe repeated or subchronic antagonist regimens as a way to produce more sustained behavioural phenotypes, including social withdrawal (Neill *et al.*, 2014; Lee & Zhou, 2019). Direct within-study comparisons of acute versus repeated or chronic exposure, or of different exposure durations, are rare in the social behaviour literature, and regimen-related differences are therefore often inferred across studies. Across the small subset of studies that directly compare exposure regimens within the same experiment, findings remain mixed, with some studies suggesting regimen or timing sensitivity and others reporting little or no difference between acute and repeated schedules (Sams-Dodd, 1996; Becker & Grecksch, 2004; Li, He & Munro, 2012; Hanks *et al.*, 2013; McKibben, Reynolds & Jenkins, 2014; Cieřlik *et al.*, 2019). Accordingly, the evidence base still lacks adequately powered head-to-head comparisons that systematically manipulate regimen parameters across NMDAR antagonists, leaving an important gap in the social behaviour literature.

Hyperlocomotion is a well-documented feature of NMDAR antagonist exposure in preclinical models and is commonly interpreted as a phenotype with relevance to the positive symptoms of schizophrenia (Large, 2007; Neill *et al.*, 2010, 2014; Frohlich & Van Horn, 2014; Lee & Zhou, 2019; Benvenuti *et al.*, 2022). In the social interaction studies, NMDAR antagonists were associated with higher locomotor activity measured during the test, while reciprocal social engagement was reduced. This is relevant for interpreting social interaction assays because the test sums active social behaviour over a brief encounter in a novel arena; thus, scores depend on how animals allocate time between social contact, exploration, and other competing activities during the session (File & Hyde, 1978; File & Seth, 2003; Toth & Neumann, 2013; Kondrakiewicz *et al.*, 2019). More broadly, the potential for non-social behavioural changes to shape apparent social effects is not unique to NMDAR antagonists and has been explored in the context of psychostimulant (Deak *et al.*, 2009; řlamberová *et al.*, 2010, 2015) and sedative (File & Seth, 2003) drugs and across behavioural models where altered arousal, exploration, or motor output can shift time budget-based social endpoints (Nadler *et al.*, 2004; Silverman *et al.*, 2010; Yang *et al.*, 2011). In NMDAR antagonist models, reduced social interaction alongside increased activity makes a simple

global behavioural suppression account unlikely, being consistent with drug-induced changes in how behaviour is organised during the encounter, including more competing non-social exploration and shorter or more fragmented affiliative bouts. Accordingly, stronger inference would come from routinely pairing social outcomes with within-task locomotion and complementary ethological measures such as bout structure and partner-directed investigation and from explicitly distinguishing acute drug state effects from delayed or post-drug phenotypes when claims about social dysfunction or pharmacological rescue are made (File & Hyde, 1978; File & Seth, 2003; Nadler *et al.*, 2004; Silverman *et al.*, 2010; Yang *et al.*, 2011; Peleh *et al.*, 2019; Jabarin, Netser & Wagner, 2022).

In the social preference literature, locomotor activity did not show a clear overall effect in our synthesis, but this does not imply that non-social influences are insignificant. Preference readouts are fundamentally time budget measures of where an animal spends time in the apparatus (Nadler *et al.*, 2004; Yang *et al.*, 2011). Small shifts in thigmotaxis, side bias, or avoidance of exposed zones can alter chamber or zone occupancy and therefore alter derived sociability indices even when overall distance travelled is not consistently different across groups (Yang *et al.*, 2011; Kondrakiewicz *et al.*, 2019). For this reason, social preference protocols recommend reporting activity proxies such as chamber entries and exploration measures alongside occupancy-based sociability indices (Nadler *et al.*, 2004; Yang *et al.*, 2011), yet these measures are not consistently collected or reported; focusing on one or two preference metrics can miss non-social drivers of performance that affect interpretation (Kondrakiewicz *et al.*, 2019; Winiarski *et al.*, 2022; Jabarin *et al.*, 2022). Taken together, the absence of a clear locomotor signal in this subset may reflect both genuine task-specific expression of the phenotype and the smaller, potentially selective evidence base for within-task activity reporting in social preference studies.

Antipsychotic treatment was associated with increased social behaviour, but the accompanying locomotor pattern suggests that this should not be interpreted as selective restoration of social function in isolation. Because social interaction measures depend on animals being active in the arena (File & Hyde, 1978; File & Seth, 2003), the co-occurrence of higher social behaviour with reduced locomotor activity suggests that at least part of the apparent rescue may reflect a change

in time allocation and reduced behavioural competition during the encounter rather than a selective change in social processes. This interpretation is also consistent with the broader point that many antipsychotics can produce sedation and reduced activity in humans (Hynes *et al.*, 2020; Nomura *et al.*, 2025) and laboratory animals (Bernardi, De Souza & Palermo Neto, 1981; van der Zwaal *et al.*, 2010; McOmish *et al.*, 2012; Lian *et al.*, 2015), so state effects are a plausible contributor to time-based social outcomes when activity shifts in parallel. The antipsychotic literature is also concentrated in a small number of compounds, most commonly clozapine, haloperidol, risperidone, and olanzapine.

These findings should be interpreted alongside a previous systematic review and meta-analysis of treatments for social interaction impairments in animal models of schizophrenia, which also reported beneficial treatment effects, particularly for atypical antipsychotics (Hazani, Lavidor & Weller, 2022). However, important differences in scope and design limit direct comparison. That review synthesised treatment effects across multiple schizophrenia-related models and treatment classes, included only studies in which the model produced a social interaction deficit, and restricted outcomes to social interaction time or number of interactions. This design is appropriate for asking whether treatments reverse an established deficit, but it may favour evidence in which the expected phenotype and treatment response are already present, thereby overemphasising pharmacological responsiveness as a marker of model validity. It also does not distinguish between social interaction and social preference paradigms or evaluate locomotor activity, both of which are central to interpreting social outcomes. Our results are consistent with antipsychotic-related increases in social behaviour, but extend previous evidence by placing these effects within a more specific NMDAR antagonist framework and by showing that apparent social improvement should be interpreted cautiously.

None of this excludes pharmacologically meaningful effects on social behaviour, because antipsychotics may increase the likelihood that animals approach and maintain contact with social stimuli. Even if such effects are genuine in the assay, the modest and inconsistent clinical signal for primary negative symptoms and social dysfunction means that apparent preclinical rescue should still not be overinterpreted (Correll & Schooler, 2020; Spark *et al.*, 2022). Ultimately, basic

measures of social proximity in laboratory animals may lack construct validity to meaningfully reflect the complex negative symptoms experienced by patients with schizophrenia. This caution fits with wider meta-research on translation, which highlights gaps between preclinical and clinical practice, concerns about preclinical study rigour, and attenuation from animal to human effects, with only a small fraction of animal-tested interventions reaching regulatory approval (Ineichen *et al.*, 2024a). It also aligns with translational critiques from other disease areas, where even apparently clinically relevant outcomes can have limited translational validity and where a narrow set of experimental practices can dominate the evidence base (Berg *et al.*, 2024). In this context, the antipsychotic evidence mainly shows that the assays are pharmacologically sensitive, while the link between changes in preclinical social endpoints and clinically meaningful improvements in social functioning remains uncertain. More broadly, our findings challenge the view that NMDAR antagonist-based social behaviour assays provide strong predictive validity for the negative symptom domain of schizophrenia. Although these assays detect social behavioural changes and are responsive to clinically approved antipsychotics, this pattern does not establish that they capture clinically meaningful social dysfunction. Instead, the apparent translational gap may reflect not only true cross-species or mechanistic failure, but also the cumulative effects of publication bias, heterogeneity, limited construct validity, and non-specific behavioural changes within the preclinical evidence base.

Risk of bias assessment highlighted a familiar pattern in *in vivo* neuroscience, where key safeguards are often ignored or reported too vaguely to support confident judgment, particularly for randomisation, allocation concealment, and blinded outcome assessment (Worp *et al.*, 2010; Macleod *et al.*, 2015). Although excluding studies rated at high overall risk did not materially change the main findings, this sensitivity analysis has clear limits when most judgements fall into an “unclear” category rather than “low” risk. In that situation, the key issue is not simply a small subset of demonstrably flawed studies, but the possibility that important design and analysis practices are inconsistently implemented and selectively reported. The high prevalence of “unclear” ratings, therefore, primarily reflects reporting limitations and reinforces the need for better transparency in preclinical methods and analysis. This is particularly relevant because efforts to

improve reporting quality, including guideline-based initiatives, have not consistently delivered large improvements in practice (Leung *et al.*, 2018; Hair *et al.*, 2019), so risk of bias assessment will remain constrained unless reporting standards are more reliably adopted and enforced.

This methodological uncertainty is also evident in the inter-rater reliability results; agreement was lower for domains that are central to behavioural inference but hard to judge from published reports, notably pseudoreplication and procedural equivalence. Pseudoreplication is a recurring concern because the effective experimental unit is often clustered, for example, within dyads, cages, or litters, yet analyses and reporting may still treat individual observations as independent. This is especially relevant for reciprocal interaction designs, where behaviour arises from a coupled pair and non-independence can inflate apparent precision and produce overconfident estimates if it is not handled appropriately in the analysis (Lazic, 2010; Lazic *et al.*, 2018, 2020). Procedural equivalence is difficult to appraise retrospectively because many determinants of social behaviour depend on details that are inconsistently reported, including partner preparation and matching, handling and habituation, apparatus configuration, scoring definitions, and whether groups received the same vehicles, routes of administration, and administration to test timings apart from the intervention itself. Accordingly, disagreement for pseudoreplication and procedural equivalence likely reflects more than incomplete reporting alone. These issues are still unevenly recognised and operationalised across fields, as they are not always treated as core validity threats in routine behavioural practice, even though guidance on rigour and transparent reporting in animal research has become increasingly explicit (Lazic *et al.*, 2018; Percie du Sert *et al.*, 2020; Smith, 2020; Wilson *et al.*, 2023).

Sex representation in the available evidence is highly imbalanced, which limits how confidently the reported NMDAR antagonist effects on social outcomes can be generalised. Sex bias is a long-standing feature of preclinical neuroscience (Beery & Zucker, 2011; Shansky & Woolley, 2016). In preclinical schizophrenia research, this imbalance is sometimes justified by concerns about variability in hormonal regulation and behaviour of females (Kokras & Dalla, 2014; Raimondi, Tripp & Ostroff, 2023), and is sometimes framed as less problematic because epidemiological evidence suggests a higher incidence of this disorder in males (McGrath *et al.*, 2004; Leucht *et al.*,

2025). However, the variability rationale is not well supported across many behavioural endpoints (Prendergast, Onishi & Zucker, 2014), and policy shifts have explicitly encouraged routine consideration of sex as a biological variable rather than treating male-only sampling as the default (Clayton & Collins, 2014). In this context, the predominance of male-only datasets should be treated as a constraint on external validity and on the identification of boundary conditions. A straightforward implication for this field is that future work should more routinely include both sexes and report sex-stratified results where feasible to clarify whether social phenotypes and pharmacological sensitivity are robust across sex, strengthening the generalisability of the findings.

Species coverage is a further limitation for external validity. The evidence base in this review rests largely on rodent protocols, with a smaller zebrafish subset. Although we screened records involving other species, including primates (Mao *et al.*, 2008) and invertebrate models (Moulin *et al.*, 2023; Dunham *et al.*, 2024), these studies did not meet the inclusion criteria for controlled social interaction or social preference outcomes with extractable group-level data. As a result, the conclusions are anchored in a narrow set of species and assay traditions. A natural implication is that broader cross-species sampling, alongside deliberate introduction of heterogeneity within species and across laboratories, would help clarify which NMDAR antagonist-related social effects are robust and which are contingent on particular implementations (Richter *et al.*, 2010, 2011; Voelkl *et al.*, 2020; Carneiro *et al.*, 2023).

A limitation of this review is that the searches were completed in 2023 in a field that continues to grow, as NMDAR antagonist models remain widely used. It is therefore likely that additional eligible studies have been published since the search date. Even so, the number of included studies was large, which strengthens the synthesis and allows a broad characterisation of the evidence base available up to that point. At the same time, inference is constrained by substantial heterogeneity in methodology and reporting, much of which could not be accounted for by the moderators that could be analysed with adequate support. In parts of the dataset, sensitivity to small study effects suggests that conventional pooled estimates may overstate effects, and risk of bias assessments were often dominated by unclear judgments because key methodological details were not reported consistently. Taken together, these limitations indicate that the pooled effects summarise a

heterogeneous literature in which social behaviour is operationalised in multiple ways, rather than providing a single estimate that applies uniformly across settings.

Within these limitations, this review synthesises a large preclinical literature and summarises how NMDAR antagonism relates to social outcomes as they are most commonly assessed. By integrating findings across compounds, species, and paradigms, it provides an updated account of the consistency of reported effects and the degree of uncertainty that accompanies them. Bringing this work together also helps place individual study results in a broader context and provides a clearer basis for interpreting the wider preclinical literature on NMDAR antagonist models.

V. CONCLUSIONS

1. Across preclinical studies, NMDAR antagonists are associated with reduced social behaviour on average, but effects are highly variable across experiments and contexts;
2. In social interaction tests, the overall evidence supports a robust reduction in reciprocal social interaction, even when accounting for small study effects;
3. In social preference paradigms, estimates also suggest reduced sociability, but bias adjustment eliminates the effect, indicating that the evidence is more sensitive to small-study effects;
4. Locomotor activity provides critical interpretive context. NMDAR antagonists increase activity during social interaction tests, highlighting the importance of separating social effects from non-social performance factors;
5. By contrast, locomotor activity measured within social preference paradigms shows no clear overall change, despite an overall reduction in social preference induced by NMDAR antagonists;
6. Clinically approved antipsychotics increase social endpoints in NMDAR antagonist-exposed animals, but parallel reductions in locomotion suggest that apparent normalisation may arise from altered activity and time allocation, which limits the extent to which these findings can be interpreted as modelling clinically meaningful improvements in social functioning;

7. The evidence base is constrained by substantial heterogeneity and by incomplete reporting of core design and analysis safeguards, including randomisation, blinding, and unit of analysis decisions, as well as marked sex imbalance;
8. Future studies will be most informative if they strengthen transparency and methodological rigour, predefine primary social outcomes, routinely report within-task activity measures alongside social endpoints, and use designs and analyses that appropriately account for non-independent observations.

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Assistance from ChatGPT 5.2 (OpenAI, San Francisco, CA, USA) was used to support the optimisation of text and formatting throughout the manuscript. All outputs were critically reviewed, edited, and validated by the authors, who take full responsibility for the content and interpretation of the work.

VII. AUTHOR CONTRIBUTIONS

Conceptualization: MG-L, MBA, MA, BDA, LMB, TCC, DVM, AP-A, DAR, DJS, APH; **Data curation:** MG-L, APH; **Formal Analysis:** MG-L; **Funding acquisition:** APH; **Investigation:** MG-L, MBA, MA, BDA, LMB, TCC, DVM, AP-A, DAR, DJS, APH; **Methodology:** MG-L, MBA, MA, BDA, LMB, TCC, DVM, AP-A, DAR, DJS, APH; **Project administration:** MG-L, APH; **Resources:**

APH; **Software:** MG-L; **Supervision:** APH; **Validation:** MG-L; **Visualization:** MG-L; **Writing – original draft:** MG-L; **Writing – review & editing:** MBA, MA, BDA, LMB, TCC, DVM, AP-A, DAR, DJS, APH.

VIII. DATA ACCESSIBILITY

Data and code supporting the findings of this study are openly available via the Open Science Framework osf.io/98fau (Gallas-Lopes *et al.*, 2023a) and GitHub github.com/matheusglls/NMDA-sb (Gallas-Lopes, 2026).

IX. REFERENCES

- ADLER, C.M., MALHOTRA, A.K., ELMAN, I., GOLDBERG, T., EGAN, M., PICKAR, D. & BREIER, A. (1999) Comparison of ketamine-induced thought disorder in healthy volunteers and thought disorder in schizophrenia. *The American Journal of Psychiatry* **156**, 1646–1649.
- ALLEN, R.M. & YOUNG, S.J. (1978) Phencyclidine-induced psychosis. *The American Journal of Psychiatry* **135**, 1081–1084.
- ARAKAWA, H. (2018) Ethological approach to social isolation effects in behavioral studies of laboratory rodents. *Behavioural Brain Research* **341**, 98–108.
- BAHADORAN, Z., MIRMIRAN, P., KASHFI, K. & GHASEMI, A. (2020) Importance of Systematic Reviews and Meta-analyses of Animal Studies: Challenges for Animal-to-Human Translation.
- BECKER, A. & GRECKSCH, G. (2004) Ketamine-induced changes in rat behaviour: a possible animal model of schizophrenia. Test of predictive validity. *Progress in Neuro-Psychopharmacology and Biological Psychiatry* **28**, 1267–1277.
- BEERY, A.K. & ZUCKER, I. (2011) Sex bias in neuroscience and biomedical research. *Neuroscience and Biobehavioral Reviews* **35**, 565–572.
- BENVENUTTI, R., GALLAS-LOPES, M., MARCON, M., RESCHKE, C.R., HERRMANN, A.P. & PIATO, A. (2022) Glutamate NMDA Receptor Antagonists with Relevance to Schizophrenia: A Review of Zebrafish Behavioral Studies. *Current Neuropharmacology* **20**, 494–509.
- BERG, I., HÄRVELID, P., ZÜRRER, W.E., ROSSO, M., REICH, D.S. & INEICHEN, B.V. (2024) Which experimental factors govern successful animal-to-human translation in multiple sclerosis drug development? A systematic review and meta-analysis. *eBioMedicine* **110**. Elsevier.
- BERNARDI, M.M., DE SOUZA, H. & PALERMO NETO, J. (1981) Effects of single and long-term haloperidol administration on open field behavior of rats. *Psychopharmacology* **73**, 171–175.
- BRUNSON, J.C. & READ, Q.D. (2023) ggalluvial: Alluvial Plots in ‘ggplot2’. <https://cran.r-project.org/web/packages/ggalluvial/index.html> [accessed 10 January 2026].
- BUTTON, K.S., IOANNIDIS, J.P.A., MOKRYSZ, C., NOSEK, B.A., FLINT, J., ROBINSON, E.S.J. & MUNAFÒ, M.R. (2013) Power failure: why small sample size undermines the reliability of neuroscience. *Nature Reviews. Neuroscience* **14**, 365–376.

- BYRT, T., BISHOP, J. & CARLIN, J.B. (1993) Bias, prevalence and kappa. *Journal of Clinical Epidemiology* **46**, 423–429.
- CARNEIRO, C.F.D., DRUDE, N., HÜLSEMAN, M., COLLAZO, A. & TOELCH, U. (2023) Mapping strategies towards improved external validity in preclinical translational research. *Expert Opinion on Drug Discovery* **18**, 1273–1285.
- CATTS, V.S., LAI, Y.L., WEICKERT, C.S., WEICKERT, T.W. & CATTS, S.V. (2016) A quantitative review of the postmortem evidence for decreased cortical *N*-methyl-d-aspartate receptor expression levels in schizophrenia: How can we link molecular abnormalities to mismatch negativity deficits? *Biological Psychology* **116**, 57–67.
- CIEŚLIK, P., RADULSKA, A., PELIKANT-MAŁECKA, I., PŁOSKA, A., KALINOWSKI, L. & WIEROŃSKA, J.M. (2019) Reversal of MK-801-Induced Disruptions in Social Interactions and Working Memory with Simultaneous Administration of LY487379 and VU152100 in Mice. *International Journal of Molecular Sciences* **20**, 2781.
- CLAYTON, J.A. & COLLINS, F.S. (2014) Policy: NIH to balance sex in cell and animal studies. *Nature* **509**, 282–283. Nature Publishing Group.
- COHEN, J. (1960) A Coefficient of Agreement for Nominal Scales. *Educational and Psychological Measurement* **20**, 37–46. SAGE Publications Inc.
- CORRELL, C.U. & SCHOOLER, N.R. (2020) Negative Symptoms in Schizophrenia: A Review and Clinical Guide for Recognition, Assessment, and Treatment. *Neuropsychiatric Disease and Treatment* **16**, 519–534.
- DEAK, T., ARAKAWA, H., BEKKEDAL, M.Y.V. & PANKSEPP, J. (2009) Validation of a novel social investigation task that may dissociate social motivation from exploratory activity. *Behavioural Brain Research* **199**, 326–333.
- DUNHAM, K.E., KHALED, K.H., WEIZMAN, L. & VENTON, B.J. (2024) Microdosing ketamine in *Drosophila* does not block serotonin reuptake, but causes complex behavioral changes mediated by glutamate and serotonin receptors. *Journal of Neurochemistry* **168**, 1097–1112.
- DUNN, R.W., CORBETT, R. & FIELDING, S. (1989) Effects of 5-HT_{1A} receptor agonists and NMDA receptor antagonists in the social interaction test and the elevated plus maze. *European Journal of Pharmacology* **169**, 1–10.
- EGERTON, A., REID, L., MCGREGOR, S., COCHRAN, S.M., MORRIS, B.J. & PRATT, J.A. (2008) Subchronic and chronic PCP treatment produces temporally distinct deficits in attentional set shifting and prepulse inhibition in rats. *Psychopharmacology* **198**, 37–49.
- EGGER, M., SMITH, G.D., SCHNEIDER, M. & MINDER, C. (1997) Bias in meta-analysis detected by a simple, graphical test. *BMJ* **315**, 629–634. British Medical Journal Publishing Group.
- ELEFThERIOU, C., GIACHETTI, S., HICKSON, R., KAMNIOTI-DUMONT, L., TEMPLAAR, R., AALTONEN, A., TSOUKALA, E., KIM, N., FRYER-PETRIDIS, L., HENLEY, C., ERDEM, C., WILSON, E., MAIO, B., YE, J., PIERCE, J.C., ET AL. (2025) Better statistical reporting does not lead to statistical rigour: lessons from two decades of pseudoreplication in mouse-model studies of neurological disorders. *Molecular Autism* **16**, 30.
- FILE, S.E. & HYDE, J. R. G. (1978) Can Social Interaction Be Used to Measure Anxiety? *British Journal of Pharmacology* **62**, 19–24.
- FILE, S.E. & SETH, P. (2003) A review of 25 years of the social interaction test. *European Journal of Pharmacology* **463**, 35–53.

- FROHLICH, J. & VAN HORN, J.D. (2014) Reviewing the ketamine model for schizophrenia. *Journal of Psychopharmacology* **28**, 287–302. SAGE Publications Ltd STM.
- FUSAR-POLI, P., PAPANASTASIOU, E., STAHL, D., ROCCHETTI, M., CARPENTER, W., SHERGILL, S. & MCGUIRE, P. (2015) Treatments of Negative Symptoms in Schizophrenia: Meta-Analysis of 168 Randomized Placebo-Controlled Trials. *Schizophrenia Bulletin* **41**, 892–899.
- GALDERISI, S., MUCCI, A., BUCHANAN, R.W. & ARANGO, C. (2018) Negative symptoms of schizophrenia: new developments and unanswered research questions. *The Lancet Psychiatry* **5**, 664–677. Elsevier.
- GALLAS-LOPES, M. (2026) matheusglls/SRMA-NMDA-social-behaviour: Initial public release. Zenodo. <https://zenodo.org/records/18284730> [accessed 17 January 2026].
- GALLAS-LOPES, M., ABREU, M., ANDRADES, M., ARBO, B., BASTOS, L., CAETANO, T., MÜLLER, D., ALVES, A., STEIN, D. & ROSA, D. (2023a) Effects of NMDA antagonists on social behavior: a systematic review and meta-analysis of preclinical studies. *OSF*. OSF.
- GALLAS-LOPES, M., ABREU, M., ANDRADES, M., ARBO, B., BASTOS, L., CAETANO, T., MÜLLER, D., ROSA, D., DE CASTRO SANTOS, A. & STEIN, D. (2023b) Effects of NMDA antagonists on social behavior: a systematic review and meta-analysis of preclinical studies - Search Terms. *OSF*.
- GALLAS-LOPES, M., ABREU, M.B., ANDRADES, M., ARBO, B.D., BASTOS, L.M., CAETANO, T.C., MÜLLER, D.V., ROSA, D.A., SANTOS, A., STEIN, D.J. & HERRMANN, A.P. (2024) Effects of NMDA receptor antagonists on social behavior: a systematic review and meta-analysis of preclinical studies relevant to schizophrenia. PROSPERO. <https://www.crd.york.ac.uk/PROSPERO/view/CRD42023402129>.
- GELMAN, A. & LOKEN, E. (2013) The garden of forking paths: Why multiple comparisons can be a problem, even when there is no “fishing expedition” or “p-hacking” and the research hypothesis was posited ahead of time. *Department of Statistics, Columbia University* **348**, 3.
- GENG, Y. & PETERSON, R.T. (2019) The zebrafish subcortical social brain as a model for studying social behavior disorders. *Disease Models & Mechanisms* **12**, dmm039446.
- GIONGO, F.K., GALLAS-LOPES, M., BENVENUTTI, R., SACHETT, A., BASTOS, L.M., ROSA, A.R. & HERRMANN, A.P. (2023) Effects of Taurine in Mice and Zebrafish Behavioral Assays With Translational Relevance to Schizophrenia. *International Journal of Neuropsychopharmacology* **26**, 125–136.
- HAGAN, J.J. & JONES, D.N.C. (2005) Predicting Drug Efficacy for Cognitive Deficits in Schizophrenia. *Schizophrenia Bulletin* **31**, 830–853.
- HAIR, K., MACLEOD, M.R., SENA, E.S., SENA, E.S., HAIR, K., MACLEOD, M.R., HOWELLS, D., BATH, P., IRVINE, C., MACCALLUM, C., MORRISON, G., CLARK, A., ALVINO, G., DOHM, M., LIAO, J., ET AL. (2019) A randomised controlled trial of an Intervention to Improve Compliance with the ARRIVE guidelines (IICARus). *Research Integrity and Peer Review* **4**, 12.
- HALLER, J. & BAKOS, N. (2002) Stress-induced social avoidance: A new model of stress-induced anxiety? *Physiology & Behavior* **77**, 327–332.
- HANKS, A.N., DLUGOLENSKI, K., HUGHES, Z.A., SEYMOUR, P.A. & MAJCHERZAK, M.J. (2013) Pharmacological disruption of mouse social approach behavior: Relevance to negative symptoms of schizophrenia. *Behavioural Brain Research* **252**, 405–414.

- HARVEY, P.D., STRASSNIG, M.T. & SILBERSTEIN, J. (2019) Prediction of disability in schizophrenia: Symptoms, cognition, and self-assessment. *Journal of Experimental Psychopathology* **10**, 2043808719865693. SAGE Publications.
- HAZANI, R., LAVIDOR, M. & WELLER, A. (2022) Treatments for Social Interaction Impairment in Animal Models of Schizophrenia: A Critical Review and Meta-analysis. *Schizophrenia Bulletin* **48**, 1179–1193.
- HERZOG, D.P., MELLEMA, R.M., REMMERS, F., LUTZ, B., MÜLLER, M.B. & TRECCANI, G. (2020) Sexually Dimorphic Behavioral Profile in a Transgenic Model Enabling Targeted Recombination in Active Neurons in Response to Ketamine and (2R,6R)-Hydroxynorketamine Administration. *International Journal of Molecular Sciences* **21**, 2142.
- HIRST, J.A., HOWICK, J., ARONSON, J.K., ROBERTS, N., PERERA, R., KOSHIARIS, C. & HENEGHAN, C. (2014) The Need for Randomization in Animal Trials: An Overview of Systematic Reviews. *PLOS ONE* **9**, e98856. Public Library of Science.
- HOOIJMANS, C.R., INT'HOUT, J., RITSKES-HOITINGA, M. & ROVERS, M.M. (2014a) Meta-Analyses of Animal Studies: An Introduction of a Valuable Instrument to Further Improve Healthcare. *ILAR Journal* **55**, 418–426.
- HOOIJMANS, C.R., ROVERS, M.M., DE VRIES, R.B., LEENAARS, M., RITSKES-HOITINGA, M. & LANGENDAM, M.W. (2014b) SYRCLE's risk of bias tool for animal studies. *BMC Medical Research Methodology* **14**, 43.
- HOWES, O., FUSAR-POLI, P. & OSUGO, M. (2023) Treating negative symptoms of schizophrenia: current approaches and future perspectives. *The British Journal of Psychiatry* **223**, 332–335.
- HUNNIFORD, V.T., MONTROY, J., FERGUSON, D.A., AVEY, M.T., WEVER, K.E., MCCANN, S.K., FOSTER, M., FOX, G., LAFRENIERE, M., GHALY, M., MANNELL, S., GODWINSKA, K., GENTLES, A., SELIM, S., MACNEIL, J., ET AL. (2021) Epidemiology and reporting characteristics of preclinical systematic reviews. *PLoS Biology* **19**, e3001177.
- HYNES, C., MCWILLIAMS, S., CLARKE, M., FITZGERALD, I., FEENEY, L., TAYLOR, M., BOLAND, F. & KEATING, D. (2020) Check the effects: systematic assessment of antipsychotic side-effects in an inpatient cohort. *Therapeutic Advances in Psychopharmacology* **10**, 2045125320957119. SAGE Publications Ltd STM.
- INEICHEN, B.V., FURRER, E., GRÜNINGER, S.L., ZÜRRER, W.E. & MACLEOD, M.R. (2024a) Analysis of animal-to-human translation shows that only 5% of animal-tested therapeutic interventions obtain regulatory approval for human applications. *PLOS Biology* **22**, e3002667. Public Library of Science.
- INEICHEN, B.V., HELD, U., SALANTI, G., MACLEOD, M.R. & WEVER, K.E. (2024b) Systematic review and meta-analysis of preclinical studies. *Nature Reviews Methods Primers* **4**, 72. Nature Publishing Group.
- JABARIN, R., NETSER, S. & WAGNER, S. (2022) Beyond the three-chamber test: toward a multimodal and objective assessment of social behavior in rodents. *Molecular Autism* **13**, 41.
- JENKINS, T.A., HARTE, M.K., MCKIBBEN, C.E., ELLIOTT, J.J. & REYNOLDS, G.P. (2008) Disturbances in social interaction occur along with pathophysiological deficits following sub-chronic phencyclidine administration in the rat. *Behavioural Brain Research* **194**, 230–235.
- KOKRAS, N. & DALLA, C. (2014) Sex differences in animal models of psychiatric disorders. *British Journal of Pharmacology* **171**, 4595–4619.

- KONDRAKIEWICZ, K., KOSTECKI, M., SZADZIŃSKA, W. & KNAPSKA, E. (2019) Ecological validity of social interaction tests in rats and mice. *Genes, Brain and Behavior* **18**, e12525.
- KOREVAAR, D.A., HOOFT, L. & TER RIET, G. (2011) Systematic reviews and meta-analyses of preclinical studies: Publication bias in laboratory animal experiments. *Laboratory Animals* **45**, 225–230. SAGE Publications.
- KRAUSE, M., ZHU, Y., HUH, M., SCHNEIDER-THOMA, J., BIGHELLI, I., NIKOLAKOPOULOU, A. & LEUCHT, S. (2018) Antipsychotic drugs for patients with schizophrenia and predominant or prominent negative symptoms: a systematic review and meta-analysis. *European Archives of Psychiatry and Clinical Neuroscience* **268**, 625–639.
- LAHTI, A.C., WEILER, M.A., TAMARA MICHAELIDIS, B.A., PARWANI, A. & TAMMINGA, C.A. (2001) Effects of ketamine in normal and schizophrenic volunteers. *Neuropsychopharmacology: Official Publication of the American College of Neuropsychopharmacology* **25**, 455–467.
- LANDAUER, M.R. & BALSTER, R.L. (1982) A new test for social investigation in mice: Effects of d-amphetamine. *Psychopharmacology* **78**, 322–325.
- LANDIS, S.C., AMARA, S.G., ASADULLAH, K., AUSTIN, C.P., BLUMENSTEIN, R., BRADLEY, E.W., CRYSTAL, R.G., DARNELL, R.B., FERRANTE, R.J., FILLIT, H., FINKELSTEIN, R., FISHER, M., GENDELMAN, H.E., GOLUB, R.M., GOUDREAU, J.L., ET AL. (2012) A call for transparent reporting to optimize the predictive value of preclinical research. *Nature* **490**, 187–191. Nature Publishing Group.
- LARGE, C.H. (2007) Do NMDA receptor antagonist models of schizophrenia predict the clinical efficacy of antipsychotic drugs? *Journal of Psychopharmacology* **21**, 283–301. SAGE Publications Ltd STM.
- LAZIC, S.E. (2010) The problem of pseudoreplication in neuroscientific studies: is it affecting your analysis? *BMC neuroscience* **11**, 5.
- LAZIC, S.E., CLARKE-WILLIAMS, C.J. & MUNAFÒ, M.R. (2018) What exactly is ‘N’ in cell culture and animal experiments? *PLOS Biology* **16**, e2005282. Public Library of Science.
- LAZIC, S.E., MELLOR, J.R., ASHBY, M.C. & MUNAFÒ, M.R. (2020) A Bayesian predictive approach for dealing with pseudoreplication. *Scientific Reports* **10**, 2366. Nature Publishing Group.
- LEE, G. & ZHOU, Y. (2019) NMDAR Hypofunction Animal Models of Schizophrenia. *Frontiers in Molecular Neuroscience* **12**. Frontiers.
- LEUCHT, S., LEUCHT, C., HUH, M., CHAIMANI, A., MAVRIDIS, D., HELFER, B., SAMARA, M., RABAIOLI, M., BÄCHER, S., CIPRIANI, A., GEDDES, J.R., SALANTI, G. & DAVIS, J.M. (2017) Sixty Years of Placebo-Controlled Antipsychotic Drug Trials in Acute Schizophrenia: Systematic Review, Bayesian Meta-Analysis, and Meta-Regression of Efficacy Predictors. *The American Journal of Psychiatry* **174**, 927–942.
- LEUCHT, S., SIAFIS, S., MCGRATH, J.J., MCGORRY, P., HOWES, O.D., TAMMINGA, C., CARR, R., BIGHELLI, I., SCHNEIDER-THOMA, J., PRILLER, J. & DAVIS, J.M. (2025) Schizophrenia. *Nature Reviews Disease Primers* **11**, 83. Nature Publishing Group.
- LEUNG, V., ROUSSEAU-BLASS, F., BEAUCHAMP, G. & PANG, D.S.J. (2018) ARRIVE has not ARRIVED: Support for the ARRIVE (Animal Research: Reporting of in vivo Experiments) guidelines does not improve the reporting quality of papers in animal welfare, analgesia or anesthesia. *PLOS ONE* **13**, e0197882. Public Library of Science.
- LI, D., PAN, Q., XIAO, Y. & HU, K. (2024) Advances in the study of phencyclidine-induced schizophrenia-like animal models and the underlying neural mechanisms. *Schizophrenia* **10**, 65. Nature Publishing Group.

- LI, M., HE, W. & MUNRO, R. (2012) Differential effects of acute amphetamine and phencyclidine treatment and withdrawal from repeated amphetamine or phencyclidine treatment on social interaction and social memory in rats. *PsyCh Journal* **1**, 56–68.
- LIAN, J., DE SANTIS, M., HE, M. & DENG, C. (2015) Risperidone-induced weight gain and reduced locomotor activity in juvenile female rats: The role of histaminergic and NPY pathways. *Pharmacological Research* **95–96**, 20–26.
- MACLEOD, M.R., LAWSON McLEAN, A., KYRIAKOPOULOU, A., SERGHIU, S., DE WILDE, A., SHERRATT, N., HIRST, T., HEMBLADE, R., BAHOR, Z., NUNES-FONSECA, C., POTLURU, A., THOMSON, A., BAGINSKITAE, J., EGAN, K., VESTERINEN, H., ET AL. (2015) Risk of Bias in Reports of In Vivo Research: A Focus for Improvement. *PLoS Biology* **13**, e1002273.
- MAO, C.V., HORI, E., MAIOR, R., ONO, T. & NISHIJO, H. (2008) A Primate Model of Schizophrenia Using Chronic PCP Treatment. *Reviews in the Neurosciences* **19**.
- MCGRATH, J., SAHA, S., WELHAM, J., EL SAADI, O., MACCAULEY, C. & CHANT, D. (2004) A systematic review of the incidence of schizophrenia: the distribution of rates and the influence of sex, urbanicity, migrant status and methodology. *BMC Medicine* **2**, 13.
- MCKIBBEN, C.E., REYNOLDS, G.P. & JENKINS, T.A. (2014) Analysis of sociability and preference for social novelty in the acute and subchronic phencyclidine rat. *Journal of Psychopharmacology (Oxford, England)* **28**, 955–963.
- MCOMISH, C.E., LIRA, A., HANKS, J.B. & GINGRICH, J.A. (2012) Clozapine-induced locomotor suppression is mediated by 5-HT_{2A} receptors in the forebrain. *Neuropsychopharmacology: Official Publication of the American College of Neuropsychopharmacology* **37**, 2747–2755.
- VAN DER MIERDEN, S., HOOIJMANS, C.R., TILLEMA, A.H., REHN, S., BLEICH, A. & LEENAARS, C.H. (2022) Laboratory animals search filter for different literature databases: PubMed, Embase, Web of Science and PsycINFO. *Laboratory Animals* **56**, 279–286. SAGE Publications.
- MILLAN, M.J., FONE, K., STECKLER, T. & HORAN, W.P. (2014) Negative symptoms of schizophrenia: Clinical characteristics, pathophysiological substrates, experimental models and prospects for improved treatment. *Eur. Neuropsychopharmacol.* **24**, 645–692. M.J. Millan, Pole of Innovation in Neuropsychiatry, Institut de Recherche Servier, 125 Chemin de Ronde, 78290 Croissy-sur-Seine, Paris, France.
- MITCHELL, E.J., BRETT, R.R., ARMSTRONG, J.D., SILLITO, R.R. & PRATT, J.A. (2020) Temporal dissociation of phencyclidine: Induced locomotor and social alterations in rats using an automated homecage monitoring system – implications for the 3Rs and preclinical drug discovery. *Journal of Psychopharmacology (Oxford, England)* **34**, 709–715.
- MIYAMOTO, Y. & NITTA, A. (2014) Behavioral Phenotypes for Negative Symptoms in Animal Models of Schizophrenia. *Journal of Pharmacological Sciences* **126**, 310–320.
- MOSOLOV, S.N. & YALTONSKAYA, P.A. (2022) Primary and Secondary Negative Symptoms in Schizophrenia. *Frontiers in Psychiatry* **12**. Frontiers.
- MOULIN, T.C., STOJANOVIC, T., RAJESH, R.P., PAREEK, T., DONZELLI, L., WILLIAMS, M.J. & SCHIÖTH, H.B. (2023) Effects of Transient Administration of the NMDA Receptor Antagonist MK-801 in *Drosophila melanogaster* Activity, Sleep, and Negative Geotaxis. *Biomedicines* **11**, 192. Multidisciplinary Digital Publishing Institute.
- MUNAFÒ, M.R., NOSEK, B.A., BISHOP, D.V.M., BUTTON, K.S., CHAMBERS, C.D., PERCIE DU SERT, N., SIMONSOHN, U., WAGENMAKERS, E.-J., WARE, J.J. & IOANNIDIS, J.P.A. (2017) A manifesto for reproducible science. *Nature Human Behaviour* **1**, 0021. Nature Publishing Group.

- NADLER, J.J., MOY, S.S., DOLD, G., AND SIMMONS, N., PEREZ, A., YOUNG, N.B., BARBARO, R.P., PIVEN, J., MAGNUSON, T.R. & CRAWLEY, J.N. (2004) Automated apparatus for quantitation of social approach behaviors in mice. *Genes, Brain and Behavior* **3**, 303–314.
- NAKAGAWA, S., LAGISZ, M., O'DEA, R.E., POTTIER, P., RUTKOWSKA, J., SENIOR, A.M., YANG, Y. & NOBLE, D.W.A. (2023) orchaRd 2.0: An R package for visualising meta-analyses with orchard plots. *Methods in Ecology and Evolution* **14**, 2003–2010.
- NAKAGAWA, S. & SCHIELZETH, H. (2013) A general and simple method for obtaining R² from generalized linear mixed-effects models. *Methods in Ecology and Evolution* **4**, 133–142.
- NEILL, J.C., BARNES, S., COOK, S., GRAYSON, B., IDRIS, N.F., MCLEAN, S.L., SNIGDHA, S., RAJAGOPAL, L. & HARTE, M.K. (2010) Animal models of cognitive dysfunction and negative symptoms of schizophrenia: Focus on NMDA receptor antagonism. *Pharmacology & Therapeutics* **128**, 419–432.
- NEILL, J.C., HARTE, M.K., HADDAD, P.M., LYDALL, E.S. & DWYER, D.M. (2014) Acute and chronic effects of NMDA receptor antagonists in rodents, relevance to negative symptoms of schizophrenia: A translational link to humans. *European Neuropsychopharmacology* **24**, 822–835.
- NEWCOMER, J.W., FARBER, N.B., JEVTOVIC-TODOROVIC, V., SELKE, G., MELSON, A.K., HERSHEY, T., CRAFT, S. & OLNEY, J.W. (1999) Ketamine-Induced NMDA Receptor Hypofunction as a Model of Memory Impairment and Psychosis. *Neuropsychopharmacology* **20**, 106–118. Nature Publishing Group.
- NOMURA, N., SIAFIS, S., SCHNEIDER-THOMA, J., BRANDT, L., PARK, J., EFTHIMIOU, O., PRILLER, J., DAVIS, J.M., TAKEUCHI, H. & LEUCHT, S. (2025) The trajectory of sedative adverse events caused by antipsychotics: a meta-analysis of individual participant data from randomised, placebo-controlled, clinical trials in acute phase schizophrenia. *The Lancet Psychiatry* **12**, 266–275. Elsevier.
- Ogi A, LICITRA R, NAEF V, MARCHESE M, FRONTE B, GAZZANO A, & SANTORELLI FM (2020) Social Preference Tests in Zebrafish: A Systematic Review. *Frontiers in veterinary science* **7**, 590057. Switzerland.
- OUZZANI, M., HAMMADY, H., FEDOROWICZ, Z. & ELMAGARMID, A. (2016) Rayyan—a web and mobile app for systematic reviews. *Systematic Reviews* **5**, 210.
- PAGE, M.J., MCKENZIE, J.E., BOSSUYT, P.M., BOUTRON, I., HOFFMANN, T.C., MULROW, C.D., SHAMSEER, L., TETZLAFF, J.M., AKL, E.A., BRENNAN, S.E., CHOU, R., GLANVILLE, J., GRIMSHAW, J.M., HRÓBJARTSSON, A., LALU, M.M., ET AL. (2021a) The PRISMA 2020 statement: an updated guideline for reporting systematic reviews. *BMJ* **372**, n71. British Medical Journal Publishing Group.
- PAGE, M.J., MOHER, D., BOSSUYT, P.M., BOUTRON, I., HOFFMANN, T.C., MULROW, C.D., SHAMSEER, L., TETZLAFF, J.M., AKL, E.A., BRENNAN, S.E., CHOU, R., GLANVILLE, J., GRIMSHAW, J.M., HRÓBJARTSSON, A., LALU, M.M., ET AL. (2021b) PRISMA 2020 explanation and elaboration: updated guidance and exemplars for reporting systematic reviews. *BMJ* **372**, n160. British Medical Journal Publishing Group.
- PELEH, T., IKE, K.G.O., WAMS, E.J., LEBOS, E.P. & HENGERER, B. (2019) The reverse translation of a quantitative neuropsychiatric framework into preclinical studies: Focus on social interaction and behavior. *Neuroscience and Biobehavioral Reviews* **97**, 96–111.
- PERCIE DU SERT, N., HURST, V., AHLUWALIA, A., ALAM, S., AVEY, M.T., BAKER, M., BROWNE, W.J., CLARK, A., CUTHILL, I.C., DIRNAGL, U., EMERSON, M., GARNER, P., HOLGATE, S.T., HOWELLS, D.W., KARP,

- N.A., ET AL. (2020) The ARRIVE guidelines 2.0: Updated guidelines for reporting animal research. *PLoS biology* **18**, e3000410.
- POMAROL-CLOTET, E., HONEY, G.D., MURRAY, G.K., CORLETT, P.R., ABSALOM, A.R., LEE, M., MCKENNA, P.J., BULLMORE, E.T. & FLETCHER, P.C. (2006) Psychological effects of ketamine in healthy volunteers. Phenomenological study. *The British Journal of Psychiatry: The Journal of Mental Science* **189**, 173–179.
- PRENDERGAST, B.J., ONISHI, K.G. & ZUCKER, I. (2014) Female mice liberated for inclusion in neuroscience and biomedical research. *Neuroscience & Biobehavioral Reviews* **40**, 1–5.
- PUSTEJOVSKY, J.E., PEKOFISKY, S. & ZHANG, J. (2025) clubSandwich: Cluster-Robust (Sandwich) Variance Estimators with Small-Sample Corrections. <https://cran.r-project.org/web/packages/clubSandwich/index.html> [accessed 10 January 2026].
- RAIMONDI, G.M., TRIPP, R. & OSTROFF, L. (2023) The Track-by-Day method for monitoring the rodent estrous cycle. *Current protocols* **3**, e747.
- RAWAT, R., TUNC-OZCAN, E., MCGUIRE, T.L., PENG, C.-Y. & KESSLER, J.A. (2022) Ketamine activates adult-born immature granule neurons to rapidly alleviate depression-like behaviors in mice. *Nature Communications* **13**, 2650.
- RICHTER, S.H., GARNER, J.P., AUER, C., KUNERT, J. & WÜRBEL, H. (2010) Systematic variation improves reproducibility of animal experiments. *Nature Methods* **7**, 167–168. Nature Publishing Group.
- RICHTER, S.H., GARNER, J.P., ZIPSER, B., LEWEJOHANN, L., SACHSER, N., TOUMA, C., SCHINDLER, B., CHOURBAJI, S., BRANDWEIN, C., GASS, P., STIPDONK, N. VAN, HARST, J. VAN DER, SPRUIJT, B., VÖIKAR, V., WOLFER, D.P., ET AL. (2011) Effect of Population Heterogenization on the Reproducibility of Mouse Behavior: A Multi-Laboratory Study. *PLOS ONE* **6**, e16461. Public Library of Science.
- RILEY, R.D., HIGGINS, J.P.T. & DEEKS, J.J. (2011) Interpretation of random effects meta-analyses. *BMJ* **342**, d549. British Medical Journal Publishing Group.
- RITSKES-HOITINGA, M. & POUND, P. (2022) The role of systematic reviews in identifying the limitations of preclinical animal research, 2000–2022: part 1. *Journal of the Royal Society of Medicine* **115**, 186–192.
- ROSSO, M., HERRERA, A., WÜRBEL, H. & VOELKL, B. (2024) Evidence of HARKing in mouse behavioural tests of anxiety. *Royal Society Open Science* **11**, 231744.
- SABE, M., ZHAO, N., CRIPPA, A. & KAISER, S. (2021) Antipsychotics for negative and positive symptoms of schizophrenia: dose-response meta-analysis of randomized controlled acute phase trials. *NPJ Schizophrenia* **7**, 43.
- SAMS-DODD, F. (1996) Phencyclidine-induced stereotyped behaviour and social isolation in rats: a possible animal model of schizophrenia. *Behavioural Pharmacology* **7**, 3–23.
- SAVERINO C & GERLAI R (2008) The social zebrafish: behavioral responses to conspecific, heterospecific, and computer animated fish. *Behavioural brain research* **191**, 77–87. Netherlands.
- SEILLIER, A. & GIUFFRIDA, A. (2016) Disruption of social cognition in the sub-chronic PCP rat model of schizophrenia: Possible involvement of the endocannabinoid system. *European Neuropsychopharmacology: The Journal of the European College of Neuropsychopharmacology* **26**, 298–309.

- SENA, E.S., WOPR, H.B. VAN DER, BATH, P.M.W., HOWELLS, D.W. & MACLEOD, M.R. (2010) Publication Bias in Reports of Animal Stroke Studies Leads to Major Overstatement of Efficacy. *PLOS Biology* **8**, e1000344. Public Library of Science.
- SHANSKY, R.M. & WOOLLEY, C.S. (2016) Considering Sex as a Biological Variable Will Be Valuable for Neuroscience Research. *The Journal of Neuroscience: The Official Journal of the Society for Neuroscience* **36**, 11817–11822.
- SILVERMAN, J.L., YANG, M., LORD, C. & CRAWLEY, J.N. (2010) Behavioural phenotyping assays for mouse models of autism. *Nature reviews. Neuroscience* **11**, 490–502.
- SIMMONS, J.P., NELSON, L.D. & SIMONSOHN, U. (2011) False-Positive Psychology: Undisclosed Flexibility in Data Collection and Analysis Allows Presenting Anything as Significant. *Psychological Science* **22**, 1359–1366. SAGE Publications Inc.
- ŠLAMBEROVÁ, R., MIKULECKÁ, A., MACÚCHOVÁ, E., HREBIČKOVÁ, I., ŠEVČÍKOVÁ, M., NOHEJLOVÁ, K. & POMETLOVÁ, M. (2015) Effects of psychostimulants on social interaction in adult male rats. *Behavioural Pharmacology* **26**, 776–785.
- ŠLAMBEROVÁ, R., MIKULECKÁ, A., POMETLOVÁ, M., SCHUTOVÁ, B., HRUBÁ, L. & DEYKUN, K. (2010) The effect of methamphetamine on social interaction of adult male rats. *Behavioural Brain Research* **214**, 423–427.
- SMITH, A.J. (2020) Guidelines for planning and conducting high-quality research and testing on animals. *Laboratory Animal Research* **36**, 21.
- SPARK, D.L., FORNITO, A., LANGMEAD, C.J. & STEWART, G.D. (2022) Beyond antipsychotics: a twenty-first century update for preclinical development of schizophrenia therapeutics. *Translational Psychiatry* **12**, 147. Nature Publishing Group.
- STANLEY, T. (2008) Meta-Regression Methods for Detecting and Estimating Empirical Effects in the Presence of Publication Selection*. *Oxford Bulletin of Economics and Statistics* **70**, 103–127. Department of Economics, University of Oxford.
- STANLEY, T.D. & DOUCOLIAGOS, H. (2014) Meta-regression approximations to reduce publication selection bias. *Research Synthesis Methods* **5**, 60–78.
- STERNE, J.A.C., SUTTON, A.J., IOANNIDIS, J.P.A., TERRIN, N., JONES, D.R., LAU, J., CARPENTER, J., RÜCKER, G., HARBORD, R.M., SCHMID, C.H., TETZLAFF, J., DEEKS, J.J., PETERS, J., MACASKILL, P., SCHWARZER, G., ET AL. (2011) Recommendations for examining and interpreting funnel plot asymmetry in meta-analyses of randomised controlled trials. *BMJ* **343**, d4002. British Medical Journal Publishing Group.
- TOTH, I. & NEUMANN, I.D. (2013) Animal models of social avoidance and social fear. *Cell and Tissue Research* **354**, 107–118.
- TOWNSEND, H.G.G., OSTERRIEDER, K., JELINSKI, M.D., MORCK, D.W., WALDNER, C.L., COX, W.R., GERDTS, V., POTTER, A.A., BABIUK, L.A. & CROSS, J.C. (2025) A call to action to address critical flaws and bias in laboratory animal experiments and preclinical research. *Scientific Reports* **15**, 30745. Nature Publishing Group.
- VESTERINEN, H.M., SENA, E.S., EGAN, K.J., HIRST, T.C., CHUROLOV, L., CURRIE, G.L., ANTONIC, A., HOWELLS, D.W. & MACLEOD, M.R. (2014) Meta-analysis of data from animal studies: a practical guide. *Journal of Neuroscience Methods* **221**, 92–102.
- VIECHTBAUER, W. (2010) Conducting Meta-Analyses in R with the metafor Package. *Journal of Statistical Software* **36**, 1–48.

- VOELKL, B., ALTMAN, N.S., FORSMAN, A., FORSTMEIER, W., GUREVITCH, J., JARIC, I., KARP, N.A., KAS, M.J., SCHIELZETH, H., VAN DE CASTEELE, T. & WÜRBEL, H. (2020) Reproducibility of animal research in light of biological variation. *Nature Reviews Neuroscience* **21**, 384–393. Nature Publishing Group.
- VRAJOVÁ, M., ŠTASTNÝ, F., HORÁČEK, J., LOCHMAN, J., ŠERÝ, O., PEKOVÁ, S., KLASCHKA, J. & HÖSCHL, C. (2010) Expression of the Hippocampal NMDA Receptor GluN1 Subunit and Its Splicing Isoforms in Schizophrenia: Postmortem Study. *Neurochemical Research* **35**, 994–1002.
- WEE, B.E.F., FRANCIS, T.J., LEE, C.Y., LEE, J.M. & DOHANICH, G.P. (1995) Mate preference and avoidance in female rats following treatment with scopolamine. *Physiology & Behavior* **58**, 97–100.
- WEICKERT, C.S., FUNG, S.J., CATTS, V.S., SCHOFIELD, P.R., ALLEN, K.M., MOORE, L.T., NEWELL, K.A., PELLE, D., HUANG, X.-F., CATTS, S.V. & WEICKERT, T.W. (2013) Molecular evidence of N-methyl-D-aspartate receptor hypofunction in schizophrenia. *Molecular Psychiatry* **18**, 1185–1192.
- WILSON, C.A. & KOENIG, J.I. (2014) Social interaction and social withdrawal in rodents as readouts for investigating the negative symptoms of schizophrenia. *European neuropsychopharmacology: the journal of the European College of Neuropsychopharmacology* **24**, 759–773.
- WILSON, E., RAMAGE, F.J., WEVER, K.E., SENA, E.S., MACLEOD, M.R. & CURRIE, G.L. (2023) Designing, conducting, and reporting reproducible animal experiments.
- WINIARSKI, M., KONDRAKIEWICZ, L., KONDRAKIEWICZ, K., JĘDRZEJEWSKA-SZMEK, J., TURZYŃSKI, K., KNAPSKA, E. & MEYZA, K. (2022) Social deficits in BTBR T+ Itpr3tf/J mice vary with ecological validity of the test. *Genes, Brain, and Behavior* **21**, e12814.
- WORP, H.B. VAN DER, HOWELLS, D.W., SENA, E.S., PORRITT, M.J., REWELL, S., O'COLLINS, V. & MACLEOD, M.R. (2010) Can Animal Models of Disease Reliably Inform Human Studies? *PLOS Medicine* **7**, e1000245. Public Library of Science.
- YANG, M., SILVERMAN, J.L. & CRAWLEY, J.N. (2011) Automated Three-Chambered Social Approach Task for Mice. *Current protocols in neuroscience / editorial board, Jacqueline N. Crawley ... [et al.]* **CHAPTER 8**, Unit-8.26.
- YANG, Y., MACLEOD, M., PAN, J., LAGISZ, M. & NAKAGAWA, S. (2023) Advanced methods and implementations for the meta-analyses of animal models: Current practices and future recommendations. *Neuroscience & Biobehavioral Reviews* **146**, 105016.
- VAN DER ZWAAL, E.M., LUIJENDIJK, M.C.M., EVERS, S.S., LA FLEUR, S.E. & ADAN, R.A.H. (2010) Olanzapine affects locomotor activity and meal size in male rats. *Pharmacology, Biochemistry, and Behavior* **97**, 130–137.
- ZWETSLOOT, P.-P., VAN DER NAALD, M., SENA, E.S., HOWELLS, D.W., INTHOUT, J., DE GROOT, J.A., CHAMULEAU, S.A., MACLEOD, M.R. & WEVER, K.E. (2017) Standardized mean differences cause funnel plot distortion in publication bias assessments. *eLife* **6**, e24260. eLife Sciences Publications, Ltd.

FIGURE LEGENDS

Figure 1. Flow diagram of the literature search and study selection process.

Figure 2. Risk of Bias summary across domains in the included studies. Bars represent the proportion of studies rated as low, unclear, or high risk of bias for each domain, based on SYRCLE's risk of bias tool with additional domains for pseudoreplication and procedural equivalence. Domains with a higher proportion of low-risk judgments indicate stronger methodological rigour, whereas domains dominated by unclear or high risk suggest potential limitations due to insufficient reporting or methodological concerns. The "Overall" category reflects the study-level risk of bias assessment.

Figure 3. Effects of NMDA receptor antagonists on social interaction in laboratory animals.

(A) Alluvial plot showing the distribution of effect sizes across species and NMDA antagonists. Flow width represents the number of effect sizes contributing to each category. (B) Orchard plot of the overall pooled effect of NMDA receptor antagonists on social interaction. (C) Orchard plots showing effects stratified by NMDA antagonist. (D) Orchard plots showing effects stratified by species. Filled circles represent individual effect sizes scaled by study precision, and open circles indicate pooled point estimates with 95% confidence intervals (thick horizontal lines) and prediction intervals (thin horizontal lines). k denotes the number of effect sizes, with the number of independent studies shown in parentheses.

Figure 4. Effects of NMDA receptor antagonists on social preference in laboratory animals.

(A) Alluvial plot showing the distribution of effect sizes across species and NMDA antagonists. Flow width represents the number of effect sizes contributing to each category. (B) Orchard plot of the overall pooled effect of NMDA receptor antagonists on social preference. (C) Orchard plots showing effects stratified by NMDA antagonist. (D) Orchard plots showing effects stratified by species. Filled circles represent individual effect sizes scaled by study precision, and open circles indicate pooled point estimates with 95% confidence intervals (thick horizontal lines) and prediction intervals (thin horizontal lines). k denotes the number of effect sizes, with the number of independent studies shown in parentheses.

Figure 5. Effects of antipsychotics on NMDA receptor antagonist–induced alterations in social interaction in laboratory animals. (A) Alluvial plot showing the distribution of effect sizes across species, NMDA receptor antagonists, and antipsychotics. Flow width represents the number of effect sizes contributing to each category. (B) Orchard plot of the overall pooled effect of antipsychotics on NMDA antagonist–induced changes in social interaction. Filled circles represent individual effect sizes scaled by study precision, and the open circle indicates the pooled point estimate with 95% confidence intervals (thick horizontal lines) and prediction interval (thin horizontal lines). k denotes the number of effect sizes, with the number of independent studies shown in parentheses. (C) Evidence map of effect sizes by NMDA antagonist and antipsychotic. Circle size represents the number of effect sizes, and colour intensity indicates the mean Hedges' g . (D) Evidence map of effect sizes by species and antipsychotic, with circle size indicating the number of effect sizes and colour representing the mean Hedges' g .

Figure 6. Effects of antipsychotics on NMDA receptor antagonist–induced alterations in social preference in laboratory animals. (A) Alluvial plot showing the distribution of effect sizes across species, NMDA receptor antagonists, and antipsychotics. Flow width represents the number of effect sizes contributing to each category. (B) Orchard plot of the overall pooled effect of antipsychotics on NMDA antagonist–induced changes in social preference. Filled circles represent individual effect sizes scaled by study precision, and the open circle indicates the pooled point estimate with 95% confidence intervals (thick horizontal lines) and prediction interval (thin horizontal lines). k denotes the number of effect sizes, with the number of independent studies shown in parentheses. (C) Evidence map of effect sizes by NMDA antagonist and antipsychotic. Circle size represents the number of effect sizes, and colour intensity indicates the mean Hedges' g . (D) Evidence map of effect sizes by species and antipsychotic, with circle size indicating the number of effect sizes and colour representing the mean Hedges' g .

Figure 1

Flow diagram of study selection

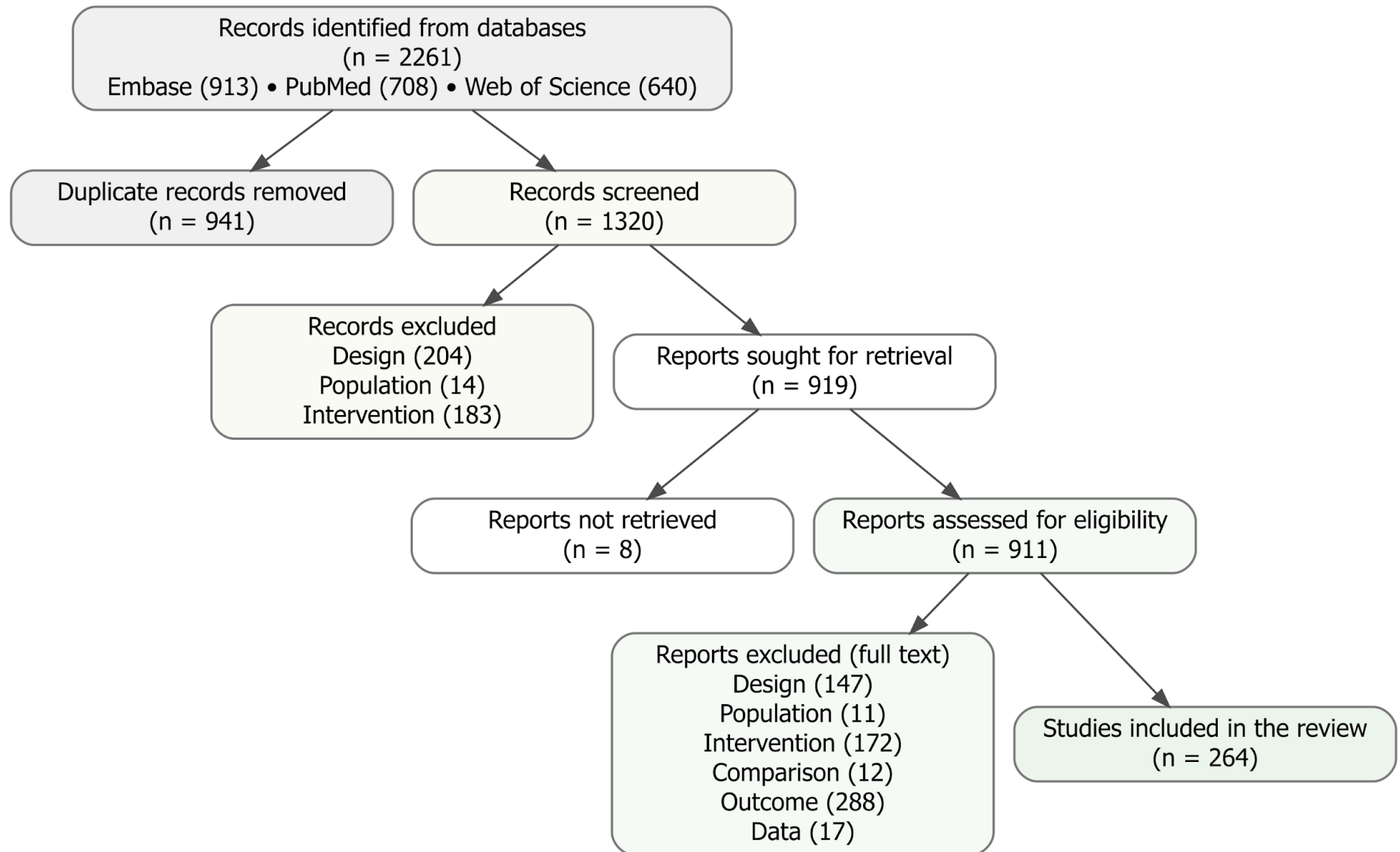


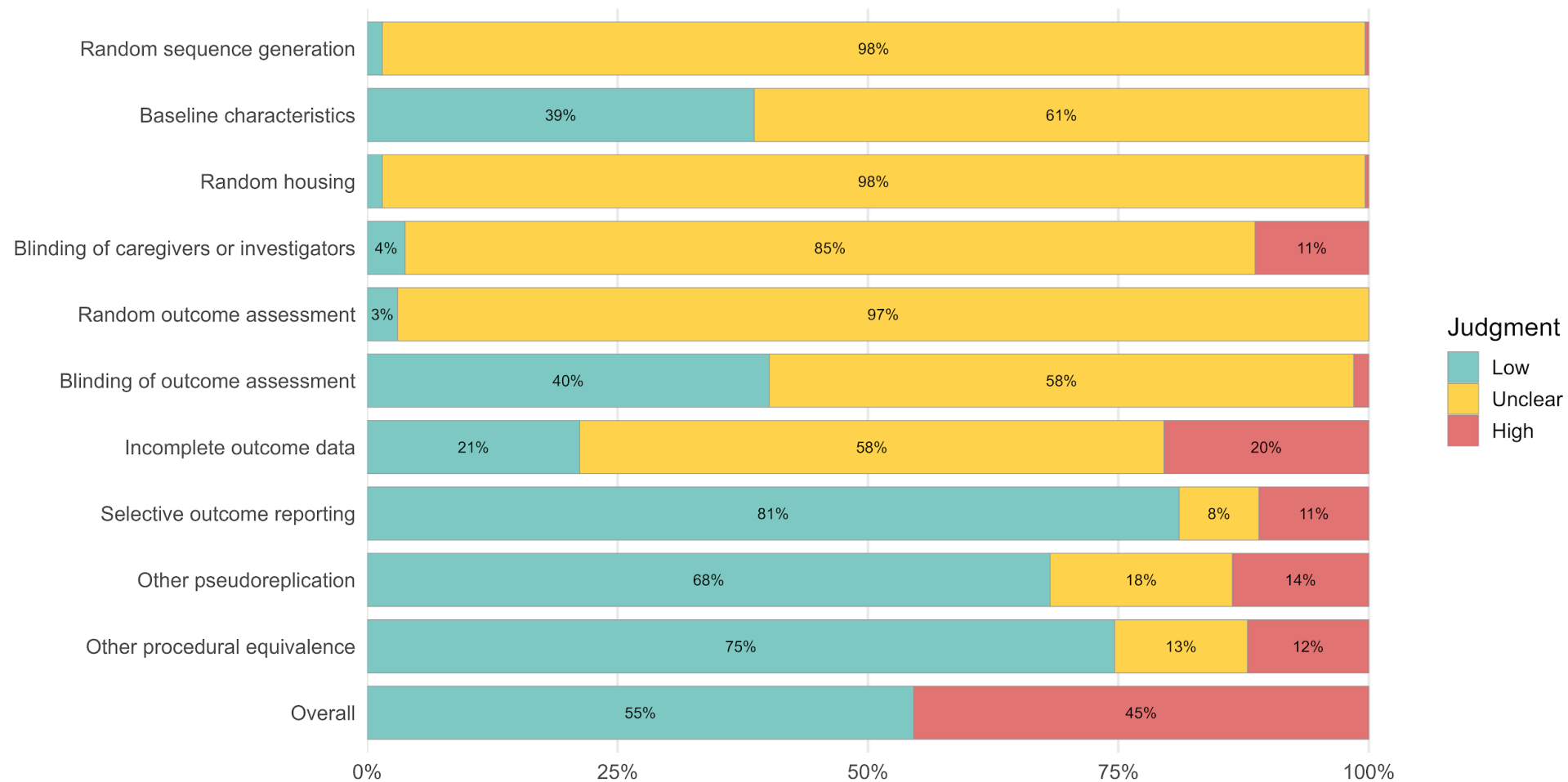
Figure 2

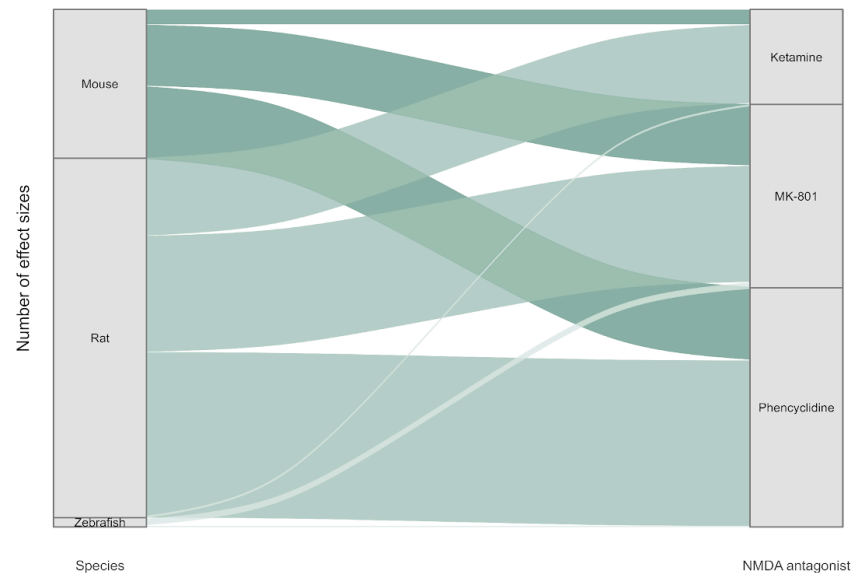
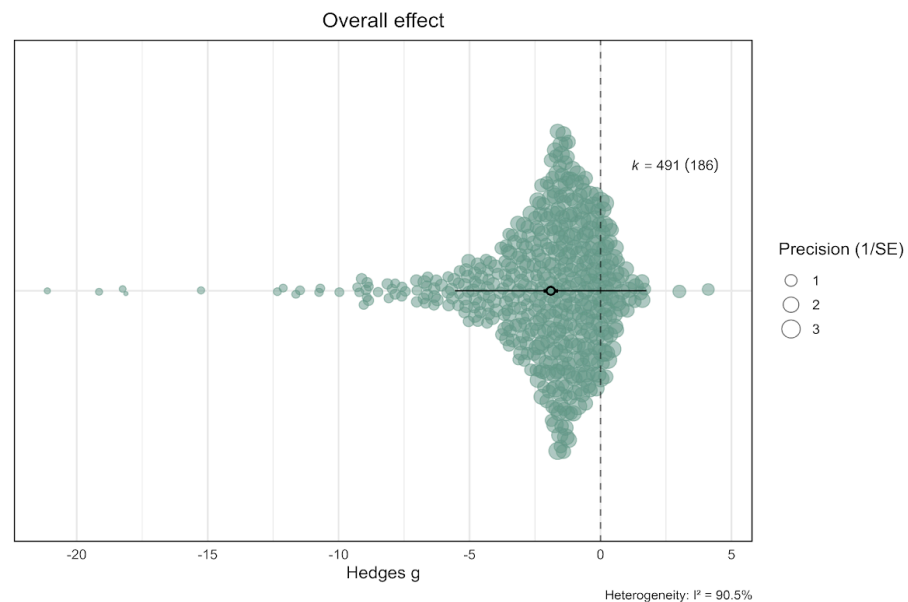
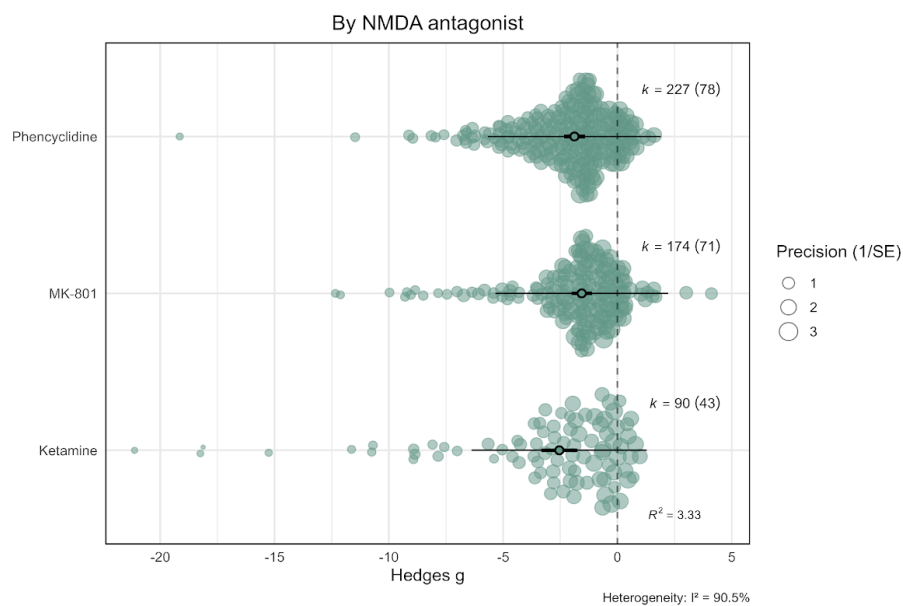
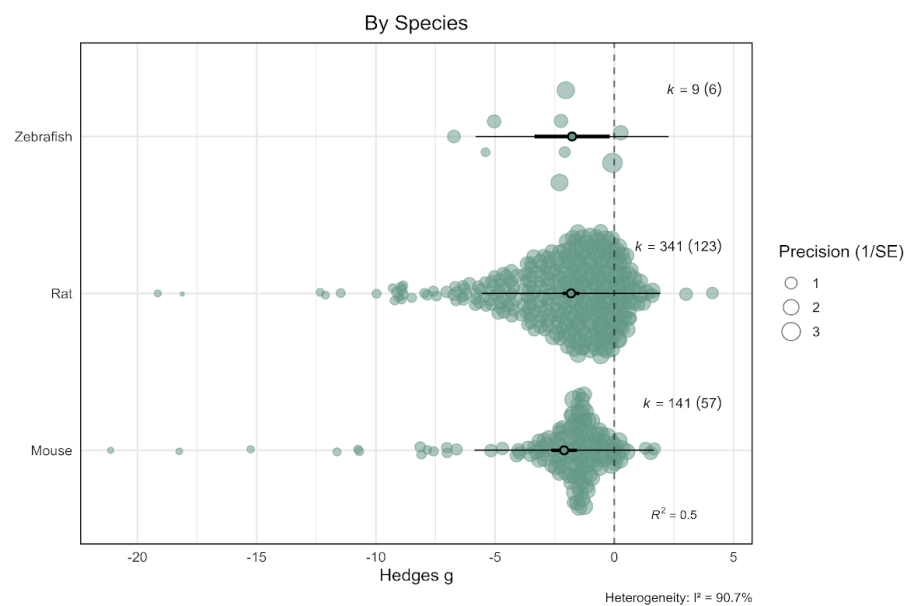
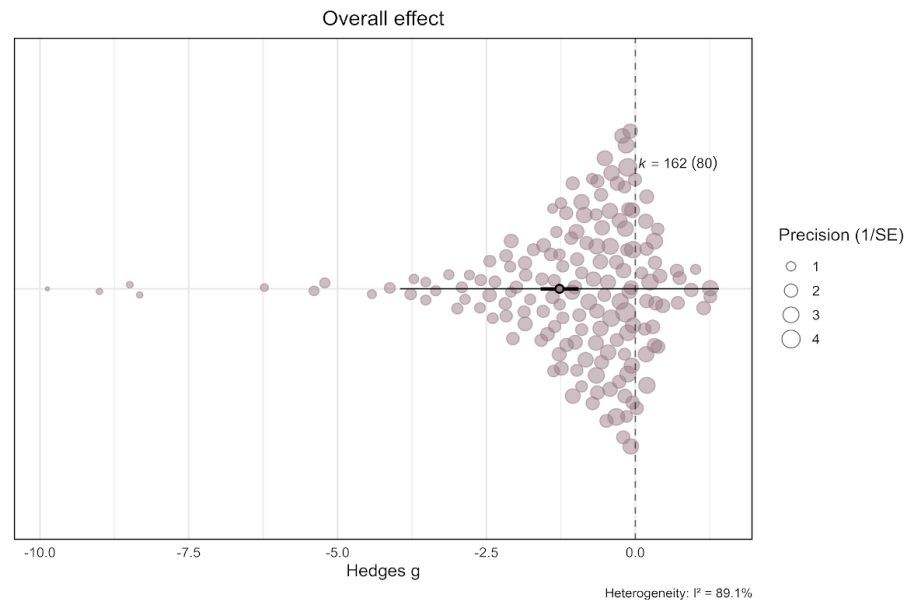
Figure 3**A****B****C****D**

Figure 4

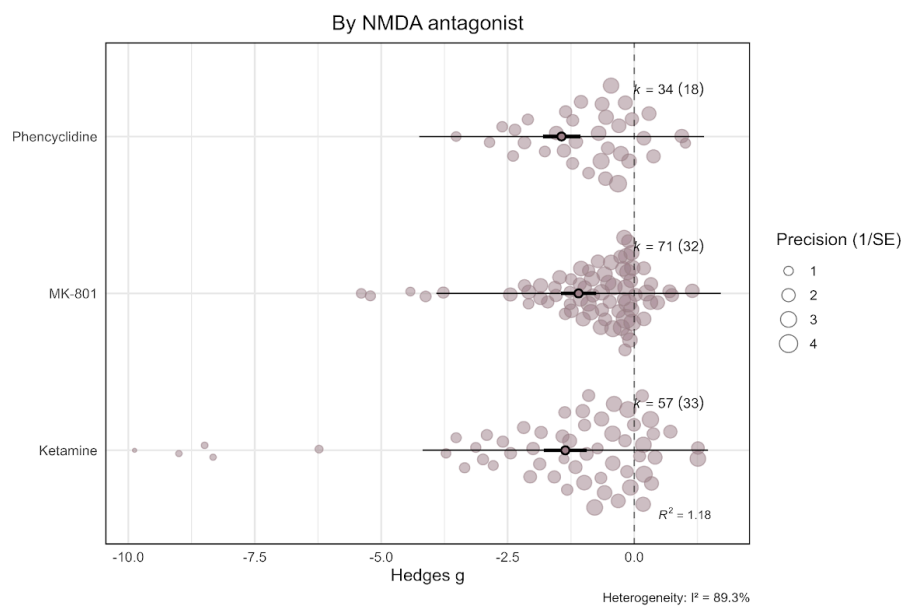
A



B



C



D

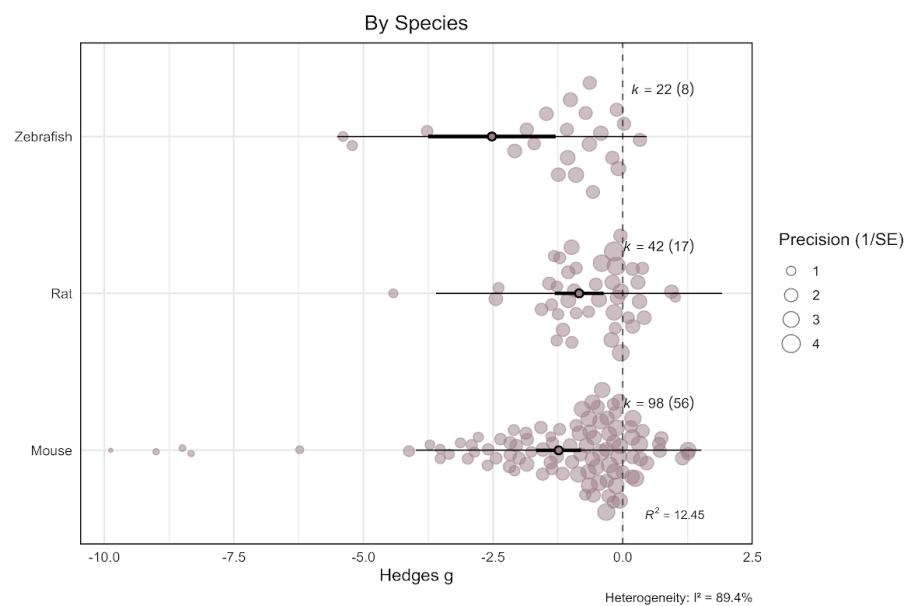


Figure 5

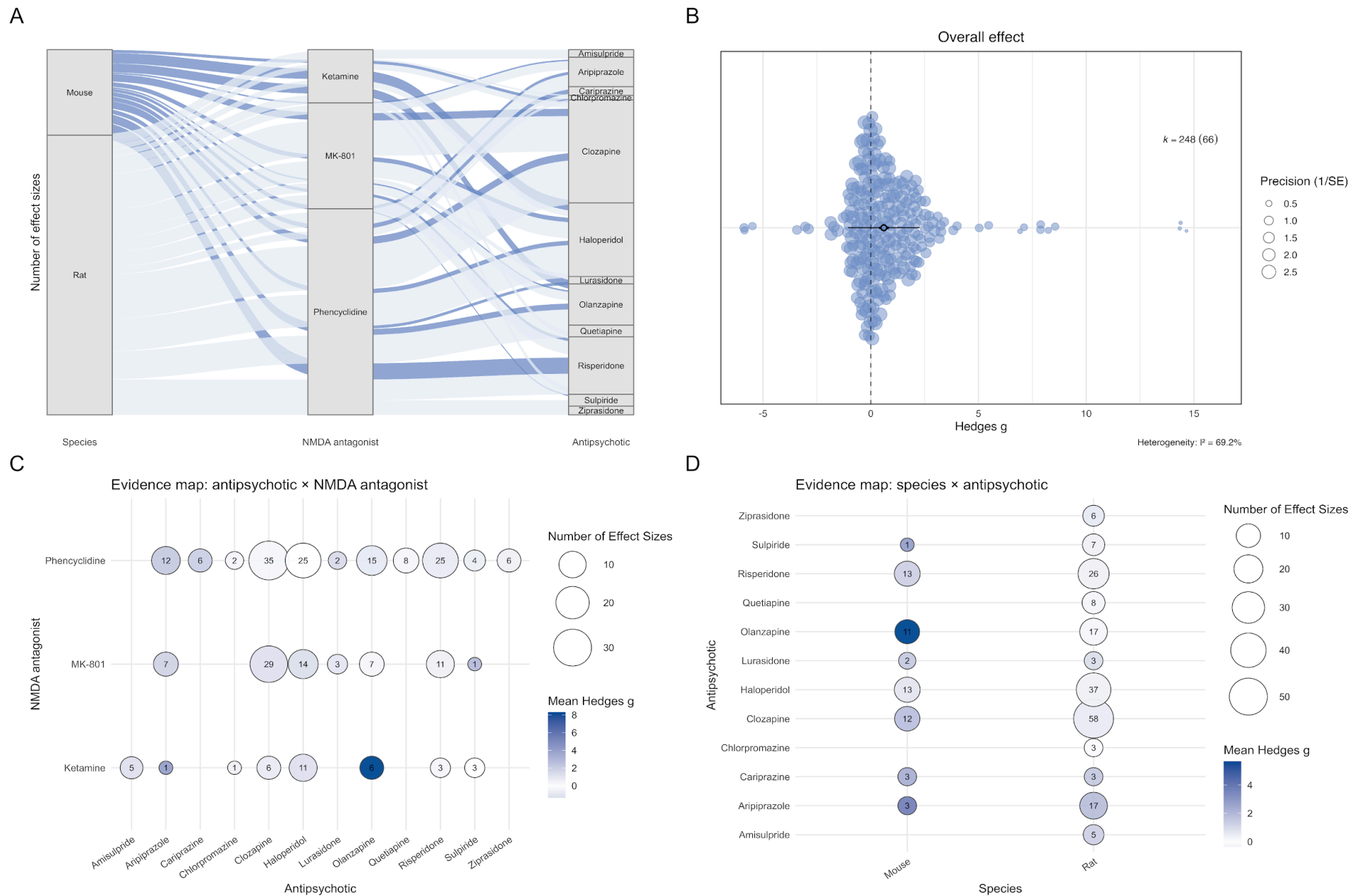
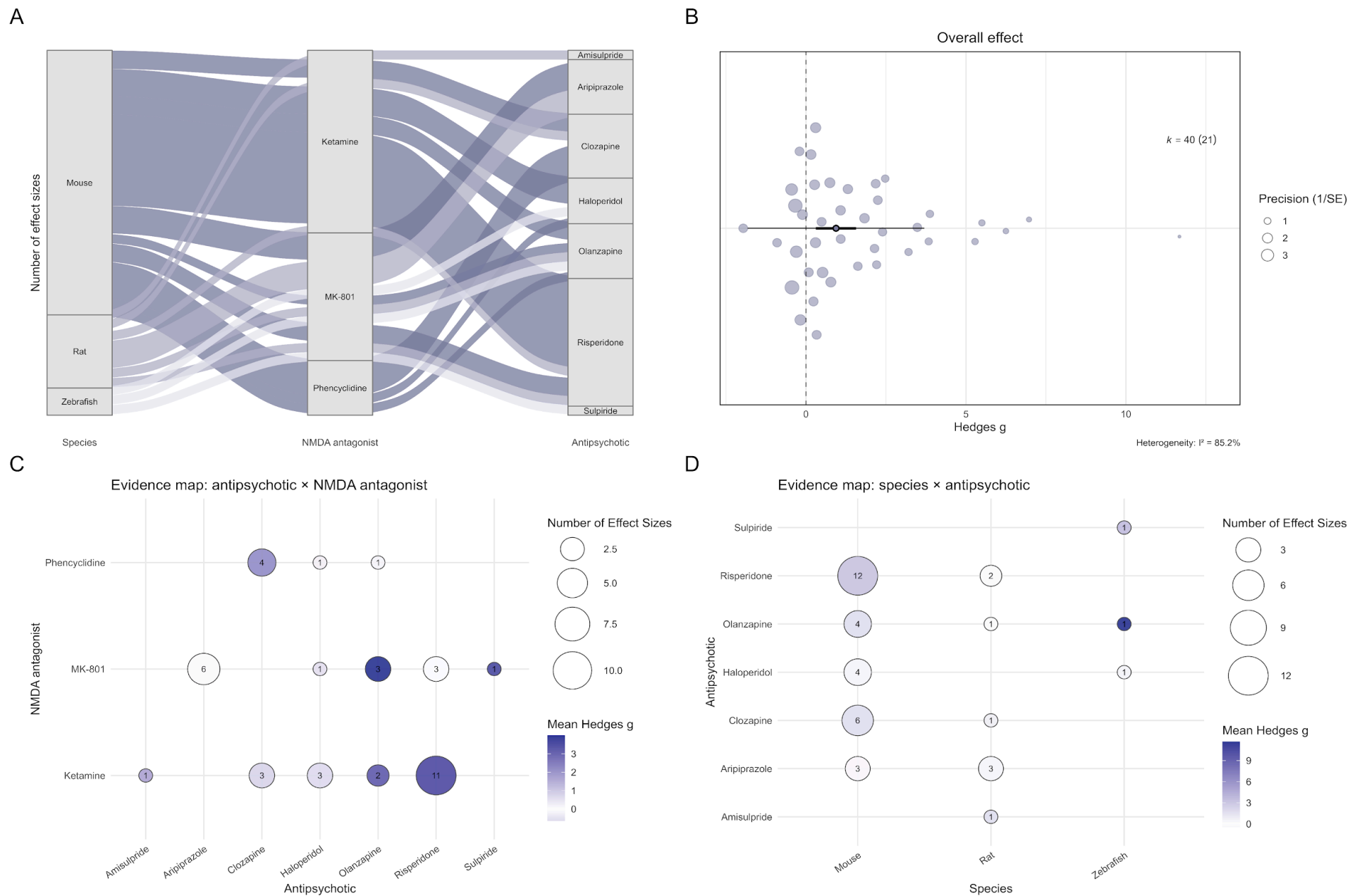


Figure 6



SUPPORTING INFORMATION

Effects of NMDA antagonists on social behaviour: a systematic review and meta-analysis of preclinical studies

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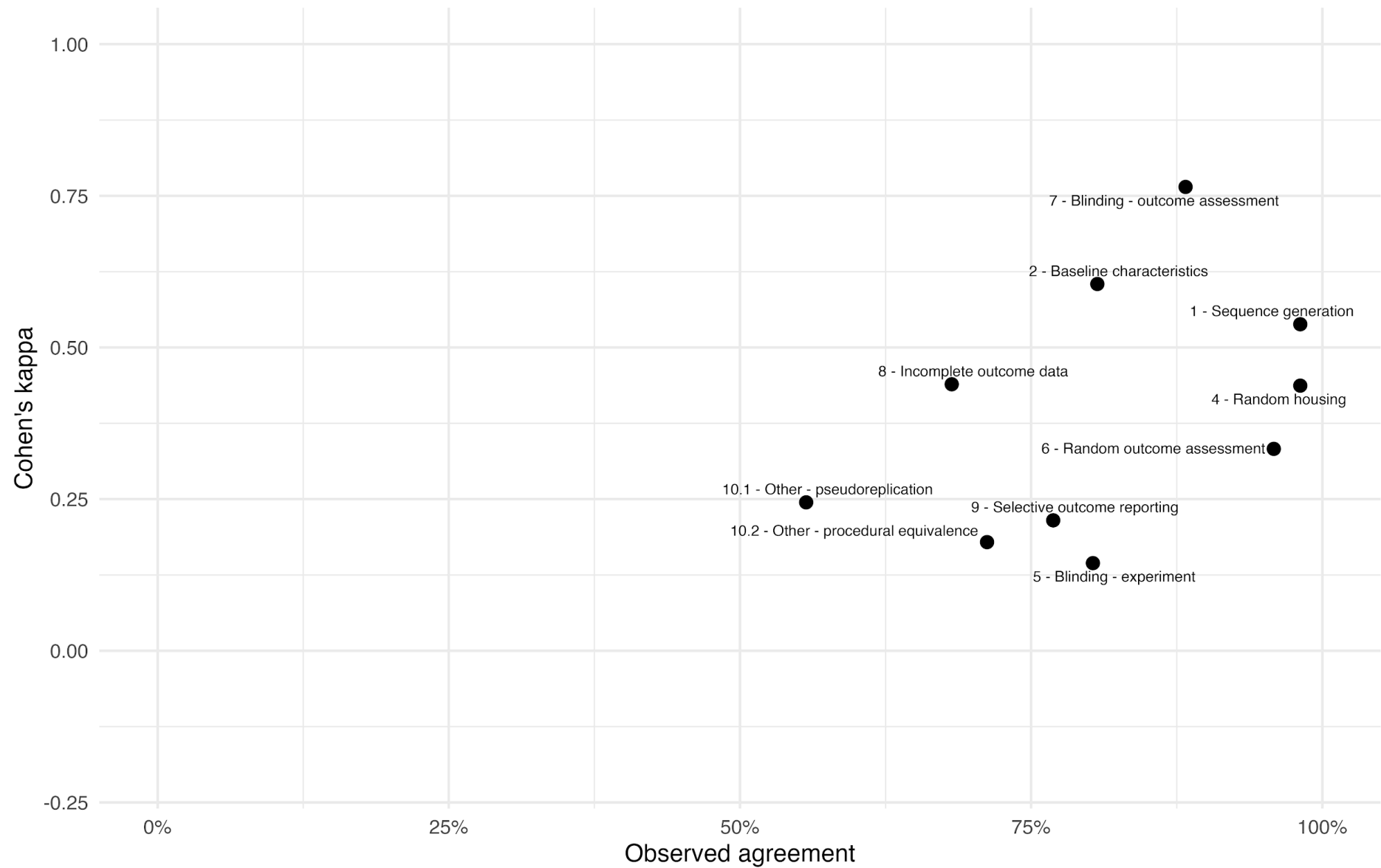
Figure S1

Figure S1. Observed agreement plotted against Cohen's kappa for each Risk of Bias domain. Each point represents one domain from the risk of bias assessment based on independent ratings of two reviewers. The x-axis shows the percentage of identical judgments between reviewers (observed agreement), whereas the y-axis shows Cohen's kappa, which corrects for agreement expected by chance.

Figure S2

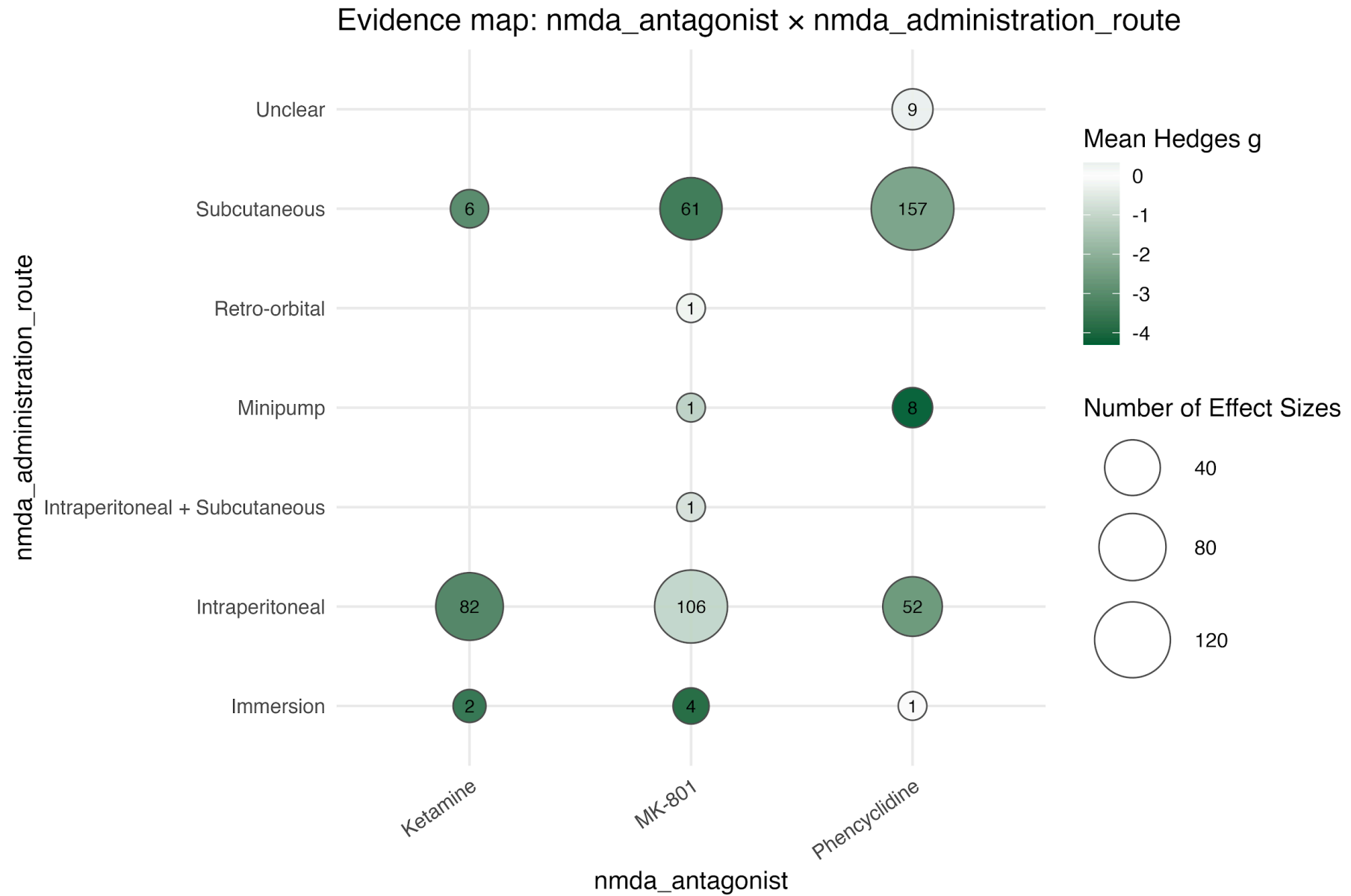


Figure S2. Evidence map of effect sizes for social interaction by NMDA antagonist and route of administration. Circle size represents the number of effect sizes, and colour intensity indicates the mean Hedges' g.

Figure S3

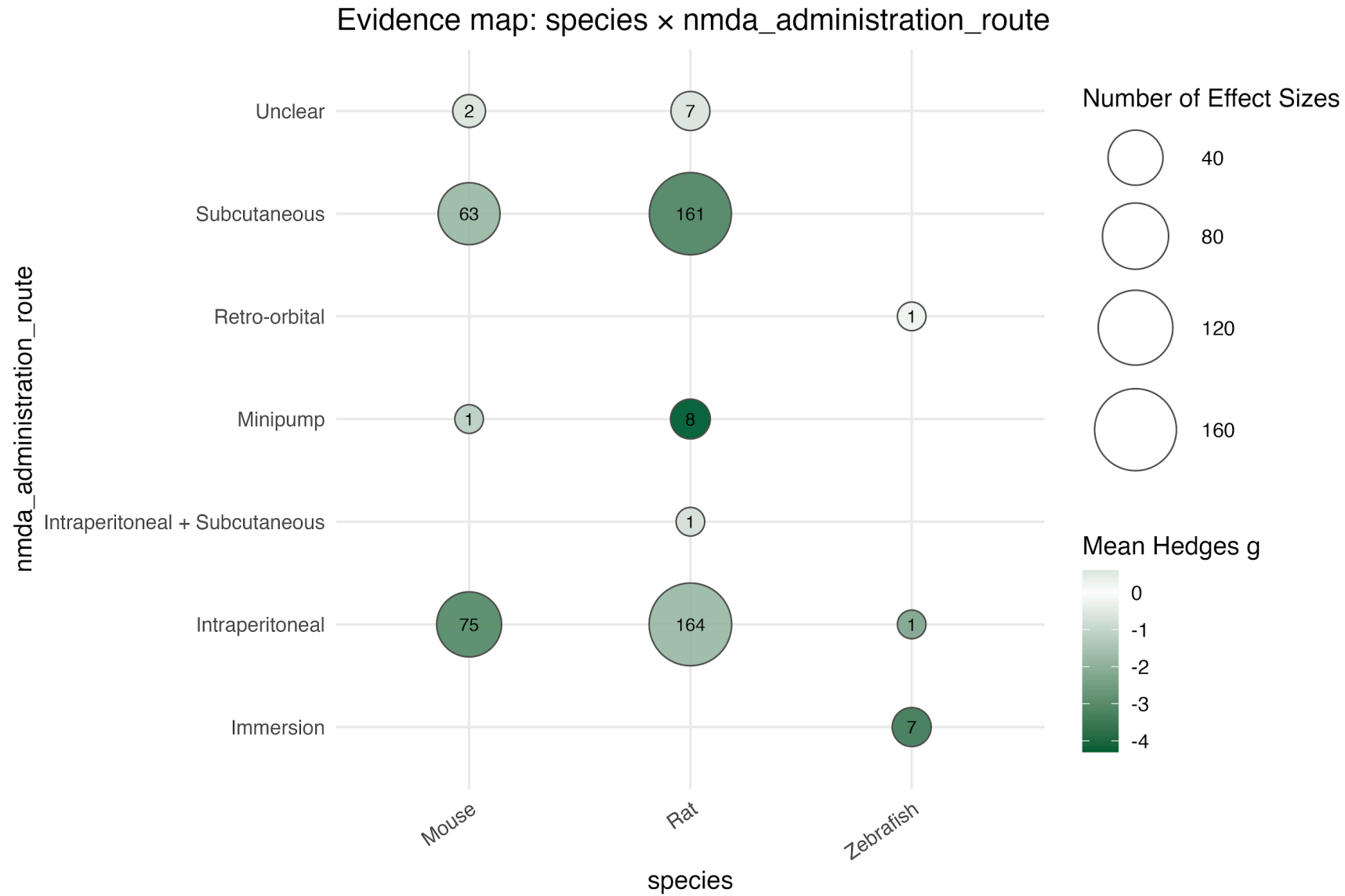


Figure S3. Evidence map of effect sizes for social interaction by species and route of administration. Circle size represents the number of effect sizes, and colour intensity indicates the mean Hedges' g.

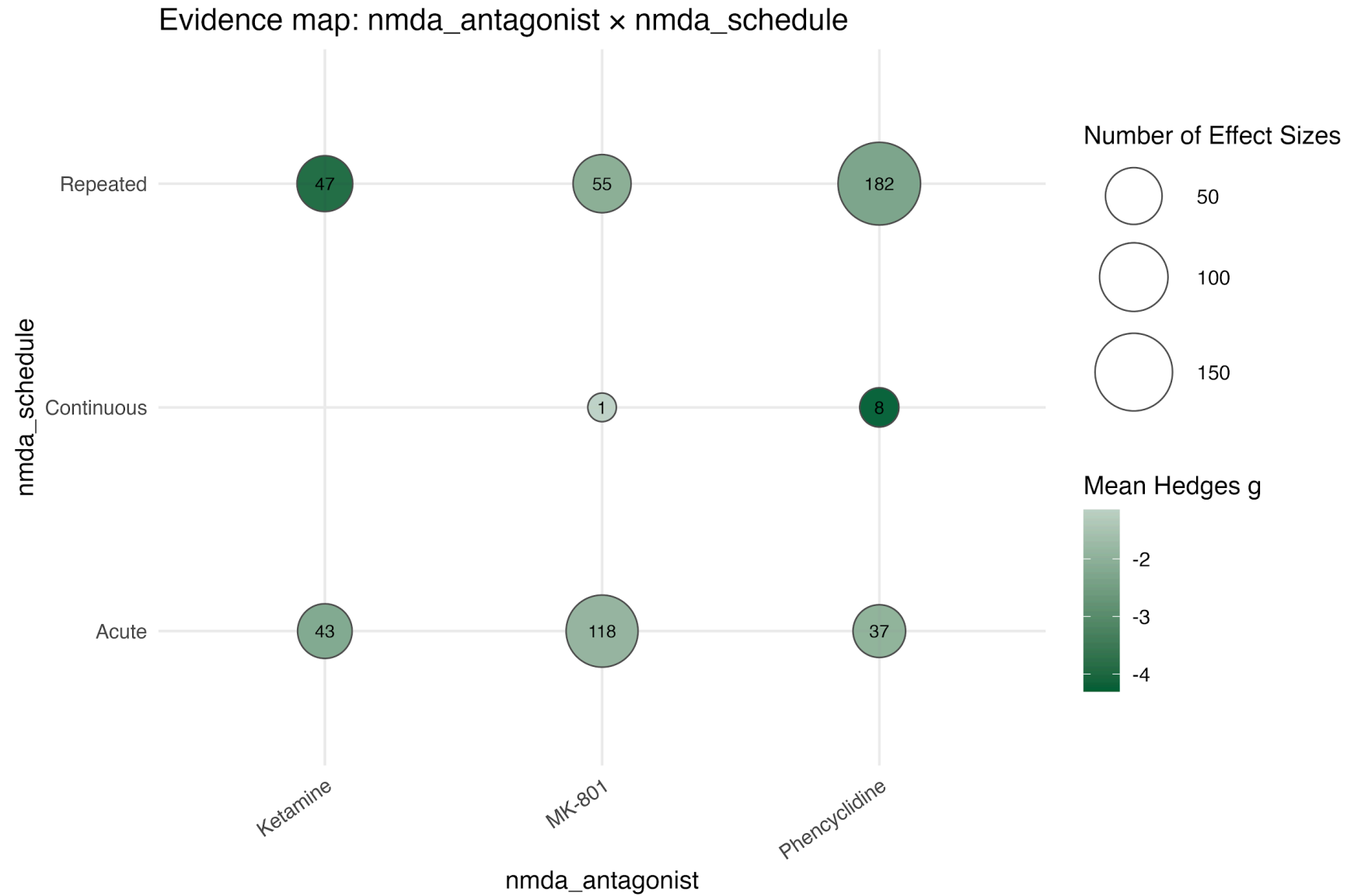
Figure S4

Figure S4. Evidence map of effect sizes for social interaction by NMDA antagonist and schedule of administration. Circle size represents the number of effect sizes, and colour intensity indicates the mean Hedges' g.

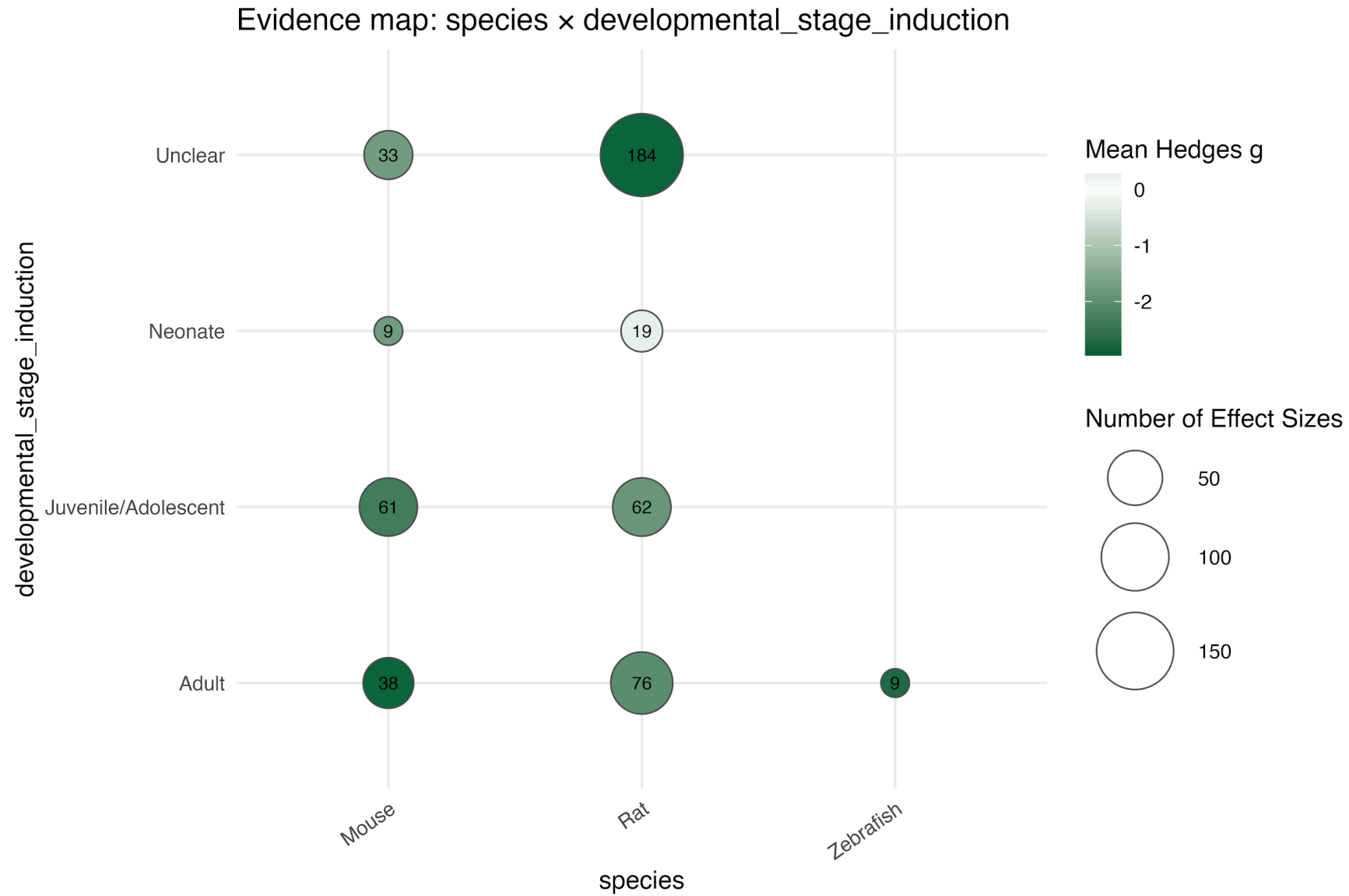
Figure S5

Figure S5. Evidence map of effect sizes for social interaction by species and developmental stage at induction. Circle size represents the number of effect sizes, and colour intensity indicates the mean Hedges' g .

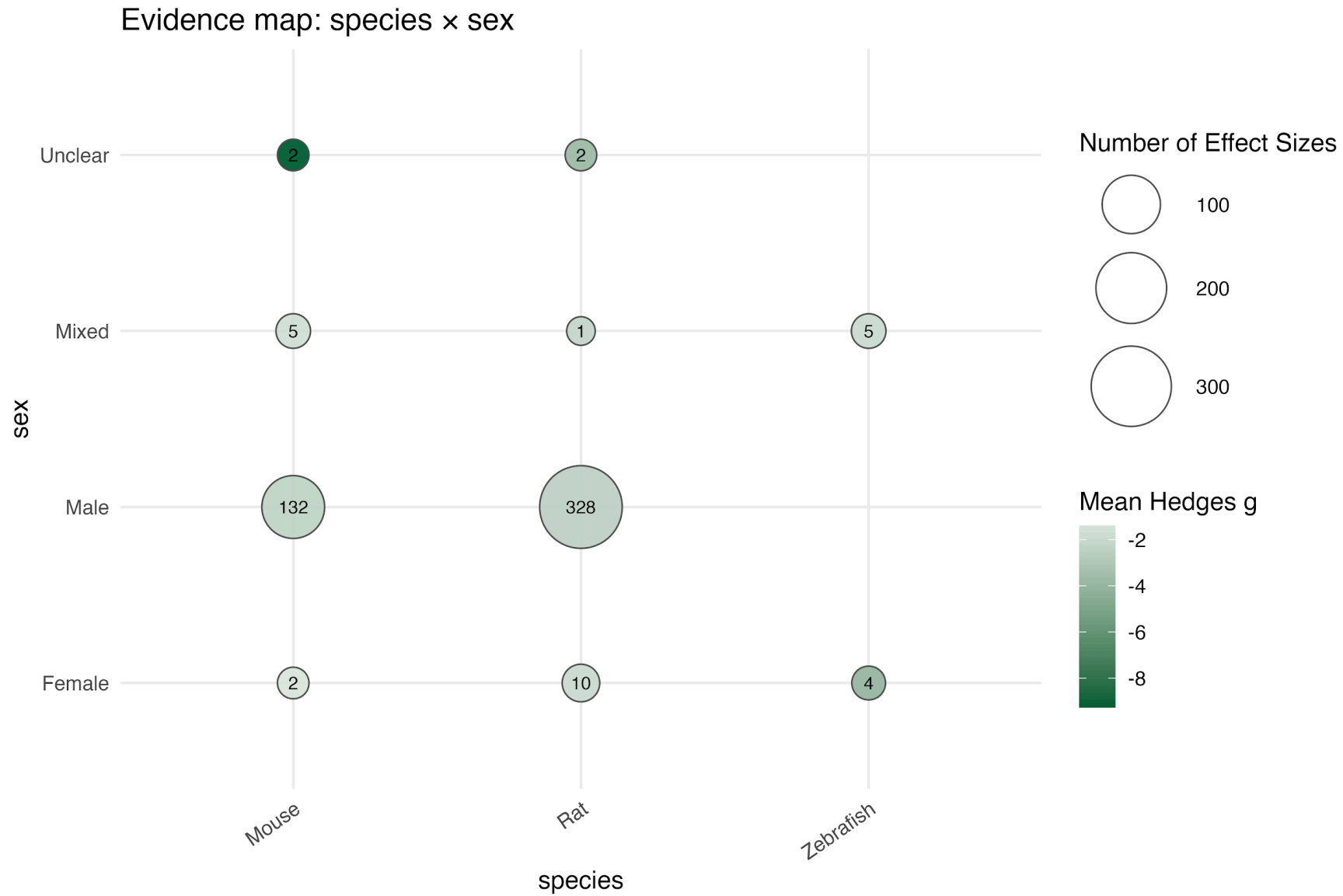
Figure S6

Figure S6. Evidence map of effect sizes for social interaction by species and sex. Circle size represents the number of effect sizes, and colour intensity indicates the mean Hedges' g .

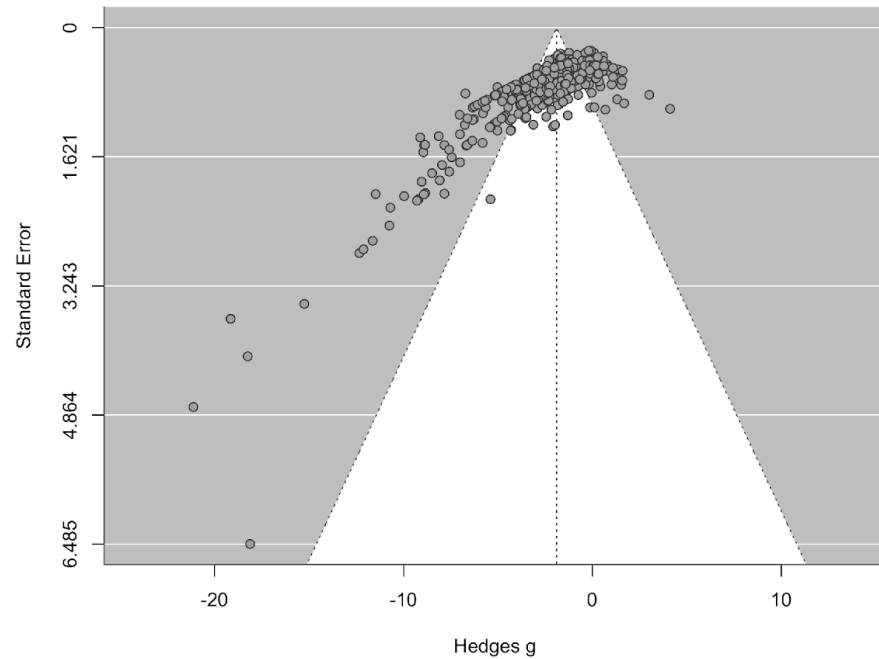
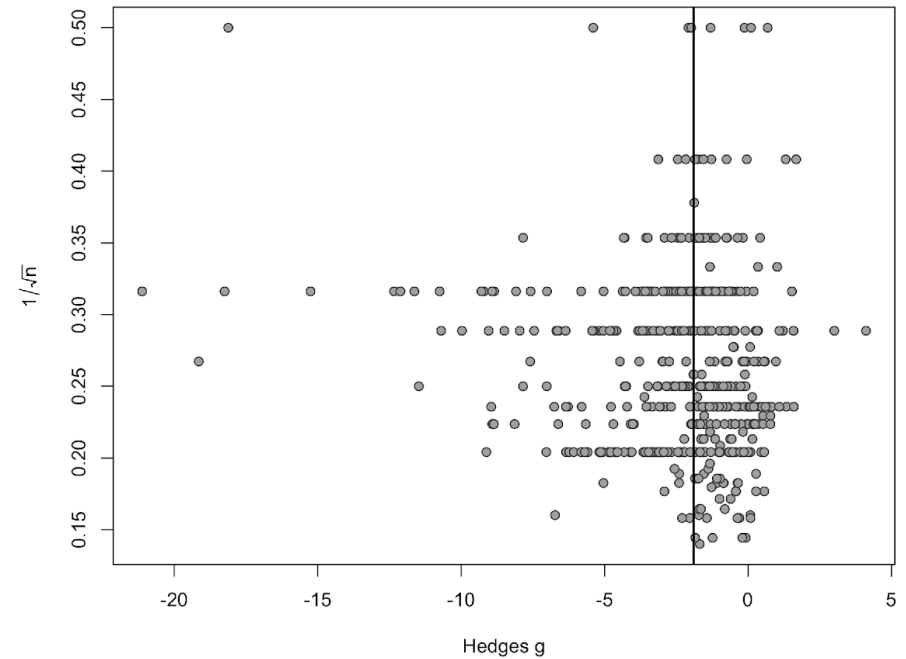
Figure S7**A****B**

Figure S7. Funnel plots of effect sizes for social interaction. (A) Funnel plot showing Hedges' g plotted against the standard error. (B) Funnel plot showing Hedges' g plotted against the inverse square root of total sample size as an alternative measure of study precision. Each point corresponds to one effect size.

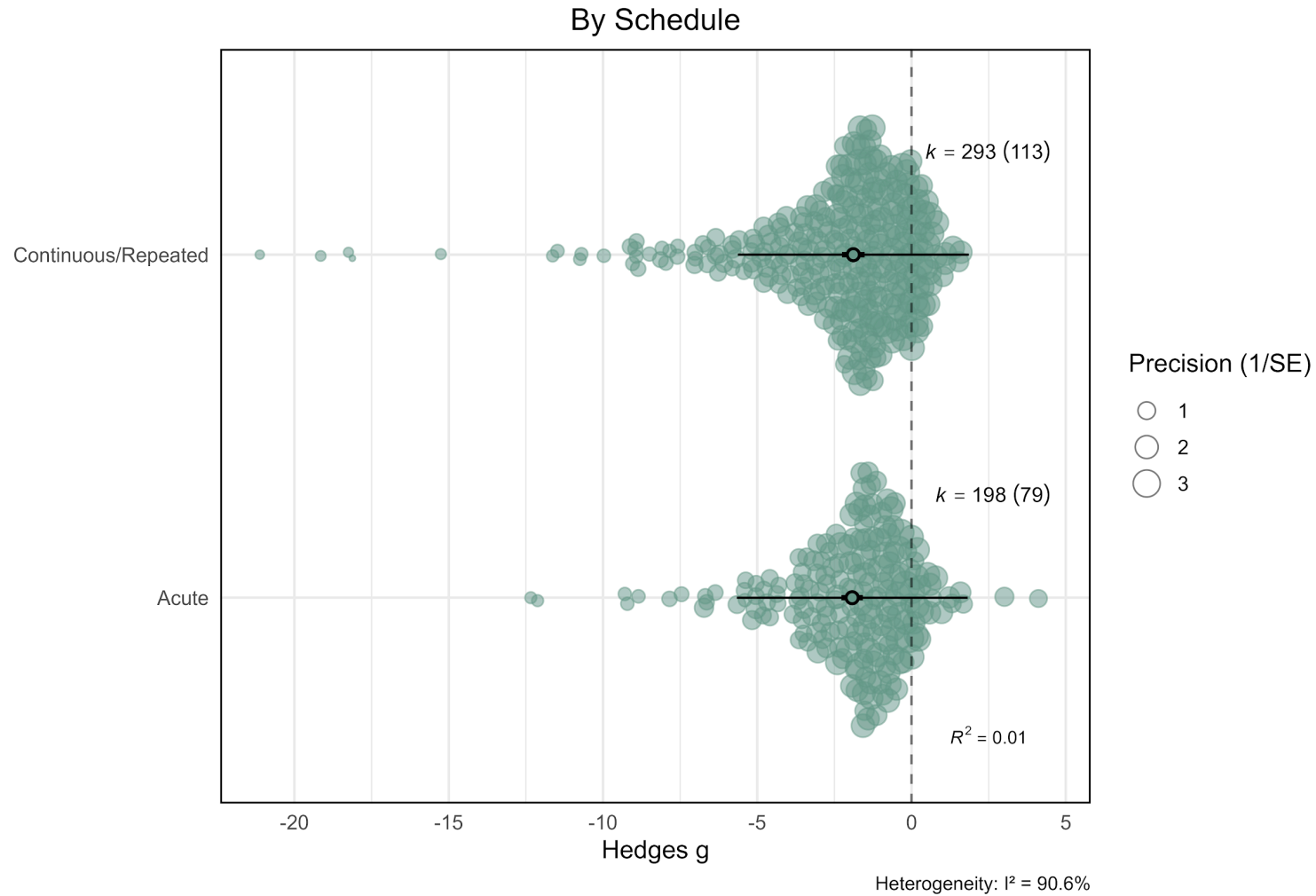
Figure S8

Figure S8. Effects of NMDA receptor antagonists on social interaction stratified by administration schedule. Orchard plot showing pooled effects for acute and continuous/repeated schedules. Filled circles represent individual effect sizes scaled by study precision, and open circles indicate pooled point estimates with 95% confidence intervals (thick horizontal lines) and prediction intervals (thin horizontal lines). k denotes the number of effect sizes, with the number of independent studies shown in parentheses.

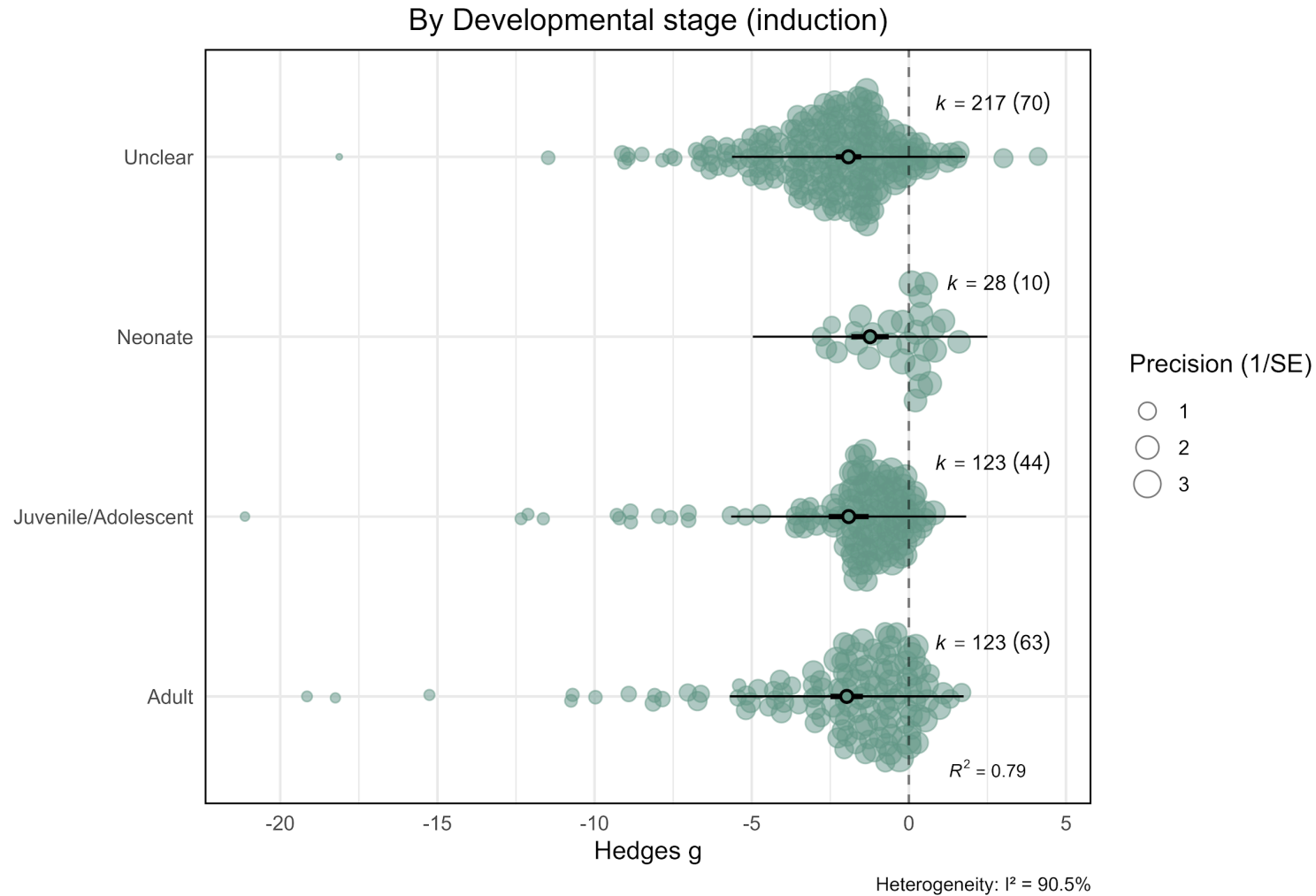
Figure S9

Figure S9. Effects of NMDA receptor antagonists on social interaction stratified by developmental stage at induction. Orchard plot showing pooled effects for adult, juvenile/adolescent, neonatal, and studies with unclear developmental stage at induction. Filled circles represent individual effect sizes scaled by study precision, and open circles indicate pooled point estimates with 95% confidence intervals (thick horizontal lines) and prediction intervals (thin horizontal lines). k denotes the number of effect sizes, with the number of independent studies shown in parentheses.

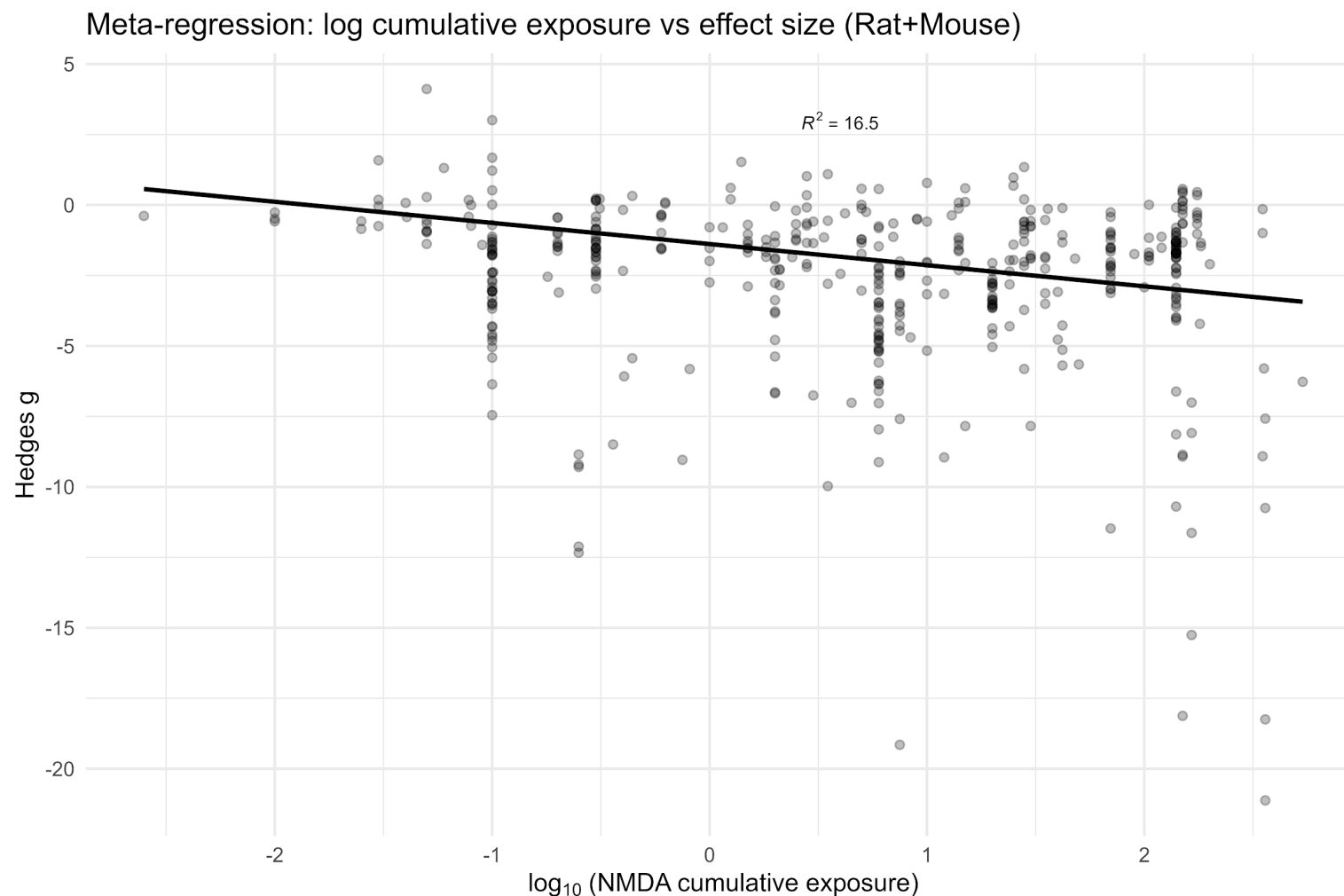
Figure S10

Figure S10. Meta-regression of log-transformed cumulative NMDA antagonist exposure and effect sizes for social interaction. The plot shows Hedges' g as a function of log₁₀-transformed cumulative NMDA antagonist exposure, including only studies conducted in rats and mice. Cumulative exposure was calculated, where data were available, as the product of dose, intervention frequency, and duration of exposure. Each point represents one effect size, and the solid line represents the fitted meta-regression slope from a multivariate random-effects model.

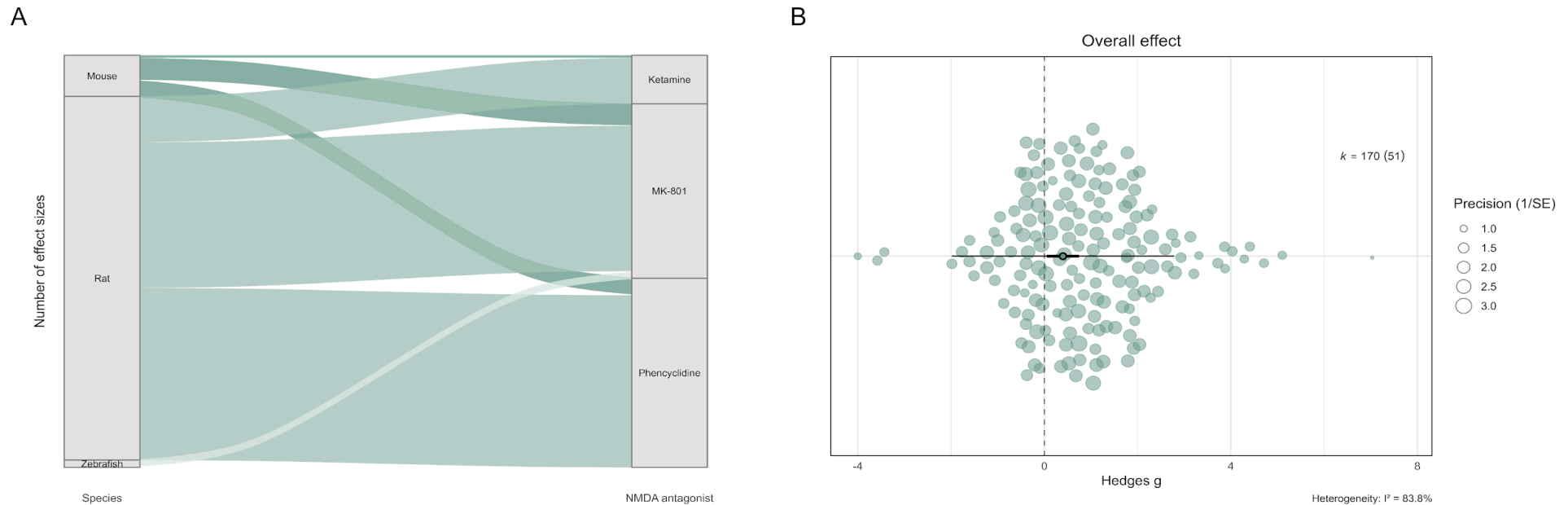
Figure S11

Figure S11. Effects of NMDA receptor antagonists on locomotor activity in experiments measuring social interaction. (A) Alluvial plot showing the distribution of effect sizes across species and NMDA receptor antagonists for studies in which locomotor activity was assessed alongside social interaction. Flow width represents the number of effect sizes contributing to each category. (B) Orchard plot of the overall pooled effect of NMDA receptor antagonists on locomotor activity. Filled circles represent individual effect sizes scaled by study precision, and the open circle indicates the pooled point estimate with 95% confidence intervals (thick horizontal lines) and prediction interval (thin horizontal lines). k denotes the number of effect sizes, with the number of independent studies shown in parentheses.

Figure S12

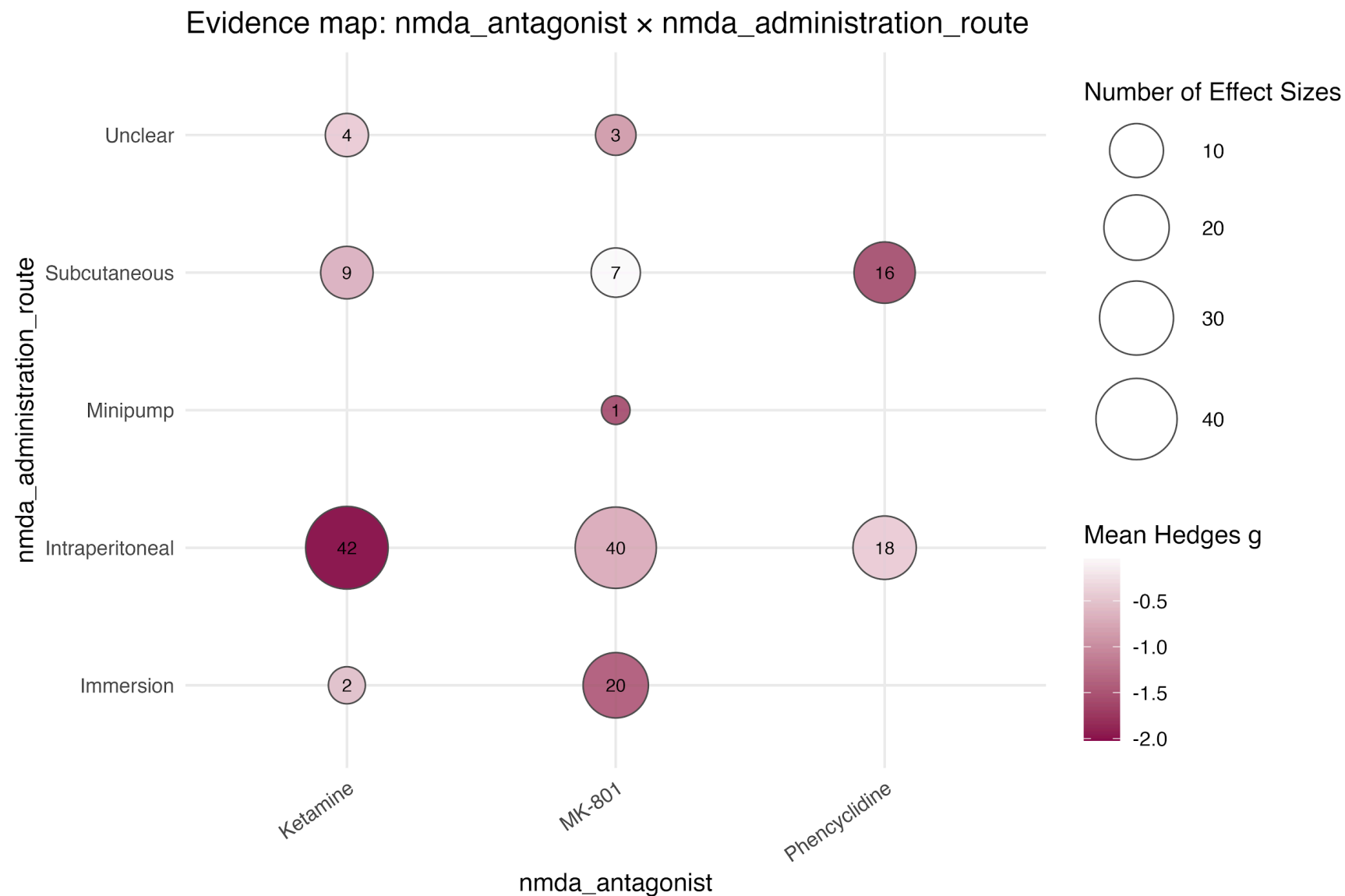


Figure S12. Evidence map of effect sizes for social preference by NMDA antagonist and route of administration. Circle size represents the number of effect sizes, and colour intensity indicates the mean Hedges' g.

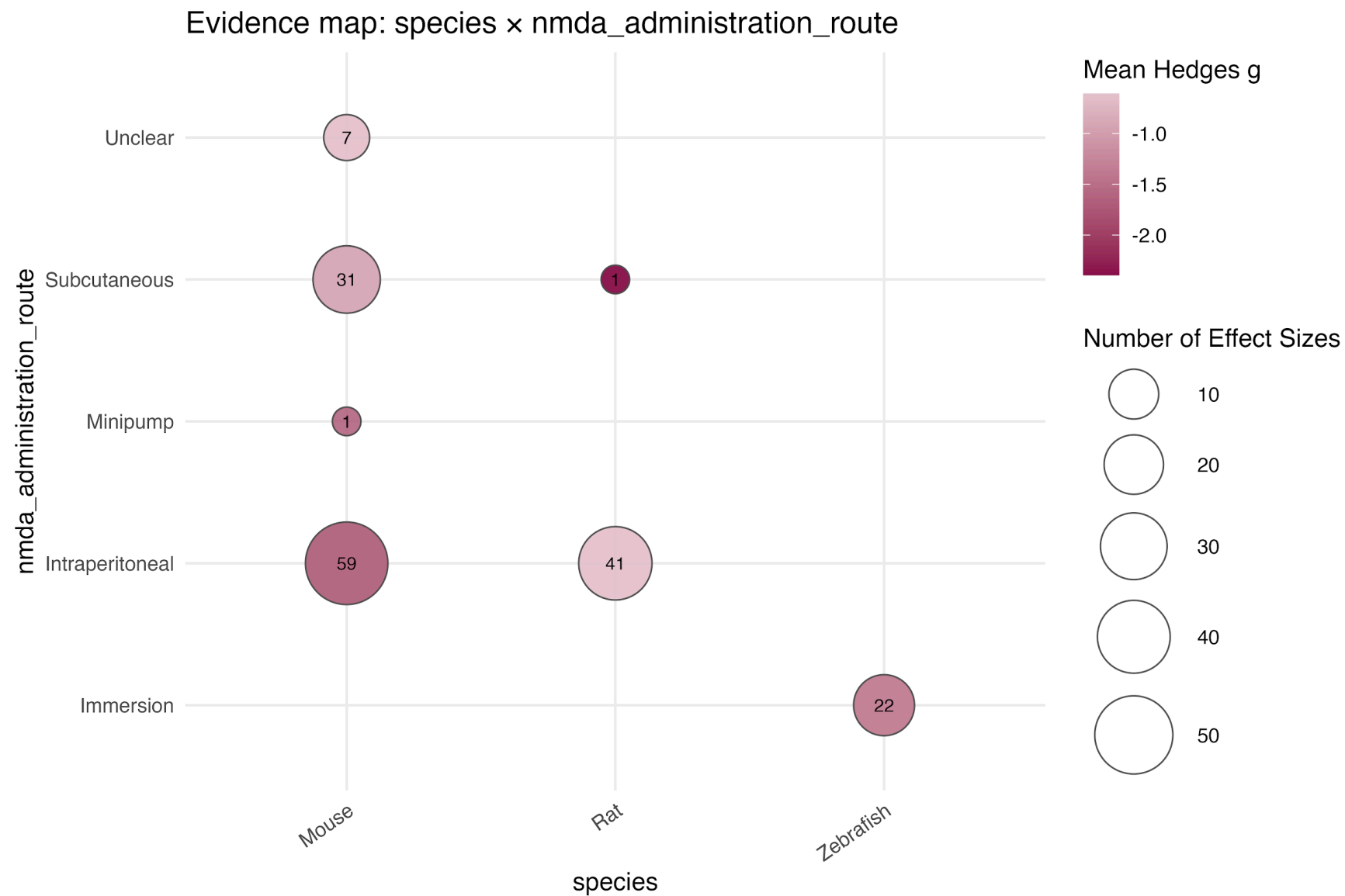
Figure S13

Figure S13. Evidence map of effect sizes for social preference by species and route of administration. Circle size represents the number of effect sizes, and colour intensity indicates the mean Hedges' g.

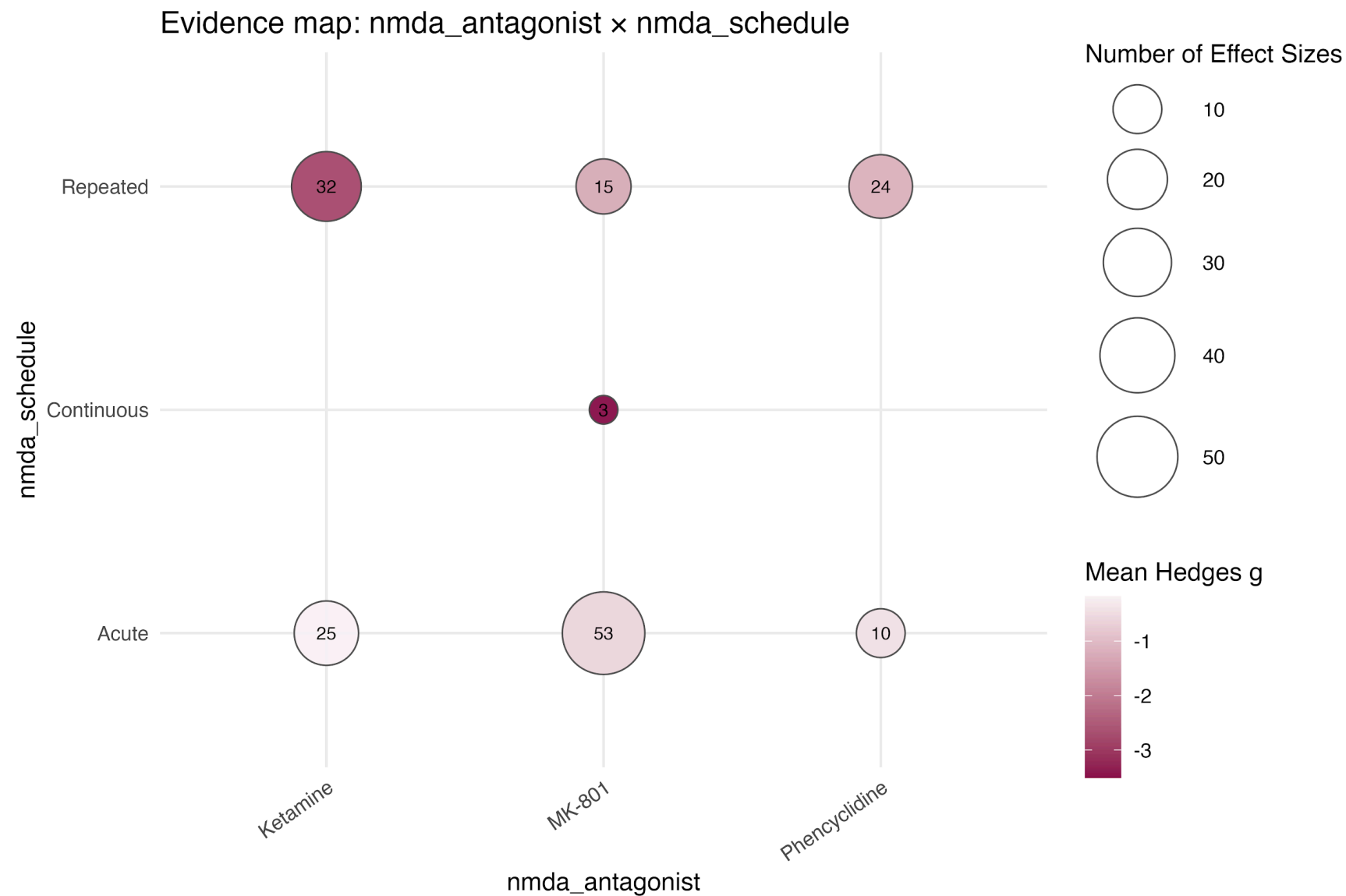
Figure S14

Figure S14. Evidence map of effect sizes for social preference by NMDA antagonist and schedule of administration. Circle size represents the number of effect sizes, and colour intensity indicates the mean Hedges' g.

Figure S15



Figure S15. Evidence map of effect sizes for social preference by species and developmental stage at induction. Circle size represents the number of effect sizes, and colour intensity indicates the mean Hedges' g.

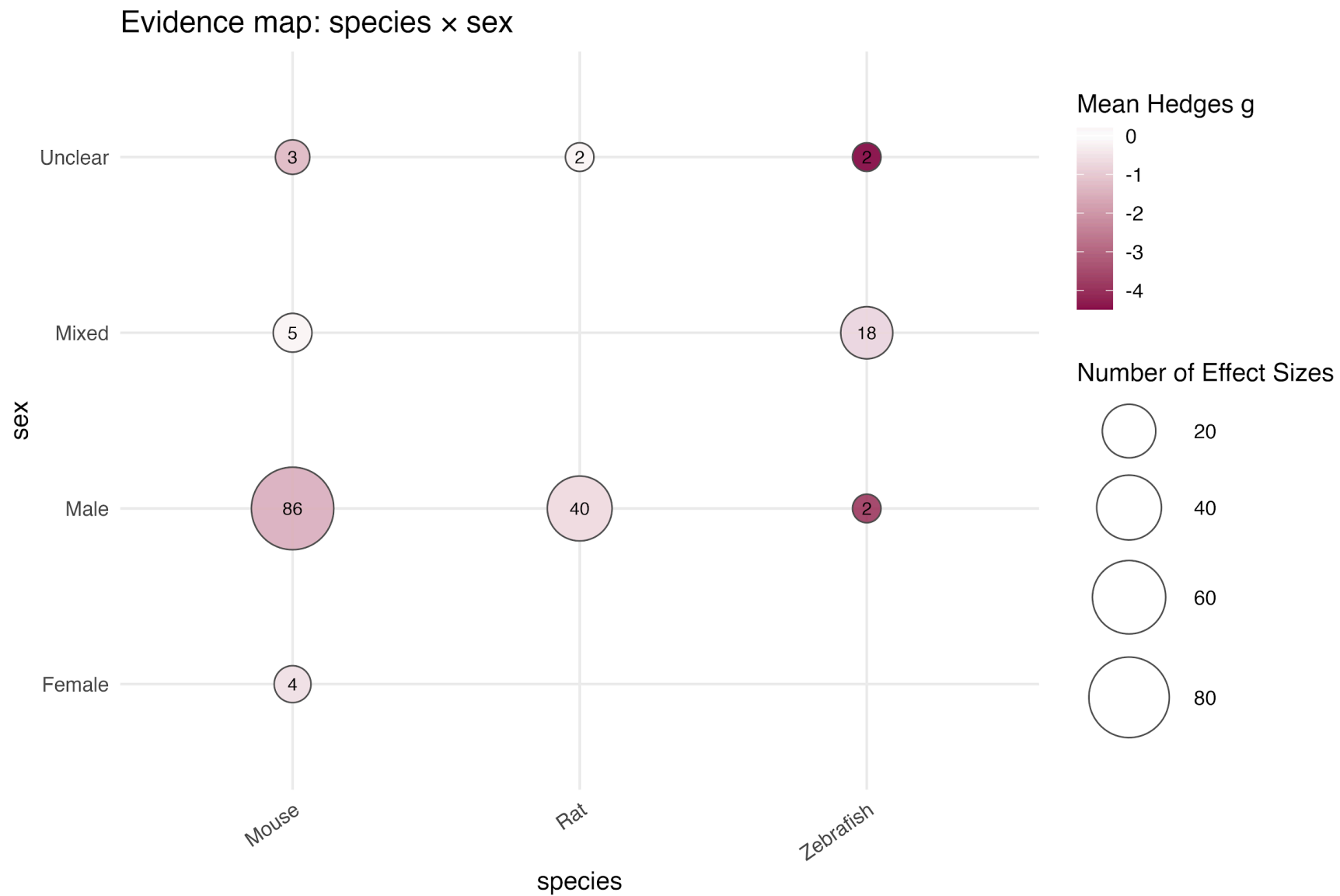
Figure S16

Figure S16. Evidence map of effect sizes for social preference by species and sex. Circle size represents the number of effect sizes, and colour intensity indicates the mean Hedges' g .

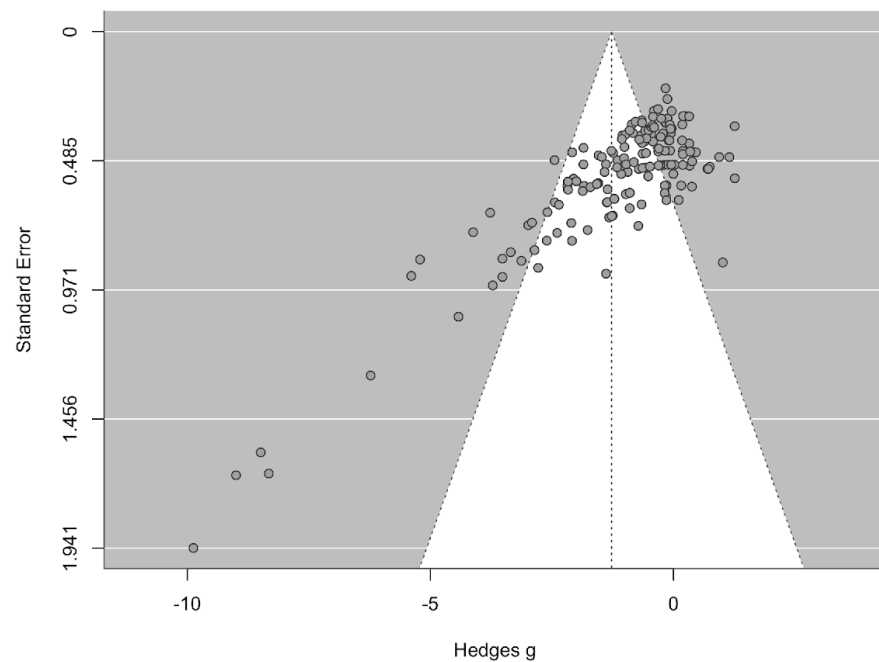
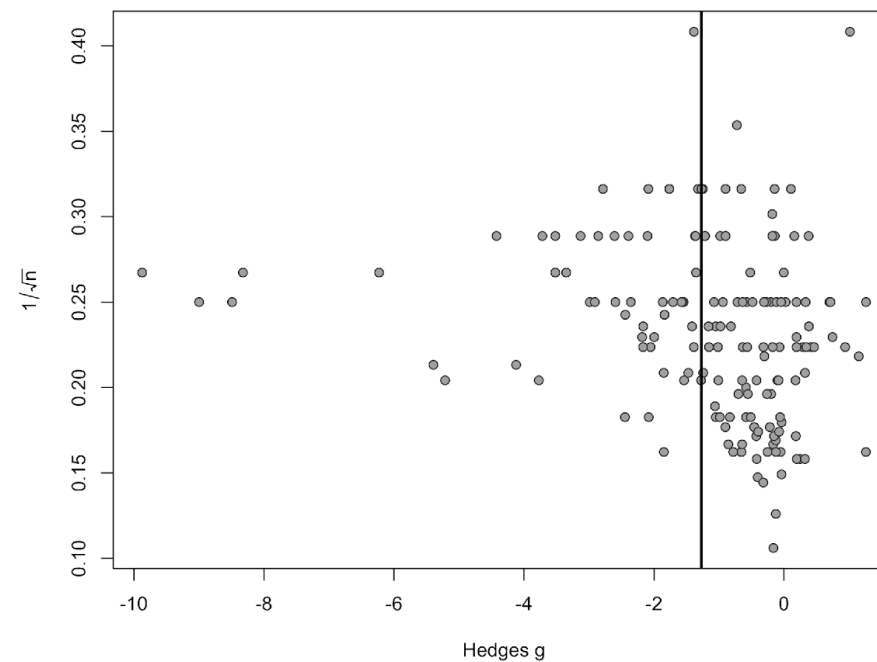
Figure S17**A****B**

Figure S17. Funnel plots of effect sizes for social preference. (A) Funnel plot showing Hedges' g plotted against the standard error. (B) Funnel plot showing Hedges' g plotted against the inverse square root of total sample size as an alternative measure of study precision. Each point corresponds to one effect size.

Figure S18

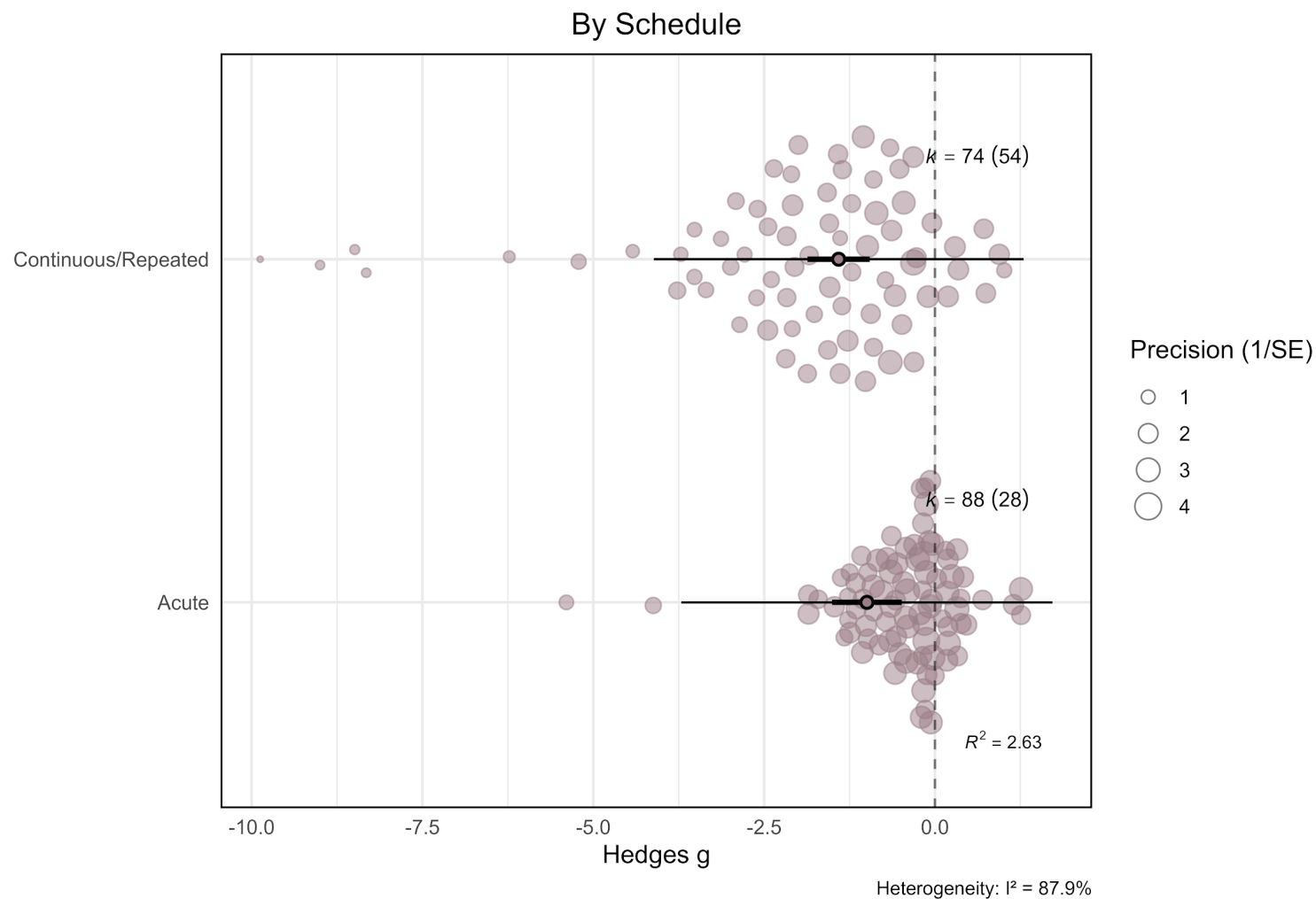


Figure S18. Effects of NMDA receptor antagonists on social preference stratified by administration schedule. Orchard plot showing pooled effects for acute and continuous/repeated schedules. Filled circles represent individual effect sizes scaled by study precision, and open circles indicate pooled point estimates with 95% confidence intervals (thick horizontal lines) and prediction intervals (thin horizontal lines). k denotes the number of effect sizes, with the number of independent studies shown in parentheses.

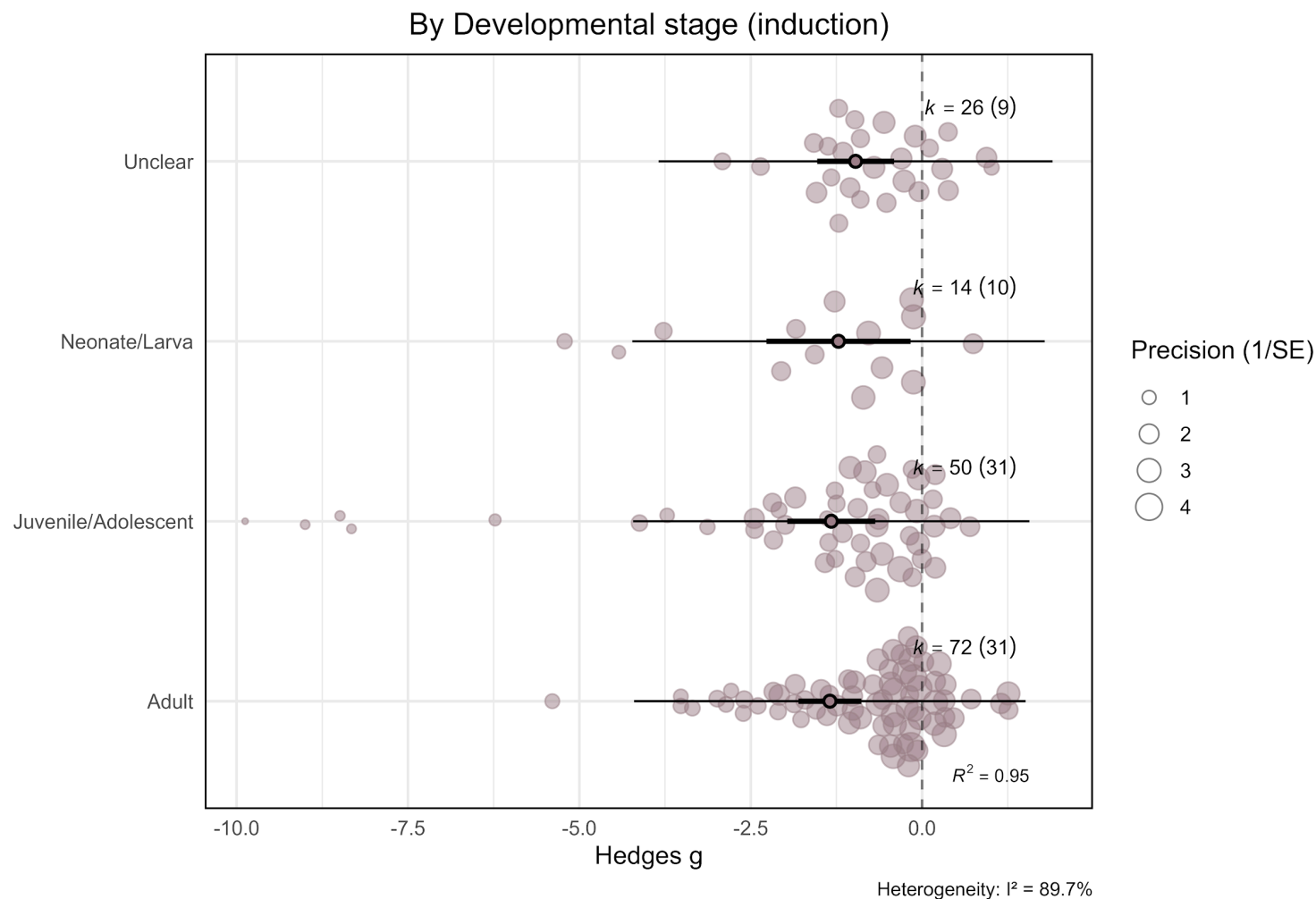
Figure S19

Figure S19. Effects of NMDA receptor antagonists on social preference stratified by developmental stage at induction. Orchard plot showing pooled effects for adult, juvenile/adolescent, neonatal/larval, and studies with unclear developmental stage at induction. Filled circles represent individual effect sizes scaled by study precision, and open circles indicate pooled point estimates with 95% confidence intervals (thick horizontal lines) and prediction intervals (thin horizontal lines). k denotes the number of effect sizes, with the number of independent studies shown in parentheses.

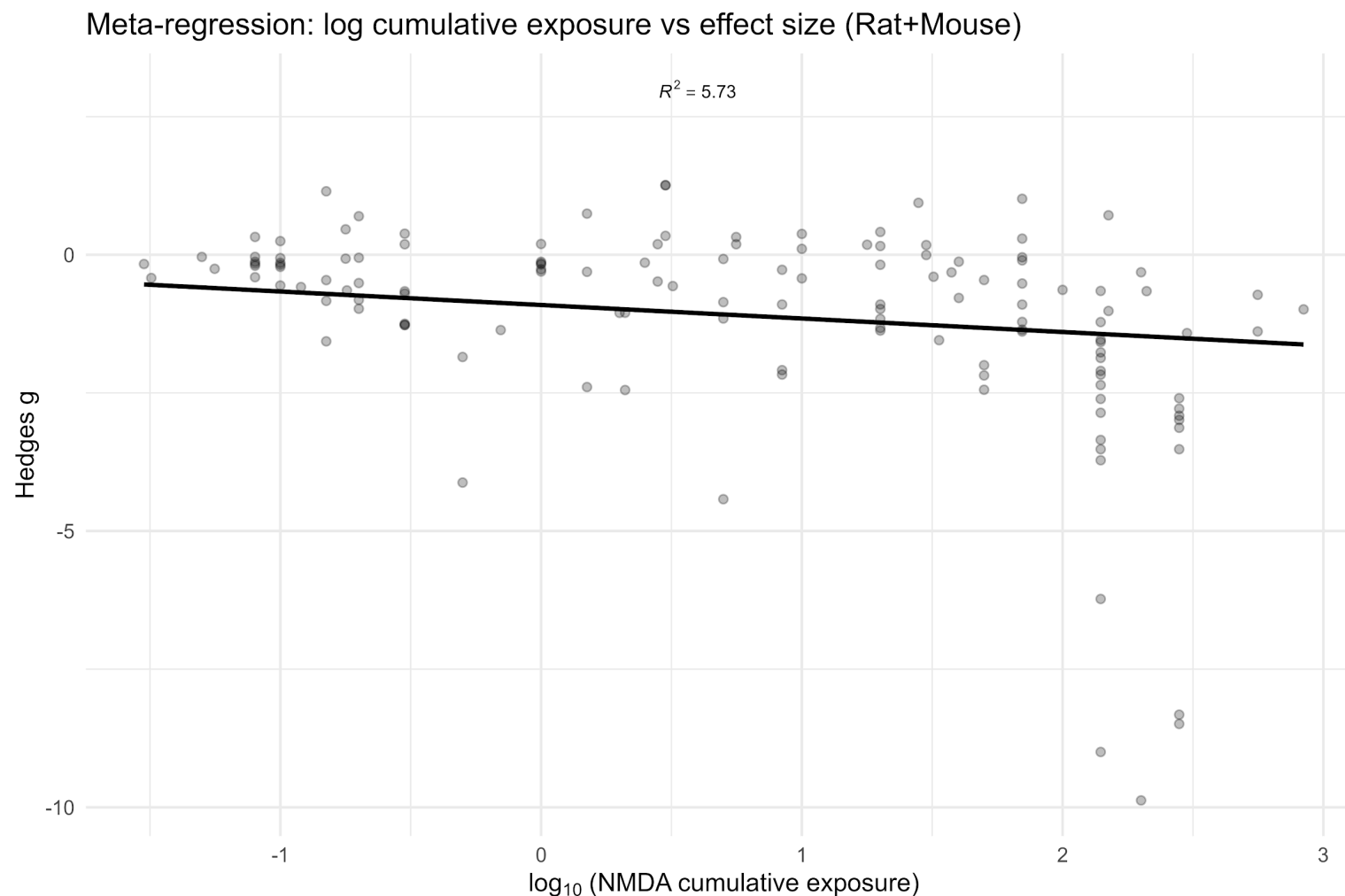
Figure S20

Figure S20. Meta-regression of log-transformed cumulative NMDA antagonist exposure and effect sizes for social preference. The plot shows Hedges' g as a function of $\log_{10}$ -transformed cumulative NMDA antagonist exposure, including only studies conducted in rats and mice. Cumulative exposure was calculated, where data were available, as the product of dose, intervention frequency, and duration of exposure. Each point represents one effect size, and the solid line represents the fitted meta-regression slope from a multivariate random-effects model.

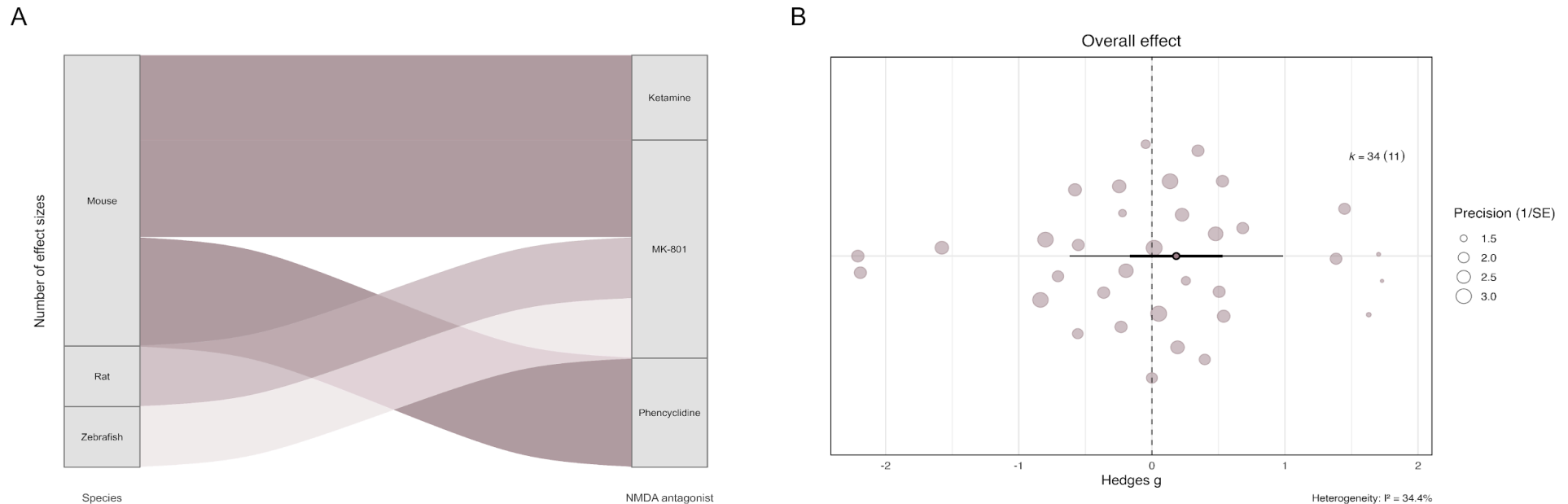
Figure S21

Figure S21. Effects of NMDA receptor antagonists on locomotor activity in experiments measuring social preference. (A) Alluvial plot showing the distribution of effect sizes across species and NMDA receptor antagonists for studies in which locomotor activity was assessed alongside social preference. Flow width represents the number of effect sizes contributing to each category. (B) Orchard plot of the overall pooled effect of NMDA receptor antagonists on locomotor activity. Filled circles represent individual effect sizes scaled by study precision, and the open circle indicates the pooled point estimate with 95% confidence intervals (thick horizontal lines) and prediction interval (thin horizontal lines). k denotes the number of effect sizes, with the number of independent studies shown in parentheses.

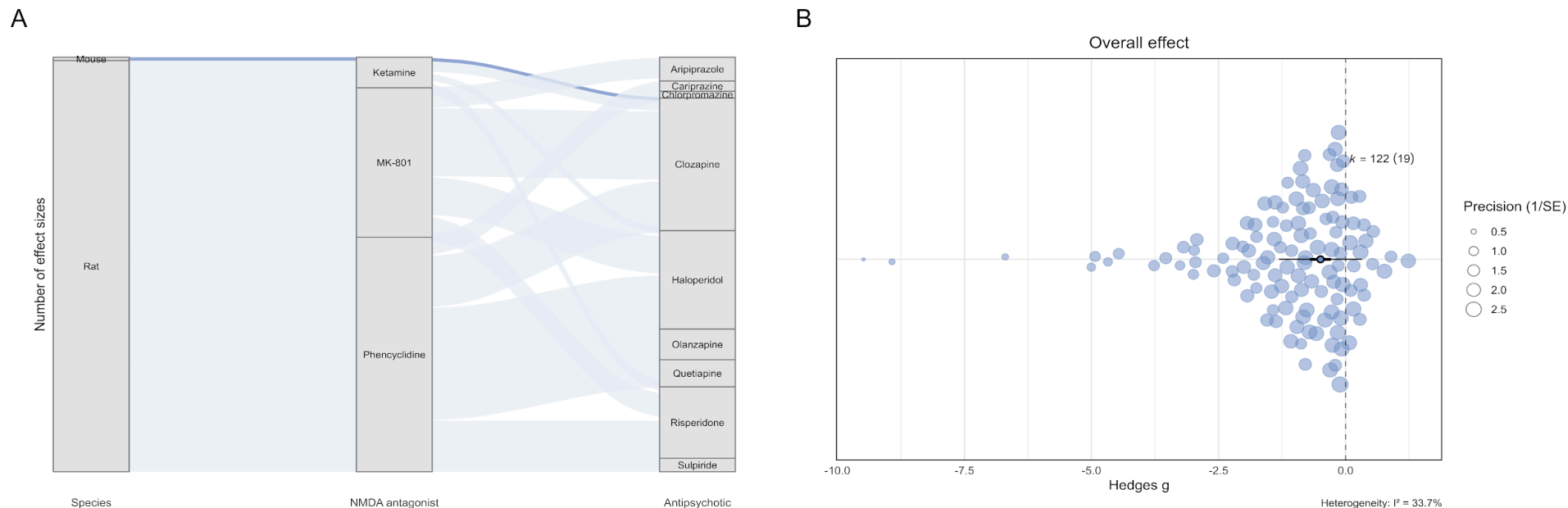
Figure S22

Figure S22. Effects of antipsychotics on NMDA receptor antagonist–induced locomotor alterations in social interaction experiments. (A) Alluvial plot showing the distribution of effect sizes across species, NMDA receptor antagonists, and antipsychotics for experiments in which locomotor activity was assessed alongside social interaction. Flow width represents the number of effect sizes contributing to each category. (B) Orchard plot of the overall pooled effect of antipsychotics on NMDA antagonist–induced locomotor alterations. Filled circles represent individual effect sizes scaled by study precision, and the open circle indicates the pooled point estimate with 95% confidence intervals (thick horizontal lines) and prediction interval (thin horizontal lines). k denotes the number of effect sizes, with the number of independent studies shown in parentheses.

3.2 CAPÍTULO 2

Ketamine-induced NMDA receptor hypofunction alters social and locomotor behavior in adult zebrafish

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Ketamine-induced NMDA receptor hypofunction alters social and locomotor behavior in adult zebrafish

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ABSTRACT

Background: NMDA receptor antagonists, such as ketamine, are widely used to model schizophrenia-related phenotypes in preclinical studies. While zebrafish have emerged as a promising model organism for neuropsychiatric research, few studies have characterized their behavioral responses to repeated ketamine exposure.

Methods: Three independent experiments were conducted to evaluate the acute, repeated and sustained behavioral effects of ketamine in adult zebrafish. In Experiment I, fish were exposed to 10, 20, or 40 mg/L ketamine once daily for five days, and submitted to the social preference (SPT) and open tank (OTT) tests on days 1 and 5, and re-exposed and re-tested on day 7 after a 48-hour washout. Experiments II and III assessed whether behavioral changes persisted following 5- or 14-day exposure protocols, with testing conducted 48 hours after the final treatment.

Results: Ketamine induced robust, concentration-dependent alterations in Experiment I: it reduced social interaction and increased locomotor activity in the SPT on all experimental days, while increased rotational behavior in the OTT on days 1 and 5. These effects did not intensify over repeated exposure and were not sustained after a 48-hour washout in either protocol (Experiments II and III).

Conclusions: The results support the utility of zebrafish for modeling acute behavioral responses to NMDA receptor antagonism, capturing features of schizophrenia-like phenotypes. However, no evidence of behavioral sensitization or lasting disruption was observed, diverging from rodent studies. Future studies should incorporate antipsychotic validation, neurochemical analyses, and alternative exposure strategies to further develop zebrafish as a translational model for psychiatric research.

Keywords: Ketamine; NMDA receptor hypofunction; Social withdrawal; Hyperlocomotion; Stereotyped swimming; Schizophrenia-like phenotypes

1 INTRODUCTION

2 Animal models that recapitulate schizophrenia-like behavioral phenotypes are essential for uncovering the
3 disorder's mechanisms and advancing translational research in psychiatry. Among the pharmacological
4 tools used in preclinical studies, non-competitive N-methyl-D-aspartate receptor (NMDAR) antagonists
5 such as MK-801, phencyclidine, and ketamine are widely employed to induce behavioral alterations that
6 reflect core symptom domains of schizophrenia [1–5]. By mimicking NMDAR hypofunction, these
7 compounds contribute to dopaminergic hyperactivity in subcortical regions and evoke transient behavioral
8 changes that resemble the positive, negative, and cognitive symptoms associated with the disorder [2, 5–
9 8].

10 Rodent models have played a central role in preclinical schizophrenia research. However, a substantial
11 number of pharmacological candidates that showed promise in these models have failed in clinical trials,
12 highlighting a critical gap in translational success [9, 10]. These failures raise concerns about the external
13 validity of such models, particularly their ability to predict complex human outcomes. In response, there is
14 growing recognition of the need for alternative models that can enhance translational relevance and reduce
15 species-specific biases. One such strategy involves testing conserved behavioral and pharmacological
16 effects across diverse species [11–14]. Among non-rodent models, zebrafish (*Danio rerio* Hamilton, 1882)
17 have emerged as a valuable organism in neurobehavioral research due to their well-characterized nervous
18 system, evolutionary conservation of key neurotransmitter pathways, and suitability for high-throughput
19 experimental designs [15–17]. These features make zebrafish a promising platform for cross-species
20 validation and for improving the predictive value of preclinical findings.

21 Previous work from our group demonstrated that the NMDAR antagonist MK-801 (dizocilpine) disrupts
22 social behavior and induces hyperlocomotion in adult zebrafish, supporting the model's relevance for
23 studying schizophrenia-like phenotypes [18, 19]. Compared to MK-801, the behavioral effects of ketamine
24 in zebrafish are less consistently described. Although ketamine is widely used in preclinical models of
25 schizophrenia, studies in zebrafish vary considerably in design and outcomes tested, with limited
26 agreement on its impact across social, locomotor, and exploratory domains [1].

27 The effects of ketamine exposure on zebrafish behavior have been most commonly examined after acute
28 exposures [1, 20–22], and far less is known about how behavior changes with repeated administration, or
29 about whether these effects persist once treatment has ended. In rodents, repeated exposure to NMDAR
30 antagonists like ketamine results in progressively heightened behavioral responses or long-lasting
31 changes that remain after drug clearance [23–28]. These findings suggest that both the temporal dynamics
32 and the cumulative impact of ketamine exposure are important factors in shaping behavioral outcomes.
33 Whether zebrafish show similar patterns is still unknown.

34 This study aimed to characterize schizophrenia-like behavioral effects of ketamine in adult zebrafish across
35 three experimental protocols. Experiment I examined how behavioral responses evolved over repeated
36 exposures and test sessions to assess the potential development of sensitization. Experiment II evaluated
37 whether ketamine-induced behavioral alterations persisted after a short-term (5-day) exposure followed
38 by a 48-hour washout period. Experiment III extended this investigation by assessing whether a longer-
39 term (14-day) exposure regimen would lead to detectable effects after the same washout interval.
40 Together, these protocols allowed for the systematic assessment of both short- and long-term
41 consequences of ketamine exposure across different behavioral domains.

42 MATERIALS AND METHODS

43 This study was conducted in accordance with ethical guidelines and approved by the institutional animal
44 ethics committee of the Universidade Federal do Rio Grande do Sul (approval: #35525/2018). The
45 reporting of this study follows the ARRIVE guidelines for animal research transparency and rigor [29]. A

completed ARRIVE Essential 10 checklist is available at osf.io/yqv6a. The experimental protocols for all three experiments were preregistered on the Open Science Framework, specifying hypotheses, dependent variables, conditions, analyses, outlier exclusion criteria, and sample size calculation [30] (Experiment I: osf.io/r7xdy [31]; Experiment II: osf.io/kjf3r [32]; Experiment III: osf.io/vkcft [33]). Assistance from ChatGPT-4o (OpenAI, San Francisco, CA, USA) was used to support the optimization of text and formatting throughout the manuscript. All outputs were critically reviewed, edited, and validated by the authors, who take full responsibility for the content and interpretation of the work.

Animals, housing, and husbandry

A total of 224 adult wild-type short-fin zebrafish, aged between 3 and 4 months, were used across three separate experiments: 128 animals in Experiment I, and 48 animals each in Experiments II and III. Of these, 217 completed the full experimental protocol and behavioral assessments (Experiment I: 122; Experiment II: 48; Experiment III: 47) – missing data is due to the death of 7 fish during the course of the experiments (full details in the “Statistical analysis” section). The final sample included 111 females and 106 males, with the following sex distribution: Experiment I – 62 females, 60 males; Experiment II – 25 females, 23 males; Experiment III – 24 females, 23 males. As no sex-specific effects were found in Experiment I, data from male and female animals were combined in all analyses. Animals were sourced from a commercial supplier (Delphis, Porto Alegre, RS, Brazil). All animals were maintained at the zebrafish facility of the Instituto de Ciências Básicas da Saúde (Universidade Federal do Rio Grande do Sul) under a controlled 14:10 hour light/dark cycle (lights on at 07:00). Prior to the onset of experiments, fish were allowed to acclimate to laboratory conditions for at least two weeks. They were group-housed in 16 L glass tanks (40 × 20 × 24 cm) enriched with small PVC tubes and kept at a maximum density of two animals per liter. Tanks were filled with dechlorinated tap water and equipped with continuous aeration and mechanical, biological, and chemical filtration systems (Altamar, Jacaré, SP, Brazil). Water quality parameters were maintained within acceptable ranges for zebrafish: temperature at 27 ± 2 °C, pH 6.8 ± 0.3 , and conductivity between 500–800 $\mu\text{S}/\text{cm}$. Fish were fed twice daily with a combination of *Artemia salina* and commercial flake food (Poytara, Araraquara, SP, Brazil). During the experimental phase, animals were redistributed into smaller aquariums (18 × 18 × 18 cm) according to the treatment group. Each group was housed separately, and all tanks were kept under identical conditions within the same facility, including lighting, temperature, and vertical level. Tanks were coded to maintain caregiver blinding regarding treatment allocation. There were two housing tanks for each treatment group in all experiments. Physical barriers were used to prevent visual contact between groups, both in housing aquariums and during exposure to treatments. At the conclusion of the experiments, animals were euthanized using hypothermic shock. This procedure involved immersion in water cooled to 2–4 °C for at least two minutes after cessation of opercular movements, followed by decapitation to ensure death. Sex was confirmed post-mortem via examination of the gonads.

Drugs

Ketamine solutions were prepared by diluting a 10% (v/v) injectable formulation (Syntec, Tamboré, SP, Brazil) in water maintained under the same conditions as the housing tanks. In Experiment I, concentrations of 10, 20, and 40 mg/L were used. In Experiments II and III, only the 40 mg/L concentration was used. These concentrations were selected based on previous studies reported in the literature [1, 21, 22]. Fish were exposed to the treatment solutions by immersion in a beaker for 20 minutes. The 20-minute immersion period was chosen following established protocols reported in the literature and pilot experiments conducted by our group using another NMDA receptor antagonist in zebrafish [1, 18, 21, 22]. Control animals were handled in the same way but immersed in ketamine-free water under identical conditions.

Experimental Design

1 This study included three independent experiments, each conducted with a separate set of animals. The
2 design and timeline for each experiment are illustrated in Figure 1. Animals were allocated to the
3 experimental groups using block randomization, ensuring balance across presumed sex and home tank
4 origin. Because zebrafish sex cannot be confirmed without gonadal inspection, group allocation was
5 counterbalanced based on external sex characteristics at the time of assignment. To minimize potential
6 biases, the order of behavioral testing was determined using block randomization, with the sequence of
7 randomized blocks generated via [random.org](https://www.random.org), taking into account the test apparatuses to ensure even
8 distribution across treatment conditions. The experimental unit in all analyses was a single animal, with
9 each fish contributing one independent data point to the statistical comparisons. Both the experimenters
10 conducting the behavioral assessments and those responsible for transferring the animals to the test
11 apparatus were blinded to group allocation.

12 Experiment I

13 This experiment investigated the behavioral effects of repeated ketamine exposure at different
14 concentrations. A total of 128 adult zebrafish were randomly assigned to one of four treatment groups (n
15 = 32): control (17 females, 15 males), ketamine 10 mg/L (16 females, 16 males), ketamine 20 mg/L (15
16 females, 17 males), and ketamine 40 mg/L (16 females, 16 males). Missing data in the analysis reflect
17 animal mortality as detailed in the “Statistical analysis” section. Animals received a 20-minute exposure to
18 their assigned treatment once per day from experimental days 1 to 5. For each session, fish were
19 transferred in groups of four to beakers containing 400 mL of the appropriate solution. After the washout
20 period on day 6 (no handling or exposure), a final treatment was administered on day 7. This administration
21 schedule was informed by prior research in mice, which demonstrated behavioral sensitization following
22 five consecutive days of ketamine exposure and a subsequent 48-hours washout period [27]. Following
23 each exposure, animals were returned to their respective home tanks. Behavioral testing included both
24 the social preference test (SPT) and the open tank test (OTT), conducted in sequence on days 1, 5, and
25 7 immediately after drug exposure. This experiment was designed to test the hypothesis that repeated
26 exposure to ketamine reduces social interaction (as measured by time spent in the interaction zone during
27 the SPT) and increases rotations in the OTT.

28 Experiment II

29 Building on the findings from Experiment I, the 40 mg/L ketamine concentration was selected for further
30 investigation in Experiment II. This experiment evaluated whether behavioral changes induced by
31 ketamine persist after treatment ends, comparing the effects of a single exposure to those of repeated
32 administration. A total of 48 adult zebrafish were randomly assigned to one of three treatment groups (n
33 = 16): control (11 females, 5 males), single exposure (7 females, 9 males), and repeated exposure (7
34 females, 9 males). Animals in the repeated group received a 20-minute exposure to 40 mg/L ketamine
35 once daily from experimental days 1 to 5. The single exposure group was treated with water from days 1
36 to 4 and exposed to 40 mg/L ketamine only on day 5. Control animals were exposed to water throughout
37 the same period. For each exposure session, fish were transferred in groups of four to beakers containing
38 400 mL of the assigned solution. On day 6, animals remained in their home tanks without intervention.
39 Behavioral testing was carried out on day 7, 48 hours after the final exposure. All animals were evaluated
40 in both the SPT and the OTT, in that order. This experiment was designed to test the hypothesis that
41 repeated ketamine exposure induces sustained behavioral alterations as compared to a single exposure.

42 Experiment III

43 Following the results of Experiment II, a new experiment was designed to assess whether a longer
44 exposure period would lead to sustained behavioral effects. A total of 48 adult zebrafish were randomly
45 assigned to one of three treatment groups (n = 16): control (9 females, 7 males), single exposure (5
46 females, 11 males), and repeated exposure (10 females, 6 males). Missing data in the analysis reflect

animal mortality as detailed in the “Statistical analysis” section. Animals in the repeated group received a 20-minute exposure to 40 mg/L ketamine once daily from experimental days 1 to 14. The single exposure group was treated with water from days 1 to 13 and received a single 40 mg/L ketamine exposure on day 14. Control animals were exposed to water throughout the same period. For each exposure session, fish were transferred in groups of four to beakers containing 400 mL of the assigned solution. On day 15, animals remained in their home tanks without intervention. Behavioral testing was carried out on day 16, 48 hours after the final exposure. All animals were evaluated in both the SPT and the OTT, in that order. This experiment was designed to test the hypothesis that repeated ketamine exposure induces sustained behavioral alterations as compared to a single exposure.

Behavioral tests

The SPT and OTT were conducted sequentially to assess behavioral outcomes. Testing took place during the light phase, between 7:00 a.m. and 1:00 p.m. Animals were recorded from the front view during the SPT and from above during the OTT. Behavioral data were analyzed using ANY-maze® tracking software (version 7.20; Wood Dale, IL, USA). All researchers involved in analyzing the recordings were blinded to group allocation.

Social preference test

The SPT was used to assess social affiliation in zebrafish [18, 19]. After treatment, each animal was individually placed in a rectangular tank (30 × 10 × 15 cm) positioned between two adjacent tanks (15 × 10 × 13 cm each). One side tank contained a group of 10 conspecifics (social stimulus), while the other was left empty (neutral stimulus). All tanks were filled with water under housing conditions to a height of 10 cm. The location of the social stimulus (left or right) was alternated across subjects to control for side preference. Water in the setup was replaced between tests. For analysis, the test tank was virtually divided lengthwise into three equal zones: the interaction zone (nearest to the social stimulus), the neutral zone (nearest to the empty tank), and a central intermediate zone. Video recordings lasted 7 minutes, consisting of a 2-minute habituation period followed by 5 minutes of data collection. The primary outcome was time spent in the interaction zone. Additional variables included total distance traveled, maximum speed, number of line crossings, percentage of distance traveled in the interaction zone, and time spent immobile.

Open tank test

The OTT was conducted immediately after the SPT to evaluate locomotor activity and stereotyped swimming behavior [18]. Each fish was placed in the center of a circular white arena (24 cm in diameter, 8 cm high) filled with 2 cm of water under standard conditions. Recordings were captured from an overhead view for a 5-minute session. The arena was refilled with clean water between trials. For behavioral analysis, the arena was virtually divided into two zones: a center zone (12 cm in diameter) and a peripheral zone. The primary variable of interest was the number of rotations, which served as a marker of repetitive or stereotyped movement. Additional outcomes included total distance traveled, maximum speed, time spent in the center zone, percentage of distance traveled in the center, absolute turn angle, and time spent immobile.

Statistical analysis

For Experiment I, the planned sample size was 48 animals per group (total = 192), based on a power of 0.88 to detect a medium effect size ($f = 0.3$, $\alpha = 0.05$). After completing two batches (64 animals each), testing was halted due to consistent results and effect sizes greater than expected. The final sample size was 32 per group (total = 128). A sensitivity analysis indicated that, for post hoc comparisons between two groups ($n = 32$ each), the design had sufficient power ($1 - \beta = 0.88$, $\alpha = 0.05$) to detect a minimum effect size of $d = 0.79$. Since most of the observed effect sizes exceeded this threshold, the study was adequately powered to detect the effects identified in pairwise comparisons. No outliers were excluded. Six animals

died during the experiment: three from the 40 mg/L group (two females during exposure on day 3; one male one hour after exposure on day 4), one male in the control group (bitten by a tankmate on day 5), and one male from each the 10 and 20 mg/L groups (after testing on day 5). For Experiments II and III, the initial plan was 32 animals per group (total = 96), based on a power of 0.94 to detect a large effect ($f = 0.4$, $\alpha = 0.05$). After one batch (48 animals), testing was discontinued due to the absence of clear treatment effects and in order to minimize animal use. The final sample size was 16 per group. One male from the control group died during Experiment III; no deaths occurred in Experiment II. Sample sizes were estimated using G*Power (version 3.1; Düsseldorf, Germany). Effect sizes (Cohen's d) and 95% confidence intervals were calculated using R (version 4.3.1), RStudio (version 2023.12.1+402; Boston, MA, USA), and the meta package (version 8.0-1) [34]. In Experiment I, exploratory subgroup analyses across test days were conducted using a random-effects model. Behavioral data were analyzed in GraphPad Prism (version 10.1.0; Boston, MA, USA). One-way ANOVA followed by Bonferroni post hoc tests was used for parametric outcomes; immobility time was analyzed using the Kruskal–Wallis test followed by Dunn's correction. Results are reported as mean $\pm$ SD for normally distributed outcomes and as median with interquartile range for nonparametric data (as determined by inspecting the Q–Q plots). A p-value below 0.05 was considered statistically significant.

Deviations from registered protocols

In Experiment I, the preregistered sample size was 48 animals per group (total = 192), but data collection was stopped after 32 animals per group due to consistent results and larger-than-expected effects. The planned analysis involved a two-way ANOVA with treatment group and experimental day as factors. However, because individual animals could not be reliably tracked across days, each day was analyzed independently using one-way ANOVA to avoid inflating the sample size. Additionally, "percent distance in the interaction zone" and "percent distance in the center" were not preregistered outcomes but were included post hoc in Experiment I for consistency with Experiments II and III. In Experiments II and III, although 32 animals per group were planned, testing was halted after one batch (16 per group) due to a lack of apparent treatment effects and to reduce unnecessary animal use, in accordance with the 3Rs principle [35]. Across all experiments, immobility time was analyzed using Kruskal–Wallis followed by Dunn's test instead of ANOVA, as this outcome violated parametric assumptions.

RESULTS

Experiment I

Ketamine administration led to consistent, concentration-dependent behavioral changes across tests. In the SPT, all concentrations reduced time spent in the interaction zone and percent distance in the interaction zone, indicating decreased social preference. At the same time, ketamine increased total distance traveled, maximum speed, and number of line crossings, reflecting a pattern of hyperlocomotion. In the OTT, ketamine also increased number of rotations, distance traveled, and maximum speed, especially at higher concentrations, while the lower concentration increased time spent in the center and percent distance in the center, an anxiolytic-like response typically interpreted as reduced thigmotaxis in an unfamiliar environment. Subgroup analyses revealed no significant differences across testing days for any outcome, suggesting that repeated ketamine exposure did not result in behavioral sensitization. The following sections provide a more detailed account of these results.

Social preference test

Figure 2 shows the effects of different ketamine concentrations (10, 20, and 40 mg/L) in the SPT across three test sessions (days 1, 5, and 7). A significant effect of treatment was observed on each test day for the primary outcome, with ketamine reducing time spent in the interaction zone (day 1: $F(3,124) = 32.96$, $p < 0.0001$; day 5: $F(3,120) = 13.80$, $p < 0.0001$; day 7: $F(3,118) = 12.73$, $p < 0.0001$; Figure 2A). Bonferroni post hoc comparisons showed that all ketamine concentrations significantly reduced social preference

1 compared to the control group on all three days. Effect sizes are presented in Figure 2B. On day 1, effect
2 sizes were directly proportional to ketamine concentration, indicating a concentration-dependent increase
3 in effect magnitude: -1.21 (95% CI [-1.75, -0.68]) for 10 mg/L, -1.76 [-2.33, -1.18] for 20 mg/L, and -2.87 [-
4 3.57, -2.16] for 40 mg/L. This pattern persisted on day 5 (-0.57 [-1.07, -0.06] for 10 mg/L; -0.90 [-1.42, -
5 0.38] for 20 mg/L; -1.71 [-2.31, -1.12] for 40 mg/L) and day 7 (-0.76 [-1.27, -0.24] for 10 mg/L; -0.84 [-1.36,
6 -0.32] for 20 mg/L; -1.71 [-2.31, -1.12] for 40 mg/L). No significant differences were found across testing
7 days in the subgroup analysis ($p = 0.2760$).

8 Ketamine also significantly increased total distance traveled during the test sessions (Figure 2C). An effect
9 of treatment was detected on all days (day 1: $F(3,124) = 40.58$, $p < 0.0001$; day 5: $F(3,120) = 28.25$,
10 $p < 0.0001$; day 7: $F(3,118) = 31.25$, $p < 0.0001$). Post hoc tests indicated significant increases at all
11 concentrations on day 1, at 20 and 40 mg/L on day 5, and again for all groups on day 7. Effect sizes are
12 shown in Figure 2D. On each day, effect sizes increased with concentration: 0.91 [0.40, 1.43] for 10 mg/L,
13 1.92 [1.32, 2.51] for 20 mg/L, and 2.37 [1.73, 3.02] for 40 mg/L on day 1; 0.53 [0.03, 1.04] for 10 mg/L, 1.08
14 [0.55, 1.61] for 20 mg/L, and 2.12 [1.48, 2.75] for 40 mg/L on day 5; and 1.11 [0.58, 1.65] for 10 mg/L, 1.37
15 [0.82, 1.93] for 20 mg/L, and 2.12 [1.49, 2.76] for 40 mg/L on day 7. No significant differences were found
16 across testing days in the subgroup analysis ($p = 0.7371$).

17 A significant main effect of treatment was also observed for maximum swimming speed (Figure 2E), with
18 increases detected on each test day (day 1: $F(3,124) = 20.31$, $p < 0.0001$; day 5: $F(3,120) = 5.82$, $p = 0.001$;
19 day 7: $F(3,118) = 12.81$, $p < 0.0001$). Post hoc analysis showed that only the 20 and 40 mg/L groups
20 significantly differed from controls across all days, with no significant effects at 10 mg/L. As shown in Figure
21 2F, effect sizes increased with concentration: 0.59 [0.09, 1.09] for 10 mg/L, 1.31 [0.77, 1.85] for 20 mg/L,
22 and 2.07 [1.46, 2.68] for 40 mg/L on day 1; 0.46 [-0.04, 0.96] for 10 mg/L, 0.74 [0.22, 1.25] for 20 mg/L,
23 and 1.26 [0.71, 1.82] for 40 mg/L on day 5; and 0.60 [0.09, 1.11] for 10 mg/L, 1.09 [0.55, 1.62] for 20 mg/L,
24 and 1.38 [0.82, 1.95] for 40 mg/L on day 7. No significant differences were found across testing days in
25 the subgroup analysis ($p = 0.5640$).

26 Line crossings, another index of locomotor activity, was also increased following ketamine exposure
27 (Figure 2G) on all days (day 1: $F(3,124) = 51.21$, $p < 0.0001$; day 5: $F(3,120) = 35.98$, $p < 0.0001$; day 7:
28 $F(3,118) = 39.69$, $p < 0.0001$). All ketamine groups differed from controls on days 1 and 7, while only the
29 20 and 40 mg/L concentrations were significant on day 5. Effect sizes are shown in Figure 2H. On each
30 test day, values increased with concentration: 1.27 [0.73, 1.81] for 10 mg/L, 2.37 [1.72, 3.01] for 20 mg/L,
31 and 3.05 [2.32, 3.77] for 40 mg/L on day 1; 0.70 [0.19, 1.21] for 10 mg/L, 1.20 [0.66, 1.73] for 20 mg/L, and
32 2.57 [1.88, 3.27] for 40 mg/L on day 5; and 1.06 [0.53, 1.60] for 10 mg/L, 1.38 [0.83, 1.94] for 20 mg/L, and
33 2.55 [1.86, 3.24] for 40 mg/L on day 7. No significant differences were found across testing days in the
34 subgroup analysis ($p = 0.5838$).

35 Treatment also significantly decreased the percent distance traveled within the interaction zone
36 (Supplementary Figure 1A). An effect was observed on all test days (day 1: $F(3,124) = 28.64$, $p < 0.0001$;
37 day 5: $F(3,120) = 15.12$, $p < 0.0001$; day 7: $F(3,118) = 7.67$, $p = 0.0001$). Post hoc analysis confirmed that
38 all concentrations significantly reduced this metric compared to controls on each day. Effect sizes are
39 shown in Supplementary Figure 1B. Effect magnitudes increased with concentration, reflected by more
40 negative values: -1.13 [-1.66, -0.60] for 10 mg/L, -1.67 [-2.24, -1.10] for 20 mg/L, and -2.44 [-3.10, -1.79]
41 for 40 mg/L on day 1; -0.75 [-1.26, -0.24] for 10 mg/L, -1.10 [-1.64, -0.57] for 20 mg/L, and -1.75 [-2.35, -
42 1.15] for 40 mg/L on day 5; and -0.56 [-1.07, -0.06] for 10 mg/L, -0.56 [-1.07, -0.06] for 20 mg/L, and -1.22
43 [-1.77, -0.66] for 40 mg/L on day 7. No significant differences were found across testing days in the
44 subgroup analysis ($p = 0.0770$).

45 No significant effect of ketamine was observed on time spent immobile during the social preference test
46 (Supplementary Figure 1C). Kruskal–Wallis tests yielded non-significant results on all test days (day 1:
47 $H = 7.09$, $p = 0.0692$; day 5: $H = 3.02$, $p = 0.3879$; day 7: $H = 1.77$, $p = 0.6223$). Effect sizes are presented

in Supplementary Figure 1D. On day 1, values were -0.36 [-0.85, 0.14] for 10 mg/L, -0.30 [-0.79, 0.19] for 20 mg/L, and -0.17 [-0.66, 0.32] for 40 mg/L. Day 5 values were 0.00 [-0.49, 0.50] for 10 mg/L, -0.30 [-0.80, 0.19] for 20 mg/L, and -0.43 [-0.95, 0.08] for 40 mg/L. On day 7, effect sizes were -0.39 [-0.90, 0.11] for 10 mg/L, -0.42 [-0.92, 0.08] for 20 mg/L, and 0.15 [-0.36, 0.65] for 40 mg/L. No significant differences were found across testing days in the subgroup analysis ($p = 0.9385$).

Open tank test

Figure 3 summarizes the results of the OTT, assessing multiple behavioral outcomes in response to different ketamine concentrations (10, 20, and 40 mg/L) across three sessions (days 1, 5, and 7). For the primary outcome, number of rotations, a significant effect of treatment was observed on day 1 ($F(3,124) = 5.32$, $p = 0.0018$) and day 5 ($F(3,120) = 9.61$, $p < 0.0001$), but not on day 7 ($F(3,118) = 1.73$, $p = 0.1637$; Figure 3A). Post hoc comparisons indicated significant increases in rotations for animals treated with 20 and 40 mg/L on day 1, and for 40 mg/L on day 5. Effect sizes and 95% confidence intervals are shown in Figure 3B. On day 1, values were: 0.15 [-0.34, 0.64] for 10 mg/L, 0.68 [0.18, 1.19] for 20 mg/L, and 0.94 [0.42, 1.46] for 40 mg/L. Day 5 showed a similar trend, with effect sizes of 0.22 [-0.27, 0.72] for 10 mg/L, 0.38 [-0.12, 0.87] for 20 mg/L, and 1.15 [0.60, 1.70] for 40 mg/L. On day 7, effect sizes were 0.21 [-0.29, 0.71] for 10 mg/L, -0.02 [-0.52, 0.47] for 20 mg/L, and 0.63 [0.11, 1.15] for 40 mg/L. Effect sizes increased with ketamine concentration across all days. No significant differences were found across testing days in the subgroup analysis ($p = 0.4911$).

For distance traveled, a significant effect of treatment was observed on day 1 ($F(3,124) = 21.81$, $p < 0.0001$), day 5 ($F(3,120) = 19.72$, $p < 0.0001$), and day 7 ($F(3,118) = 11.86$, $p < 0.0001$; Figure 3C). Post hoc comparisons revealed significant increases in distance for animals treated with 20 and 40 mg/L on day 1, and only 40 mg/L on days 5 and 7. Effect sizes and 95% confidence intervals are displayed in Figure 3D. On day 1, values were: -0.25 [-0.74, 0.24] for 10 mg/L, 0.81 [0.30, 1.32] for 20 mg/L, and 1.46 [0.91, 2.02] for 40 mg/L. On day 5, effect sizes were -0.07 [-0.56, 0.43] for 10 mg/L, 0.33 [-0.17, 0.83] for 20 mg/L, and 1.30 [0.74, 1.86] for 40 mg/L. On day 7, values were -0.14 [-0.64, 0.36] for 10 mg/L, -0.01 [-0.50, 0.49] for 20 mg/L, and 0.98 [0.44, 1.52] for 40 mg/L. The magnitude of the effect sizes increased with ketamine concentration on each day. No significant differences were found across testing days in the subgroup analysis ($p = 0.7904$).

For maximum speed, a significant effect of treatment was found on day 1 ($F(3,124) = 9.34$, $p < 0.0001$) and day 5 ($F(3,120) = 4.97$, $p = 0.0028$), but not on day 7 ($F(3,118) = 2.31$, $p = 0.0799$; Figure 3E). Post hoc analyses revealed significant increases in maximum speed only for the 40 mg/L group on days 1 and 5. Effect sizes and 95% confidence intervals are presented in Figure 3F. On day 1, values were: -0.11 [-0.60, 0.38] for 10 mg/L, 0.65 [0.14, 1.15] for 20 mg/L, and 0.95 [0.43, 1.46] for 40 mg/L. On day 5, values were: -0.38 [-0.87, 0.12] for 10 mg/L, 0.16 [-0.33, 0.66] for 20 mg/L, and 0.63 [0.11, 1.15] for 40 mg/L. On day 7, effect sizes were -0.23 [-0.73, 0.27] for 10 mg/L, 0.04 [-0.46, 0.54] for 20 mg/L, and 0.50 [-0.01, 1.02] for 40 mg/L. No significant differences were found across testing days in the subgroup analysis ($p = 0.5732$).

For time spent in the center, a significant effect of treatment was found on day 1 ($F(3,124) = 2.78$, $p = 0.0439$) and day 5 ($F(3,120) = 3.65$, $p = 0.0146$), but not on day 7 ($F(3,118) = 1.84$, $p = 0.1441$; Figure 3G). Post hoc comparisons indicated significantly increased center time only for animals treated with 10 mg/L on days 1 and 5. Effect sizes and 95% confidence intervals are provided in Figure 3H. On day 1, values were: 0.87 [0.36, 1.39] for 10 mg/L, 0.40 [-0.10, 0.89] for 20 mg/L, and 0.44 [-0.06, 0.93] for 40 mg/L. On day 5, effect sizes were 0.74 [0.23, 1.25] for 10 mg/L, 0.18 [-0.31, 0.68] for 20 mg/L, and 0.49 [-0.03, 1.00] for 40 mg/L. By day 7, effects diminished: 0.50 [-0.01, 1.01] for 10 mg/L, 0.26 [-0.24, 0.76] for 20 mg/L, and 0.07 [-0.43, 0.58] for 40 mg/L. The magnitude of the effect was highest for 10 mg/L across all days. Increased center time is interpreted as an anxiolytic effect, and was only observed at the lowest concentration. No significant differences were found across testing days in the subgroup analysis ($p =$

0.3194).

For percent distance in the center, a significant effect of treatment was detected on day 1 ($F(3,124) = 6.01$, $p = 0.0007$) and day 5 ($F(3,120) = 3.18$, $p = 0.0266$), but not on day 7 ($F(3,118) = 1.57$, $p = 0.2015$; Supplementary Figure 2A). Post hoc analysis revealed significant increases only for animals treated with 10 mg/L on days 1 and 5. Effect sizes and 95% confidence intervals are shown in Supplementary Figure 2B. On day 1, effect sizes were: 1.03 [0.51, 1.55] for 10 mg/L, 0.59 [0.09, 1.09] for 20 mg/L, and 0.45 [-0.04, 0.95] for 40 mg/L. Day 5 values were: 0.71 [0.20, 1.22] for 10 mg/L, 0.18 [-0.32, 0.67] for 20 mg/L, and 0.43 [-0.08, 0.94] for 40 mg/L. By day 7, effect sizes decreased: 0.44 [-0.06, 0.95] for 10 mg/L, 0.25 [-0.25, 0.75] for 20 mg/L, and 0.04 [-0.47, 0.55] for 40 mg/L. Increased percent distance in center is interpreted as anxiolytic-like behavior, observed only at the lowest ketamine concentration. No significant differences were found across testing days in the subgroup analysis ($p = 0.1010$).

For absolute turn angle, no significant effects of treatment were observed on day 1 ($F(3,124) = 0.62$, $p = 0.6036$), day 5 ($F(3,120) = 1.11$, $p = 0.3485$), or day 7 ($F(3,118) = 0.56$, $p = 0.6395$; Supplementary Figure 2C). No group significantly differed from controls on any day. Effect sizes are shown in Supplementary Figure 2D. On day 1: -0.25 [-0.74, 0.24] for 10 mg/L, 0.03 [-0.46, 0.52] for 20 mg/L, and -0.07 [-0.56, 0.42] for 40 mg/L. On day 5: -0.02 [-0.51, 0.48] for 10 mg/L, -0.37 [-0.86, 0.13] for 20 mg/L, and -0.15 [-0.66, 0.36] for 40 mg/L. On day 7: -0.08 [-0.58, 0.41] for 10 mg/L, -0.27 [-0.77, 0.23] for 20 mg/L, and -0.02 [-0.53, 0.48] for 40 mg/L. No significant differences were found across testing days in the subgroup analysis ($p = 0.8217$).

No significant effect of ketamine was observed on time spent immobile during the open tank test on any of the three experimental days (day 1: $H = 7.68$, $p = 0.053$; day 5: $H = 6.73$, $p = 0.0812$; day 7: $H = 3.63$, $p = 0.3038$; Supplementary Figure 2E). Effect sizes and 95% confidence intervals are shown in Supplementary Figure 2F. On day 1: -0.24 [-0.73, 0.25] for 10 mg/L, -0.29 [-0.78, 0.21] for 20 mg/L, and -0.36 [-0.86, 0.13] for 40 mg/L. On day 5: -0.27 [-0.77, 0.22] for 10 mg/L, -0.29 [-0.79, 0.21] for 20 mg/L, and -0.48 [-1.00, 0.03] for 40 mg/L. On day 7: -0.43 [-0.93, 0.08] for 10 mg/L, -0.30 [-0.80, 0.20] for 20 mg/L, and -0.44 [-0.95, 0.07] for 40 mg/L. No significant differences were found across testing days in the subgroup analysis ($p = 0.2612$).

Experiment II

Ketamine administration at 40 mg/L failed to produce sustained behavioral alterations following either acute or repeated exposure. In the SPT, neither a single administration on day 5 nor repeated treatment over 5 consecutive days affected time spent in the interaction zone, percent distance in the interaction zone, or locomotor-related measures such as distance traveled, line crossings, maximum speed, or time immobile. Likewise, in the OTT, no significant effects were observed for number of rotations, distance traveled, time spent in the center, percent distance in the center, absolute turn angle, or time immobile. A significant reduction in maximum speed was detected only in the group receiving repeated administration, suggesting a limited and transient effect. Overall, the absence of robust behavioral changes after a 48-hour washout indicates that neither acute nor repeated ketamine exposure at this concentration produced sustained effects. The following sections provide a more detailed account of these results.

Social preference test

Figure 4 shows the effects of single and repeated ketamine exposure (40 mg/L) on behavior in the SPT. In this protocol, ketamine was administered daily for five consecutive days, and behavioral testing was conducted 48 hours after the final concentration. For the primary outcome, time spent in the interaction zone, no significant effect of treatment was observed ($F(2,45) = 0.40$, $p = 0.6701$; Figure 4A). Effect sizes are shown in Figure 4G: -0.32 [-1.02, 0.37] for the single exposure group and -0.11 [-0.81, 0.58] for the repeated group. For distance traveled, there was also no significant treatment effect ($F(2,45) = 0.60$, $p = 0.5524$; Figure 4B), with effect sizes of 0.25 [-0.45, 0.94] and 0.42 [-0.28, 1.12] for single and repeated

exposures, respectively. No effect of treatment was detected for maximum speed ($F(2,45) = 0.22$, $p = 0.8066$; Figure 4C), and effect sizes were 0.13 [-0.56, 0.83] for single and -0.10 [-0.79, 0.59] for repeated exposure. Similarly, no significant effect was found for the number of line crossings ($F(2,45) = 0.09$, $p = 0.9137$; Figure 4D), with effect sizes of 0.11 [-0.59, 0.80] and -0.05 [-0.74, 0.65], respectively. For percent distance in the interaction zone ($F(2,45) = 0.09$, $p = 0.9172$; Figure 4E), results were non-significant, with effect sizes of -0.10 [-0.80, 0.59] for single and 0.05 [-0.65, 0.74] for repeated exposure. Finally, for time immobile, no significant difference was found ($H = 0.325$, $p = 0.8501$; Figure 4F), and the effect sizes were 0.31 [-0.39, 1.01] for single and 0.12 [-0.58, 0.81] for repeated exposure.

Open tank test

Figure 5 presents the results of the OTT conducted after single or repeated ketamine exposure (40 mg/L). In this protocol, ketamine was administered daily for five consecutive days, and behavioral testing was conducted 48 hours after the final concentration. For the primary outcome, number of rotations, no significant differences were detected between groups ($F(2,45) = 0.52$, $p = 0.5994$; Figure 5A), with effect sizes of 0.12 [-0.57, 0.82] and 0.33 [-0.37, 1.03] for the single and repeated groups, respectively. For distance traveled, results were not significant ($F(2,45) = 0.36$, $p = 0.6991$; Figure 5B), and the effect sizes were -0.11 [-0.80, 0.58] for single and 0.18 [-0.51, 0.88] for repeated exposure. Maximum speed showed a significant treatment effect ($F(2,45) = 3.61$, $p = 0.0352$; Figure 5C), and post hoc comparisons indicated a significant decrease in the repeated ketamine group relative to control. Effect sizes were -0.71 [-1.42, 0.01] for single and -0.82 [-1.55, -0.10] for repeated exposure. For time spent in the center, no significant effect was found ($F(2,45) = 1.51$, $p = 0.2316$; Figure 5D), and the corresponding effect sizes were 0.45 [-0.25, 1.16] and 0.13 [-0.56, 0.83]. Likewise, no treatment effect was observed for percent distance in the center ($F(2,45) = 1.39$, $p = 0.2608$; Figure 5E), with effect sizes of 0.39 [-0.31, 1.09] and -0.23 [-0.92, 0.47], respectively. No differences were found for absolute turn angle ($F(2,45) = 0.35$, $p = 0.7091$; Figure 5F), with effect sizes of 0.03 [-0.66, 0.73] and 0.29 [-0.41, 0.99]. Lastly, for time immobile, no significant difference was observed ($H = 0.302$, $p = 0.8597$; Figure 5G), and effect sizes were 0.00 [-0.69, 0.69] and -0.36 [-1.06, 0.34] for single and repeated exposure, respectively.

Experiment III

Ketamine administration at 40 mg/L did not produce sustained behavioral effects following either single exposure on day 14 or repeated administration over 14 consecutive days. In the SPT, no differences were observed in time spent in the interaction zone, distance traveled, maximum speed, line crossings, or percent distance in the interaction zone 48 hours after treatment, with the exception of increased time immobile in the repeated group. In the OTT, no treatment effects were detected across any of the outcomes, including number of rotations, distance traveled, maximum speed, time in the center, percent distance in the center, absolute turn angle, and time immobile. Together, these findings indicate a lack of persistent behavioral alterations after ketamine exposure under this protocol. The following sections provide a more detailed account of these results.

Social preference test

Figure 6 shows the effects of single and repeated ketamine exposure (40 mg/L) on behavior in the SPT. In this protocol, ketamine was administered daily for 14 consecutive days, and behavioral testing was conducted 48 hours after the final concentration. For the primary outcome, time spent in the interaction zone, no significant effect of treatment was observed ($F(2,44) = 1.56$, $p = 0.2214$; Figure 6A). Effect sizes were 0.48 [-0.23, 1.20] for the single exposure group and 0.54 [-0.17, 1.26] for the repeated group (Figure 6G). For distance traveled, no significant differences were found ($F(2,44) = 1.95$, $p = 0.1539$; Figure 6B), with effect sizes of -0.57 [-1.29, 0.15] and -0.59 [-1.31, 0.14] for the single and repeated groups, respectively. Similarly, no significant treatment effect was observed for maximum speed ($F(2,44) = 0.10$, $p = 0.9049$; Figure 6C), with effect sizes of -0.12 [-0.83, 0.58] for single and 0.02 [-0.68, 0.73] for repeated

exposure. Line crossings did not differ significantly across groups ($F(2,44) = 1.86$, $p = 0.1678$; Figure 6D), with effect sizes of $-0.53 [-1.25, 0.19]$ for single and $-0.58 [-1.30, 0.14]$ for repeated treatment. For percent distance in the interaction zone, results were non-significant ($F(2,44) = 1.09$, $p = 0.3444$; Figure 6E), and effect sizes were $0.33 [-0.38, 1.04]$ and $0.50 [-0.22, 1.22]$, respectively. Finally, for time immobile, a significant treatment effect was observed ($H = 6.788$, $p = 0.0336$; Figure 6F), with post hoc comparisons indicating significantly greater immobility in the repeated group compared to controls. The effect sizes were $-0.01 [-0.71, 0.69]$ for single and $0.39 [-0.32, 1.11]$ for repeated ketamine exposure (Figure 6G), suggesting an increase in immobility following chronic treatment.

Open tank test

Figure 7 presents the results of the OTT following single or repeated ketamine exposure (40 mg/L) over 14 days, with testing conducted after a 48-hour washout. For the primary outcome, number of rotations, no significant treatment effects were found ($F(2,44) = 0.06$, $p = 0.9458$; Figure 7A), and effect sizes were $0.11 [-0.60, 0.81]$ and $0.05 [-0.66, 0.75]$ for single and repeated groups, respectively (Figure 7H). No significant differences were observed for distance traveled ($F(2,44) = 0.01$, $p = 0.9939$; Figure 7B), with effect sizes of $0.04 [-0.67, 0.74]$ and $0.02 [-0.68, 0.73]$. Maximum speed also did not differ across groups ($F(2,44) = 2.00$, $p = 0.1475$; Figure 7C), and effect sizes were $-0.71 [-1.43, 0.02]$ for single and $-0.49 [-1.21, 0.22]$ for repeated exposure. For time spent in the center of the tank, no treatment effects were observed ($F(2,44) = 0.09$, $p = 0.9146$; Figure 7D), and effect sizes were $0.14 [-0.57, 0.84]$ and $0.12 [-0.58, 0.83]$. Percent distance in the center also showed no significant differences ($F(2,44) = 0.62$, $p = 0.5407$; Figure 7E), with effect sizes of $-0.25 [-0.96, 0.46]$ and $-0.41 [-1.12, 0.30]$, respectively. No significant treatment effect was found for absolute turn angle ($F(2,44) = 0.46$, $p = 0.6368$; Figure 7F), with effect sizes of $0.27 [-0.44, 0.98]$ and $0.33 [-0.38, 1.04]$. Finally, time immobile did not differ significantly across groups ($H = 2.585$, $p = 0.2746$; Figure 7G), and effect sizes were $-0.45 [-1.16, 0.27]$ for single and $-0.37 [-1.09, 0.34]$ for repeated exposure (Figure 7H).

DISCUSSION

The present findings reinforce the potential of zebrafish as a translational model for investigating schizophrenia pathophysiology and screening early-stage antipsychotic candidates, while also showing relevant differences following repeated ketamine exposure as compared to rodents. Literature reviews highlight discrepancies regarding the effects of NMDA antagonists on social and locomotor parameters in zebrafish, often reflecting variation in compound type, dosing protocols, and behavioral assays [1]. In our study, ketamine induced acute, concentration-dependent behavioral alterations in adult zebrafish that were robust and reproducible. In the SPT, ketamine reduced social interaction and increased locomotor activity across all concentrations, while in the OTT it led to more selective effects, including transient increases in rotational behavior and exploratory movement. Despite repeated exposures, no behavioral sensitization emerged, and ketamine-induced changes dissipated following a 48-hour washout in both 5- and 14-day protocols. If anything, the OTT findings from Experiment I may point to a potential tolerance effect, as certain behaviors observed on day 1 diminished over subsequent experimental days. However, it remains unclear whether this reflects a true drug tolerance effect or merely results from repeated testing. These results suggest that, under the exposure conditions used here, ketamine's effects are reversible and restricted to the acute phase. Although no persistent alterations were observed, we cannot rule out the possibility that zebrafish might display chronic or progressive phenotypes under alternative experimental designs.

Experiment I revealed a clear, concentration-dependent profile of behavioral alterations following ketamine exposure, consistent with schizophrenia-relevant phenotypes. In the SPT, zebrafish showed a robust decrease in social preference across all concentrations, indicated by reduced time and percent distance in the interaction zone. This phenotype parallels the social withdrawal modeled as a negative symptom in rodent studies and is consistent with effects of NMDA receptor antagonists in mammals [36–41]. Alongside

reduced social interaction, ketamine produced hyperlocomotion, evidenced by increased distance traveled, swimming speed, and zone crossings. Increased locomotor activity following NMDA receptor antagonism has been widely used as a behavioral readout for the positive symptom dimension of schizophrenia in animal models [2, 42, 43]. These behavioral changes are in line with findings from other NMDA antagonists in zebrafish, which similarly induce both social withdrawal and hyperlocomotion [18, 19, 44–47]. Together, the results highlight the reproducibility and sensitivity of the SPT for detecting schizophrenia-relevant phenotypes in this model.

The OTT revealed a more selective profile. Rotational behavior, often interpreted as a marker of stereotypy or repetitive movement, increased transiently at higher concentrations but was not sustained beyond early sessions. Similar increases in rotational behavior have also been reported in previous studies of ketamine exposure, indicating a reproducible effect on repetitive motor activity [21, 22, 48]. Absolute turn angle remained unaffected, and ketamine's impact on general locomotor outcomes in the OTT was milder than in the SPT, emphasizing the influence of test context. Previous findings indicate that NMDA antagonist-induced hyperlocomotion is modulated by environmental novelty, with stronger effects typically observed in unfamiliar settings or in response to novel stimuli [18, 46]. In our study, hyperlocomotion following ketamine exposure was greater in the SPT compared to the OTT, likely due to the contextual salience of the task, in which the animal is placed adjacent to a compartment containing ten conspecifics that may enhance arousal or motivational drive. Since the OTT was always conducted after the SPT, we cannot rule out the possibility that this difference reflects a testing order effect. However, prior experiments using MK-801 in which SPT and OTT were conducted in separate groups of animals rather than sequentially in the same individuals still revealed hyperlocomotion in the SPT but no change in distance traveled in the OTT, suggesting that the SPT may be inherently more sensitive to detecting locomotor alterations [18]. Similarly, studies using ketamine in the novel tank test, a context involving only exposure to a new arena, have also failed to detect hyperlocomotion, further supporting the idea that the expression of this phenotype is highly dependent on the salience of the testing environment [22, 49].

At the lowest concentration (10 mg/L), ketamine elicited an anxiolytic-like response, reflected by greater center time and distance in the OTT. Other studies using ketamine have also reported anxiolytic-like effects, but typically at higher concentrations than those tested here (20–60 mg/L), particularly in the novel tank test and light-dark test [20, 22]. However, these findings require further characterization, as ketamine-induced changes in locomotor activity may confound anxiety-related outcomes. In rodents, ketamine has also been shown to produce anxiolytic- or antidepressant-like effects [50–52], though some studies suggest this may only emerge in animals previously exposed to stress protocols [53, 54]. Whether a similar stress-dependent effect exists in zebrafish remains to be tested.

Unlike the study that inspired Experiment I [27], repeated ketamine exposure did not induce behavioral sensitization. Instead, responses remained consistent across test days, contrasting with rodent models where repeated NMDA receptor blockade frequently produces progressively amplified behavioral effects [26, 28, 55–57]. This is in line with previous zebrafish research showing no evidence of tolerance or sensitization in circling behavior following five days of ketamine exposure, suggesting a stable behavioral profile over repeated administrations [48].

Experiments II and III assessed behavioral persistence after 5-day and 14-day exposure schedules, each followed by a 48-hour washout. In both cases, zebrafish behavior returned to baseline, with no lasting effects observed in social interaction, locomotion, or repetitive activity. A small reduction in maximum speed following 5-day exposure and increase in immobility was detected following 14-day exposure, but overall, the results suggest that ketamine's behavioral effects are transient in adult zebrafish under these conditions.

Studies examining the sustained effects of ketamine in zebrafish remain scarce. One study using the novel tank test reported that repeated exposure to ketamine (40 mg/L for seven days, with testing 24 hours later)

1 did not alter general locomotor or anxiety-like behavior but increased the number of full-body rotations
2 [49]. However, this test was conducted using a front-facing camera, which may limit the accurate detection
3 of rotational behavior compared to our approach. In our study, rotations were assessed in the OTT using
4 a top-down view, allowing for more precise measurement of body orientation. Our findings also differ from
5 rodent models, where repeated ketamine exposure often results in sustained disruptions in behavior [25,
6 58].

7 Zebrafish show sustained neurogenesis and regenerative ability in the central nervous system from early
8 development through adulthood [59–62]. This may have contributed to the lack of sensitization or long-
9 term behavioral effects observed in our study, as the animals may recover from repeated acute exposures.
10 Supporting this, other studies have also found no lasting damage when ketamine was administered during
11 early developmental stages [63]. In rodents, repeated exposure to NMDA receptor antagonists like
12 ketamine causes lasting impairment in cortical inhibitory circuits by triggering a redox-sensitive pathway.
13 Repeated dosing increases IL-6 levels in the brain, which activates NADPH oxidase (*Nox2*) and leads to
14 oxidative stress that disrupts the function of parvalbumin-expressing interneurons [23, 64]. These effects
15 do not occur after a single exposure and appear only after repeated challenges. While reversible in adult
16 rodents, they become permanent when the exposure occurs during development [64, 65]. Zebrafish may
17 be less sensitive to these changes, potentially requiring more frequent or intense exposure to produce
18 similar long-term effects.

19 The NMDA receptor is highly conserved across vertebrate species, including zebrafish [45, 66, 67].
20 Subunit composition, binding affinity, and overall receptor function are similar enough to support
21 comparable acute effects across species. Therefore, differences in behavioral outcomes are unlikely to
22 result from receptor pharmacology alone. One possible explanation involves species-specific differences
23 in ketamine metabolism. Although zebrafish express cytochrome P450 enzymes involved in drug
24 processing [68, 69], little is known about how ketamine is metabolized in this species. It is unclear how
25 quickly the drug is metabolized, which metabolites are generated, and whether any of them contribute to
26 behavioral effects. These factors may influence the limited duration of responses observed in this species.
27 Investigating ketamine's pharmacokinetics in zebrafish will be important for interpreting its behavioral
28 impact and for evaluating the model's relevance to other systems.

29 The findings presented here highlight the utility of zebrafish for modeling behavioral changes induced by
30 NMDA receptor antagonists. However, to improve the translational relevance of this model, future studies
31 should include pharmacological validation using clinically effective antipsychotic drugs and incorporate
32 neurochemical or physiological analyses to clarify underlying mechanisms. Expanding the behavioral
33 assessments to cover cognitive and affective domains, alongside systematic variation in exposure
34 conditions such as dosing schedules, developmental timing, and environmental context, will allow for a
35 more comprehensive evaluation of induced phenotypes.

36 The lack of sensitization or lasting behavioral effects likely reflects the specific exposure protocols used,
37 rather than an inherent limitation of the zebrafish model. Variations in exposure timing, duration, dose, or
38 the inclusion of environmental stressors may be necessary to produce more persistent effects. The
39 zebrafish offers significant experimental flexibility, and its potential to model chronic or progressive features
40 of psychiatric disorders warrants further investigation.

41 In summary, ketamine induced robust, context-dependent, and reversible behavioral alterations in adult
42 zebrafish that parallel core symptom dimensions of schizophrenia. The use of complementary behavioral
43 assays enabled a multidimensional assessment of drug effects, reinforcing the model's potential for acute
44 psychotomimetic screening. While the current exposure protocols did not produce sensitization or long-
45 lasting behavioral changes, these findings establish a strong foundation for future studies aimed at
46 optimizing the zebrafish model for translational neuropsychiatric research. With further refinement and
47 validation, zebrafish hold considerable promise as a scalable platform for mechanistic discovery and early-

phase therapeutic screening in drug development.

AUTHOR CONTRIBUTIONS

Conceptualization: M.G.-L., D.V.M., and A.P.H. **Data curation:** M.G.-L. and A.P.H. **Formal analysis:** M.G.-L. and A.P.H. **Funding acquisition:** A.P. and A.P.H. **Investigation:** M.G.-L., D.V.M., T.S.-B., L.M.B., S.Z.B., and S.M.B. **Methodology:** M.G.-L., D.V.M., and A.P.H. **Project administration:** M.G.-L. and A.P.H. **Resources:** A.P. and A.P.H. **Supervision:** M.G.-L. and A.P.H. **Validation:** M.G.-L. and A.P.H. **Visualization:** M.G.-L. and A.P.H. **Writing - original draft:** M.G.-L. **Writing - review & editing:** D.V.M., T.S.-B., L.M.B., S.Z.B., S.M.B., A.P., and A.P.H.

INTEREST STATEMENT

The authors state that there are no conflicts of interest related to this work.

DATA AVAILABILITY

All data supporting this study are publicly accessible via the Open Science Framework (osf.io/67y45) [30].

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REFERENCES

- Benvenuti, R., Gallas-Lopes, M., Marcon, M., Reschke, C. R., Herrmann, A. P., and Piato, A. (2022) Glutamate NMDA Receptor Antagonists with Relevance to Schizophrenia: A Review of Zebrafish Behavioral Studies. *Curr. Neuropharmacol.* **20**, 494–509
- Jones, C., Watson, D., and Fone, K. (2011) Animal models of schizophrenia. *Br. J. Pharmacol.* **164**, 1162–1194
- Large, C. H. (2007) Do NMDA receptor antagonist models of schizophrenia predict the clinical efficacy of antipsychotic drugs? *J. Psychopharmacol. (Oxf.)* **21**, 283–301
- Neill, J. C., Barnes, S., Cook, S., Grayson, B., Idris, N. F., McLean, S. L., Snigdha, S., Rajagopal, L., and Harte, M. K. (2010) Animal models of cognitive dysfunction and negative symptoms of schizophrenia: Focus on NMDA receptor antagonism. *Pharmacol. Ther.* **128**, 419–432
- Winship, I. R., Dursun, S. M., Baker, G. B., Balista, P. A., Kandratavicius, L., Maia-de-Oliveira, J. P., Hallak, J., and Howland, J. G. (2019) An Overview of Animal Models Related to Schizophrenia. *Can. J. Psychiatry* **64**, 5–17
- Coyle, J. T. (2012) NMDA receptor and schizophrenia: a brief history. *Schizophr. Bull.* **38**, 920–926
- Frohlich, J. and Van Horn, J. D. (2014) Reviewing the ketamine model for schizophrenia. *J. Psychopharmacol. (Oxf.)* **28**, 287–302
- Gunduz-Bruce, H. (2009) The acute effects of NMDA antagonism: From the rodent to the human brain. *Brain Res. Rev.* **60**, 279–286
- Spark, D. L., Fornito, A., Langmead, C. J., and Stewart, G. D. (2022) Beyond antipsychotics: a twenty-first century update for preclinical development of schizophrenia therapeutics. *Transl. Psychiatry* **12**, 147
- Uliana, D. L., Zhu, X., Gomes, F. V., and Grace, A. A. (2022) Using animal models for the studies of schizophrenia and depression: The value of translational models for treatment and prevention. *Front. Behav. Neurosci.* **16**
- Carneiro, C. F. D., Drude, N., Hülsemann, M., Collazo, A., and Toelch, U. (2023) Mapping strategies towards improved external validity in preclinical translational research. *Expert Opin. Drug Discov.* **18**, 1273–1285
- Keifer, J. and Summers, C. H. (2016) Putting the “Biology” Back into “Neurobiology”: The Strength of Diversity in Animal Model Systems for Neuroscience Research. *Front. Syst. Neurosci.* **10**

13. Maximino, C., Silva, R. X. do C., da Silva, S. de N. dos S., Rodrigues, L. do S. dos S., Barbosa, H., de Carvalho, T. S., Leão, L. K. R., Lima, M. G., Oliveira, K. R. M., and Herculano, A. M. (2015) Non-mammalian models in behavioral neuroscience: consequences for biological psychiatry. *Front. Behav. Neurosci.* **9**
14. Spanagel, R. (2022) Ten Points to Improve Reproducibility and Translation of Animal Research. *Front. Behav. Neurosci.* **16**, 869511
15. Fontana, B. D., Mezzomo, N. J., Kalueff, A. V., and Rosemberg, D. B. (2018) The developing utility of zebrafish models of neurological and neuropsychiatric disorders: A critical review. *Exp. Neurol.* **299**, 157–171
16. MacRae, C. A. and Peterson, R. T. (2015) Zebrafish as tools for drug discovery. *Nat. Rev. Drug Discov.* **14**, 721–731
17. Stewart, A. M., Braubach, O., Spitsbergen, J., Gerlai, R., and Kalueff, A. V. (2014) Zebrafish models for translational neuroscience research: from tank to bedside. *Trends Neurosci.* **37**, 264–278
18. Benvenuti, R., Gallas-Lopes, M., Sachett, A., Marcon, M., Strogulski, N. R., Reis, C. G., Chitolina, R., Piato, A., and Herrmann, A. P. (2021) How do zebrafish (*Danio rerio*) respond to MK-801 and amphetamine? Relevance for assessing schizophrenia-related endophenotypes in alternative model organisms. *J. Neurosci. Res.* **99**, 2844–2859
19. Giongo, F. K., Gallas-Lopes, M., Benvenuti, R., Sachett, A., Bastos, L. M., Rosa, A. R., and Herrmann, A. P. (2023) Effects of Taurine in Mice and Zebrafish Behavioral Assays With Translational Relevance to Schizophrenia. *Int. J. Neuropsychopharmacol.* **26**, 125–136
20. De Campos, E. G., Bruni, A. T., and De Martinis, B. S. (2015) Ketamine induces anxiolytic effects in adult zebrafish: A multivariate statistics approach. *Behav. Brain Res.* **292**, 537–546
21. Michelotti, P., Quadros, V. A., Pereira, M. E., and Rosemberg, D. B. (2018) Ketamine modulates aggressive behavior in adult zebrafish. *Neurosci. Lett.* **684**, 164–168
22. Riehl, R., Kyzar, E., Allain, A., Green, J., Hook, M., Monnig, L., Rhymes, K., Roth, A., Pham, M., Razavi, R., DiLeo, J., Gaikwad, S., Hart, P., and Kalueff, A. V. (2011) Behavioral and physiological effects of acute ketamine exposure in adult zebrafish. *Neurotoxicol. Teratol.* **33**, 658–667
23. Behrens, M. M., Ali, S. S., Dao, D. N., Lucero, J., Shekhtman, G., Quick, K. L., and Dugan, L. L. (2007) Ketamine-induced loss of phenotype of fast-spiking interneurons is mediated by NADPH-oxidase. *Science* **318**, 1645–1647
24. Chatterjee, M., Verma, R., Ganguly, S., and Palit, G. (2012) Neurochemical and molecular characterization of ketamine-induced experimental psychosis model in mice. *Neuropharmacology* **63**, 1161–1171
25. Chatterjee, M., Ganguly, S., Srivastava, M., and Palit, G. (2011) Effect of ‘chronic’ versus ‘acute’ ketamine administration and its ‘withdrawal’ effect on behavioural alterations in mice: Implications for experimental psychosis. *Behav. Brain Res.* **216**, 247–254
26. Galvanho, J. P., Manhães, A. C., Carvalho-Nogueira, A. C. C., Silva, J. de M., Filgueiras, C. C., and Abreu-Villaça, Y. (2020) Profiling of behavioral effects evoked by ketamine and the role of 5HT2 and D2 receptors in ketamine-induced locomotor sensitization in mice. *Prog. Neuropsychopharmacol. Biol. Psychiatry* **97**, 109775
27. Kokkinou, M., Irvine, E. E., Bonsall, D. R., Natesan, S., Wells, L. A., Smith, M., Glegola, J., Paul, E. J., Tossell, K., Veronese, M., Khadayate, S., Dedic, N., Hopkins, S. C., Ungless, M. A., Withers, D. J., and Howes, O. D. (2021) Reproducing the dopamine pathophysiology of schizophrenia and approaches to ameliorate it: a translational imaging study with ketamine. *Mol. Psychiatry* **26**, 2562–2576
28. Trujillo, K. A. and Heller, C. Y. (2020) Ketamine sensitization: Influence of dose, environment, social isolation and treatment interval. *Behav. Brain Res.* **378**, 112271
29. Percie du Sert, N., Hurst, V., Ahluwalia, A., Alam, S., Avey, M. T., Baker, M., Browne, W. J., Clark, A., Cuthill, I. C., Dirnagl, U., Emerson, M., Garner, P., Holgate, S. T., Howells, D. W., Karp, N. A., Lazic, S. E., Lidster, K., MacCallum, C. J., Macleod, M., Pearl, E. J., Petersen, O. H., Rawle, F., Reynolds, P., Rooney, K., Sena, E. S., Silberberg, S. D., Steckler, T., and Würbel, H. (2020) The ARRIVE guidelines 2.0: Updated guidelines for reporting animal research. *PLoS Biol.* **18**, e3000410
30. Gallas-Lopes, M., Müller, D., Buss, T., Bastos, L., Bruck, S., Becker, S., Piato, A., and Herrmann, A. (2023) Behavioral profiling of ketamine exposure in adult zebrafish modeling schizophrenia-related phenotypes.
31. Gallas-Lopes, M., Müller, D. V., and Herrmann, A. P. (2023) Assessment of behavioral sensitization to repeated ketamine exposure in adult zebrafish - Experiment Preregistration.
32. Gallas-Lopes, M., Müller, D. V., and Herrmann, A. P. (2023) Evaluation of sustained behavioral

- effects following 5-day ketamine exposure in adult zebrafish - Experiment Preregistration.
33. Gallas-Lopes, M., Müller, D. V., and Herrmann, A. P. (2024) Evaluation of sustained behavioral effects following 14-day ketamine exposure in adult zebrafish - Experiment Preregistration.
34. Balduzzi, S., Rücker, G., and Schwarzer, G. (2019) How to perform a meta-analysis with R: a practical tutorial. *Evid. Based Ment. Health* **22**, 153–160
35. Sneddon, L. U., Halsey, L. G., and Bury, N. R. (2017) Considering aspects of the 3Rs principles within experimental animal biology. *J. Exp. Biol.* **220**, 3007–3016
36. Becker, A., Peters, B., Schroeder, H., Mann, T., Huether, G., and Grecksch, G. (2003) Ketamine-induced changes in rat behaviour: A possible animal model of schizophrenia. *Prog. Neuropsychopharmacol. Biol. Psychiatry* **27**, 687–700
37. Becker, A. and Grecksch, G. (2004) Ketamine-induced changes in rat behaviour: a possible animal model of schizophrenia. Test of predictive validity. *Prog. Neuropsychopharmacol. Biol. Psychiatry* **28**, 1267–1277
38. Gururajan, A., Taylor, D. A., and Malone, D. T. (2010) Current pharmacological models of social withdrawal in rats: relevance to schizophrenia. *Behav. Pharmacol.* **21**, 690
39. Koros, E., Rosenbrock, H., Birk, G., Weiss, C., and Sams-Dodd, F. (2007) The selective mGlu5 receptor antagonist MTEP, similar to NMDA receptor antagonists, induces social isolation in rats. *Neuropsychopharmacol. Off. Publ. Am. Coll. Neuropsychopharmacol.* **32**, 562–576
40. Neill, J. C., Harte, M. K., Haddad, P. M., Lydall, E. S., and Dwyer, D. M. (2014) Acute and chronic effects of NMDA receptor antagonists in rodents, relevance to negative symptoms of schizophrenia: A translational link to humans. *Eur. Neuropsychopharmacol.* **24**, 822–835
41. Popik, P., Cyrano, E., Piotrowska, D., Holuj, M., Golebiowska, J., Malikowska-Racia, N., Potasiewicz, A., and Nikiforuk, A. (2024) Effects of ketamine on rat social behavior as analyzed by DeepLabCut and SimBA deep learning algorithms. *Front. Pharmacol.* **14**
42. Adell, A., Jiménez-Sánchez, L., López-Gil, X., and Romón, T. (2012) Is the Acute NMDA Receptor Hypofunction a Valid Model of Schizophrenia? *Schizophr. Bull.* **38**, 9–14
43. Geyer, M. A. and Ellenbroek, B. (2003) Animal behavior models of the mechanisms underlying antipsychotic atypicality. *Prog. Neuropsychopharmacol. Biol. Psychiatry* **27**, 1071–1079
44. Franscescon, F., Müller, T., Bertoncello, K., and Rosemberg, D. (2020) Neuroprotective role of taurine on MK-801-induced memory impairment and hyperlocomotion in zebrafish. *Neurochem. Int.* **135**
45. Menezes, F. P., Kist, L. W., Bogo, M. R., Bonan, C. D., and Da Silva, R. S. (2015) Evaluation of Age-Dependent Response to NMDA Receptor Antagonism in Zebrafish. *Zebrafish* **12**, 137–143
46. Tran, S., Muraleetharan, A., Fulcher, N., Chatterjee, D., and Gerlai, R. (2016) MK-801 increases locomotor activity in a context-dependent manner in zebrafish. *Behav. Brain Res.* **296**, 26–29
47. Zimmermann, F. F., Gaspary, K. V., Siebel, A. M., and Bonan, C. D. (2016) Oxytocin reversed MK-801-induced social interaction and aggression deficits in zebrafish. *Behav. Brain Res.* **311**, 368–374
48. Zakhary, S. M., Ayubcha, D., Ansari, F., Kamran, K., Karim, M., Leheste, J. R., Horowitz, J. M., and Torres, G. (2011) A behavioral and molecular analysis of ketamine in zebrafish. *Synap. N. Y. N* **65**, 160–167
49. Michelotti, P., Franscescon, F., Müller, T. E., Rosemberg, D. B., and Pereira, M. E. (2022) Ketamine acutely impairs memory consolidation and repeated exposure promotes stereotyped behavior without changing anxiety- and aggression-like parameters in adult zebrafish. *Physiol. Behav.* **247**, 113708
50. Engin, E., Treit, D., and Dickson, C. T. (2009) Anxiolytic- and antidepressant-like properties of ketamine in behavioral and neurophysiological animal models. *Neuroscience* **161**, 359–369
51. Garcia, L. S. B., Comim, C. M., Valvassori, S. S., Réus, G. Z., Stertz, L., Kapczinski, F., Gavioli, E. C., and Quevedo, J. (2009) Ketamine treatment reverses behavioral and physiological alterations induced by chronic mild stress in rats. *Prog. Neuropsychopharmacol. Biol. Psychiatry* **33**, 450–455
52. Tornese, P., Sala, N., Bonini, D., Bonifacino, T., La Via, L., Milanese, M., Treccani, G., Seguíni, M., Ieraci, A., Mingardi, J., Nyengaard, J. R., Calza, S., Bonanno, G., Wegener, G., Barbon, A., Popoli, M., and Musazzi, L. (2019) Chronic mild stress induces anhedonic behavior and changes in glutamate release, BDNF trafficking and dendrite morphology only in stress vulnerable rats. The rapid restorative action of ketamine. *Neurobiol. Stress* **10**, 100160
53. Fitzgerald, P. J., Yen, J. Y., and Watson, B. O. (2019) Stress-sensitive antidepressant-like effects of ketamine in the mouse forced swim test. *PLOS ONE* **14**, e0215554
54. Polis, A. J., Fitzgerald, P. J., Hale, P. J., and Watson, B. O. (2019) Rodent ketamine depression-related research: Finding patterns in a literature of variability. *Behav. Brain Res.* **376**, 112153
55. Trujillo, K. A., Zamora, J. J., and Warmoth, K. P. (2008) Increased Response to Ketamine Following

- 1 Treatment at Long Intervals: Implications for Intermittent Use. *Biol. Psychiatry* **63**, 178–183
- 2 56. Uchihashi, Y., Kuribara, H., Morita, T., and Fujita, T. (1993) The repeated administration of ketamine
- 3 induces an enhancement of its stimulant action in mice. *Jpn. J. Pharmacol.* **61**, 149–151
- 4 57. Wiley, J. L., Evans, R. L., Grainger, D. B., and Nicholson, K. L. (2008) PRECLINICAL STUDY: Age-
- 5 dependent differences in sensitivity and sensitization to cannabinoids and ‘club drugs’ in male
- 6 adolescent and adult rats. *Addict. Biol.* **13**, 277–286
- 7 58. Jeevakumar, V., Driskill, C., Paine, A., Sobhanian, M., Vakil, H., Morris, B., Ramos, J., and Kroener,
- 8 S. (2015) Ketamine administration during the second postnatal week induces enduring
- 9 schizophrenia-like behavioral symptoms and reduces parvalbumin expression in the medial prefrontal
- 10 cortex of adult mice. *Behav. Brain Res.* **282**, 165–175
- 11 59. Kishimoto, N., Shimizu, K., and Sawamoto, K. (2012) Neuronal regeneration in a zebrafish model of
- 12 adult brain injury. *Dis. Model. Mech.* **5**, 200–209
- 13 60. Kizil, C., Kaslin, J., Kroehne, V., and Brand, M. (2012) Adult neurogenesis and brain regeneration in
- 14 zebrafish. *Dev. Neurobiol.* **72**, 429–461
- 15 61. Schmidt, R., Strähle, U., and Scholpp, S. (2013) Neurogenesis in zebrafish – from embryo to adult.
- 16 *Neural Develop.* **8**, 3
- 17 62. Zambusi, A. and Ninkovic, J. (2020) Regeneration of the central nervous system-principles from brain
- 18 regeneration in adult zebrafish. *World J. Stem Cells* **12**, 8–24
- 19 63. Santos, C., Valentim, A. M., Félix, L., Balça-Silva, J., and Pinto, M. L. (2023) Longitudinal effects of
- 20 ketamine on cell proliferation and death in the CNS of zebrafish. *NeuroToxicology* **97**, 78–88
- 21 64. Behrens, M. M., Ali, S. S., and Dugan, L. L. (2008) Interleukin-6 mediates the increase in NADPH-
- 22 oxidase in the ketamine model of schizophrenia. *J. Neurosci. Off. J. Soc. Neurosci.* **28**, 13957–13966
- 23 65. Behrens, M. M. and Sejnowski, T. J. (2009) Does schizophrenia arise from oxidative dysregulation
- 24 of parvalbumin-interneurons in the developing cortex? *Neuropharmacology* **57**, 193–200
- 25 66. Cox, J. A., Kucenas, S., and Voigt, M. M. (2005) Molecular characterization and embryonic
- 26 expression of the family of N-methyl-D-aspartate receptor subunit genes in the zebrafish. *Dev. Dyn.*
- 27 *Off. Publ. Am. Assoc. Anat.* **234**, 756–766
- 28 67. Zoodsma, J. D., Gomes, C. I., Sirotkin, H. I., and Wollmuth, L. P. (2024) Behavioral Assays Dissecting
- 29 NMDA Receptor Function in Zebrafish. In *NMDA Receptors: Methods and Protocols* (Burnashev, N.
- 30 and Szepietowski, P., eds) pp. 243–255, Springer US, New York, NY
- 31 68. de Souza Anselmo, C., Sardela, V. F., de Sousa, V. P., and Pereira, H. M. G. (2018) Zebrafish (*Danio*
- 32 *rerio*): A valuable tool for predicting the metabolism of xenobiotics in humans? *Comp. Biochem.*
- 33 *Physiol. Part C Toxicol. Pharmacol.* **212**, 34–46
- 34 69. Saad, M., Cavanaugh, K., Verbueken, E., Pype, C., Casteleyn, C., Van Ginneken, C., and Van
- 35 Cruchten, S. (2016) Xenobiotic metabolism in the zebrafish: a review of the spatiotemporal
- 36 distribution, modulation and activity of Cytochrome P450 families 1 to 3. *J. Toxicol. Sci.* **41**, 1–11

Figure legends

Figure 1. Overview of the experimental design evaluating the behavioral effects of ketamine exposure in zebrafish. Experiment I: fish were exposed via immersion to water or ketamine (10, 20, or 40 mg/L) for 20 minutes daily over 5 days, with re-exposure on day 7. Behavioral tests (SPT, OTT) were conducted on days 1, 5, and 7. Experiment II: fish were exposed to water, a single ketamine immersion (40 mg/L on day 5), or repeated immersions (days 1–5). Tests were conducted on day 7. Experiment III: fish were exposed to water, a single immersion (day 14), or repeated daily immersions (days 1–14) of 40 mg/L ketamine. Testing occurred on day 16. SPT = social preference test; OTT = Open tank test.

Figure 2. Effects of exposure to different concentrations of ketamine on the behavior of adult zebrafish in the social preference test. (A) Time spent in the interaction zone, (C) total distance traveled, (E) maximum speed, and (G) number of line crossings, measured on days 1, 5, and 7. (B, D, F, H) Corresponding effect sizes (Cohen's d) for each outcome, with test for subgroup differences across assessment days. One-way ANOVA followed by Bonferroni post hoc test. Data are presented as mean $\pm$ SD. Effect sizes are expressed as Cohen's d with 95% confidence intervals. Subgroup differences tested using a random-effects model. Sample sizes: day 1 – Control = 32, Ket 10 = 32, Ket 20 = 32, Ket 40 = 32; day 5 – Control = 31, Ket 10 = 32, Ket 20 = 32, Ket 40 = 29; day 7 – Control = 31, Ket 10 = 31, Ket 20 = 31, Ket 40 = 29. Ket 10 = ketamine 10 mg/L; Ket 20 = ketamine 20 mg/L; Ket 40 = ketamine 40 mg/L.

Figure 3. Effects of exposure to different concentrations of ketamine on the behavior of adult zebrafish in the open tank test. (A) Number of rotations, (C) total distance traveled, (E) maximum speed, and (G) time spent in the center zone, measured on days 1, 5, and 7. (B, D, F, H) Corresponding effect sizes (Cohen's d) for each outcome, with test for subgroup differences across assessment days. One-way ANOVA followed by Bonferroni post hoc test. Data are presented as mean $\pm$ SD. Effect sizes are expressed as Cohen's d with 95% confidence intervals. Subgroup differences tested using a random-effects model. Sample sizes: day 1 – Control = 32, Ket 10 = 32, Ket 20 = 32, Ket 40 = 32; day 5 – Control = 31, Ket 10 = 32, Ket 20 = 32, Ket 40 = 29; day 7 – Control = 31, Ket 10 = 31, Ket 20 = 31, Ket 40 = 29. Ket 10 = ketamine 10 mg/L; Ket 20 = ketamine 20 mg/L; Ket 40 = ketamine 40 mg/L.

Figure 4. Effects of single (day 5) or repeated (days 1–5) ketamine exposures on the behavior of adult zebrafish in the social preference test. (A) Time spent in the interaction zone, (B) total distance traveled, (C) maximum speed, (D) number of line crossings, (E) percent of distance traveled in the interaction zone, and (F) time spent immobile. (G) Effect sizes (Cohen's d) for the single and repeated exposure groups across the outcomes assessed. One-way ANOVA followed by Bonferroni post hoc test was used for A–E. Kruskal–Wallis test followed by Dunn's post hoc was used for F. Data are presented as mean $\pm$ SD for A–E, and as median with interquartile range for F. Effect sizes are expressed as Cohen's d with 95% confidence intervals. Sample sizes: Control = 16, Single = 16, Repeated = 16. Single = single ketamine exposure (40 mg/L) on day 5; Repeated = daily ketamine exposures (40 mg/L) from day 1 to 5.

Figure 5. Effects of single (day 5) or repeated (days 1–5) ketamine exposures on the behavior of adult zebrafish in the open tank test. (A) Number of rotations, (B) total distance traveled, (C) maximum speed, (D) time spent in the center zone, (E) percent of distance traveled in the center zone, (F) absolute turn angle, and (G) time spent immobile. (H) Effect sizes (Cohen's d) for the single and repeated exposure groups across the outcomes assessed. One-way ANOVA followed by Bonferroni post hoc test was used for A–F and G. Kruskal–Wallis test followed by Dunn's post hoc was used for G. Data are presented as mean $\pm$ SD for A–F and G, and as median with interquartile range for G. Effect sizes are expressed as Cohen's d with 95% confidence intervals. Sample sizes: Control = 16, Single = 16, Repeated = 16. Single = single ketamine exposure (40 mg/L) on day 5; Repeated = daily ketamine exposures (40 mg/L) from day 1 to 5.

Figure 6. Effects of single (day 14) or repeated (days 1–14) ketamine exposures on the behavior of adult zebrafish in the social preference test. (A) Time spent in the interaction zone, (B) total distance traveled, (C) maximum speed, (D) number of line crossings, (E) percent of distance traveled in the interaction zone, and (F) time spent immobile. (G) Effect sizes (Cohen's d) for the single and repeated exposure groups across the outcomes assessed. One-way ANOVA followed by Bonferroni post hoc test was used for A–E. Kruskal–Wallis test followed by Dunn's post hoc was used for F. Data are presented as mean $\pm$ SD for A–E, and as median with interquartile range for F. Effect sizes are expressed as Cohen's d with 95% confidence intervals. Sample sizes: Control = 15, Single = 16, Repeated = 16. Single = single ketamine exposure (40 mg/L) on day 14; Repeated = daily ketamine exposures (40 mg/L) from day 1 to 14.

Figure 7. Effects of single (day 14) or repeated (days 1–14) ketamine exposures on the behavior of adult zebrafish in the open tank test. (A) Number of rotations, (B) total distance traveled, (C) maximum speed, (D) time spent in the center zone, (E) percent of distance traveled in the center zone, (F) absolute turn angle, and (G) time spent immobile. (H) Effect sizes (Cohen's d) for the single and repeated exposure groups across the outcomes assessed. One-way ANOVA followed by Bonferroni post hoc test was used for A–F. Kruskal–Wallis test followed by Dunn's post hoc was used for G. Data are presented as mean $\pm$ SD for A–F, and as median with interquartile range for G. Effect sizes are expressed as Cohen's d with 95% confidence intervals. Sample sizes: Control = 15, Single = 16, Repeated = 16. Single = single ketamine exposure (40 mg/L) on day 14; Repeated = daily ketamine exposures (40 mg/L) from day 1 to 14.

Figure 1

Experiment I

Days	1	2	3	4	5	6	7
Treatment (20 min)							
Control							
Ketamine 10 mg/L							
Ketamine 20 mg/L							
Ketamine 40 mg/L							
Tests	SPT				SPT		SPT
	OTT				OTT		OTT

Experiment II

Days	1	2	3	4	5	6	7
Treatment (20 min)							
Control							
Single exposure							
Repeated exposure							
Tests						SPT	
						OTT	

Experiment III

Days	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16
Treatment (20 min)																
Control																
Single exposure																
Repeated exposure																
Tests																SPT
																OTT

Figure 2

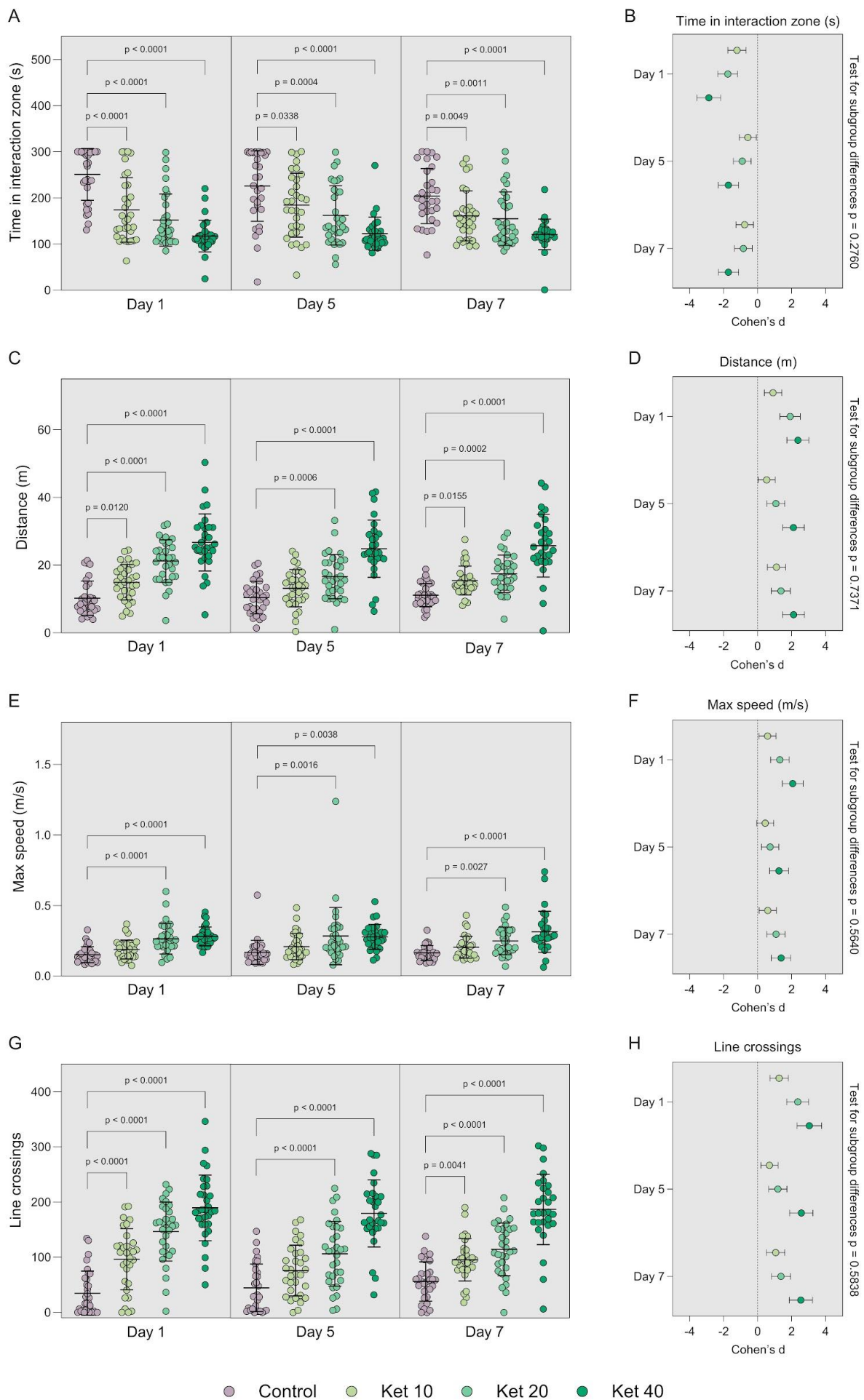


Figure 3

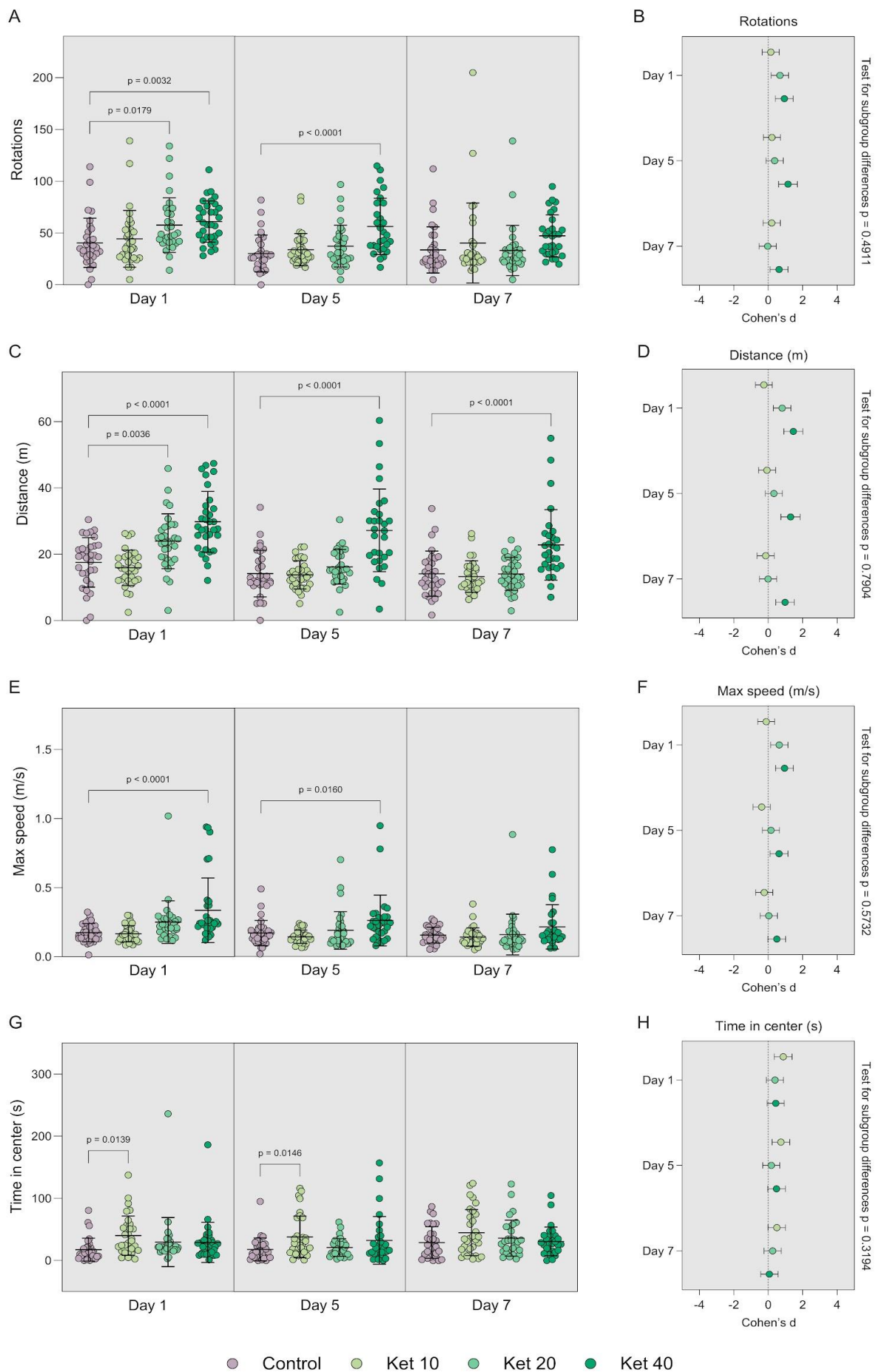


Figure 4

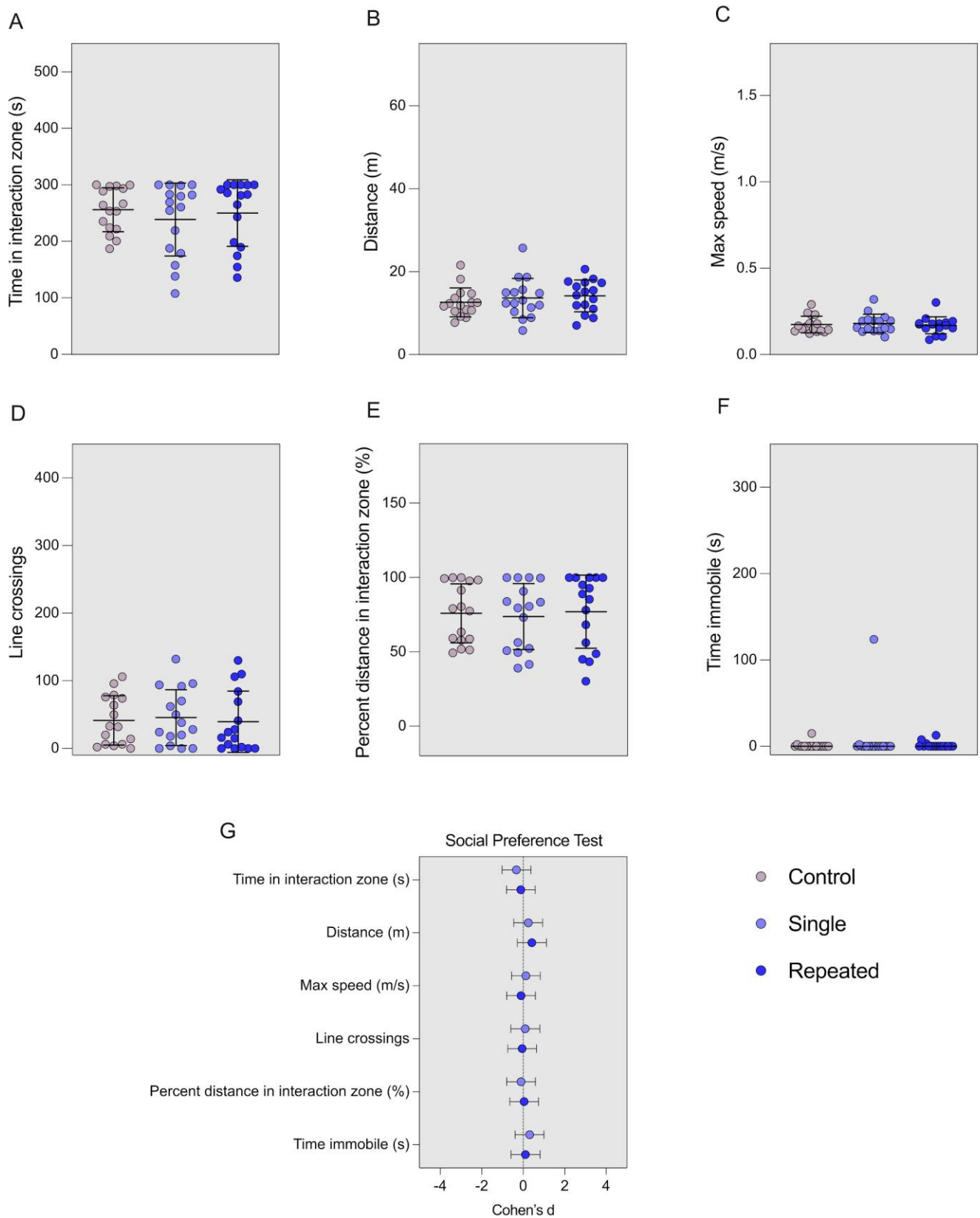


Figure 5

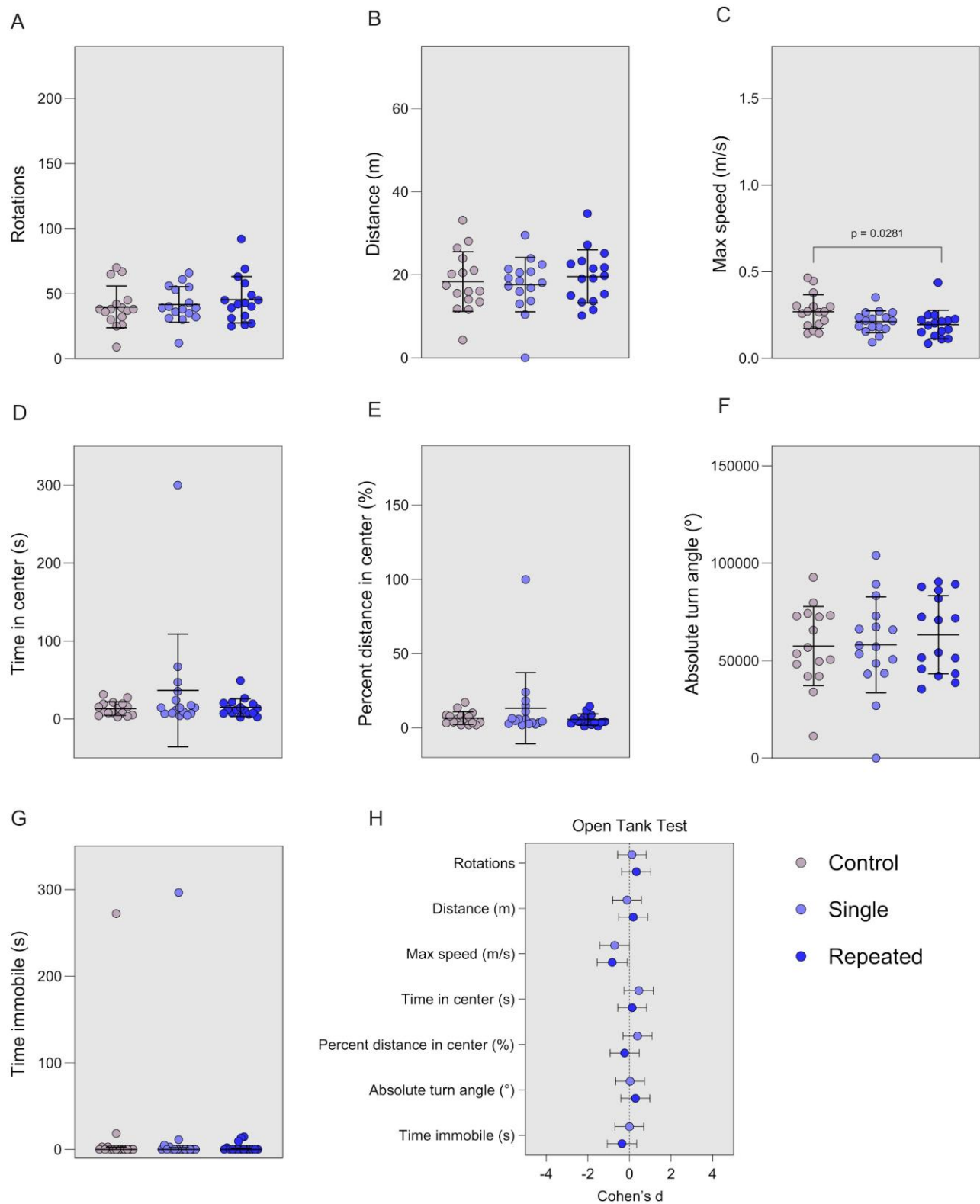


Figure 6

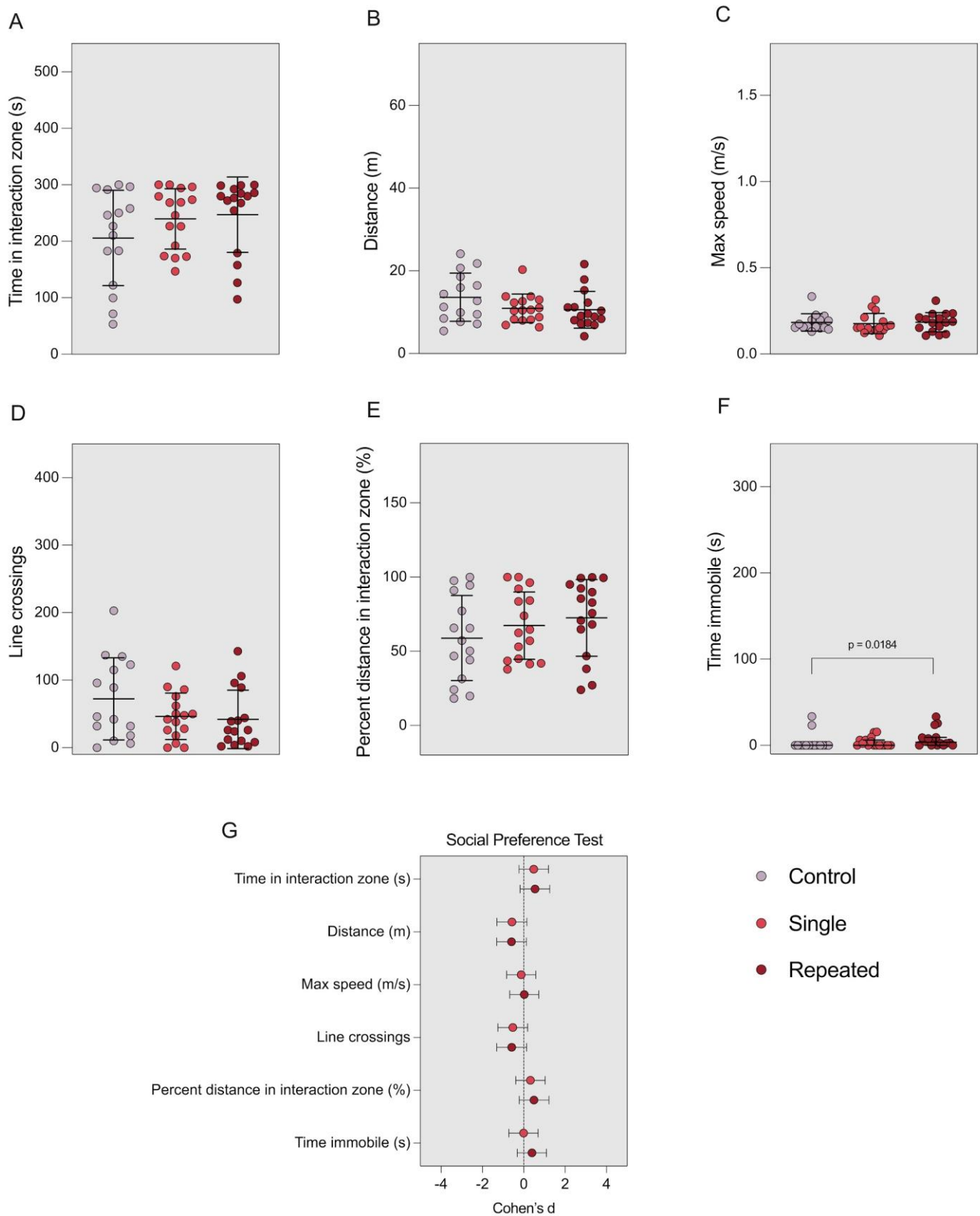
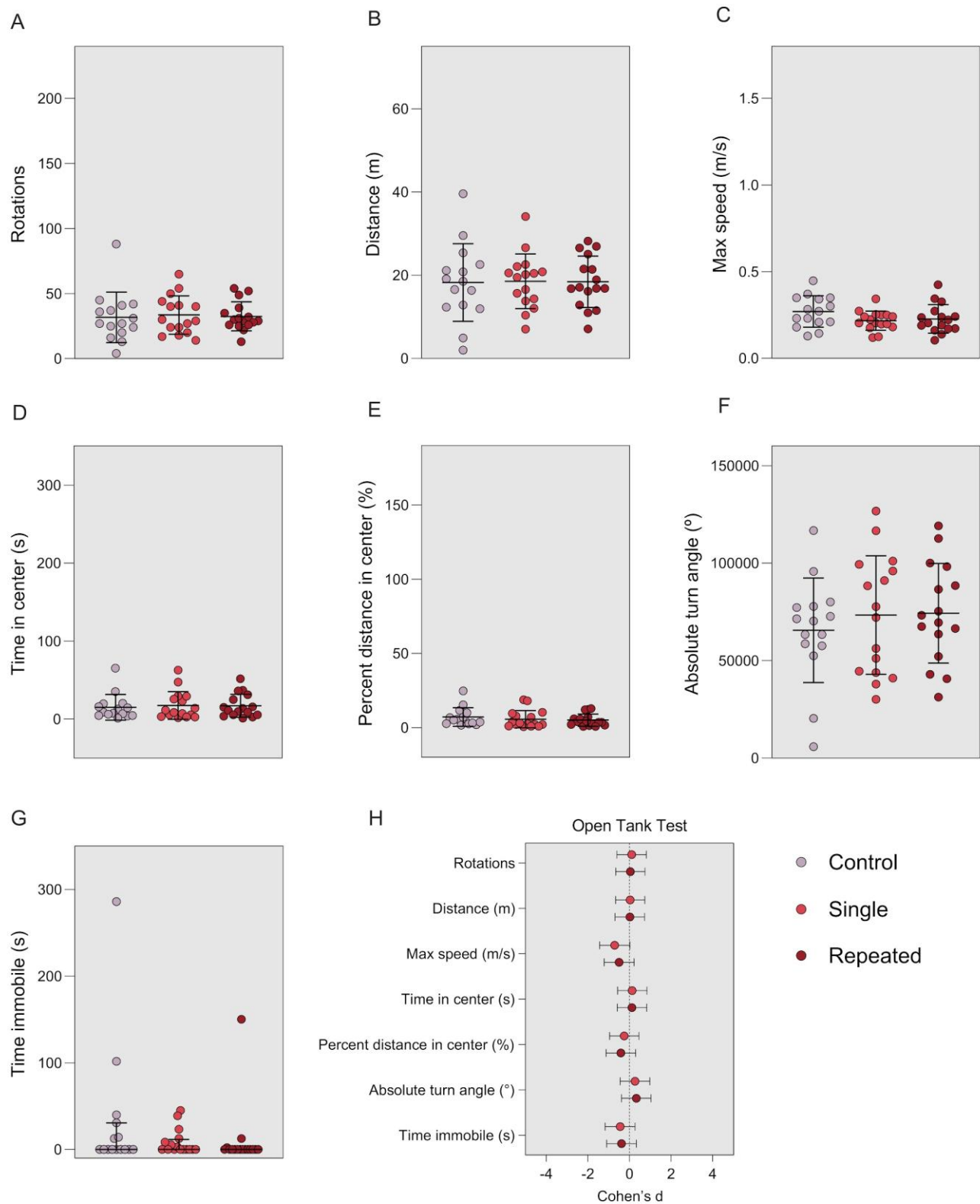


Figure 7



Supplementary Files

Ketamine-induced NMDA receptor hypofunction alters social and locomotor behavior in adult zebrafish

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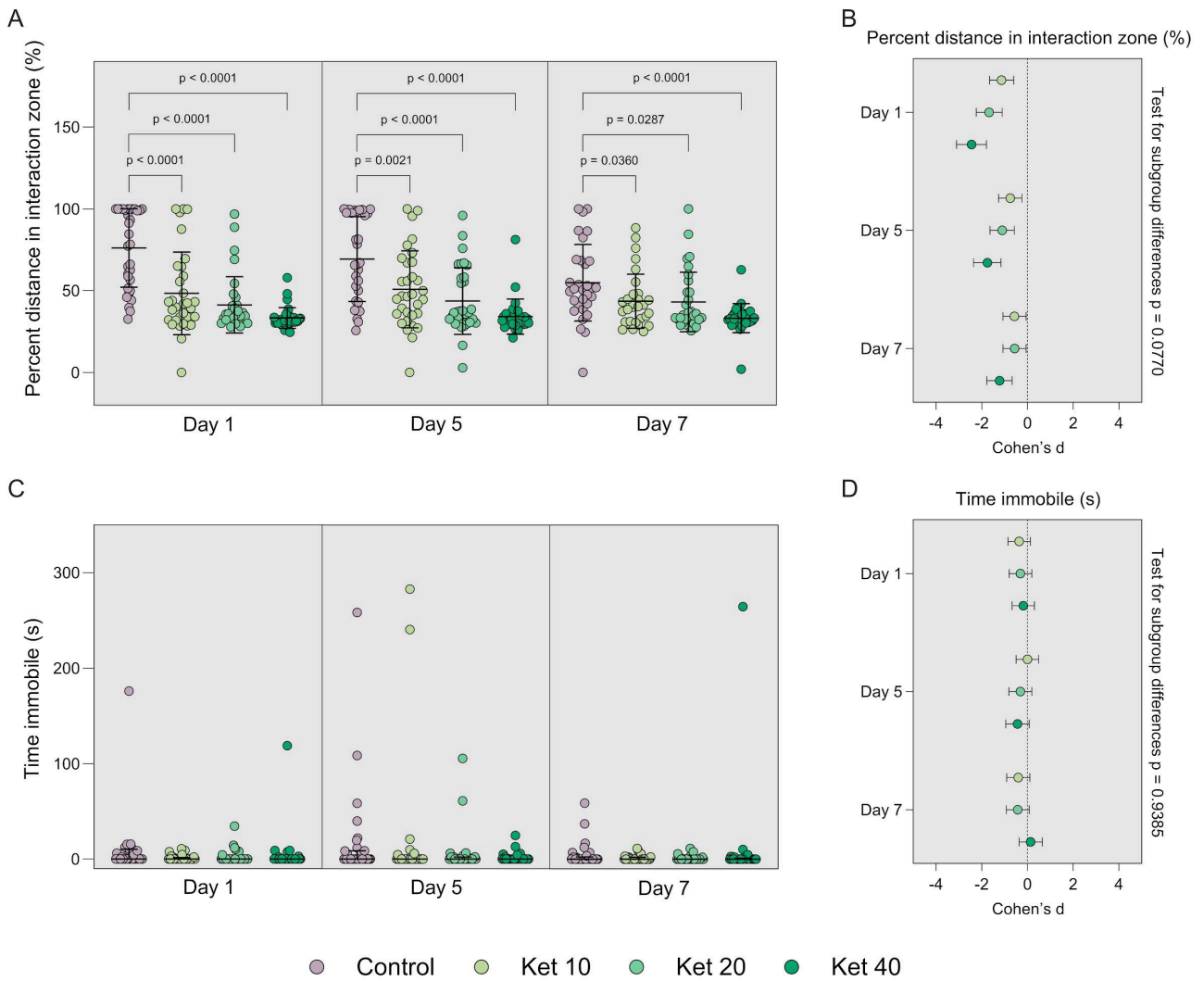
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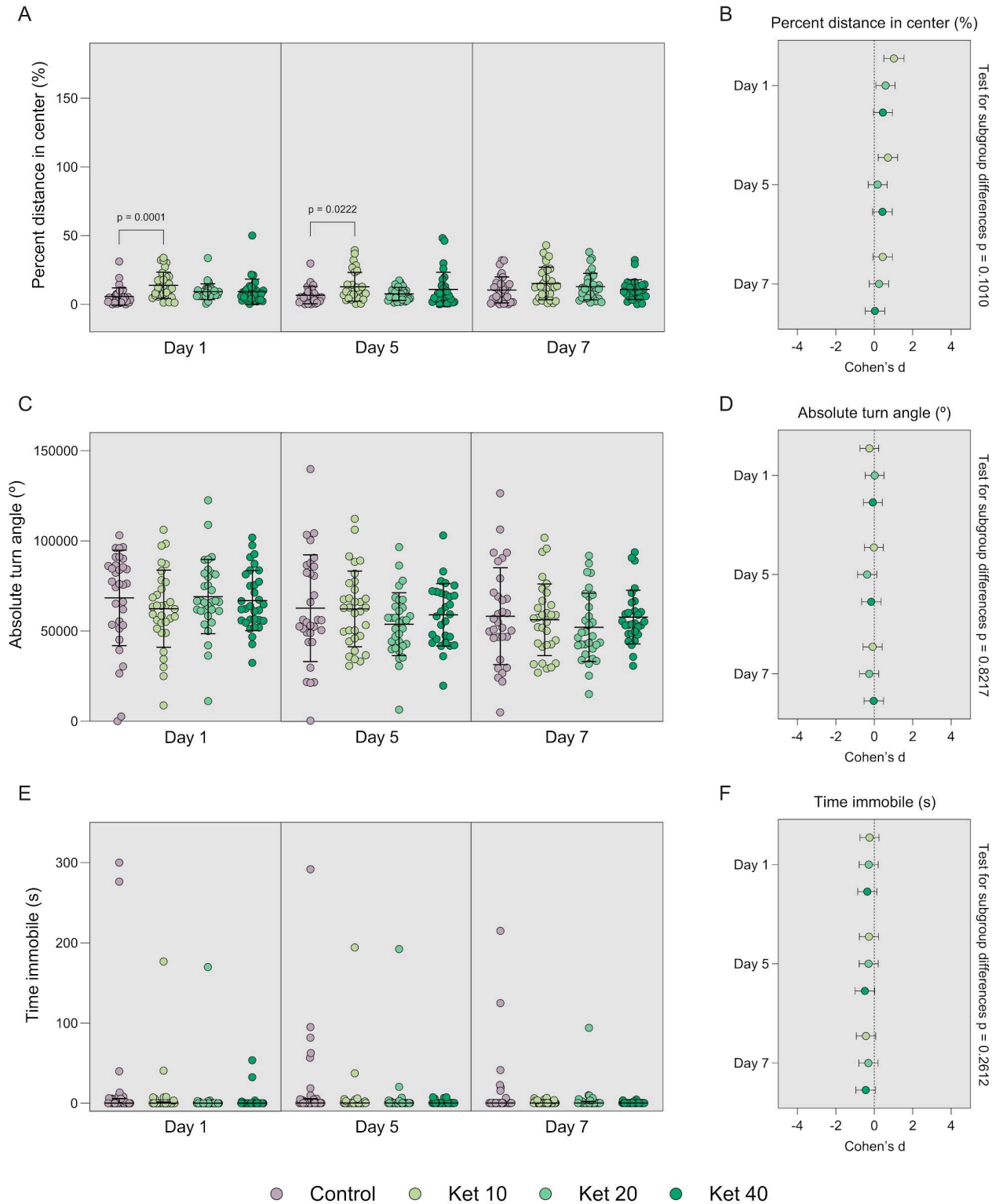
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Supplementary Figure 1



Supplementary Figure 1. Additional behavioral outcomes from the social preference test in adult zebrafish exposed to different concentrations of ketamine. (A) Percent of distance traveled in the interaction zone and (C) time spent immobile, measured on days 1, 5, and 7. (B, D) Corresponding effect sizes (Cohen's d) for each outcome, with test for subgroup differences across assessment days. One-way ANOVA followed by Bonferroni post hoc test was used for A; Kruskal–Wallis test followed by Dunn's post hoc was used for C. Data are presented as mean $\pm$ SD for A, and as median with interquartile range for C. Effect sizes are expressed as Cohen's d with 95% confidence intervals. Subgroup differences tested using a random-effects model. Sample sizes: day 1 – Control = 32, Ket 10 = 32, Ket 20 = 32, Ket 40 = 32; day 5 – Control = 31, Ket 10 = 32, Ket 20 = 32, Ket 40 = 29; day 7 – Control = 31, Ket 10 = 31, Ket 20 = 31, Ket 40 = 29. Ket 10 = ketamine 10 mg/L; Ket 20 = ketamine 20 mg/L; Ket 40 = ketamine 40 mg/L.

Supplementary Figure 2



Supplementary Figure 2. Additional behavioral outcomes from the open tank test in adult zebrafish exposed to different concentrations of ketamine. (A) Percent of distance traveled in the center zone, (C) absolute turn angle, and (E) time spent immobile, measured on days 1, 5, and 7. (B, D, F) Corresponding effect sizes (Cohen's d) for each outcome, with test for subgroup differences across assessment days. One-way ANOVA followed by Bonferroni post hoc test was used for A and C; Kruskal–Wallis test followed by Dunn's post hoc was used for E. Data are presented as mean $\pm$ SD for A and C, and as median with interquartile range for E. Effect sizes are expressed as Cohen's d with 95% confidence intervals. Subgroup differences tested using a random-effects model. Sample sizes: day 1 – Control = 32, Ket 10 = 32, Ket 20 = 32, Ket 40 = 32; day 5 – Control = 31, Ket 10 = 32, Ket 20 = 32, Ket 40 = 29; day 7 – Control = 31, Ket 10 = 31, Ket 20 = 31, Ket 40 = 29. Ket 10 = ketamine 10 mg/L; Ket 20 = ketamine 20 mg/L; Ket 40 = ketamine 40 mg/L.

4 DISCUSSÃO

A presente tese teve como objetivo principal avaliar os efeitos de antagonistas do receptor NMDA sobre o comportamento social em animais de laboratório e discutir sua relevância para modelar fenótipos relacionados aos sintomas negativos da esquizofrenia. Os dois estudos desenvolvidos indicam que esses fármacos induzem alterações sociais detectáveis em organismos modelo. A revisão sistemática e meta-análise mostrou que o antagonismo NMDA reduz medidas de interação social e de preferência social em diferentes espécies. O estudo experimental, por sua vez, mostrou que a cetamina reduz medidas sociais em peixe-zebra adulto. No entanto, esse efeito ocorreu junto a alterações locomotoras, não se intensificou após exposições repetidas, indicando ausência de sensibilização comportamental, e não persistiu após um período sem exposição. Em conjunto, esses achados sustentam a relevância dos antagonistas NMDA para investigar alterações sociais em modelos animais, mas também mostram que esses desfechos precisam ser interpretados considerando o teste utilizado e comportamentos com potencial confundidor da medida social.

Esse ponto é particularmente relevante porque os sintomas negativos da esquizofrenia permanecem entre os mais difíceis de tratar e de modelar experimentalmente. Embora antipsicóticos sejam eficazes para reduzir sintomas positivos, seus efeitos sobre sintomas negativos primários, funcionamento social e cognição são limitados (Correll; Schooler, 2020; Fusar-Poli et al., 2015; Huhn et al., 2019; McCutcheon; Keefe; McGuire, 2023). Assim, modelos animais voltados a esses domínios devem ser interpretados com cautela. Medidas de comportamento social em animais são úteis porque permitem quantificar respostas observáveis, como aproximação, permanência próxima a coespecíficos, interação direta ou coesão entre indivíduos (Toth; Neumann, 2013; Wilson; Koenig, 2014). No entanto, essas medidas não correspondem diretamente à associalidade, avolia ou anedonia observadas em pacientes. Elas funcionam como aproximações experimentais de componentes do comportamento social, e não como equivalentes completos dos sintomas negativos.

No primeiro estudo, a síntese da literatura mostrou que MK-801, cetamina e PCP estão associados à redução do comportamento social em animais de laboratório. Esse achado sustenta a ideia de que a hipofunção NMDA interfere em

processos comportamentais relevantes para o domínio social. Ao mesmo tempo, a magnitude dos efeitos variou significativamente entre estudos, paradigmas e espécies, indicando que este fenômeno não é uniforme. Essa interpretação é reforçada pelo fato de que a literatura também inclui estudos sem alteração detectável ou até com aumento de medidas de sociabilidade após a exposição a antagonistas NMDA. Parte dessa variação parece estar relacionada à intensidade da exposição farmacológica, já que a meta-regressão mostrou associação entre maior dose cumulativa de antagonistas NMDA e maior redução da interação social em roedores. Essa relação, no entanto, não foi observada de forma clara para preferência social. Essa heterogeneidade era esperada, considerando que os modelos baseados em antagonistas NMDA diferem quanto ao fármaco utilizado, dose, via de administração, duração da exposição, intervalo entre tratamento e teste, idade dos animais e tipo de desfecho social avaliado (Benvenutti et al., 2022; Lee; Zhou, 2019; Neill et al., 2010). Além dessas diferenças experimentais, a revisão também evidenciou limitações no relato metodológico de parte da literatura pré-clínica. Informações sobre randomização, cegamento, cálculo amostral, definição da unidade experimental e manejo de dados nem sempre são descritas de forma suficiente nos estudos com animais. Também foram observados indícios de efeitos de estudos pequenos, compatíveis com viés de publicação ou superestimação de efeitos em estudos com menor precisão, sobretudo nas análises de preferência social. Esse ponto é importante porque dificulta avaliar até que ponto a variação entre os estudos reflete diferenças biológicas reais, diferenças de protocolo ou limitações de desenho e relato experimental.

A análise dos desfechos locomotores reforça essa interpretação. Em muitos testes sociais, o animal precisa explorar o aparato, deslocar-se até o estímulo social e distribuir seu tempo entre zonas com ou sem coespecíficos. Por isso, alterações motoras podem modificar medidas sociais. Essa limitação já havia sido apontada em discussões sobre testes de sociabilidade em roedores, nos quais mudanças na exploração, ansiedade, resposta à novidade ou atividade locomotora podem interferir no tempo de interação ou de permanência em uma zona social (Toth; Neumann, 2013; Wilson; Koenig, 2014; Yang; Silverman; Crawley, 2011). Esse problema também foi observado nos resultados da revisão sistemática e meta-análise. Os efeitos sociais dos antagonistas NMDA frequentemente apareceram junto a alterações locomotoras.

Esse aspecto também é importante para interpretar os efeitos de antipsicóticos nos modelos animais. Em estudos pré-clínicos, a prevenção ou reversão de alterações sociais por antipsicóticos costuma ser usada como evidência de validade preditiva. No entanto, essa interpretação deve ser limitada pelo que se observa na clínica. Se os antipsicóticos disponíveis têm efeito pequeno ou inconsistente sobre sintomas negativos primários, a reversão de um déficit social em animais não deve ser automaticamente entendida como evidência de restauração de uma função social equivalente à humana (Correll; Schooler, 2020; Fusar-Poli et al., 2015; Krause et al., 2018). No primeiro estudo, a melhora de medidas sociais em animais tratados com antipsicóticos apareceu acompanhada por mudanças locomotoras. Isso sugere que parte do efeito pode refletir alteração da atividade geral, sedação ou redistribuição do tempo no aparato e não necessariamente uma melhora seletiva da sociabilidade.

O segundo estudo caracterizou os efeitos da cetamina em peixes-zebra adultos e ajudou a avaliar essa questão em uma espécie complementar aos roedores. A cetamina reduziu medidas sociais, aumentou a atividade locomotora no teste de preferência social e alterou o padrão de nado no teste de tanque aberto. Esses resultados indicam que o peixe-zebra adulto é sensível aos efeitos comportamentais agudos da hipofunção de receptores NMDA. Esse achado está de acordo com a literatura que descreve o peixe-zebra como um modelo vertebrado com sistemas neuroquímicos conservados e repertórios comportamentais quantificáveis de interesse para a esquizofrenia e outros transtornos psiquiátricos (Benvenuti et al., 2022; Kalueff; Stewart; Gerlai, 2014; Langova et al., 2020; Stewart et al., 2014). No entanto, como a redução das medidas sociais ocorreu junto ao aumento da locomoção, os dados não sustentam uma interpretação simples de déficit social seletivo. Esse achado corrobora aqueles encontrados na meta-análise.

A ausência de sensibilização comportamental após exposições repetidas à cetamina e a ausência de efeitos persistentes após o período sem exposição também são achados relevantes. Em roedores, protocolos repetidos com antagonistas NMDA são frequentemente utilizados para investigar alterações mais duradouras, incluindo mudanças comportamentais que permanecem após efeito agudo do fármaco (Frohlich; Van Horn, 2014; Lee; Zhou, 2019; Neill et al., 2010). Esse padrão, porém, não foi observado em peixe-zebra adulto nas condições testadas. Isso não invalida o modelo, mas delimita sua utilidade. Os resultados

sustentam o uso do peixe-zebra para investigar respostas agudas aos antagonistas NMDA. Por outro lado, os protocolos empregados não forneceram evidência de que a exposição repetida produza sensibilização ou alterações persistentes após 48 horas sem tratamento.

Essa diferença em relação a parte da literatura em roedores pode ter várias explicações. Diferenças entre espécies, vias de administração, absorção por imersão, metabolismo, concentração efetiva no sistema nervoso e características dos testes comportamentais podem influenciar os resultados. Outro aspecto a considerar é que o sistema nervoso central do peixe-zebra adulto apresenta capacidade neurogênica e regenerativa mais ampla do que a observada em mamíferos, com múltiplas zonas neurogênicas distribuídas pelo encéfalo e respostas de reparo mais robustas após lesão (Ghaddar et al., 2021; Kizil, 2018; Kizil et al., 2012). Embora essa característica não explique diretamente os resultados observados, pois não foi avaliada neste estudo, ela pode ser relevante para interpretar diferenças entre espécies na persistência e reversibilidade de alterações comportamentais após insultos ao sistema nervoso central. Além disso, o comportamento social do peixe-zebra é estruturado de forma diferente do comportamento social de roedores. A espécie apresenta uma tendência robusta a se aproximar de coespecíficos e formar cardumes, característica que sustenta o uso de testes de preferência social em peixe-zebra. Nesse tipo de paradigma, porém, a resposta social é inferida a partir da distribuição espacial do animal na arena e da permanência próxima ao estímulo social, que permanece fisicamente separado por barreiras transparentes (Barba-Escobedo; Gould, 2012; Miller; Gerlai, 2007). Por isso, alterações no padrão de nado, na exploração do ambiente ou na locomoção podem interferir na medida social obtida no teste. Estudos futuros poderiam combinar esse tipo de abordagem com medidas de coesão social. Essa combinação ajudaria a avaliar se os efeitos observados refletem uma alteração mais ampla do comportamento social ou se dependem do tipo de paradigma utilizado.

Considerados em conjunto, os dois estudos reforçam a utilidade dos modelos baseados em antagonismo NMDA para investigar alterações sociais associadas a fenótipos relacionados à esquizofrenia. Essa utilidade se apoia na relevância da hipofunção NMDA para a fisiopatologia proposta da esquizofrenia, uma vez que alterações nesse sistema podem afetar o equilíbrio entre neurotransmissão glutamatérgica, GABAérgica e dopaminérgica em circuitos envolvidos no transtorno

(Coyle, 2012; Gonzalez-Burgos; Lewis, 2012; Howes; Shatalina, 2022; Moghaddam; Javitt, 2012). Ao mesmo tempo, os achados reforçam que esses modelos não devem ser interpretados como reproduções completas dos sintomas negativos.

A coexistência de alterações sociais e locomotoras limita a leitura desses modelos como medidas seletivas de sintomas negativos e exige cautela ao usar a reversão por antipsicóticos como evidência de validade preditiva. Em peixes-zebra adultos, a cetamina produziu efeitos agudos, mas não persistentes, sobre medidas sociais e locomotoras, sugerindo maior utilidade do modelo para avaliar respostas agudas à hipofunção NMDA do que alterações sustentadas. Em conjunto, esses resultados reforçam que dados de interação ou preferência social devem ser analisados junto a possíveis confundidores comportamentais, especialmente locomoção. Também indicam que diferenças entre espécies, protocolos e formas de mensurar o comportamento social precisam ser consideradas antes de extrapolar os achados pré-clínicos para sintomas negativos ou para a eficácia de intervenções farmacológicas.

5 CONCLUSÕES

A presente tese mostrou que antagonistas do receptor NMDA reduzem medidas de comportamento social em animais de laboratório. A revisão sistemática e meta-análise indicou que esse efeito ocorre em paradigmas de interação social e de preferência social, embora com variação entre estudos, espécies, fármacos e protocolos experimentais. Os resultados também mostraram que alterações sociais induzidas por antagonistas NMDA devem ser interpretadas junto a outros parâmetros comportamentais. Em parte da literatura pré-clínica, os efeitos sobre interação ou preferência social ocorrem em paralelo a alterações locomotoras, o que limita a interpretação desses desfechos como medidas seletivas de sociabilidade ou como equivalentes diretos de sintomas negativos.

A análise dos efeitos de antipsicóticos reforçou essa necessidade de cautela. Embora esses fármacos possam atenuar alterações sociais induzidas por antagonistas NMDA em modelos animais, esses efeitos frequentemente aparecem acompanhados por mudanças locomotoras. Assim, a reversão de déficits sociais por antipsicóticos não deve ser interpretada, isoladamente, como evidência de validade preditiva para sintomas negativos.

No estudo experimental, a cetamina induziu alterações sociais e locomotoras agudas em peixe-zebra adulto. Esses efeitos incluíram redução de medidas sociais, aumento da atividade locomotora no teste de preferência social e alterações no padrão de nado no teste de tanque aberto. No entanto, não foram observadas evidências de sensibilização comportamental após exposições repetidas nem persistência dos efeitos após 48 horas sem exposição.

Em conjunto, os achados apoiam o uso do peixe-zebra como modelo complementar para investigar respostas comportamentais agudas à hipofunção NMDA. Ao mesmo tempo, indicam que sua utilidade depende do desfecho avaliado, do protocolo de exposição e da análise conjunta de possíveis confundidores comportamentais, especialmente locomoção.

Assim, esta tese contribui para o refinamento da interpretação de desfechos sociais em modelos animais de esquizofrenia. Os resultados sustentam que modelos baseados em antagonismo NMDA são úteis para investigar alterações sociais em animais, mas que sua relevância translacional depende da forma como esses efeitos são medidos, controlados e relacionados aos domínios clínicos da esquizofrenia.

REFERÊNCIAS

- AMERICAN PSYCHIATRIC ASSOCIATION. **Diagnostic and Statistical Manual of Mental Disorders: DSM-5-TR**. 5th ed., text rev. ed. Washington, DC: American Psychiatric Association Publishing, 2022.
- BARBA-ESCOBEDO, P. A.; GOULD, G. G. Visual social preferences of lone zebrafish in a novel environment: strain and anxiolytic effects. **Genes, Brain, and Behavior**, v. 11, n. 3, p. 366–373, abr. 2012.
- BEASLEY, C. L.; REYNOLDS, G. P. Parvalbumin-immunoreactive neurons are reduced in the prefrontal cortex of schizophrenics. **Schizophrenia Research**, v. 24, n. 3, p. 349–355, 11 abr. 1997.
- BENVENUTTI, Radharani *et al.* How do zebrafish (*Danio rerio*) respond to MK-801 and amphetamine? Relevance for assessing schizophrenia-related endophenotypes in alternative model organisms. **Journal of Neuroscience Research**, v. 99, n. 11, p. 2844–2859, 2021a.
- BENVENUTTI, Radharani *et al.* Swimming in the maze: An overview of maze apparatuses and protocols to assess zebrafish behavior. **Neuroscience and Biobehavioral Reviews**, v. 127, p. 761–778, ago. 2021b.
- BENVENUTTI, Radharani *et al.* Glutamate NMDA Receptor Antagonists with Relevance to Schizophrenia: A Review of Zebrafish Behavioral Studies. **Current Neuropharmacology**, v. 20, n. 3, p. 494–509, 4 mar. 2022.
- CACHAT, Jonathan *et al.* Measuring behavioral and endocrine responses to novelty stress in adult zebrafish. **Nature Protocols**, v. 5, n. 11, p. 1786–1799, nov. 2010.
- CARBON, Maren *et al.* Tardive dyskinesia risk with first- and second-generation antipsychotics in comparative randomized controlled trials: a meta-analysis. **World Psychiatry: Official Journal of the World Psychiatric Association**, v. 17, n. 3, p. 330–340, out. 2018.
- CARNEIRO, Clarissa F. D. *et al.* Mapping strategies towards improved external validity in preclinical translational research. **Expert Opinion on Drug Discovery**, v. 18, n. 11, p. 1273–1285, 2 nov. 2023.
- CHAN, Joe Kwun Nam *et al.* Life expectancy and years of potential life lost in people with mental disorders: a systematic review and meta-analysis. **EClinicalMedicine**, v. 65, p. 102294, nov. 2023.
- COGNATO, Giana de P. *et al.* Y-Maze memory task in zebrafish (*Danio rerio*): the role of glutamatergic and cholinergic systems on the acquisition and consolidation periods. **Neurobiology of Learning and Memory**, v. 98, n. 4, p. 321–328, nov. 2012.
- CORRELL, Christoph U. *et al.* Mortality in people with schizophrenia: a systematic review and meta-analysis of relative risk and aggravating or attenuating factors. **World Psychiatry: Official Journal of the World Psychiatric Association**, v. 21, n. 2, p. 248–271, jun. 2022.

CORRELL, Christoph U.; SCHOOLER, Nina R. Negative Symptoms in Schizophrenia: A Review and Clinical Guide for Recognition, Assessment, and Treatment. **Neuropsychiatric Disease and Treatment**, v. 16, p. 519–534, 21 fev. 2020.

COYLE, Joseph T. NMDA receptor and schizophrenia: a brief history. **Schizophrenia Bulletin**, v. 38, n. 5, p. 920–926, set. 2012.

COYLE, Joseph T.; PAUL, Steven M. Novel drug treatments for schizophrenia. **Nature Reviews. Drug Discovery**, v. 25, n. 4, p. 310–330, abr. 2026.

CREESE, I.; BURT, D. R.; SNYDER, S. H. Dopamine receptor binding predicts clinical and pharmacological potencies of antischizophrenic drugs. **Science**, v. 192, n. 4238, p. 481–483, 30 abr. 1976.

DAVIES, Cathy *et al.* Prenatal and perinatal risk and protective factors for psychosis: a systematic review and meta-analysis. **The Lancet Psychiatry**, v. 7, n. 5, p. 399–410, 1 maio 2020.

FARDE, L. *et al.* Positron emission tomographic analysis of central D1 and D2 dopamine receptor occupancy in patients treated with classical neuroleptics and clozapine. Relation to extrapyramidal side effects. **Archives of General Psychiatry**, v. 49, n. 7, p. 538–544, jul. 1992.

FEIFEL, David; SHILLING, Paul D. Promise and pitfalls of animal models of schizophrenia. **Current Psychiatry Reports**, v. 12, n. 4, p. 327–334, ago. 2010.

FETT, Anne-Kathrin J. *et al.* The relationship between neurocognition and social cognition with functional outcomes in schizophrenia: A meta-analysis. **Neuroscience & Biobehavioral Reviews**, v. 35, n. 3, p. 573–588, 1 jan. 2011.

FILE, Sandra E.; SETH, Pallab. A review of 25 years of the social interaction test. **European Journal of Pharmacology**, Animal Models of Anxiety Disorders. v. 463, n. 1, p. 35–53, 28 fev. 2003.

FROHLICH, Joel; VAN HORN, John D. Reviewing the ketamine model for schizophrenia. **Journal of Psychopharmacology**, v. 28, n. 4, p. 287–302, 1 abr. 2014.

FUSAR-POLI, Paolo *et al.* Treatments of Negative Symptoms in Schizophrenia: Meta-Analysis of 168 Randomized Placebo-Controlled Trials. **Schizophrenia Bulletin**, v. 41, n. 4, p. 892–899, jul. 2015.

GHADDAR, Batoul *et al.* Cellular Mechanisms Participating in Brain Repair of Adult Zebrafish and Mammals after Injury. **Cells**, v. 10, n. 2, p. 391, 14 fev. 2021.

GONZALEZ-BURGOS, Guillermo; LEWIS, David A. NMDA receptor hypofunction, parvalbumin-positive neurons, and cortical gamma oscillations in schizophrenia. **Schizophrenia Bulletin**, v. 38, n. 5, p. 950–957, set. 2012.

GRACE, Anthony A. Dopamine system dysregulation by the ventral subiculum as the common pathophysiological basis for schizophrenia psychosis, psychostimulant abuse, and stress. **Neurotoxicity Research**, v. 18, n. 3–4, p. 367–376, nov. 2010.

GRACE, Anthony A. Dysregulation of the dopamine system in the pathophysiology of schizophrenia and depression. **Nature Reviews. Neuroscience**, v. 17, n. 8, p. 524–532, ago. 2016.

GREEN, M. F. *et al.* Neurocognitive deficits and functional outcome in schizophrenia: are we measuring the “right stuff”? **Schizophrenia Bulletin**, v. 26, n. 1, p. 119–136, 2000.

GREEN, Michael F.; HORAN, William P.; LEE, Junghee. Social cognition in schizophrenia. **Nature Reviews Neuroscience**, v. 16, n. 10, p. 620–631, out. 2015.

HAGAN, Jim J.; JONES, Declan N. C. Predicting Drug Efficacy for Cognitive Deficits in Schizophrenia. **Schizophrenia Bulletin**, v. 31, n. 4, p. 830–853, 1 jan. 2005.

HANSEN, Helene Gjervig *et al.* Clinical Recovery Among Individuals With a First-Episode Schizophrenia an Updated Systematic Review and Meta-Analysis. **Schizophrenia Bulletin**, v. 49, n. 2, p. 297–308, 15 mar. 2023.

HASHIMOTO, Takanori *et al.* Gene expression deficits in a subclass of GABA neurons in the prefrontal cortex of subjects with schizophrenia. **The Journal of Neuroscience: The Official Journal of the Society for Neuroscience**, v. 23, n. 15, p. 6315–6326, 16 jul. 2003.

HENSSLER, Jonathan *et al.* Migration and schizophrenia: meta-analysis and explanatory framework. **European Archives of Psychiatry and Clinical Neuroscience**, v. 270, n. 3, p. 325–335, abr. 2020.

HILKER, Rikke *et al.* Heritability of Schizophrenia and Schizophrenia Spectrum Based on the Nationwide Danish Twin Register. **Biological Psychiatry**, v. 83, n. 6, p. 492–498, 15 mar. 2018.

HOLM, Minna *et al.* Employment among people with schizophrenia or bipolar disorder: A population-based study using nationwide registers. **Acta Psychiatrica Scandinavica**, v. 143, n. 1, p. 61–71, jan. 2021.

HOWE, Kerstin *et al.* The zebrafish reference genome sequence and its relationship to the human genome. **Nature**, v. 496, n. 7446, p. 498–503, 25 abr. 2013.

HOWES, Oliver D. *et al.* Treatment-Resistant Schizophrenia: Treatment Response and Resistance in Psychosis (TRRIP) Working Group Consensus Guidelines on Diagnosis and Terminology. **The American Journal of Psychiatry**, v. 174, n. 3, p. 216–229, 1 mar. 2017.

HOWES, Oliver D. *et al.* New Drug Treatments for Schizophrenia: A Review of Approaches to Target Circuit Dysfunction. **Biological Psychiatry**, v. 96, n. 8, p. 638–650, 15 out. 2024.

HOWES, Oliver D.; KAPUR, Shitij. The dopamine hypothesis of schizophrenia: version III--the final common pathway. **Schizophrenia Bulletin**, v. 35, n. 3, p. 549–562, maio 2009.

HOWES, Oliver D.; SHATALINA, Ekaterina. Integrating the Neurodevelopmental and Dopamine Hypotheses of Schizophrenia and the Role of Cortical Excitation-Inhibition Balance. **Biological Psychiatry**, v. 92, n. 6, p. 501–513, 15 set. 2022.

HUHN, Maximilian *et al.* Comparative efficacy and tolerability of 32 oral antipsychotics for the acute treatment of adults with multi-episode schizophrenia: a systematic review and network meta-analysis. **Lancet**, v. 394, n. 10202, p. 939–951, 14 set. 2019.

JAVITT, D. C.; ZUKIN, S. R. Recent advances in the phencyclidine model of schizophrenia. **The American Journal of Psychiatry**, v. 148, n. 10, p. 1301–1308, out. 1991.

JEEVAKUMAR, Vivek *et al.* Ketamine administration during the second postnatal week induces enduring schizophrenia-like behavioral symptoms and reduces parvalbumin expression in the medial prefrontal cortex of adult mice. **Behavioural Brain Research**, v. 282, p. 165–175, 1 abr. 2015.

JENTSCH, J. D.; ROTH, R. H. The neuropsychopharmacology of phencyclidine: from NMDA receptor hypofunction to the dopamine hypothesis of schizophrenia. **Neuropsychopharmacology: Official Publication of the American College of Neuropsychopharmacology**, v. 20, n. 3, p. 201–225, mar. 1999.

JONES, Ca; WATSON, Djg; FONE, Kcf. Animal models of schizophrenia. **British Journal of Pharmacology**, v. 164, n. 4, p. 1162–1194, 2011.

KAAR, Stephen J. *et al.* Pre-frontal parvalbumin interneurons in schizophrenia: a meta-analysis of post-mortem studies. **Journal of Neural Transmission**, v. 126, n. 12, p. 1637–1651, dez. 2019.

KAHN, René S. *et al.* Effectiveness of antipsychotic drugs in first-episode schizophrenia and schizophreniform disorder: an open randomised clinical trial. **Lancet**, v. 371, n. 9618, p. 1085–1097, 29 mar. 2008.

KALUEFF, A. V.; STEWART, A. M.; GERLAI, R. Zebrafish as an emerging model for studying complex brain disorders. **Trends Pharmacol. Sci.**, v. 35, n. 2, p. 63–75, 2014.

KALUEFF, Allan V. *et al.* Towards a comprehensive catalog of zebrafish behavior 1.0 and beyond. **Zebrafish**, v. 10, n. 1, p. 70–86, mar. 2013.

KALUEFF, Allan V. *et al.* The Zebrafish Neurobehavioral Catalog-Volume 2 (New Addition). **Zebrafish**, v. 22, n. 4, p. 117–135, ago. 2025.

KALUEFF, Allan V.; ECHEVARRIA, David J.; STEWART, Adam Michael. Gaining translational momentum: more zebrafish models for neuroscience research. **Progress in Neuro-Psychopharmacology & Biological Psychiatry**, v. 55, p. 1–6, 3 dez. 2014.

KAPUR, S. *et al.* Relationship between dopamine D(2) occupancy, clinical response, and side effects: a double-blind PET study of first-episode schizophrenia. **The American Journal of Psychiatry**, v. 157, n. 4, p. 514–520, abr. 2000.

KARI, G.; RODECK, U.; DICKER, A. P. Zebrafish: an emerging model system for human disease and drug discovery. **Clinical Pharmacology and Therapeutics**, v. 82, n. 1, p. 70–80, jul. 2007.

KAUL, Inder *et al.* Efficacy and Safety of Xanomeline-Trospium Chloride in Schizophrenia: A Randomized Clinical Trial. **JAMA psychiatry**, v. 81, n. 8, p. 749–756, 1 ago. 2024.

KEGELES, Lawrence S. *et al.* Increased synaptic dopamine function in associative regions of the striatum in schizophrenia. **Archives of General Psychiatry**, v. 67, n. 3, p. 231–239, mar. 2010.

KIZIL, Caghan *et al.* Adult neurogenesis and brain regeneration in zebrafish. **Developmental Neurobiology**, v. 72, n. 3, p. 429–461, 2012.

KIZIL, Caghan. Mechanisms of Pathology-Induced Neural Stem Cell Plasticity and Neural Regeneration in Adult Zebrafish Brain. **Current Pathobiology Reports**, v. 6, n. 1, p. 71–77, 2018.

KRAUSE, Marc *et al.* Antipsychotic drugs for patients with schizophrenia and predominant or prominent negative symptoms: a systematic review and meta-analysis. **European Archives of Psychiatry and Clinical Neuroscience**, v. 268, n. 7, p. 625–639, 1 out. 2018.

KRYSTAL, J. H. *et al.* Subanesthetic effects of the noncompetitive NMDA antagonist, ketamine, in humans. Psychotomimetic, perceptual, cognitive, and neuroendocrine responses. **Archives of General Psychiatry**, v. 51, n. 3, p. 199–214, mar. 1994.

LAHTI, A. C. *et al.* Subanesthetic doses of ketamine stimulate psychosis in schizophrenia. **Neuropsychopharmacology: Official Publication of the American College of Neuropsychopharmacology**, v. 13, n. 1, p. 9–19, ago. 1995.

LANGOVA, Veronika *et al.* The Role of Zebrafish and Laboratory Rodents in Schizophrenia Research. **Frontiers in Psychiatry**, v. 11, p. 703, 27 ago. 2020.

LARUELLE, M. *et al.* Single photon emission computerized tomography imaging of amphetamine-induced dopamine release in drug-free schizophrenic subjects. **Proceedings of the National Academy of Sciences of the United States of America**, v. 93, n. 17, p. 9235–9240, 20 ago. 1996.

LEE, Gloria; ZHOU, Yi. NMDAR Hypofunction Animal Models of Schizophrenia. **Frontiers in Molecular Neuroscience**, v. 12, 31 jul. 2019.

LEENAARS, Cathalijn H. C. *et al.* Animal to human translation: a systematic scoping review of reported concordance rates. **Journal of Translational Medicine**, v. 17, n. 1, p. 223, 15 jul. 2019.

LEUCHT, Stefan *et al.* Antipsychotic drugs versus placebo for relapse prevention in schizophrenia: a systematic review and meta-analysis. **Lancet**, v. 379, n. 9831, p. 2063–2071, 2 jun. 2012.

LEUCHT, Stefan *et al.* Schizophrenia. **Nature Reviews Disease Primers**, v. 11, n. 1, p. 83, 27 nov. 2025.

LEWIS, David A.; HASHIMOTO, Takanori; VOLK, David W. Cortical inhibitory neurons and schizophrenia. **Nature Reviews. Neuroscience**, v. 6, n. 4, p. 312–324, abr. 2005.

LIEBERMAN, Jeffrey A. *et al.* Effectiveness of antipsychotic drugs in patients with chronic schizophrenia. **The New England Journal of Medicine**, v. 353, n. 12, p. 1209–1223, 22 set. 2005.

LIPSKA, B. K.; WEINBERGER, D. R. To model a psychiatric disorder in animals: schizophrenia as a reality test. **Neuropsychopharmacology: Official Publication of the American College of Neuropsychopharmacology**, v. 23, n. 3, p. 223–239, set. 2000.

MAASWINKEL, Hans; ZHU, Liquan; WENG, Wei. Assessing social engagement in heterogeneous groups of zebrafish: a new paradigm for autism-like behavioral responses. **PloS One**, v. 8, n. 10, p. e75955, 2013.

MARCONI, Arianna *et al.* Meta-analysis of the Association Between the Level of Cannabis Use and Risk of Psychosis. **Schizophrenia Bulletin**, v. 42, n. 5, p. 1262–1269, set. 2016.

MARDER, Stephen R.; GALDERISI, Silvana. The current conceptualization of negative symptoms in schizophrenia. **World Psychiatry: Official Journal of the World Psychiatric Association**, v. 16, n. 1, p. 14–24, fev. 2017.

MARWAHA, Steven; JOHNSON, Sonia. Schizophrenia and employment - a review. **Social Psychiatry and Psychiatric Epidemiology**, v. 39, n. 5, p. 337–349, maio 2004.

MAXIMINO, Caio; HERCULANO, Anderson Manoel. A review of monoaminergic neuropsychopharmacology in zebrafish. **Zebrafish**, v. 7, n. 4, p. 359–378, dez. 2010.

MCCUTCHEON, Robert A.; ABI-DARGHAM, Anissa; HOWES, Oliver D. Schizophrenia, Dopamine and the Striatum: From Biology to Symptoms. **Trends in Neurosciences**, v. 42, n. 3, p. 205–220, mar. 2019.

MCCUTCHEON, Robert A.; KEEFE, Richard S. E.; MCGUIRE, Philip K. Cognitive impairment in schizophrenia: aetiology, pathophysiology, and treatment. **Molecular Psychiatry**, v. 28, n. 5, p. 1902–1918, maio 2023.

MELTZER, H. Y.; MATSUBARA, S.; LEE, J. C. The ratios of serotonin₂ and dopamine₂ affinities differentiate atypical and typical antipsychotic drugs. **Psychopharmacology Bulletin**, v. 25, n. 3, p. 390–392, 1989.

MEYER, Urs; FELDON, Joram. Epidemiology-driven neurodevelopmental animal models of schizophrenia. **Progress in Neurobiology**, v. 90, n. 3, p. 285–326, mar. 2010.

MICHELOTTI, Paula *et al.* Ketamine modulates aggressive behavior in adult zebrafish. **Neuroscience Letters**, v. 684, p. 164–168, 25 set. 2018.

MICHELOTTI, Paula *et al.* Ketamine acutely impairs memory consolidation and repeated exposure promotes stereotyped behavior without changing anxiety- and aggression-like parameters in adult zebrafish. **Physiology & Behavior**, v. 247, p. 113708, 1 abr. 2022.

MILLER, Noam; GERLAI, Robert. Quantification of shoaling behaviour in zebrafish (*Danio rerio*). **Behavioural Brain Research**, v. 184, n. 2, p. 157–166, 3 dez. 2007.

MILLER, Noam; GERLAI, Robert. From schooling to shoaling: patterns of collective motion in zebrafish (*Danio rerio*). **PloS One**, v. 7, n. 11, p. e48865, 2012.

MIYAMOTO, S. *et al.* Treatments for schizophrenia: a critical review of pharmacology and mechanisms of action of antipsychotic drugs. **Molecular Psychiatry**, v. 10, n. 1, p. 79–104, jan. 2005.

MOGHADDAM, Bitá; JAVITT, Daniel. From revolution to evolution: the glutamate hypothesis of schizophrenia and its implication for treatment. **Neuropsychopharmacology: Official Publication of the American College of Neuropsychopharmacology**, v. 37, n. 1, p. 4–15, jan. 2012.

MOURI, Akihiro *et al.* Phencyclidine animal models of schizophrenia: approaches from abnormality of glutamatergic neurotransmission and neurodevelopment. **Neurochemistry International**, v. 51, n. 2–4, p. 173–184, 2007.

NAKAZAWA, Kazu; JEEVAKUMAR, Vivek; NAKAO, Kazuhito. Spatial and temporal boundaries of NMDA receptor hypofunction leading to schizophrenia. **npj Schizophrenia**, v. 3, n. 1, p. 7, 3 fev. 2017.

NASIADKA, Andrzej; CLARK, Matthew D. Zebrafish breeding in the laboratory environment. **ILAR journal**, v. 53, n. 2, p. 161–168, 2012.

NEILL, Joanna C. *et al.* Animal models of cognitive dysfunction and negative symptoms of schizophrenia: Focus on NMDA receptor antagonism. **Pharmacology & Therapeutics**, v. 128, n. 3, p. 419–432, 1 dez. 2010.

NEILL, Joanna C. *et al.* Acute and chronic effects of NMDA receptor antagonists in rodents, relevance to negative symptoms of schizophrenia: A translational link to humans. **European Neuropsychopharmacology**, Negative symptoms of schizophrenia: clinical characteristics and their measurement, experimental modelling, and opportunities for improved treatment. v. 24, n. 5, p. 822–835, 1 maio 2014.

NESTLER, Eric J.; HYMAN, Steven E. Animal models of neuropsychiatric disorders. **Nature Neuroscience**, v. 13, n. 10, p. 1161–1169, out. 2010.

NEWCOMER, John W. *et al.* Ketamine-Induced NMDA Receptor Hypofunction as a Model of Memory Impairment and Psychosis. **Neuropsychopharmacology**, v. 20, n. 2, p. 106–118, fev. 1999.

NORTON, W.; BALLY-CUIF, L. Adult zebrafish as a model organism for behavioural genetics. **BMC Neurosci.**, v. 11, 2010.

NUECHTERLEIN, Keith H. *et al.* The MATRICS Consensus Cognitive Battery, part 1: test selection, reliability, and validity. **The American Journal of Psychiatry**, v. 165, n. 2, p. 203–213, fev. 2008.

OGI, Asahi *et al.* Social Preference Tests in Zebrafish: A Systematic Review. **Frontiers in Veterinary Science**, v. 7, p. 590057, 22 jan. 2021.

OLTRABELLA, Francesca *et al.* Role of the endocannabinoid system in vertebrates: Emphasis on the zebrafish model. **Development, Growth & Differentiation**, v. 59, n. 4, p. 194–210, maio 2017.

PANULA, P. *et al.* The comparative neuroanatomy and neurochemistry of zebrafish CNS systems of relevance to human neuropsychiatric diseases. **Neurobiology of Disease**, v. 40, n. 1, p. 46–57, out. 2010.

PANULA, Pertti *et al.* Modulatory neurotransmitter systems and behavior: towards zebrafish models of neurodegenerative diseases. **Zebrafish**, v. 3, n. 2, p. 235–247, 2006.

PAOLETTI, Pierre; BELLONE, Camilla; ZHOU, Qiang. NMDA receptor subunit diversity: impact on receptor properties, synaptic plasticity and disease. **Nature Reviews. Neuroscience**, v. 14, n. 6, p. 383–400, jun. 2013.

PERCELAY, Solenn *et al.* The 3-hit animal models of schizophrenia: Improving strategy to decipher and treat the disease? **Neuroscience and Biobehavioral Reviews**, v. 157, p. 105526, fev. 2024.

PILLINGER, Toby *et al.* Comparative effects of 18 antipsychotics on metabolic function in patients with schizophrenia, predictors of metabolic dysregulation, and association with psychopathology: a systematic review and network meta-analysis. **The Lancet. Psychiatry**, v. 7, n. 1, p. 64–77, jan. 2020.

POUND, Pandora; RITSKES-HOITINGA, Merel. Is it possible to overcome issues of external validity in preclinical animal research? Why most animal models are bound to fail. **Journal of Translational Medicine**, v. 16, n. 1, p. 304, 7 nov. 2018.

POWELL, Craig M.; MIYAKAWA, Tsuyoshi. Schizophrenia-relevant behavioral testing in rodent models: a uniquely human disorder? **Biological Psychiatry**, v. 59, n. 12, p. 1198–1207, 15 jun. 2006.

QIN, Meiying *et al.* Induction of Social Behavior in Zebrafish: Live Versus Computer Animated Fish as Stimuli. **Zebrafish**, v. 11, n. 3, p. 185–197, 1 jun. 2014.

RICO, E. P. *et al.* Zebrafish neurotransmitter systems as potential pharmacological and toxicological targets. **Neurotoxicology and Teratology**, v. 33, n. 6, p. 608–617, 2011.

RIEHL, Russell *et al.* Behavioral and physiological effects of acute ketamine exposure in adult zebrafish. **Neurotoxicology and Teratology**, Special issue: Zebrafish: Current discoveries and emerging technologies for neurobehavioral toxicology and teratology. v. 33, n. 6, p. 658–667, 1 nov. 2011.

RINK, Elke; WULLIMANN, Mario F. Development of the catecholaminergic system in the early zebrafish brain: an immunohistochemical study. **Brain Research. Developmental Brain Research**, v. 137, n. 1, p. 89–100, 30 jul. 2002.

SAVERINO, Cristina; GERLAI, Robert. The social zebrafish: Behavioral responses to conspecific, heterospecific, and computer animated fish. **Behavioural brain research**, v. 191, n. 1, p. 77–87, 5 ago. 2008.

SEEMAN, P. *et al.* Antipsychotic drug doses and neuroleptic/dopamine receptors. **Nature**, v. 261, n. 5562, p. 717–719, 24 jun. 1976.

SEIBT, Kelly Juliana *et al.* Antipsychotic drugs prevent the motor hyperactivity induced by psychotomimetic MK-801 in zebrafish (*Danio rerio*). **Behavioural Brain Research**, v. 214, n. 2, p. 417–422, 25 dez. 2010.

SEIBT, Kelly Juliana *et al.* Antipsychotic drugs reverse MK-801-induced cognitive and social interaction deficits in zebrafish (*Danio rerio*). **Behavioural Brain Research**, v. 224, n. 1, p. 135–139, 10 out. 2011.

SILVA, Patricia I. M. *et al.* Stress and fear responses in the teleost pallium. **Physiology & Behavior**, v. 141, p. 17–22, 15 mar. 2015.

SISKIND, Dan; SISKIND, Victor; KISELY, Steve. Clozapine Response Rates among People with Treatment-Resistant Schizophrenia: Data from a Systematic Review and Meta-Analysis. **Canadian Journal of Psychiatry. Revue Canadienne De Psychiatrie**, v. 62, n. 11, p. 772–777, nov. 2017.

SISON, Margarette; GERLAI, Robert. Behavioral performance altering effects of MK-801 in zebrafish (*Danio rerio*). **Behavioural Brain Research**, v. 220, n. 2, p. 331–337, 7 jul. 2011.

SOLMI, Marco *et al.* Incidence, prevalence, and global burden of schizophrenia - data, with critical appraisal, from the Global Burden of Disease (GBD) 2019. **Molecular Psychiatry**, v. 28, n. 12, p. 5319–5327, dez. 2023.

SPARK, Daisy L. *et al.* Beyond antipsychotics: a twenty-first century update for preclinical development of schizophrenia therapeutics. **Translational Psychiatry**, v. 12, n. 1, p. 147, 7 abr. 2022.

SPENCE, Rowena *et al.* The behaviour and ecology of the zebrafish, *Danio rerio*. **Biological Reviews**, v. 83, n. 1, p. 13–34, 2008.

STEWART, Adam Michael *et al.* Zebrafish models for translational neuroscience research: from tank to bedside. **Trends in Neurosciences**, v. 37, n. 5, p. 264–278, 1 maio 2014.

SULLIVAN, Patrick F.; KENDLER, Kenneth S.; NEALE, Michael C. Schizophrenia as a complex trait: evidence from a meta-analysis of twin studies. **Archives of General Psychiatry**, v. 60, n. 12, p. 1187–1192, dez. 2003.

TIIHONEN, Jari *et al.* Real-World Effectiveness of Antipsychotic Treatments in a Nationwide Cohort of 29 823 Patients With Schizophrenia. **JAMA psychiatry**, v. 74, n. 7, p. 686–693, 1 jul. 2017.

TOTH, Iulia; NEUMANN, Inga D. Animal models of social avoidance and social fear. **Cell and Tissue Research**, v. 354, n. 1, p. 107–118, 1 out. 2013.

TRAYNELIS, Stephen F. *et al.* Glutamate receptor ion channels: structure, regulation, and function. **Pharmacological Reviews**, v. 62, n. 3, p. 405–496, set. 2010.

U.S. FOOD AND DRUG ADMINISTRATION. **FDA Approves Drug with New Mechanism of Action for Treatment of Schizophrenia**. Disponível em: <<https://www.fda.gov/news-events/press-announcements/fda-approves-drug-new-mechanism-action-treatment-schizophrenia>>. Acesso em: 19 maio. 2026.

VARESE, Filippo *et al.* Childhood adversities increase the risk of psychosis: a meta-analysis of patient-control, prospective- and cross-sectional cohort studies. **Schizophrenia Bulletin**, v. 38, n. 4, p. 661–671, jun. 2012.

VASSOS, Evangelos *et al.* Meta-Analysis of the Association of Urbanicity With Schizophrenia. **Schizophrenia Bulletin**, v. 38, n. 6, p. 1118–1123, nov. 2012.

VELLIGAN, Dawn I. *et al.* Why do psychiatric patients stop antipsychotic medication? A systematic review of reasons for nonadherence to medication in patients with serious mental illness. **Patient Preference and Adherence**, v. 11, p. 449–468, 2017.

WANG, C. *et al.* Long-term behavioral and neurodegenerative effects of perinatal phencyclidine administration: implications for schizophrenia. **Neuroscience**, v. 107, n. 4, p. 535–550, 2001.

WILSON, Christina A.; KOENIG, James I. Social interaction and social withdrawal in rodents as readouts for investigating the negative symptoms of schizophrenia. **European neuropsychopharmacology: the journal of the European College of Neuropsychopharmacology**, v. 24, n. 5, p. 759–773, maio 2014.

WULLIMANN, Mario F.; RINK, Elke. The teleostean forebrain: a comparative and developmental view based on early proliferation, Pax6 activity and catecholaminergic organization. **Brain Research Bulletin**, v. 57, n. 3–4, p. 363–370, 1 mar. 2002.

YANG, Mu; SILVERMAN, Jill L.; CRAWLEY, Jacqueline N. Automated Three-Chambered Social Approach Task for Mice. **Current protocols in neuroscience / editorial board, Jacqueline N. Crawley ... [et al.]**, v. CHAPTER 8, p. Unit-8.26, jul. 2011.

ZAKHARY, Sherry M. *et al.* A behavioral and molecular analysis of ketamine in zebrafish. **Synapse (New York, N.Y.)**, v. 65, n. 2, p. 160–167, fev. 2011.

ZHAN, Zhonghua; WANG, Jiaming; SHEN, Tongping. Results of the Global Burden of Disease study for schizophrenia: trends from 1990 to 2021 and projections to 2050. **Frontiers in Psychiatry**, v. 16, p. 1629032, 2025.

ZHANG, Shuxin *et al.* Global lifetime prevalence of schizophrenia: A systematic review and meta-analysis. **Molecular Psychiatry**, 23 mar. 2026.

ZIMMERMANN, Fernanda Francine *et al.* Oxytocin reversed MK-801-induced social interaction and aggression deficits in zebrafish. **Behavioural Brain Research**, v. 311, p. 368–374, 15 set. 2016.

ANEXO A – CARTA DE PUBLICAÇÃO DO CAPÍTULO 1

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U F R G S

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Comissão De Ética No Uso De Animais



CARTA DE APROVAÇÃO

Comissão De Ética No Uso De Animais analisou o projeto:

Número: 35525

Título: Estabelecimento e validação farmacológica de modelos de esquizofrenia em peixes-zebra

Vigência: 19/06/2018 à 30/09/2027

Pesquisadores:

Equipe UFRGS:

ÂNGELO LUIS STAPASSOLI PIATO - coordenador desde 19/06/2018
Ana Paula Herrmann - coordenador desde 19/06/2018
DANIELA VARISCO MÜLLER - desde 14/08/2024
MATHEUS GALLAS LOPES - desde 14/08/2024
LEONARDO MARENSI BASTOS - desde 15/08/2024
RADHARANI BENVENUTTI - Aluno de Doutorado de 19/06/2018 até 14/08/2024
FRANCIELE KICH GIONGO - Aluno de Mestrado de 19/06/2018 até 14/08/2024
Matheus Felipe Marcon - Aluno de Doutorado de 19/06/2018 até 14/08/2024
Carlos Guilherme Rosa Reis - Aluno de Doutorado de 19/06/2018 até 14/08/2024

Comissão De Ética No Uso De Animais aprovou o mesmo, em reunião realizada pelo Mconf, em seus aspectos éticos e metodológicos, para a utilização de 192 peixes-zebra de 4 meses de idade, sendo 96 machos e 96 fêmeas, provenientes do fornecedor comercial Delphis (Porto Alegre), de acordo com os preceitos das Diretrizes e Normas Nacionais e Internacionais, especialmente a Lei 11.794 de 08 de novembro de 2008, o Decreto 6899 de 15 de julho de 2009, e as normas editadas pelo Conselho Nacional de Controle da Experimentação Animal (CONCEA), que disciplinam a produção, manutenção e/ou utilização de animais do filo Chordata, subfilo Vertebrata (exceto o homem) em atividade de ensino ou pesquisa. Toda a equipe possui capacitação validada pela CEUA para manipular os animais.

Porto Alegre, Terça-Feira, 19 de Novembro de 2024



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MAITE DE MORAES VIEIRA
Coordenador da comissão de ética

ANEXO D – APROVAÇÃO CEUA DOS EXPERIMENTOS II E III DO CAPÍTULO 2



U F R G S
UNIVERSIDADE FEDERAL
DO RIO GRANDE DO SUL

PRÓ-REITORIA DE PESQUISA

Comissão De Ética No Uso De Animais



CARTA DE APROVAÇÃO

Comissão De Ética No Uso De Animais analisou o projeto:

Número: 35525

Título: Estabelecimento e validação farmacológica de modelos de esquizofrenia em peixes-zebra

Vigência: 19/06/2018 à 30/09/2027

Pesquisadores:

Equipe UFRGS:

ÂNGELO LUIS STAPASSOLI PIATO - coordenador desde 19/06/2018
Ana Paula Herrmann - coordenador desde 19/06/2018
SAMARA MACHADO BRUCK - desde 14/08/2024
SOFIA ZANINI BECKER - desde 14/08/2024
DANIELA VARISCO MÜLLER - desde 14/08/2024
MATHEUS GALLAS LOPES - desde 14/08/2024
LEONARDO MARENSI BASTOS - desde 15/08/2024
RADHARANI BENVENUTTI - Aluno de Doutorado de 19/06/2018 até 14/08/2024
FRANCIELE KICH GIONGO - Aluno de Mestrado de 19/06/2018 até 14/08/2024
Matheus Felipe Marcon - Aluno de Doutorado de 19/06/2018 até 14/08/2024
Carlos Guilherme Rosa Reis - Aluno de Doutorado de 19/06/2018 até 14/08/2024

A Comissão De Ética No Uso De Animais aprovou o mesmo, em reunião realizada pelo Mconf, em seus aspectos éticos e metodológicos, para a utilização de 192 peixes-zebra de 4 meses de idade, sendo 96 machos e 96 fêmeas, provenientes do fornecedor comercial Delphis (CNPJ 00187830/0001-70), de acordo com os preceitos das Diretrizes e Normas Nacionais e Internacionais, especialmente a Lei 11.794 de 08 de novembro de 2008, o Decreto 6899 de 15 de julho de 2009, e as normas editadas pelo Conselho Nacional de Controle da Experimentação Animal (CONCEA), que disciplinam a produção, manutenção e/ou utilização de animais do filo Chordata, subfilo Vertebrata (exceto o homem) em atividade de ensino ou pesquisa. Toda a equipe possui capacitação validada pela CEUA para manipular os animais.

Porto Alegre, Terça-Feira, 19 de Maio de 2026

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Coordenador da comissão de ética